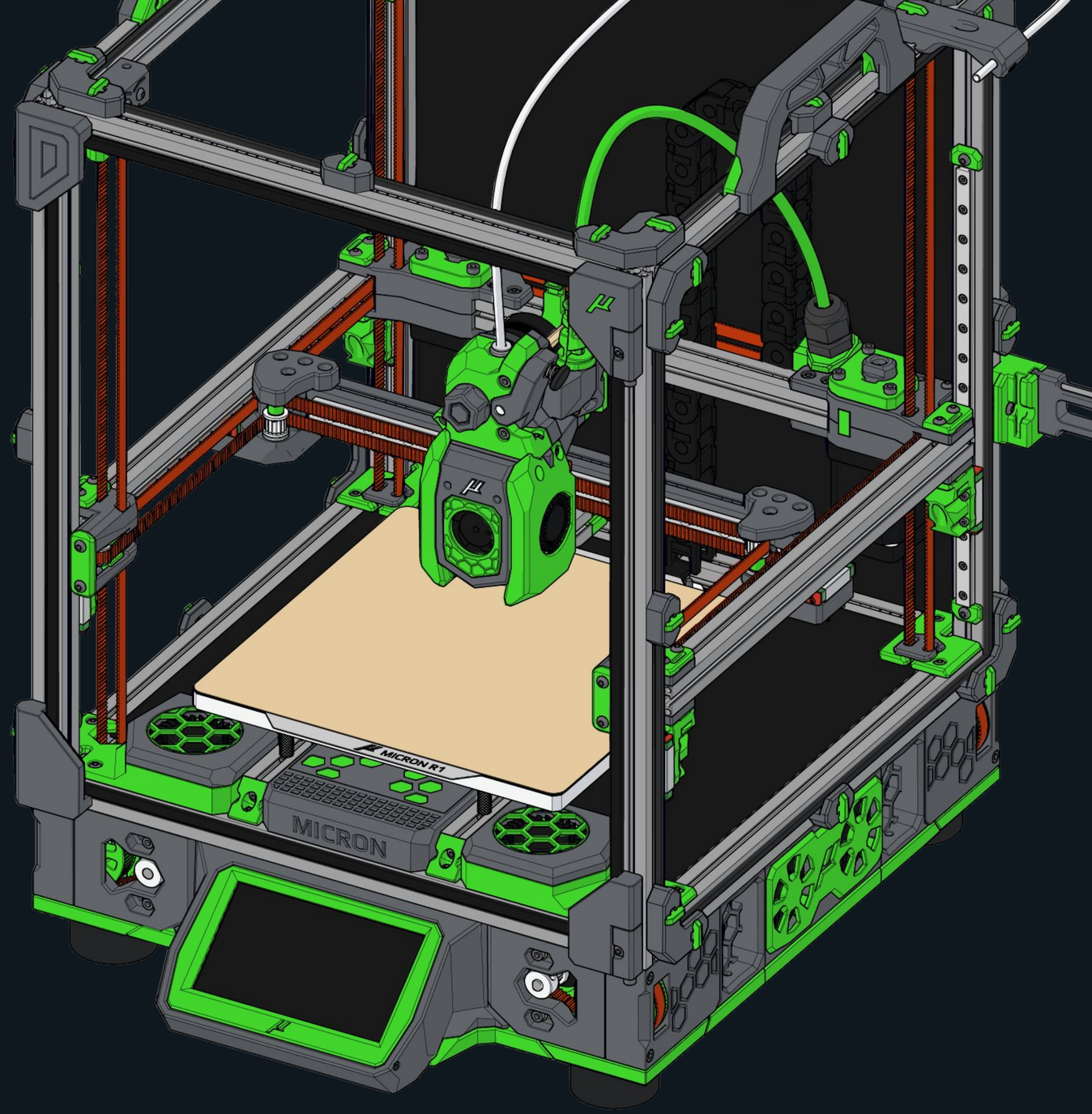
micron

R1

EVERYTHING
SMALLER
BUT THE
PRICE



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micron R1

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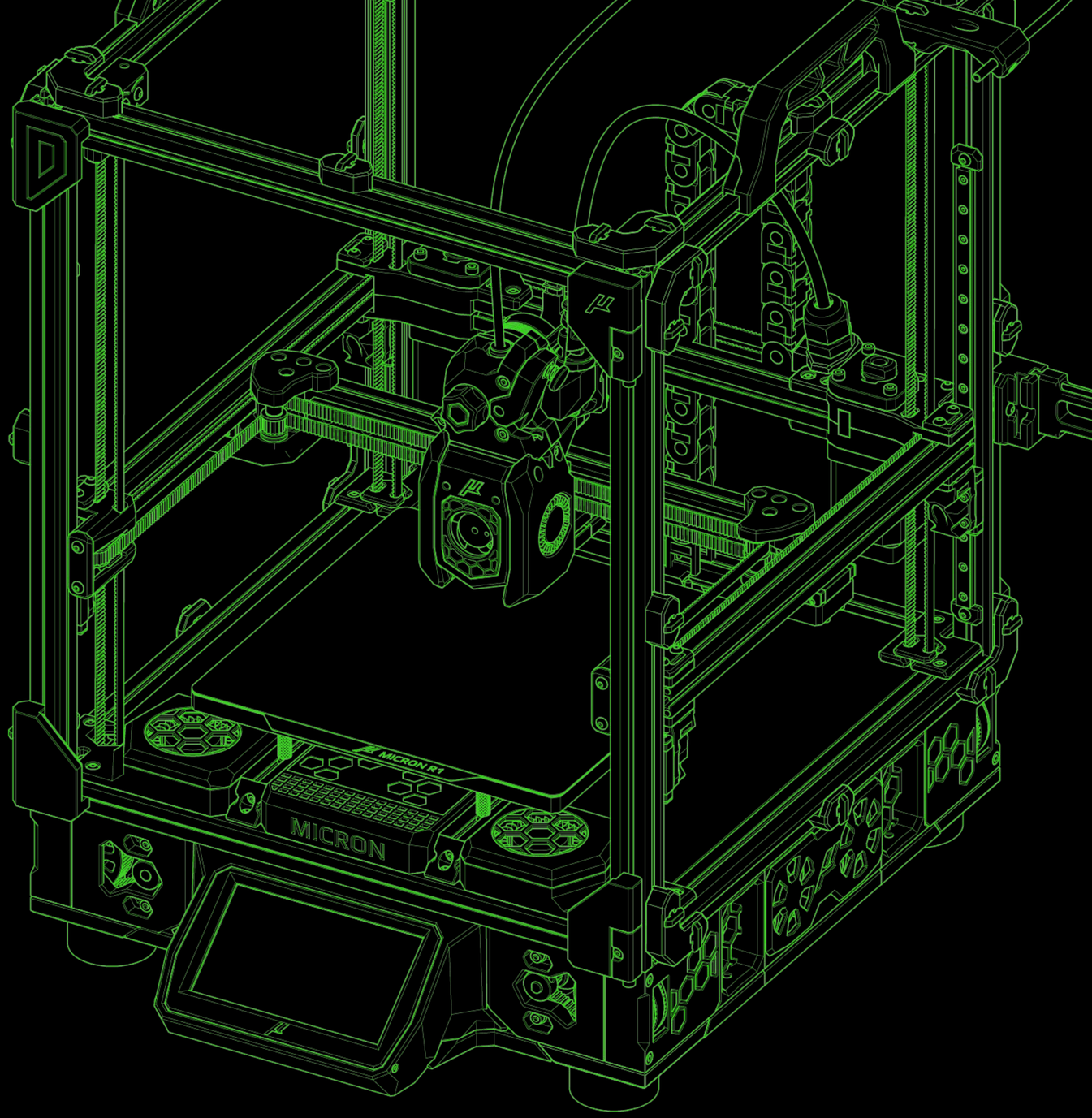


R1
micron

A/ INTRODUCTION

Originally designed by **HartK**, Micron+ is a compact, community-driven, flying gantry CoreXY printer inspired by the Voron 2.4 and supported by an active community of builders.

THE BUILD

Assembling Micron+ involves mechanical, electrical, and printed components. Familiarity with basic tools and 3D printer terminology is recommended. Refer to the BOM for the full hardware list, Github for up-to-date printed parts, and the Printed Parts Overview for recommended print settings to ensure good part strength.

FRAME

Uses 1515 aluminum extrusions to keep the machine small and stiff.

BUILD VOLUME

The current R1 variant has a build volume of 180 × 180 × 165 mm.

FLYING GANTRY

The entire XY gantry rides on belted Z rails while the bed remains fixed to the frame.

COREXY

Two motors drive crossed belts that control toolhead motion in the XY plane.

ENCLOSED

Designed to be fully enclosed and capable of printing engineering plastics like ABS and ASA.

TOOLHEAD

Micron+ is modular, allowing builders to choose from a wide range of compatible hotends, direct drive extruders, and bed probes.

B / TOOLS & PREPARATION

Before beginning assembly, review the tools and materials required for the build. Having the proper tools prepared and a clean workspace will help ensure a smooth assembly process and prevent damage to components.

REQUIRED TOOLS

- Ball-end hex drivers (2, 2.5, 3 mm)
- Wire cutter
- Wire stripper
- Crimping tool
- Measuring tape
- Threadlocker
- Grease (EP2)

RECOMMENDED TOOLS

- A flat build surface large enough for Micron+ to comfortably sit on
- A reference square
- Digital calipers

C/ PRINTED PARTS

These print settings follow vetted Voron guidelines and are recommended for reliable printed parts.

PRINTING METHOD
Fused Deposition Modeling (FDM)

MATERIAL
ABS or ASA

LAYER HEIGHT
0.2 mm

EXTRUSION WIDTH
0.4 mm (forced)

INFILL TYPE
Grid, Gyroid, Honeycomb, Triangle, or Cubic

INFILL PERCENTAGE
40%

WALL/PERIMETER COUNT
4

SOLID TOP/BOTTOM LAYERS
5

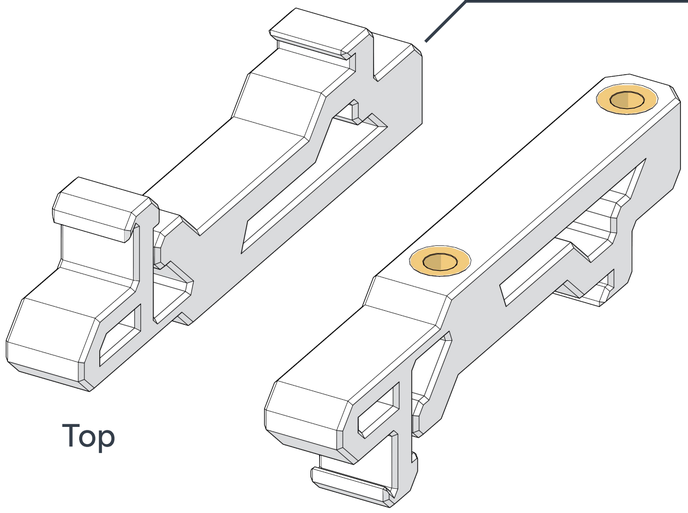
FILENAMES
Printed part filenames follow the Voron naming convention. The prefix ***[a]_*** indicates the part should be printed in the accent color. A suffix such as ***_x4*** indicates the number of copies required. For example, ***[a]_front_cover_x2.stl*** means print two copies in your accent color.

C/ PRINTED PARTS

HEATSETS - ELECTRONICS

Install the heatset inserts as shown.

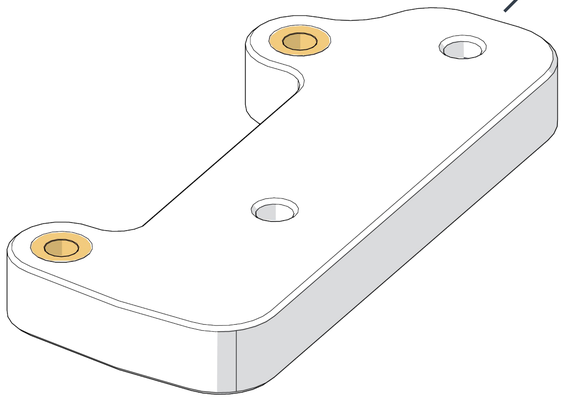
UHP_200_din_bracket_x2.stl

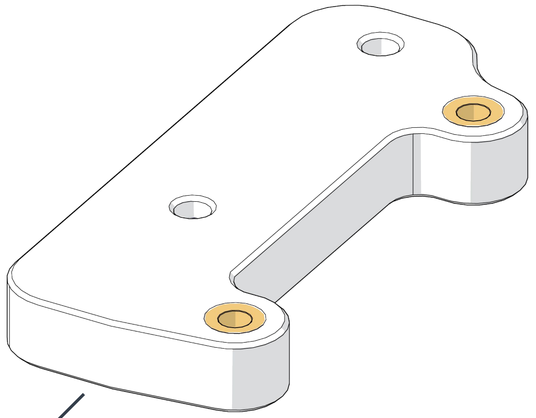


Top

Bottom

ldo_uhp200_psu_mount_a_x1.stl





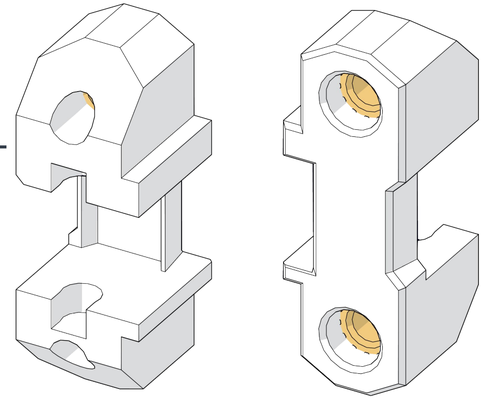
ldo_uhp200_psu_mount_b_x1.stl

C/ PRINTED PARTS

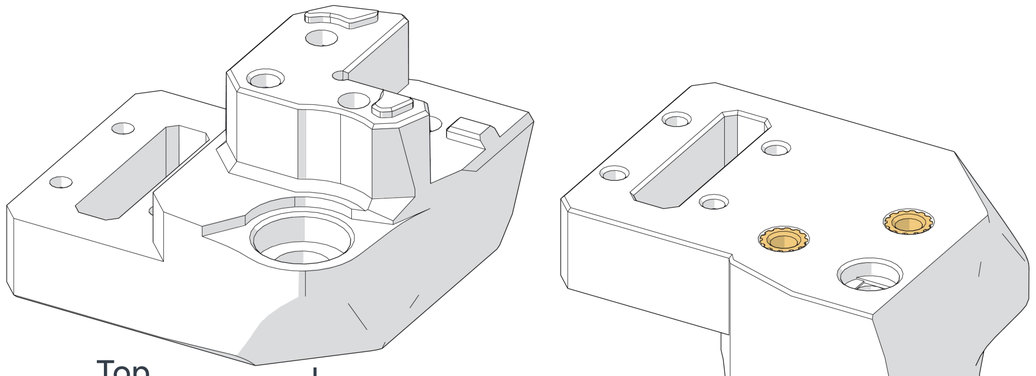
HEATSETS - GANTRY

Install the heatset inserts as shown.

[a]_idler_carrier_pinned_x2.stl



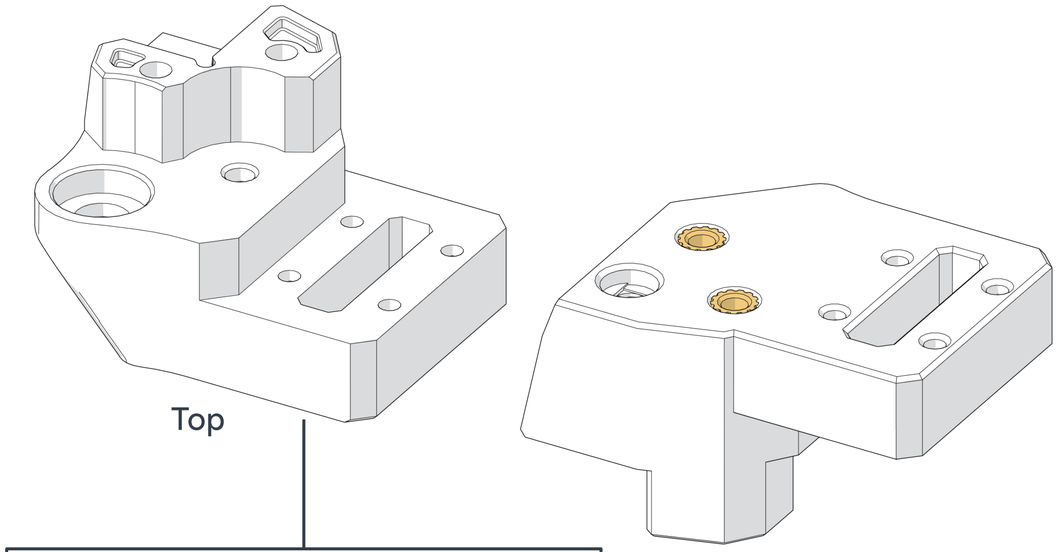
Front Back



Top

xy_joint_lower_left_live_idlers_x1.stl

Bottom



Top

xy_joint_lower_right_live_idlers_x1.stl

Bottom

D/ HARDWARE

A NOTE ON KITS

Micron+ kits are available from several manufacturers, each providing a slightly different selection of components. This manual is written specifically for the **LDO Micron+ 180 kit**, which includes many pre-selected, high-quality components designed to simplify assembly. Because of this, some parts, wiring layouts, and assembly steps may vary slightly between kits.

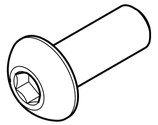
While the kit includes most of the required hardware and components, builders must still make several important hardware choices:

- Hotend
- Extruder
- Bed probe
- Printed parts

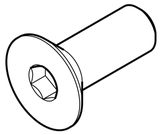
D/ HARDWARE

REFERENCE

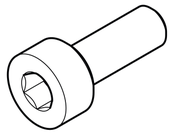
The following list identifies the most common fasteners and components to help builders recognize them during assembly.



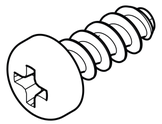
Button head cap screw (BHCS)
Low-profile screw used to fasten printed parts where clearance is limited.



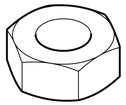
Flat head countersunk screw (FHCS)
Countersunk screw designed to sit flush with the surface.



Socket head cap screw (SHCS)
Standard hex-drive screw used for most structural connections.

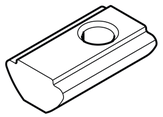


Self-tapping screw
Coarse-thread screw used to fasten directly into plastic printed parts.

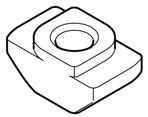


Hex nut
Standard hexagonal nut used with screws or bolts to secure components together.

Heatset insert
Brass threaded insert installed into printed parts to provide strong metal threads.

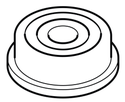


Post-install T-slot nut (T-nut)
Sliding nut that fits into the extrusion slots and allows components to be mounted.

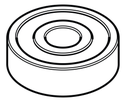


Hammerhead nut
Drop-in extrusion nut that rotates into place, commonly used for panels and accessories.

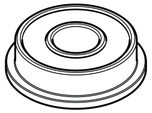
D/ HARDWARE



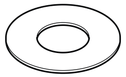
F623-2RS flanged bearing
Flanged bearing used in idler assemblies to guide belts.



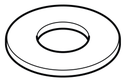
625-2RS bearing
Standard radial bearing used in motion assemblies such as belt idlers.



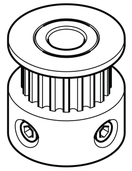
F695 flanged bearing
Large flanged bearing commonly used in the Z-axis belt path.



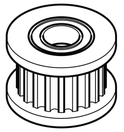
Shim
Thin spacer used to fine-tune alignment and movement of bearings and idlers.



Washer
Flat spacer used to distribute load and prevent screws from damaging surfaces.

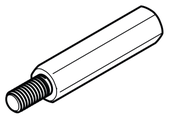


2GT 16T pulley
Timing pulley mounted to stepper motors to drive the 2GT belts for CoreXY motion.

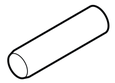


2GT 16T idler
Free-spinning pulley used to guide and redirect belts along the motion path.

Set/grub screw
Small screw used to secure pulleys or gears to motor shafts.



Standoff
Threaded spacer used to create fixed distance between components.



Shaft
Precision metal rod used as the axle for idlers and bearings in the motion system.

E / FRAME

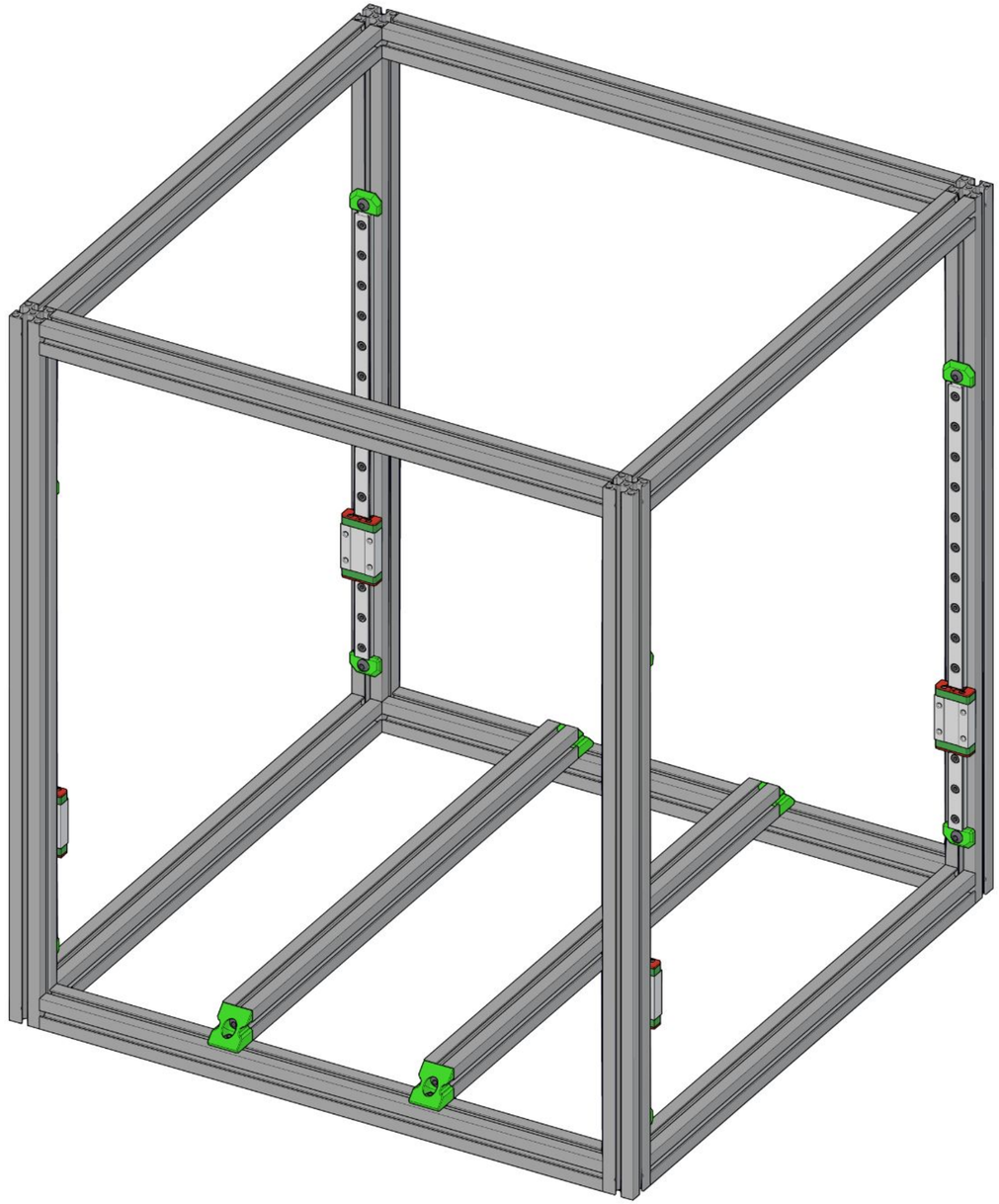
The frame is the foundation of the printer. Ensuring the extrusions are aligned and square now will help guarantee smooth motion later in the build.

HARDWARE

4	1515 A extrusions (verticals)
10	1515 B extrusions (horizontals & bed)
4	MGN7H linear rails & carriages
60	M2 × 6 SHCS
60	M2 hex nuts
4	M3 × 6 BHCS
36	M3 × 8 BHCS
30	M3 hex nuts

EXTRAS

36	M3 hex nuts for screwed-in panels (Optional)
4	M3 hex nuts for mounting LEDs (Optional)
2	M3 hex nuts for PTFE tube retainer (Optional)
2	M3 hex nuts for Raspberry Pi (Optional)

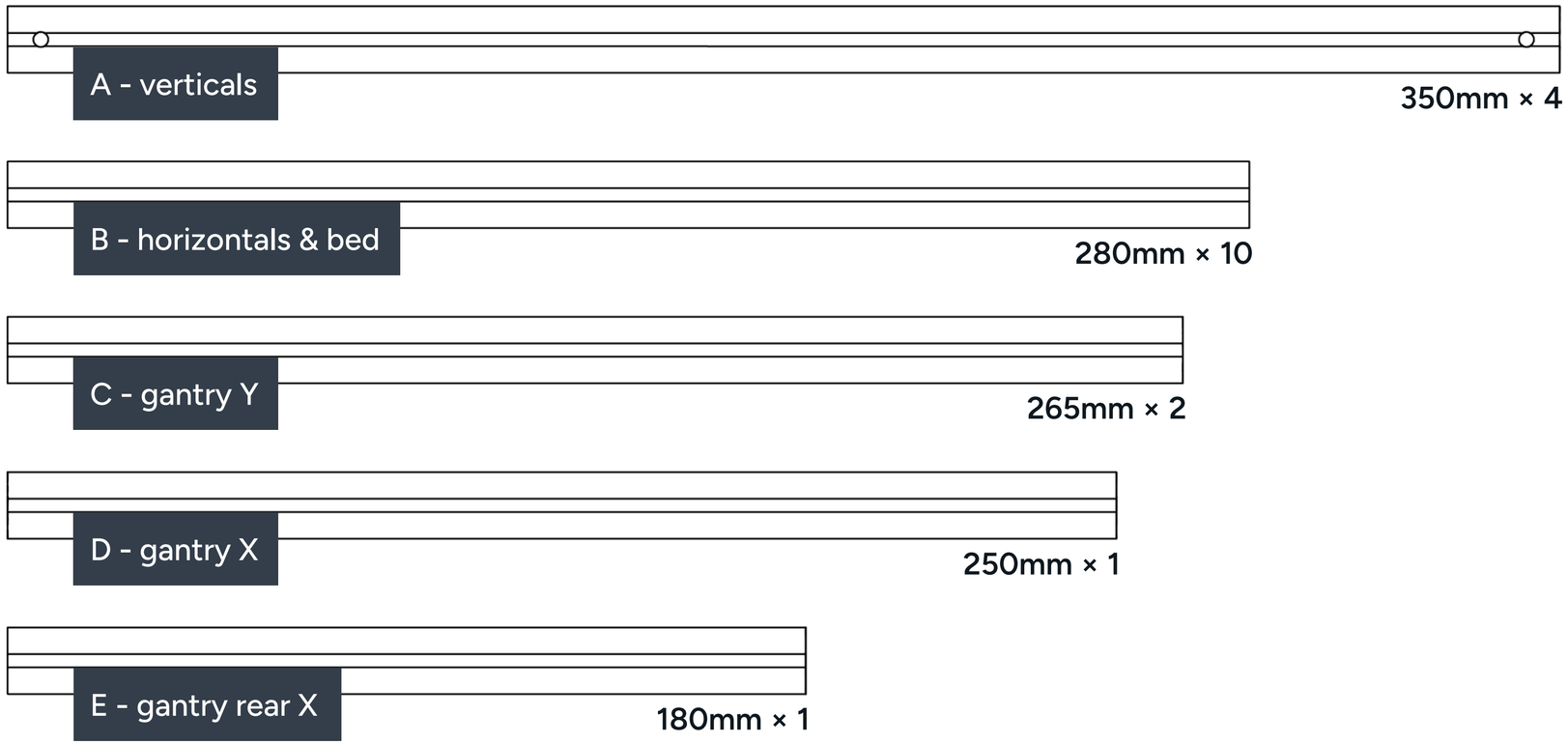


E / FRAME

EXTRUSIONS

To avoid confusion, the extrusions are identified by the labels **A–E** shown on these reference pages.

Labeling the extrusions ahead of time or keeping this reference page nearby can make assembly faster.



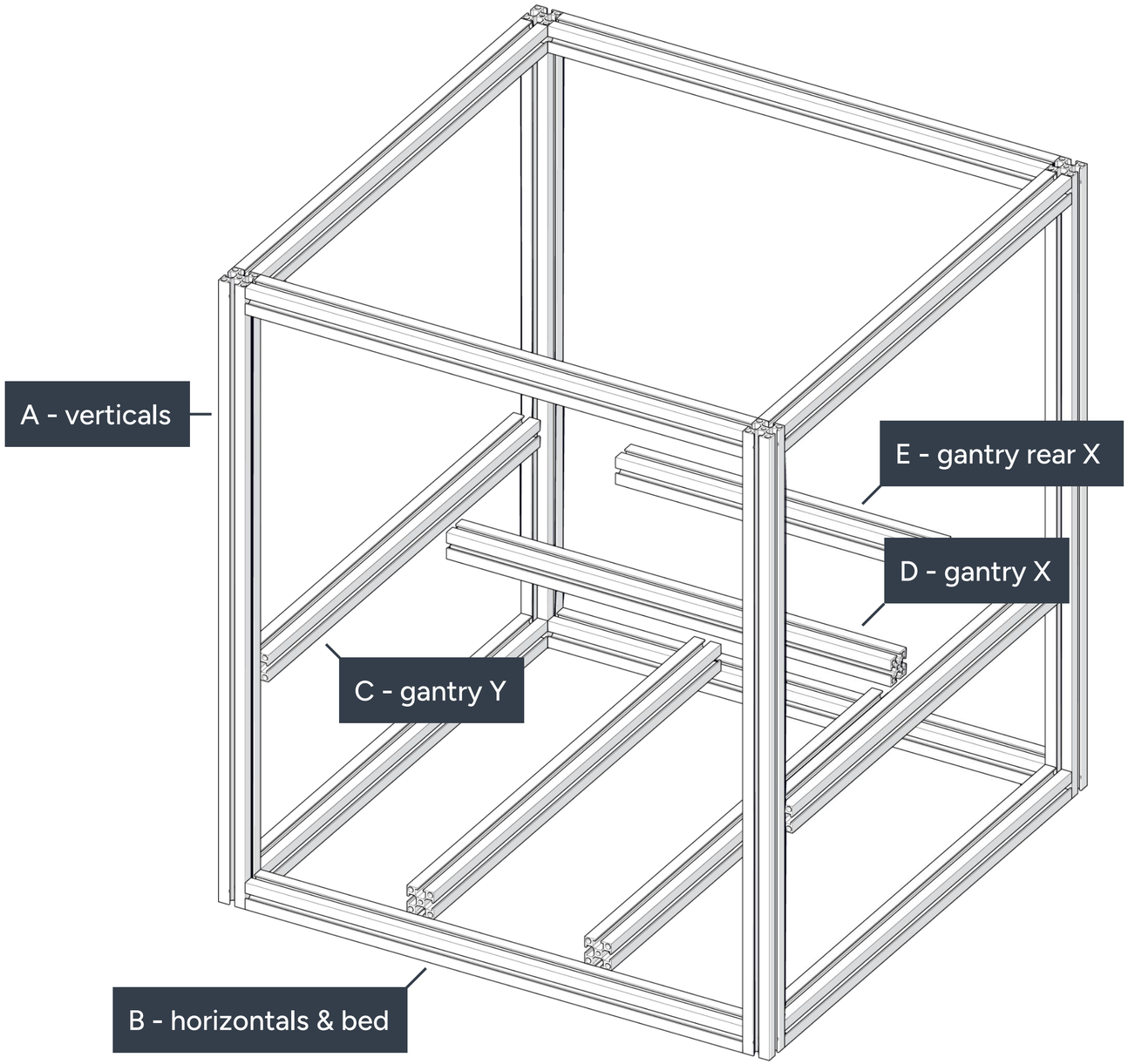
NOTE

Some kits include bed extrusions mounted with blind screws. **These 2 extrusions will be longer than the other 8 B extrusions.**

E / FRAME

EXTRUSIONS

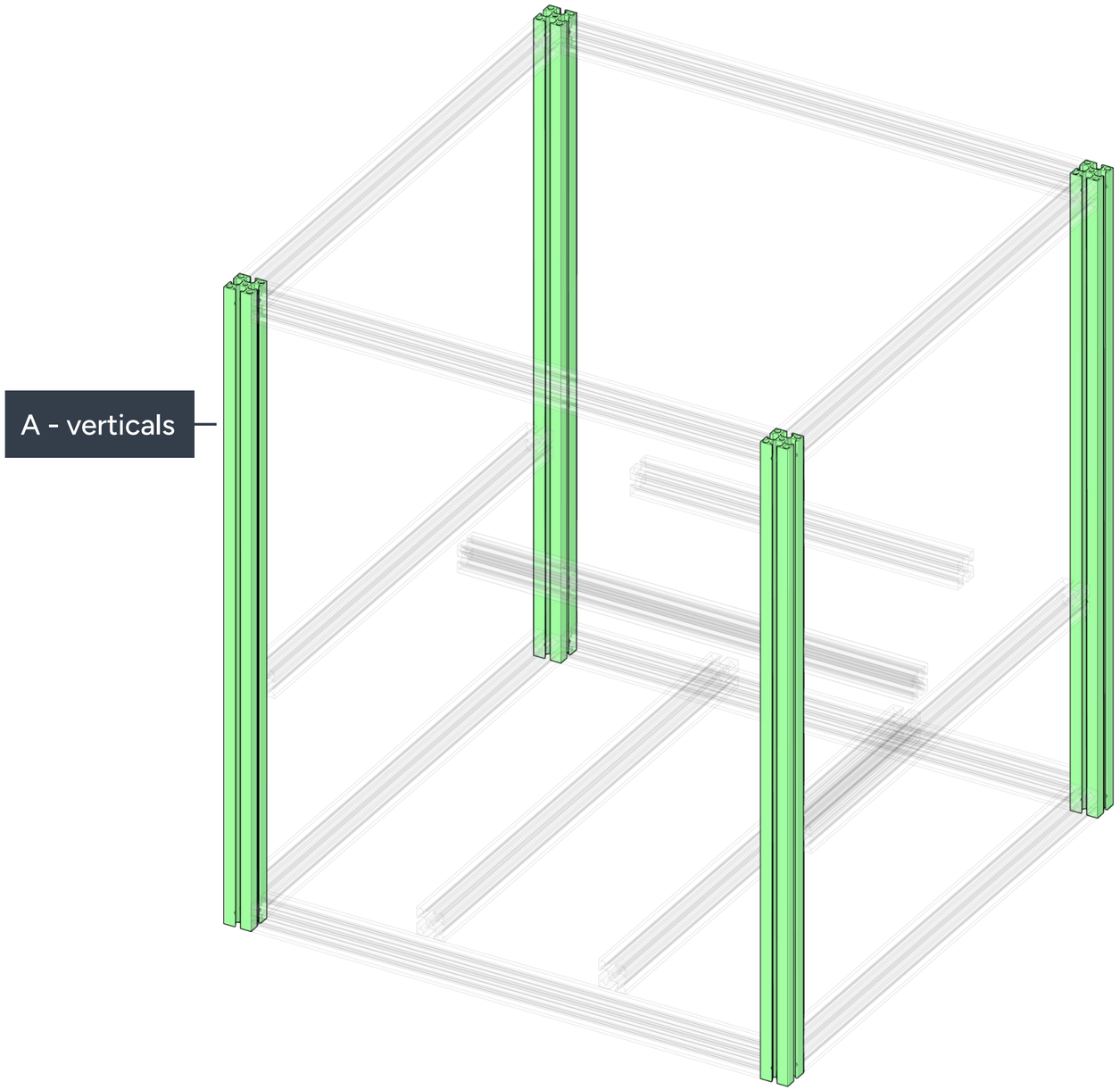
This diagram below shows where each extrusion type is used in the frame.



E / FRAME

A EXTRUSIONS

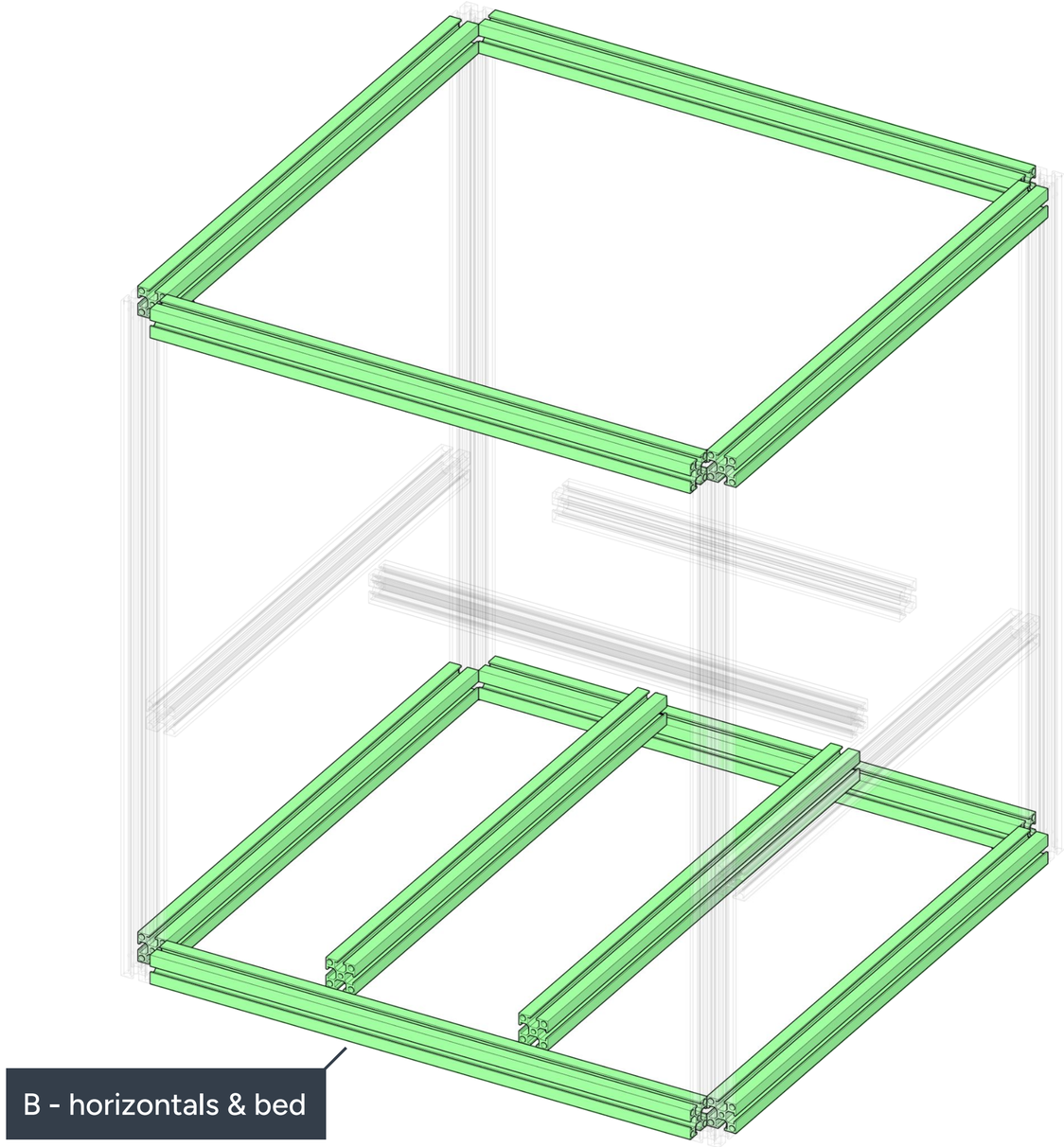
There are **four A vertical extrusions** that form the corner columns. Each has two blind holes at both ends to allow a hex driver to reach the BHCS used for blind joints.



E / FRAME

B EXTRUSIONS

There are **ten B horizontal extrusions** that form the frame and bed supports. These extrusions use button head cap screws (BHCS) in the ends for blind joints.



B - horizontals & bed

NOTE

Some kits include bed extrusions mounted with blind screws. **These 2 extrusions will be longer than the other 8 B extrusions.**

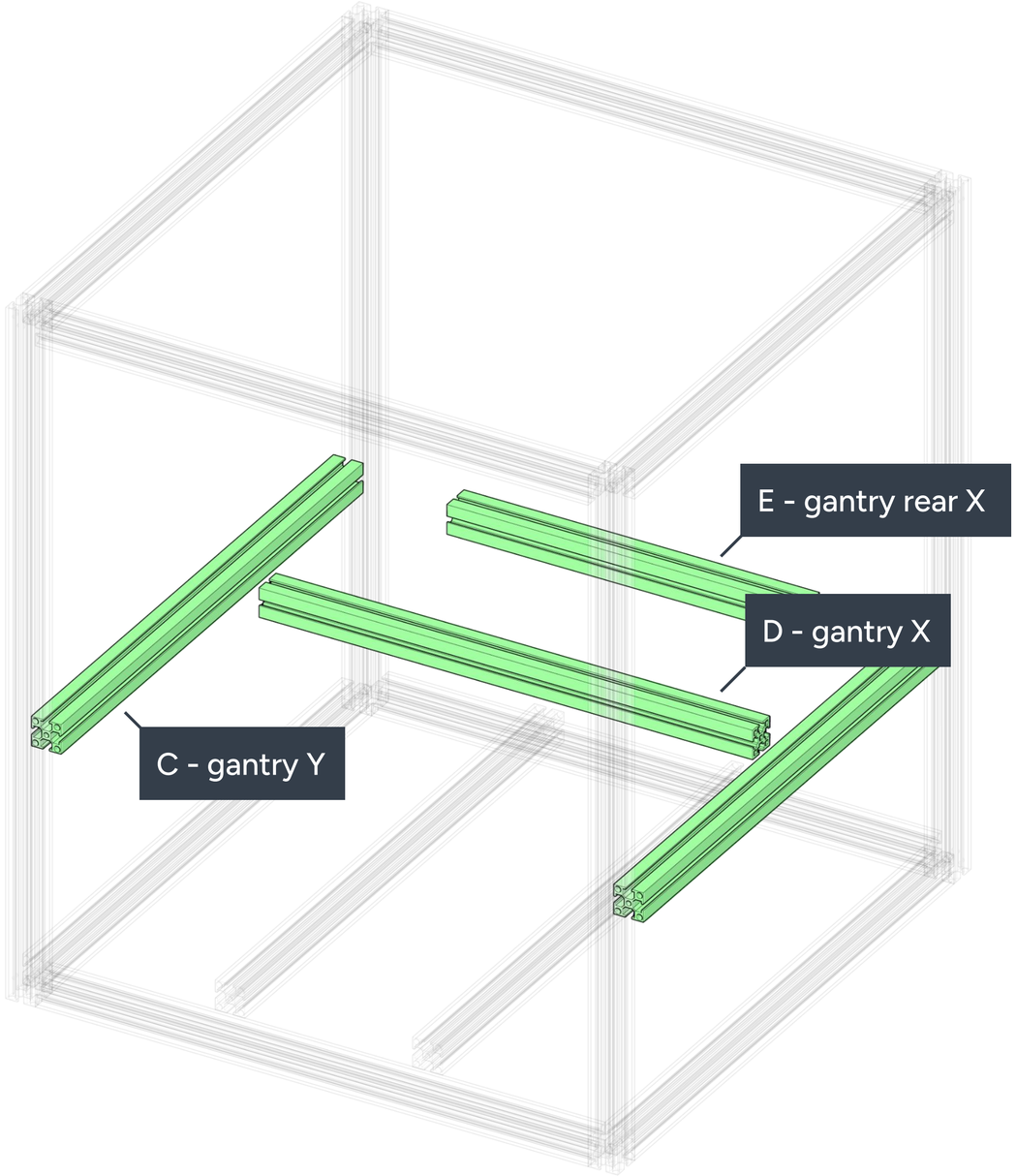
E / FRAME

C, D, E EXTRUSIONS

Two C extrusions form the gantry's Y arms. They are shorter than B extrusions to allow the flying gantry to tilt.

The **D extrusion** forms the gantry's X beam. It is shorter to fit within the frame.

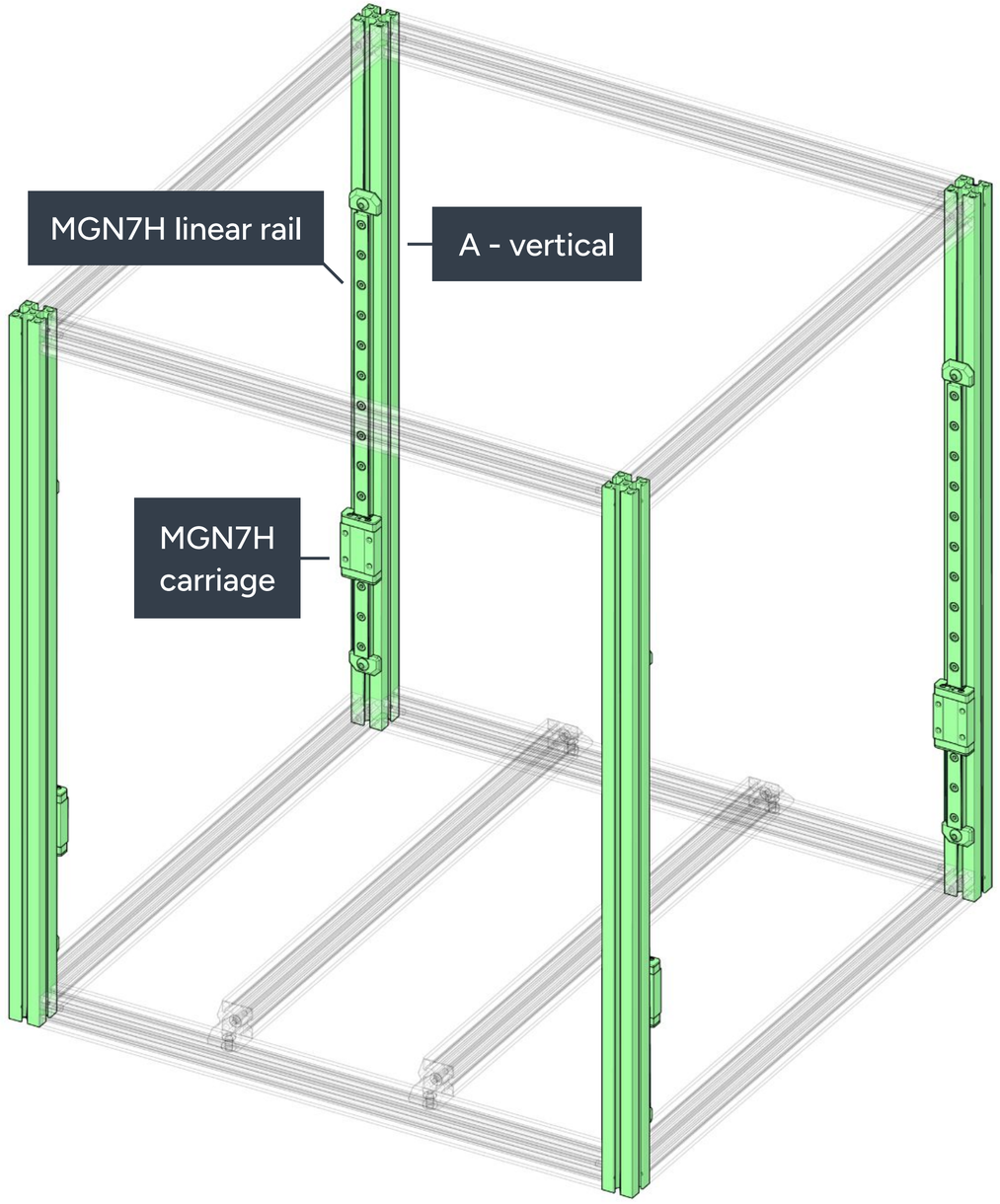
The **E extrusion** sits at the rear of the gantry X. It is the shortest to make room for the A/B drives.



E / FRAME

PREPPING A EXTRUSIONS

Each A vertical extrusion receives an **MGN7H linear rail** mounted using **M2 × 6 SHCS** into **M2 hex nuts**. The LDO kit includes **nut bars** to position the screws correctly. Printed adapters can be used instead.

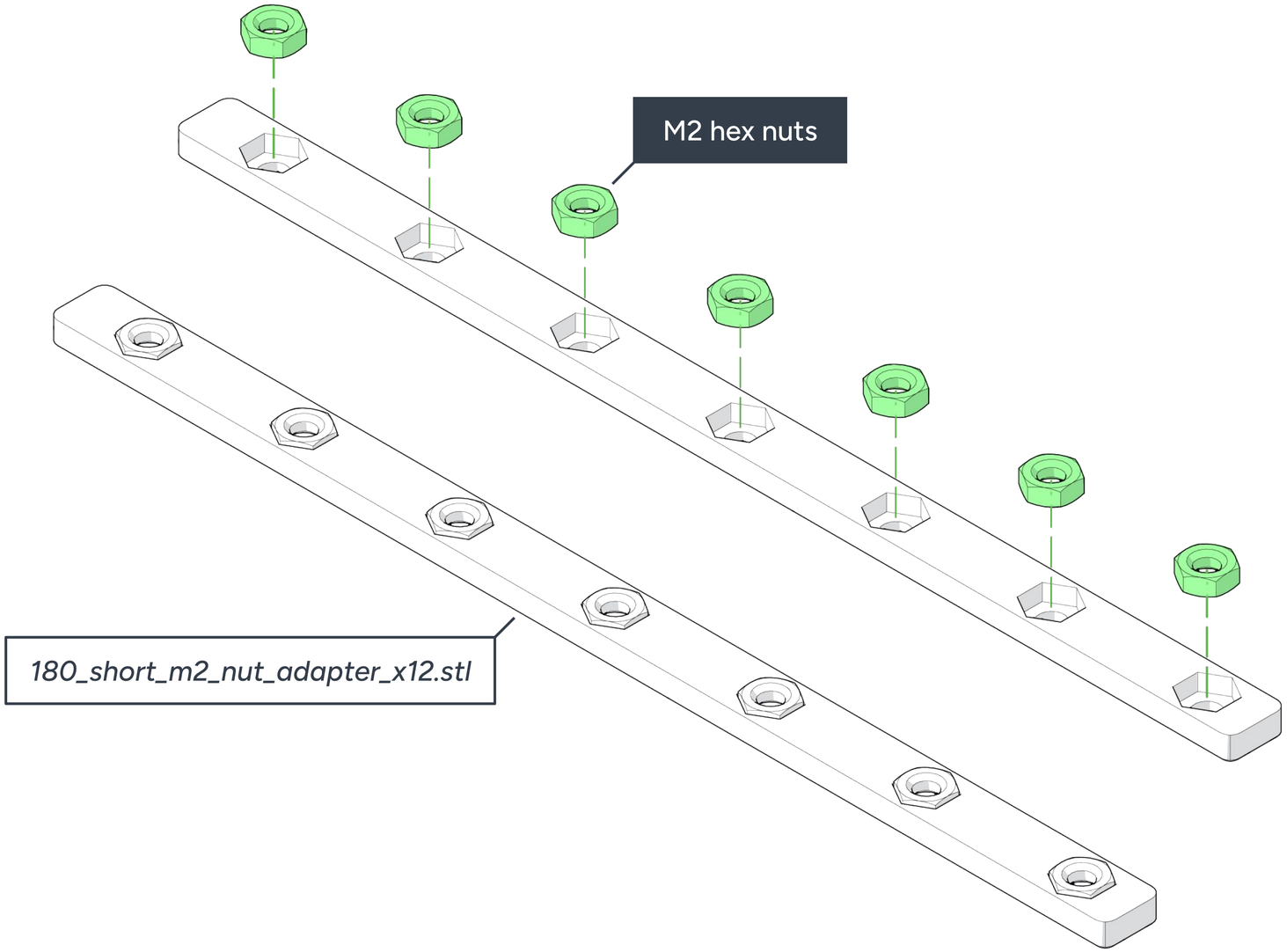


E / FRAME

PREPPING A EXTRUSIONS

Printed adapters hold **7 M2 hex nuts** each.
Two adapters are required per linear rail.

Insert the hex nuts before sliding the adapters into the extrusion slot.

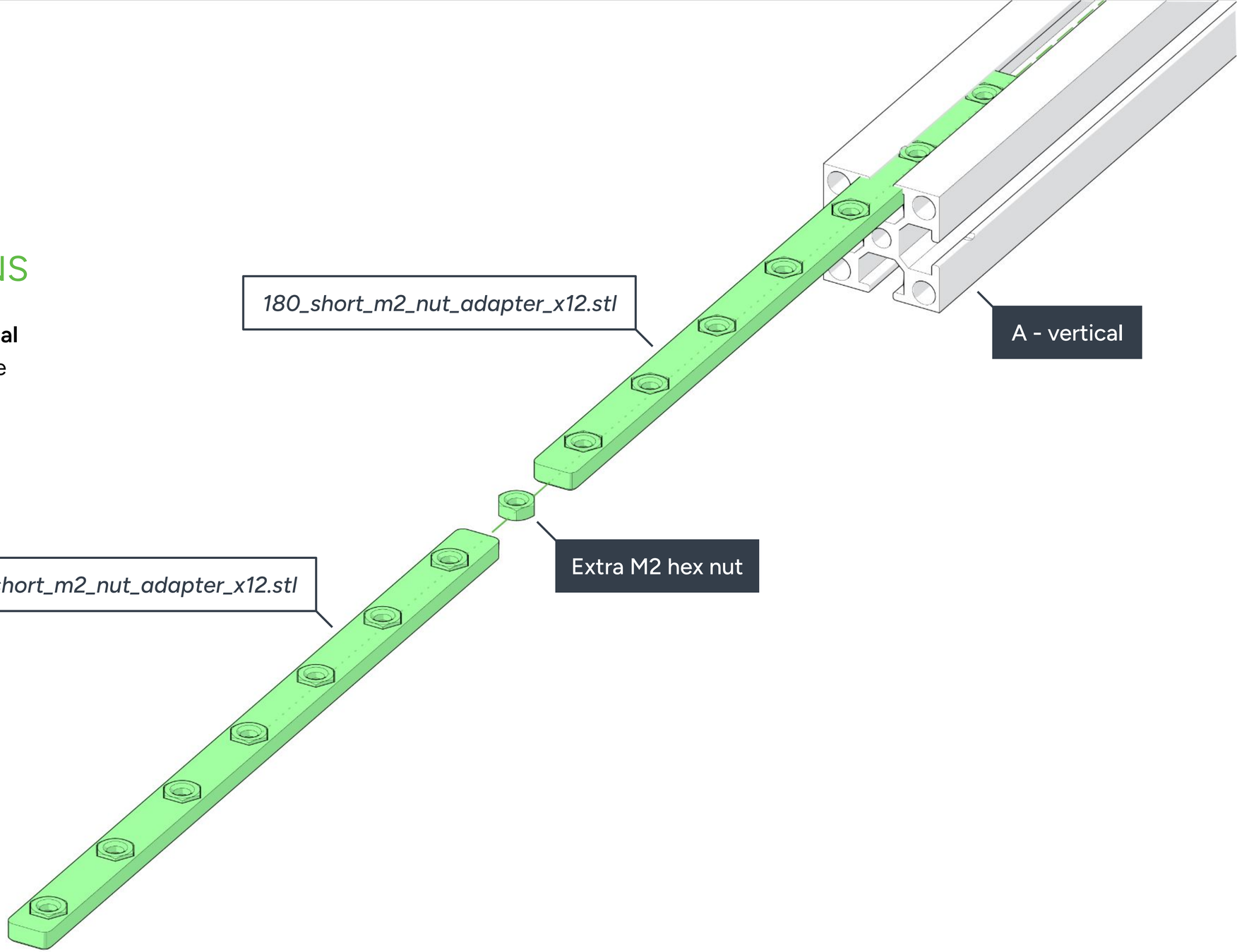


E / FRAME

PREPPING A EXTRUSIONS

Two printed adapters and **one additional M2 hex nut** are required to populate the 15 mounting holes of an **MGN7H rail.**

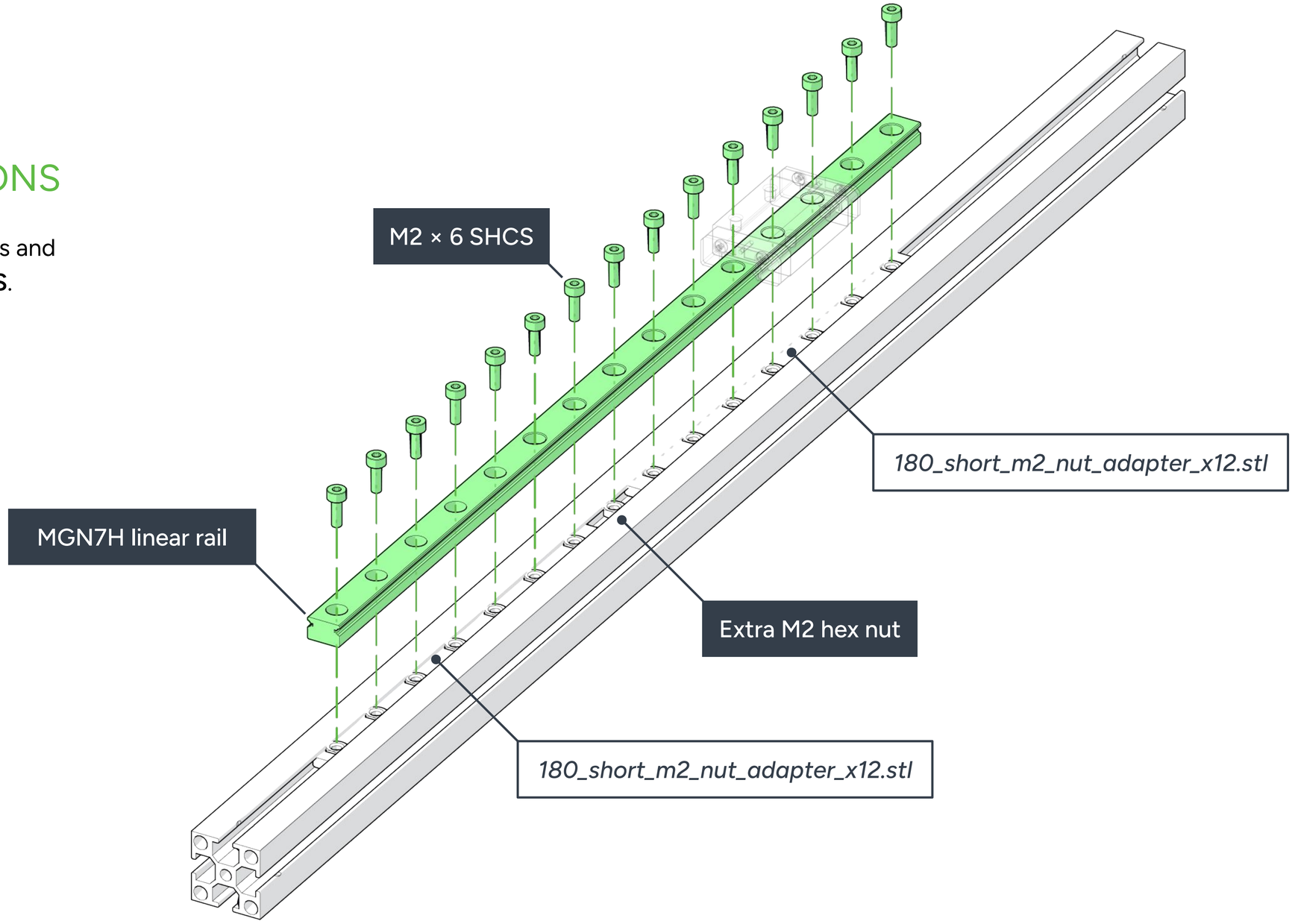
Slide them into the bottom of the **A vertical extrusion**.



E / FRAME

PREPPING A EXTRUSIONS

Line up the **MGN7H rail** over the nuts and lightly secure it with **15 M2 × 6 SHCS**.
Do not tighten them down yet.



NOTE

The linear rail is pictured without the carriage for illustrative purposes. Mind the carriage and do not let it fall off.

E / FRAME

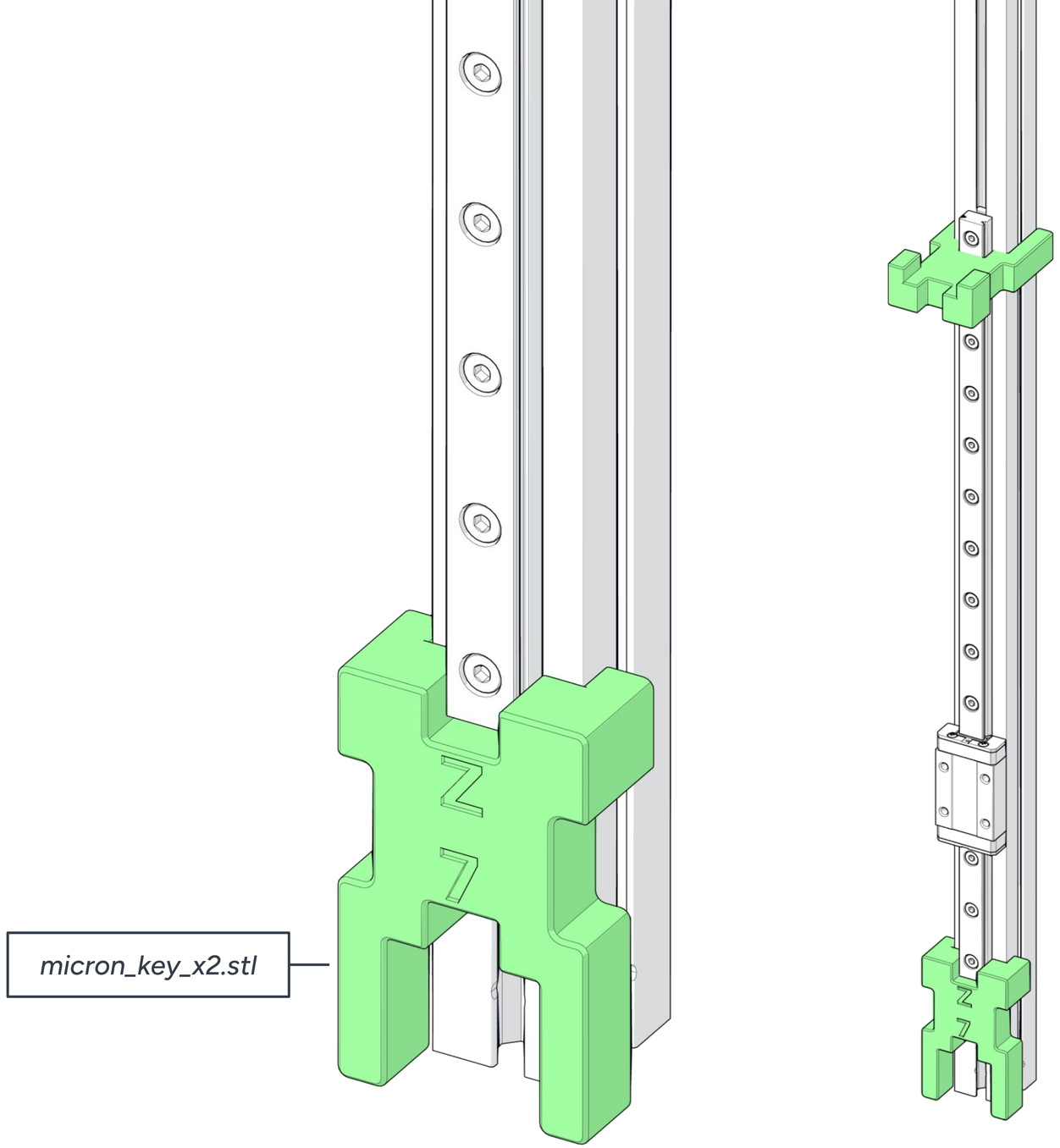
PREPPING A EXTRUSIONS

The MGN7H rail should still be able to slide with light force. It will be positioned 36 mm from the end and centered on the extrusion using two printed keys (***micron_key_x2.stl***).

Stand the A vertical extrusion upright. Using the **first key**, position the rail in the “Z” groove as shown. This sets the 36 mm spacing.

Use the **second key** to straddle the extrusion and rail at the opposite end in the “7” groove. This centers the rail on the extrusion.

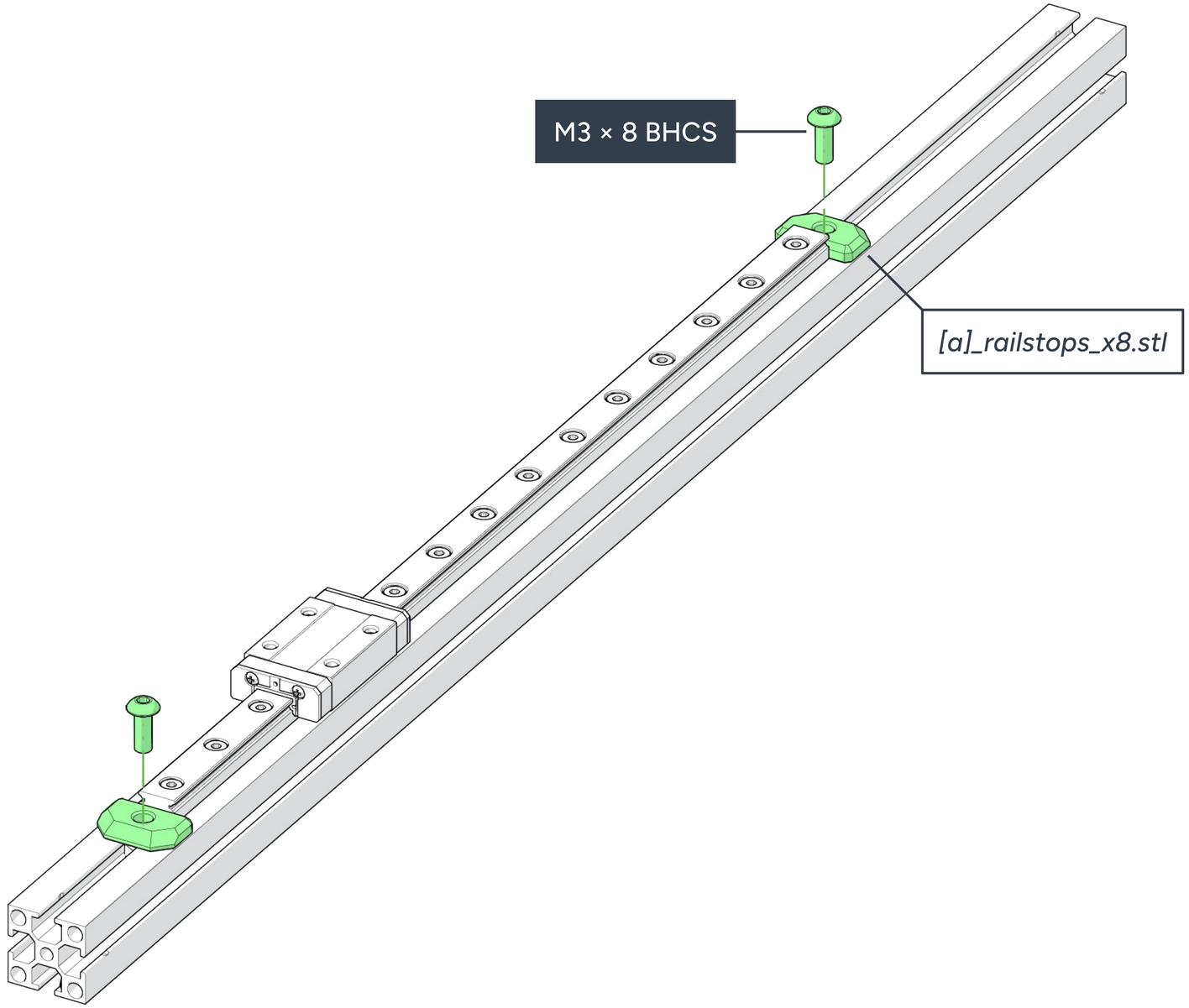
Tighten the screws.



E / FRAME

PREPPING A EXTRUSIONS

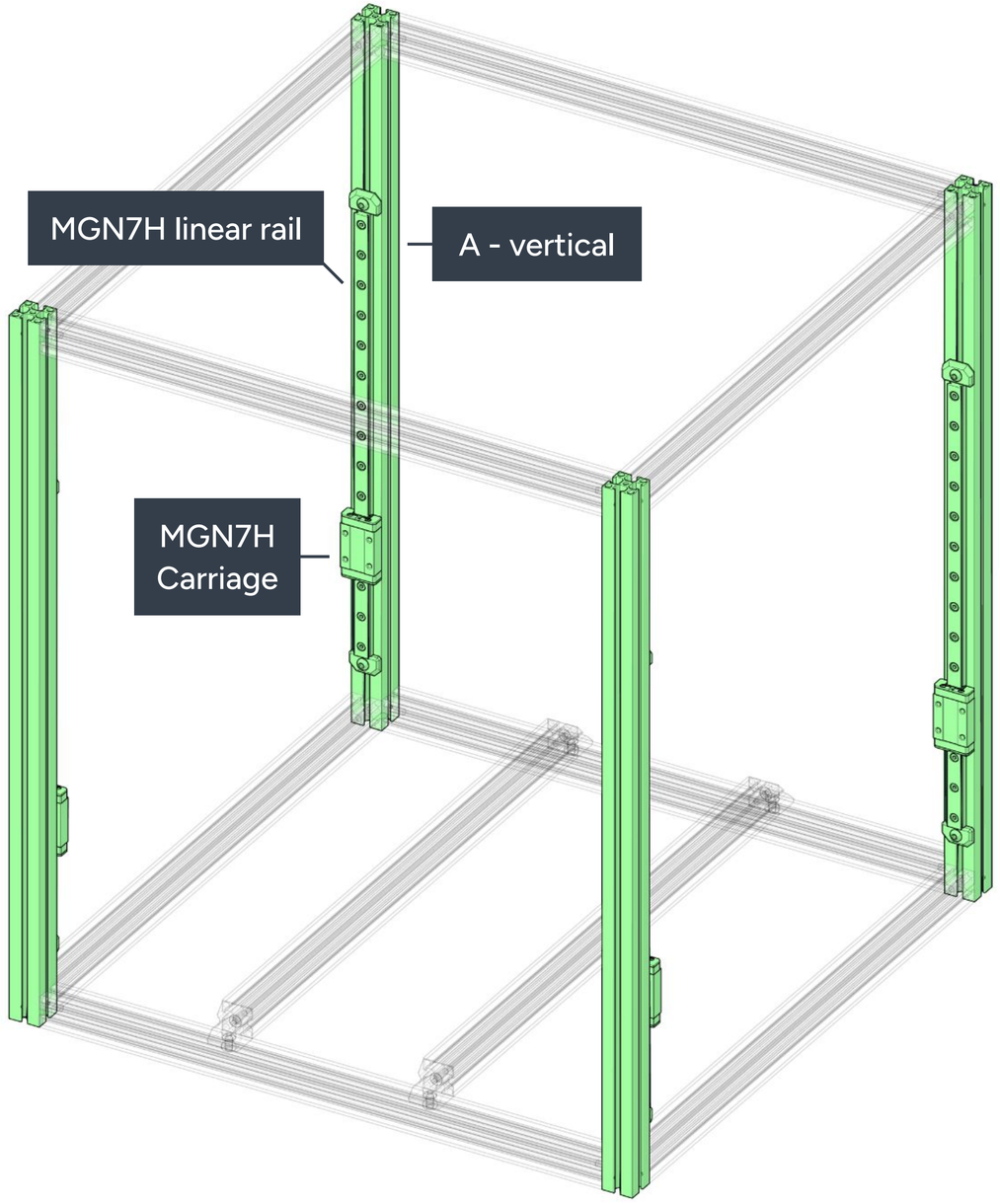
Insert **M3 hex nuts** on each end and secure the printed rail stops with **two M3 × 8 BHCS**.



E / FRAME

PREPPING A EXTRUSIONS

Once all four A vertical extrusions have been prepped, we are ready to begin the outer frame assembly.



CONCEPT

BLIND JOINT BASICS

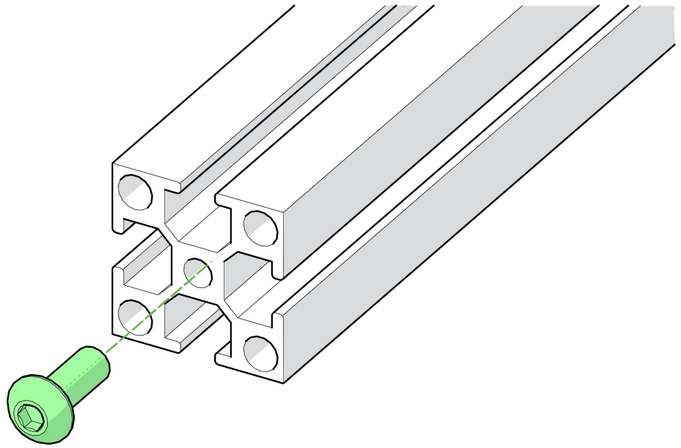
Blind joints provide a simple and rigid way to join aluminum extrusions. Thread a button head cap screw (BHCS) a few turns into the end channel of one extrusion, leaving the screw head exposed. Slide the adjoining extrusion onto the screw through its slot, then tighten through the access hole to secure the joint.

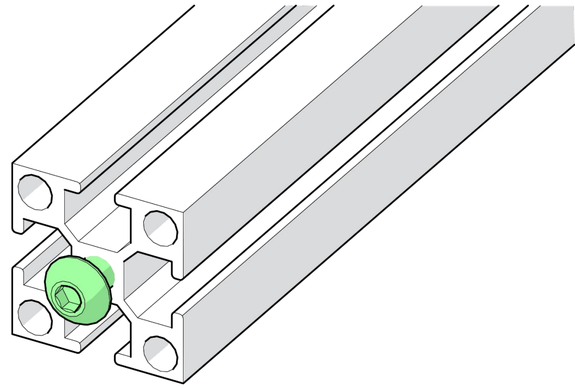
Key points:

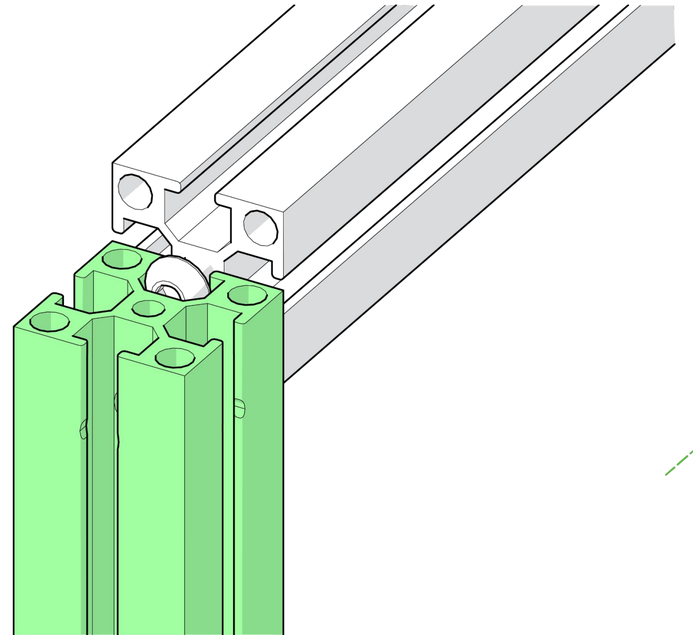
- Keep extrusions flush and aligned before applying torque
- Tighten gradually, not all at once

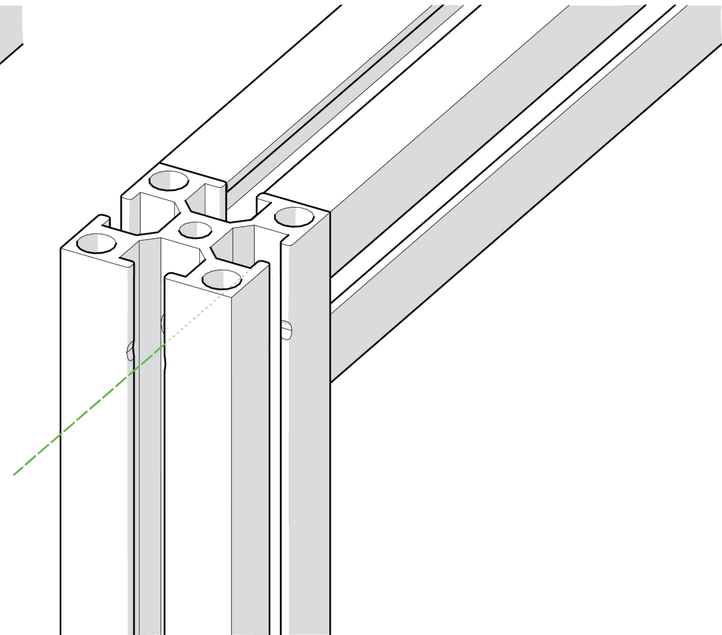
Common mistakes:

- Tightening before parts are fully seated, pulling the joint out of square
- Entering the slot misaligned, causing damage to the slot or extrusion









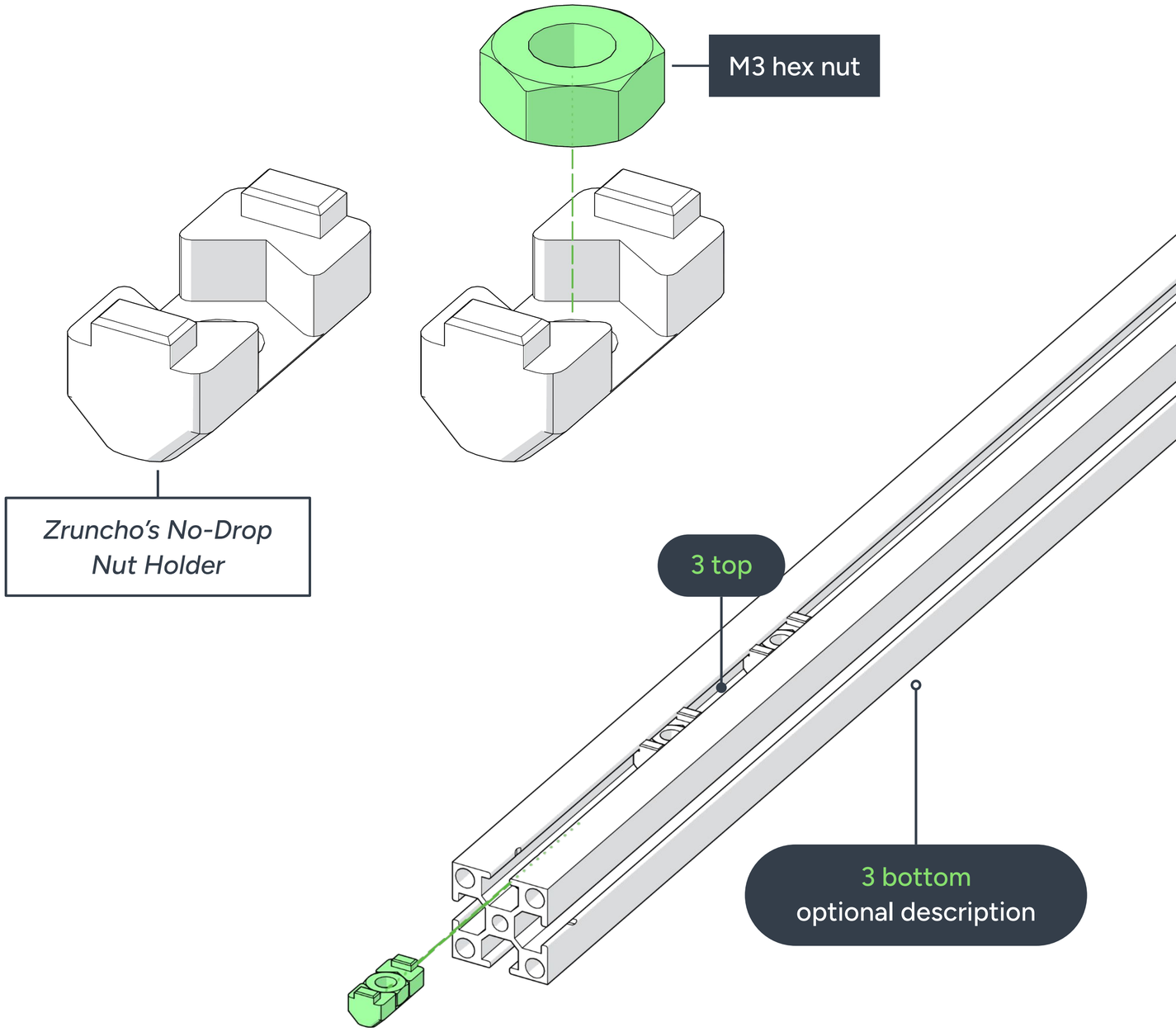
CONCEPT

PRELOADED NUTS

Once the ends of a 1515 extrusion are blocked, inserting additional nuts becomes difficult. Be sure to preload the required nuts in the correct slots as you build. Adding missing nuts later almost always requires partial disassembly.

To prevent nuts from sliding inside the extrusion, we recommend printing “no-drop” nut holders, which retain them by friction. See the Micron GitHub for available designs.

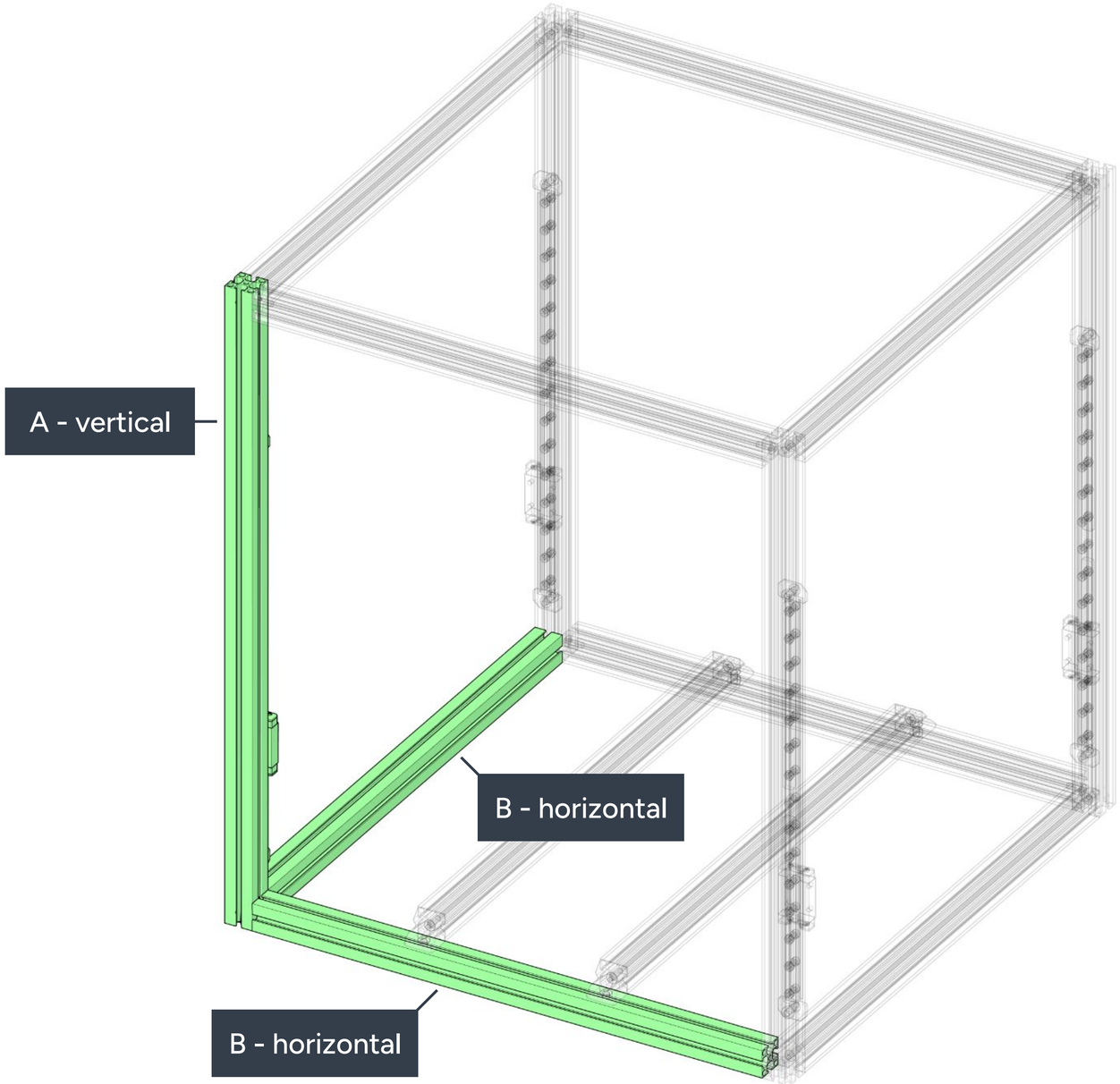
Callouts for preloaded nuts indicate both the **quantity** and the **face of the extrusion** where the nuts should be inserted. Some locations are optional. See the diagram for an example.



E / FRAME

OUTER FRAME

Start with a corner. You will need **one A vertical extrusion** and **two B horizontal extrusions**.

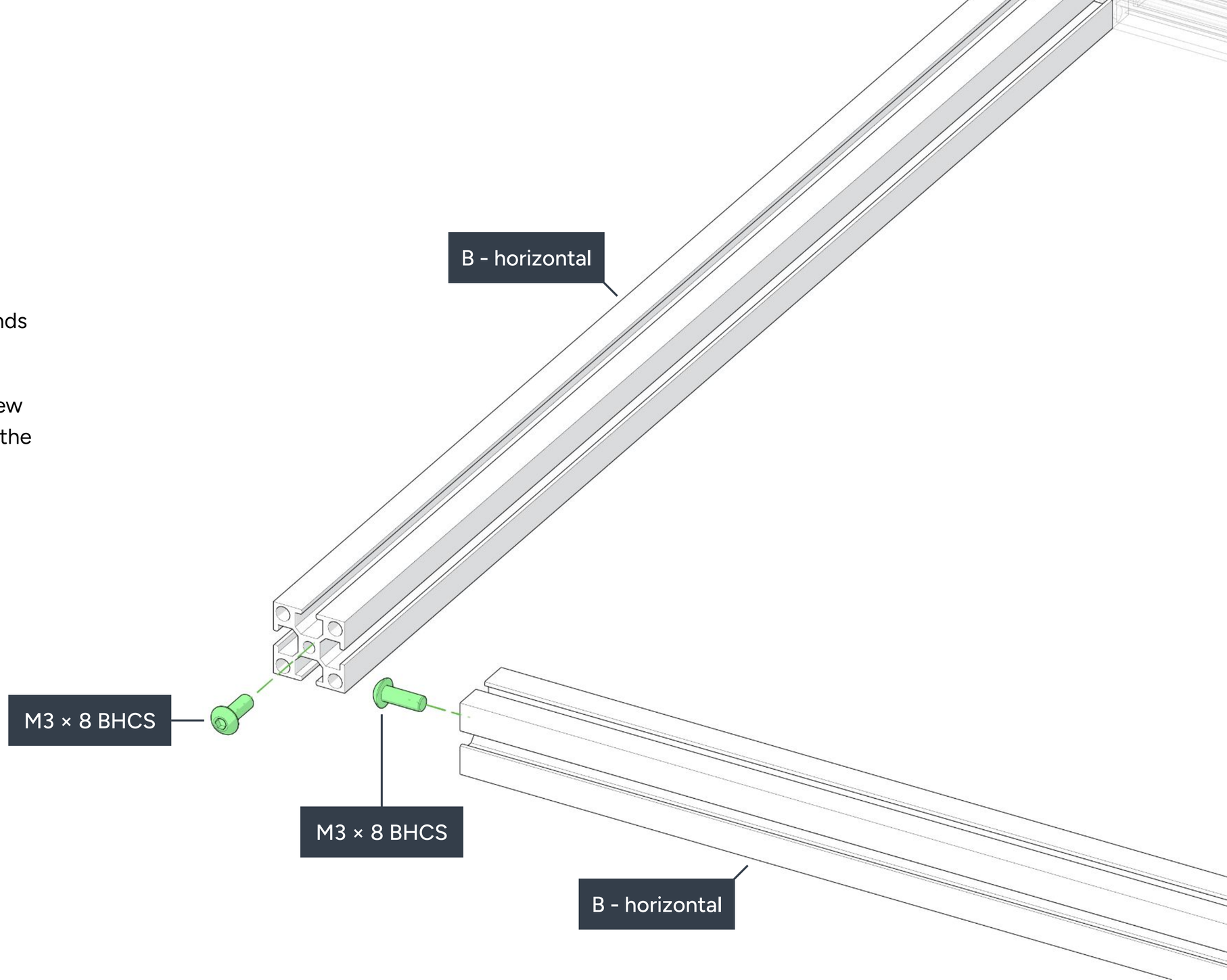


E / FRAME

OUTER FRAME

Thread an **M3 × 8 BHCS** into the ends of the two B horizontal extrusions.

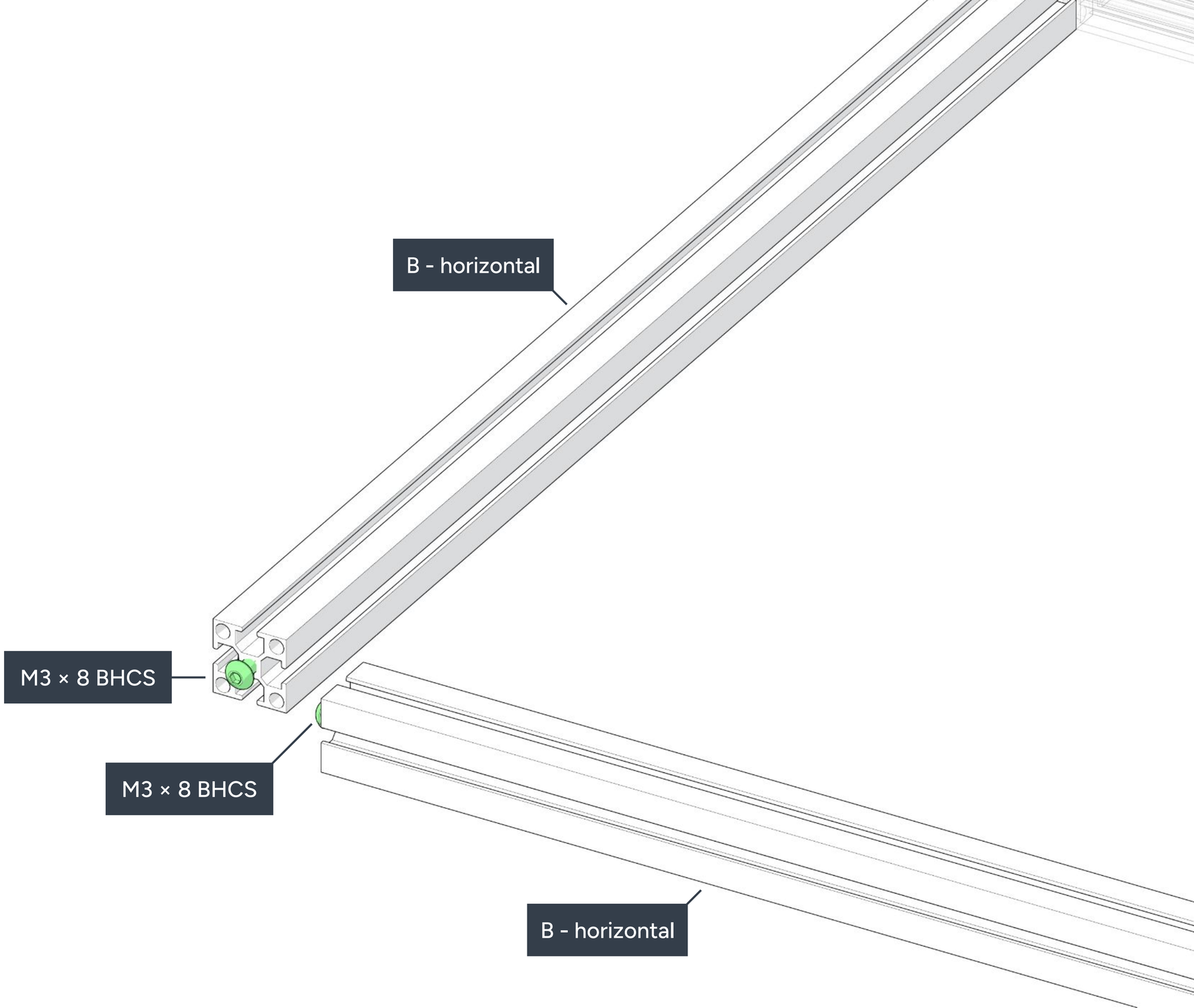
Leave enough space under the screw head so it can slide into the slot of the A extrusion.



E / FRAME

OUTER FRAME

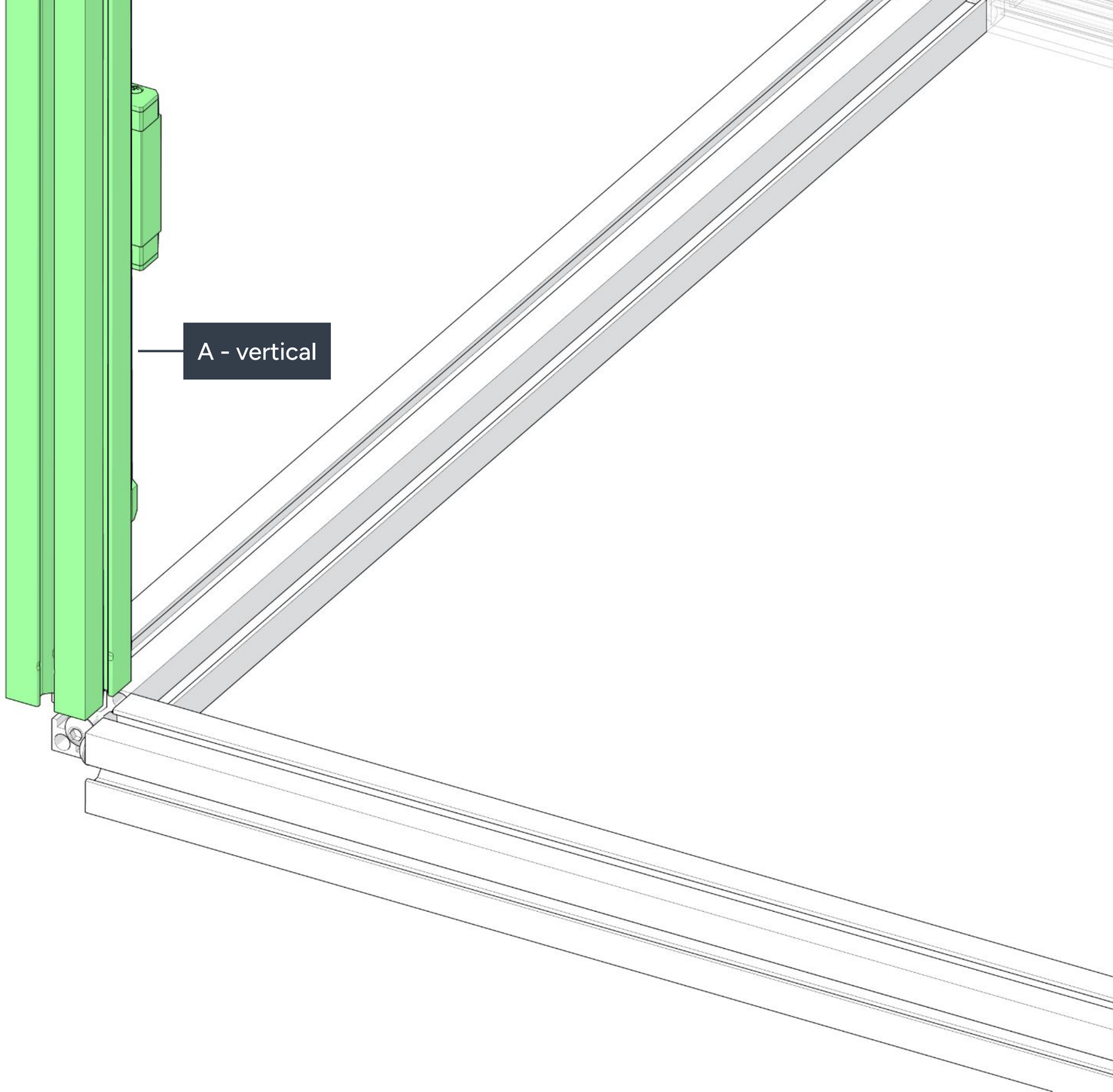
With the right clearance, the BHCS heads should fit in the channels of the incoming A extrusion.



E / FRAME

OUTER FRAME

Slide the **A vertical extrusion** onto the two B horizontal extrusions.



NOTE

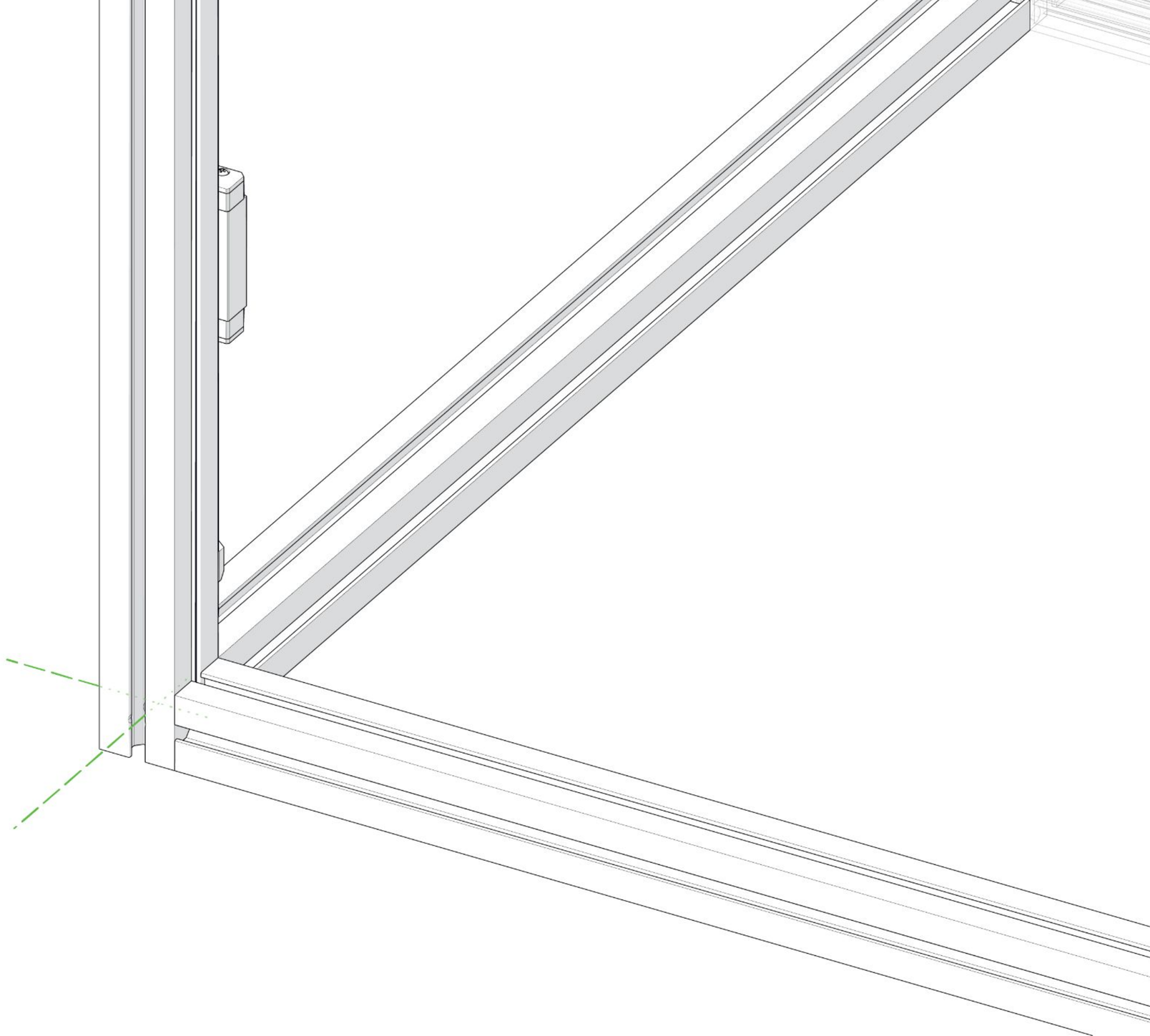
Each of your A vertical extrusions should have an MGN7H linear rail. **Make sure they are facing inwards.**

E / FRAME

OUTER FRAME

Insert the hex driver through the blind holes to reach each BHCS and tighten.

Again, it is important to build on a flat reference surface. Misalignment here will affect the rest of the build.



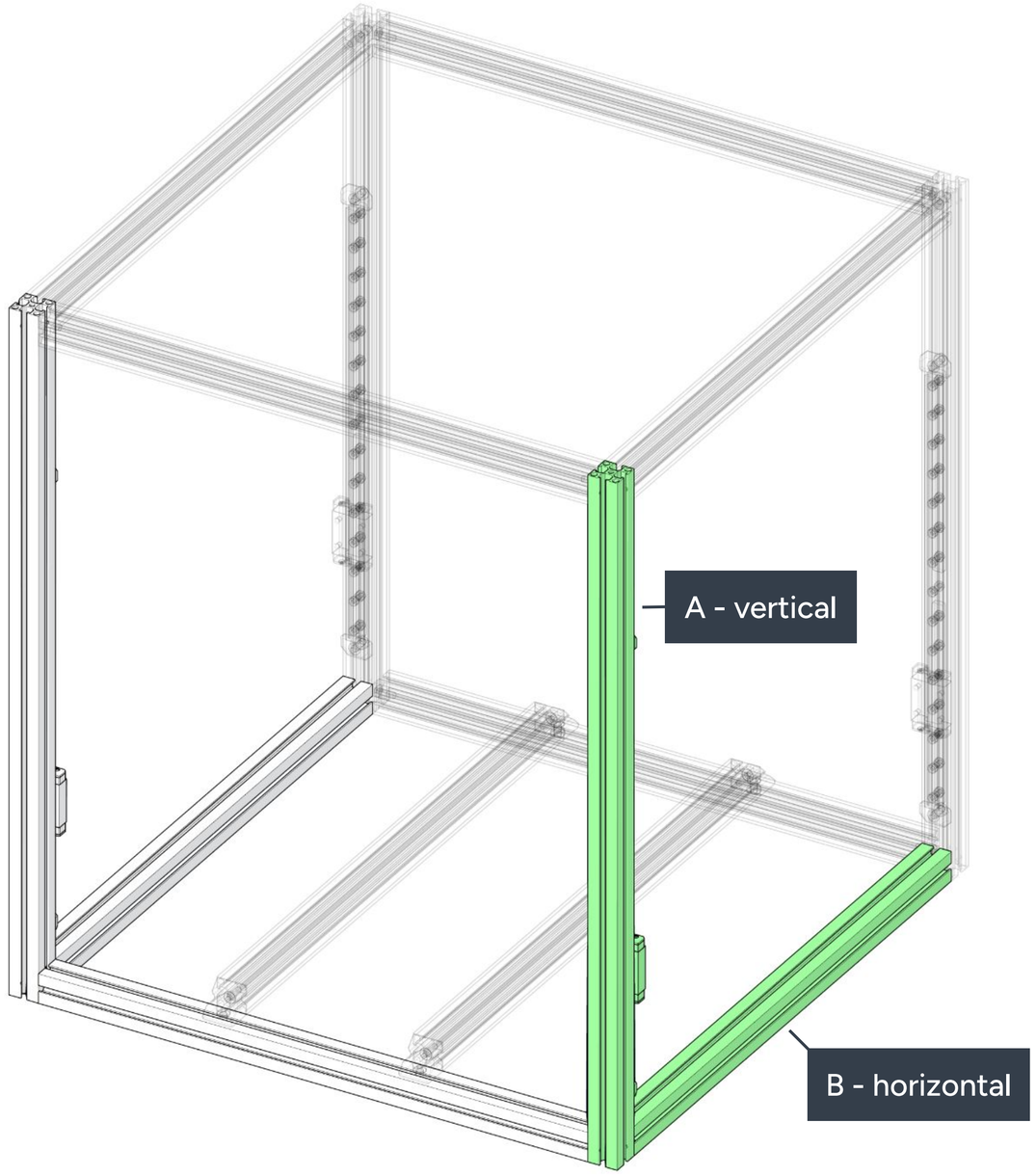
NOTE

Use the straight end of the hex driver for final tightening. Using the ball end may round out screws in blind holes.

E / FRAME

OUTER FRAME

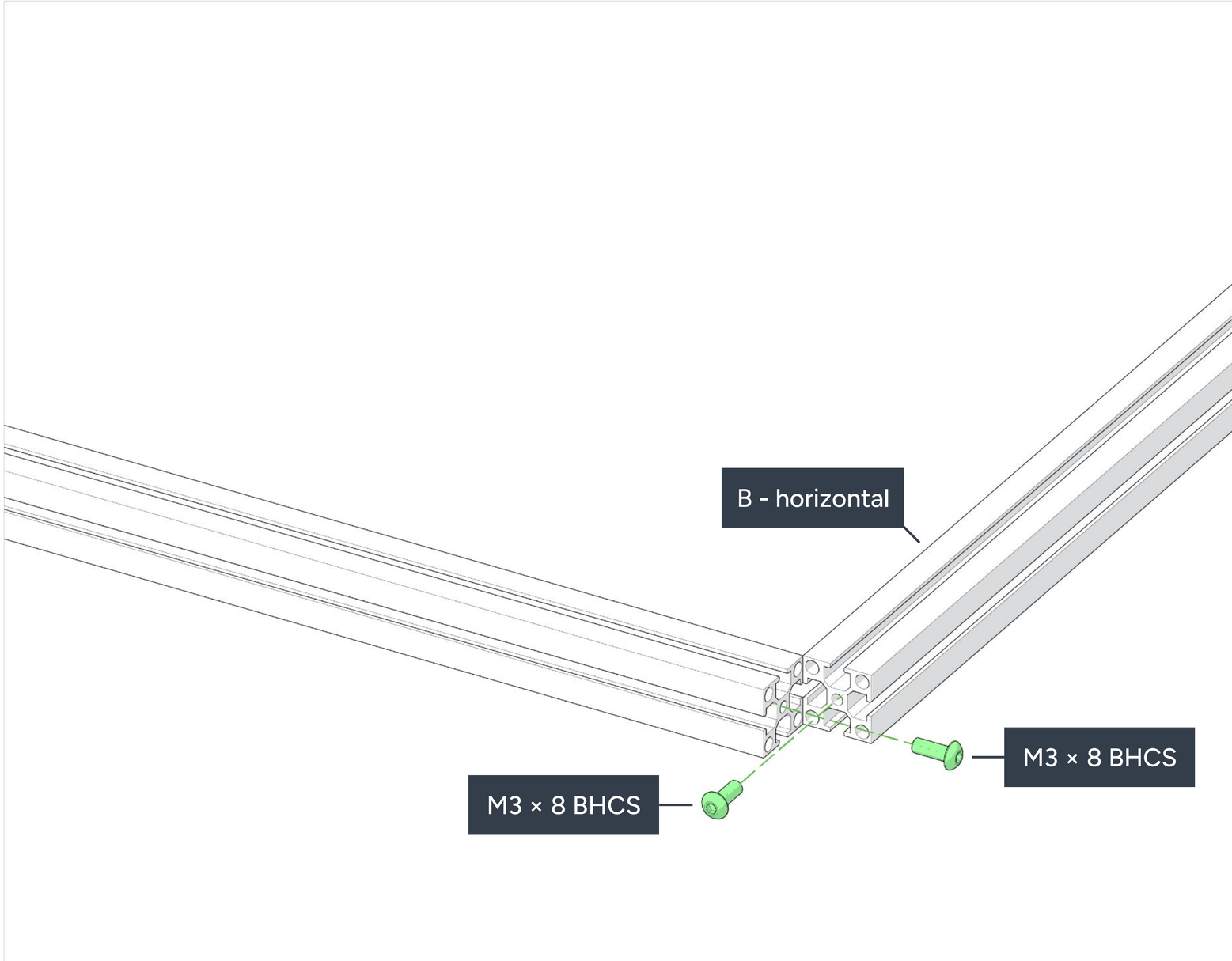
Move to the adjacent corner.
Use one **A vertical extrusion**
and one **B horizontal extrusion**.



E / FRAME

OUTER FRAME

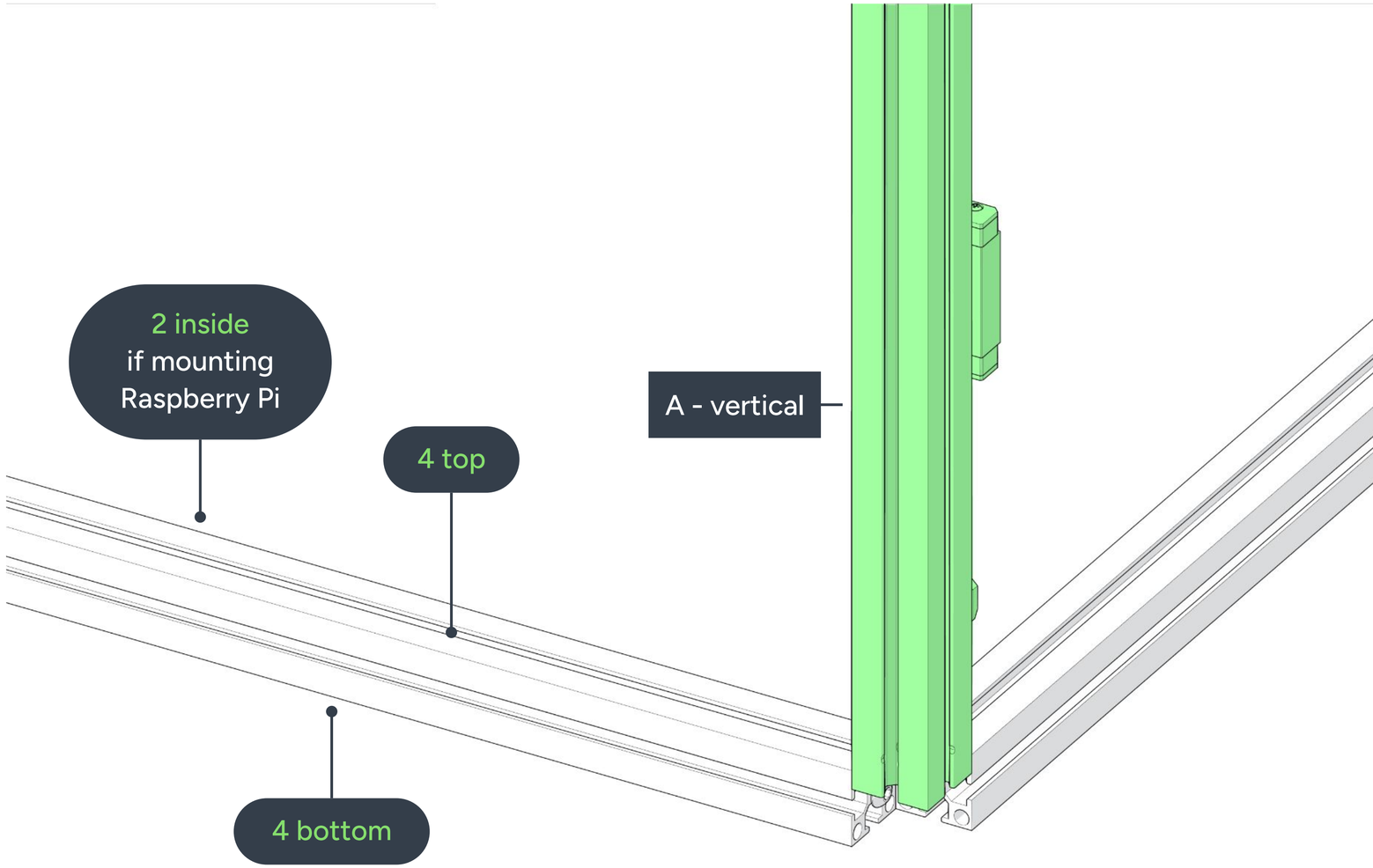
Thread an **M3 × 8 BHCS** into the end of the previous B horizontal extrusion and the new **B horizontal extrusion**. Leave them slightly proud.



E / FRAME

OUTER FRAME

Slide the second **A vertical extrusion** onto the two B horizontal extrusions.



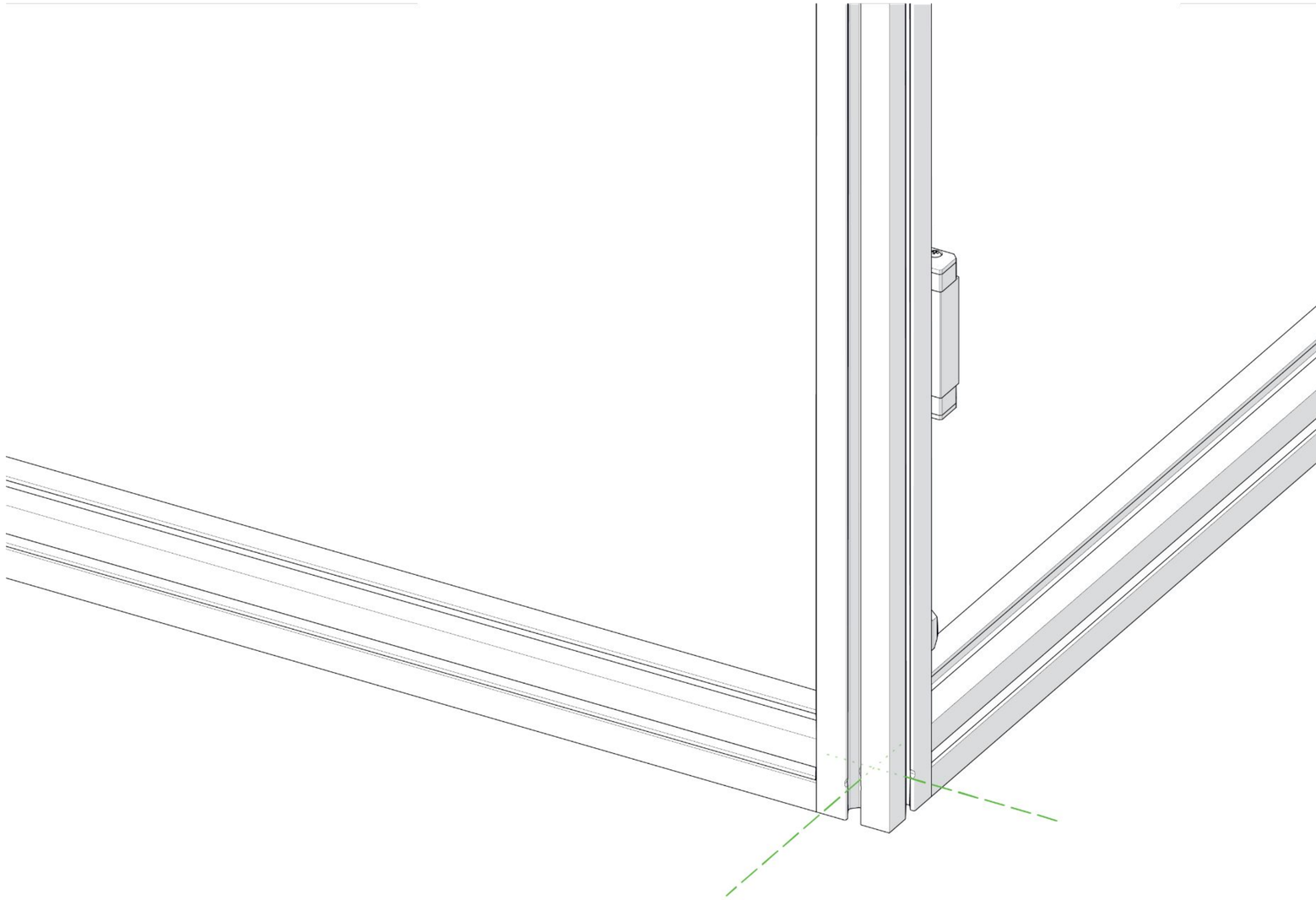
NOTE

Don’t forget the preloaded nuts.

E / FRAME

OUTER FRAME

Insert the hex driver through the blind holes to reach each BHCS and tighten.



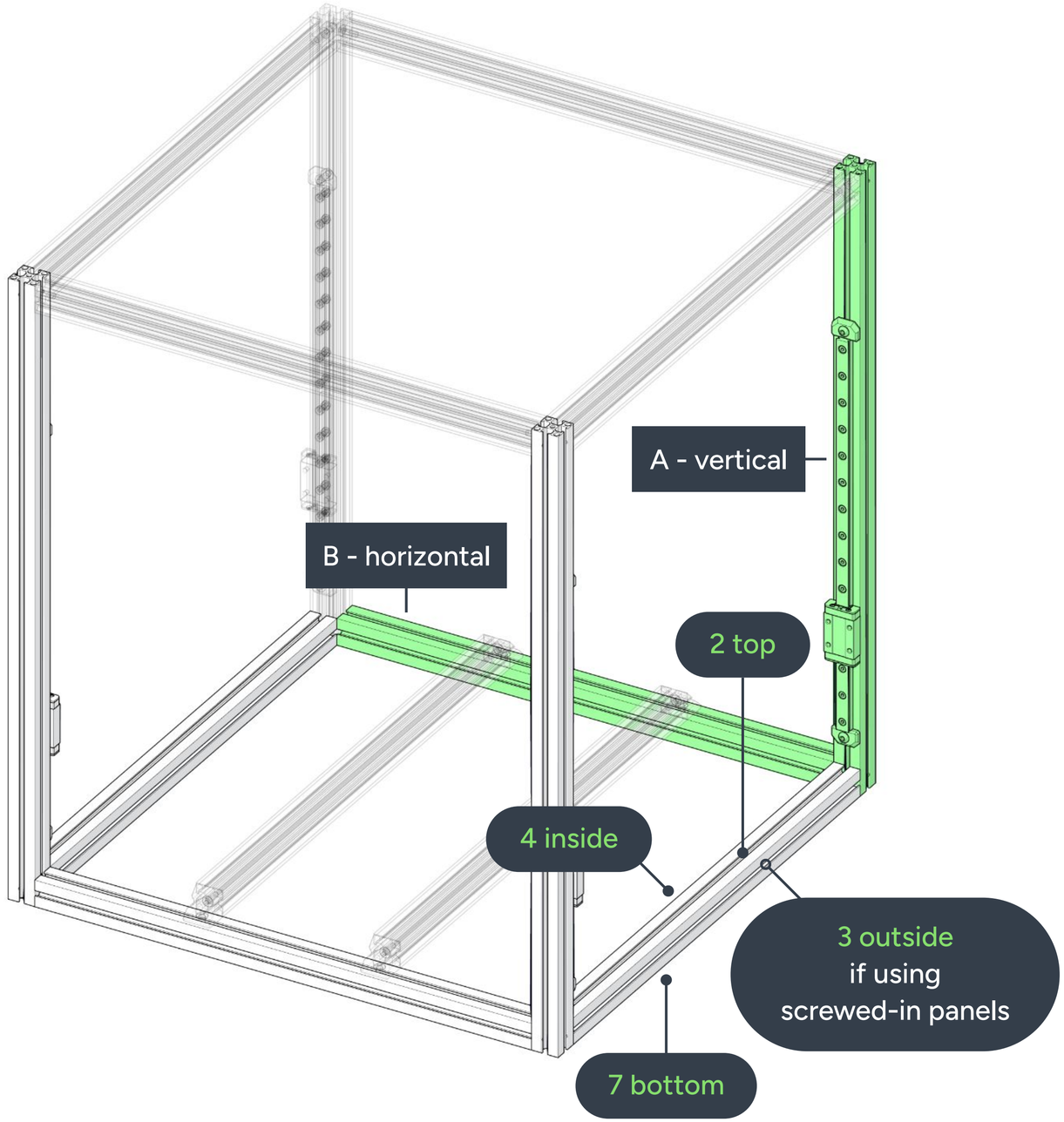
NOTE

Use the straight end of the hex driver for final tightening. Using the ball end may round out screws in blind holes.

E / FRAME

OUTER FRAME

Repeat this process until the bottom frame of four B horizontal extrusions and four A vertical extrusions is assembled.



NOTE

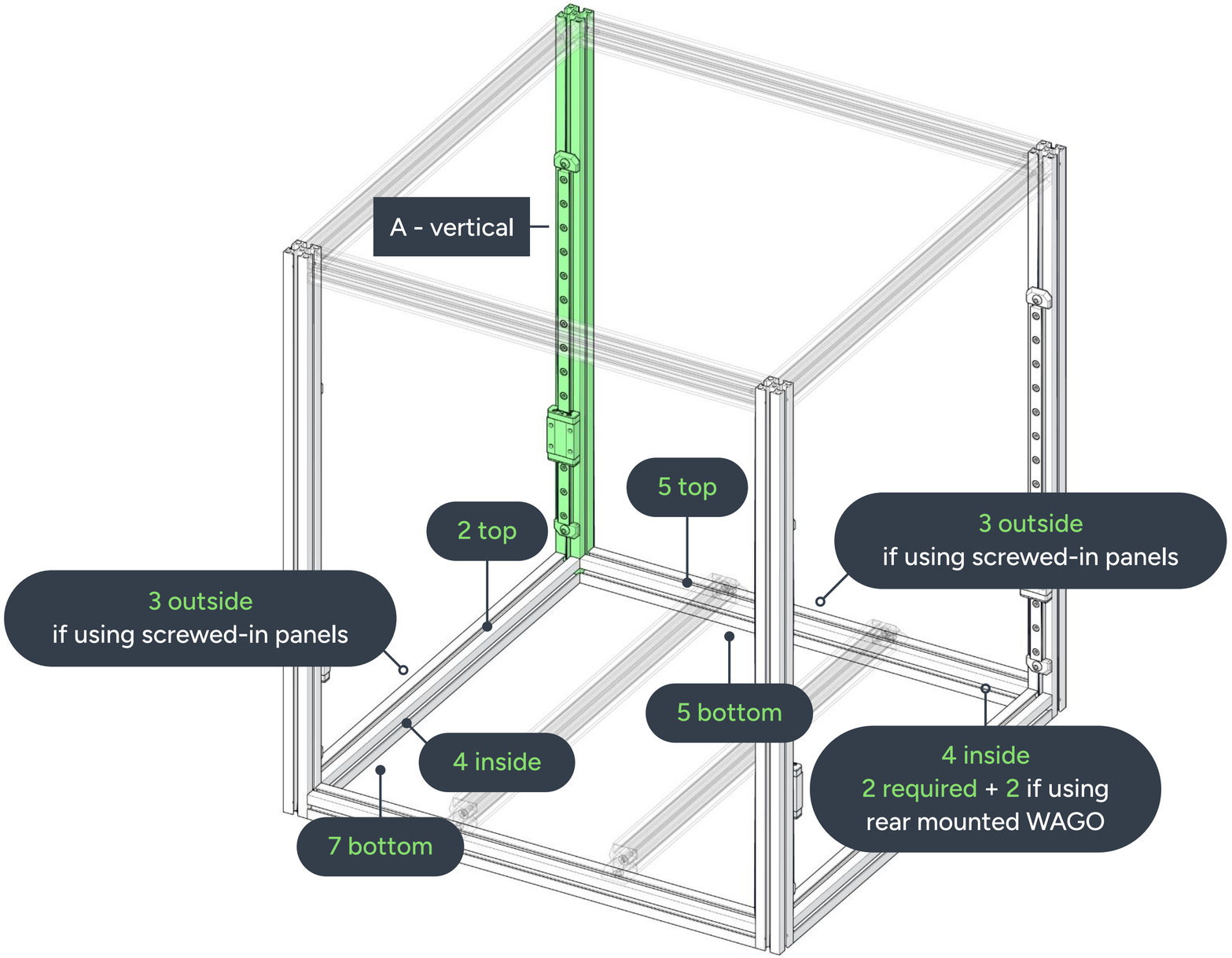
Don’t forget the preloaded nuts.

E / FRAME

OUTER FRAME

The four corners of the lower frame should now be assembled.

Check that the frame is square. The four B horizontal extrusions should sit flat on the reference surface without rocking.



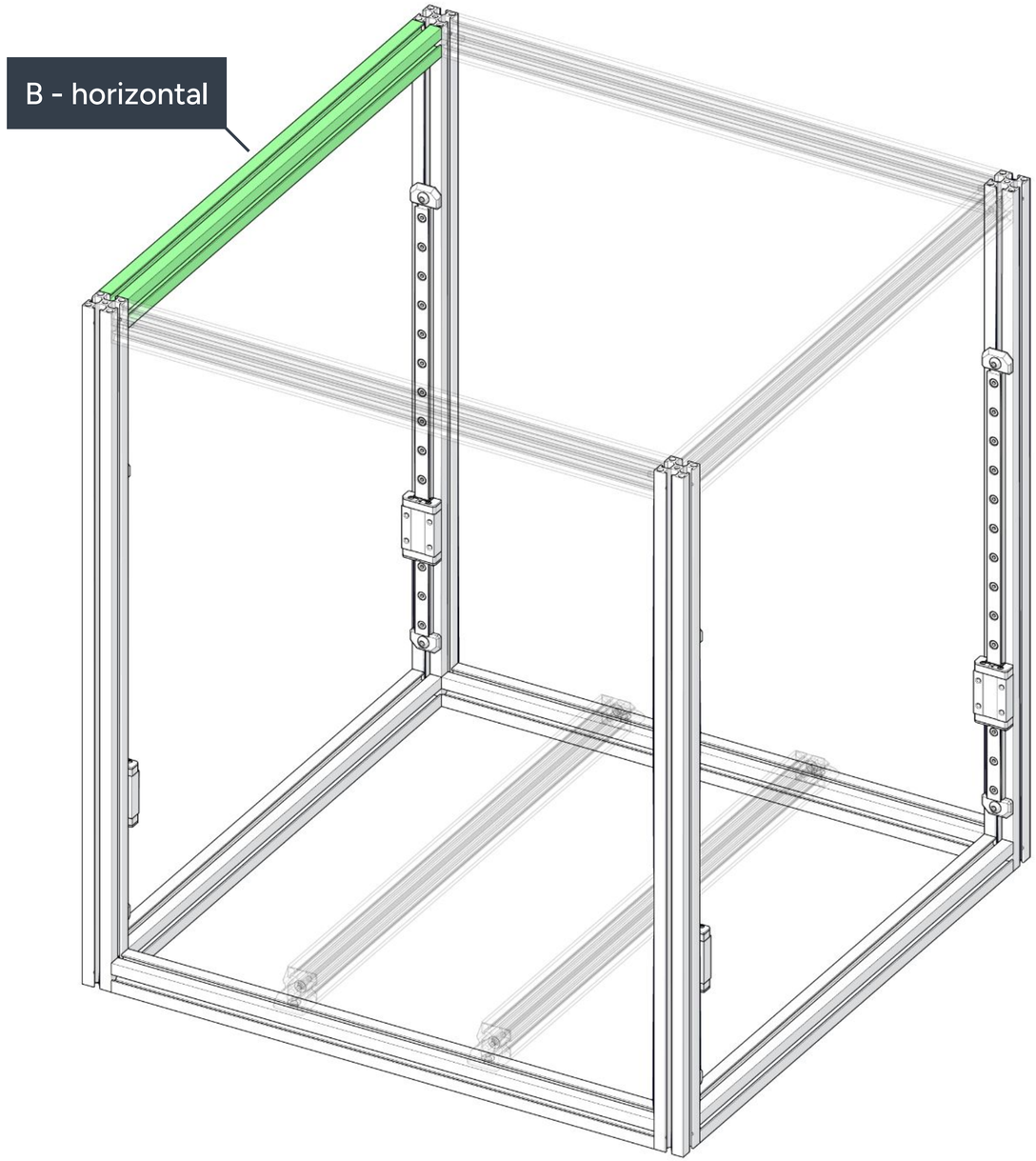
NOTE

Don’t forget the preloaded nuts.

E / FRAME

OUTER FRAME

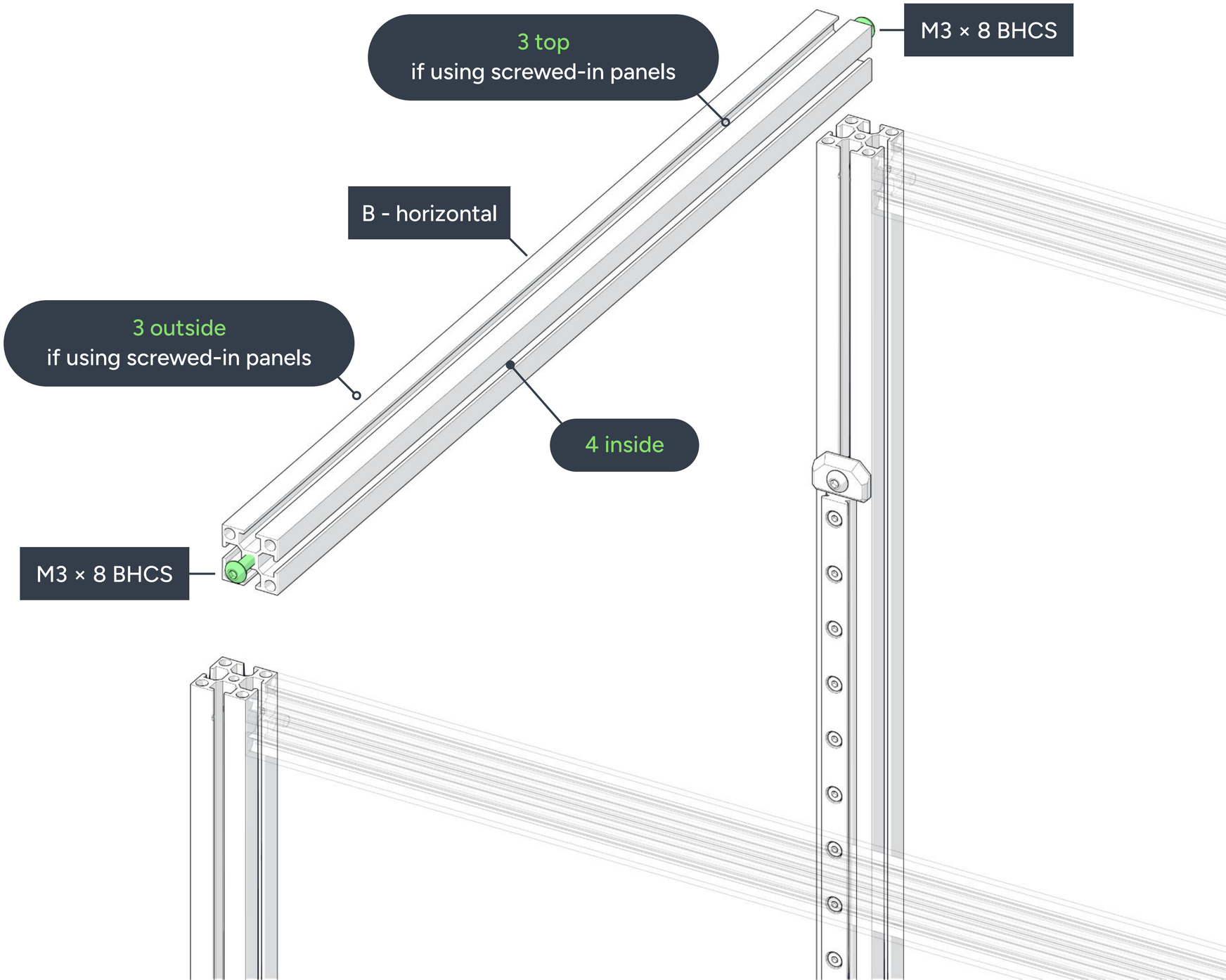
Now we will close up the top frame with more **B horizontal extrusions**.



E / FRAME

OUTER FRAME

Pre-install **two M3 × 8 BHCS** into the ends of a **B extrusion**, then slide the extrusion into place. You may need to gently spread the vertical extrusions to avoid marring.



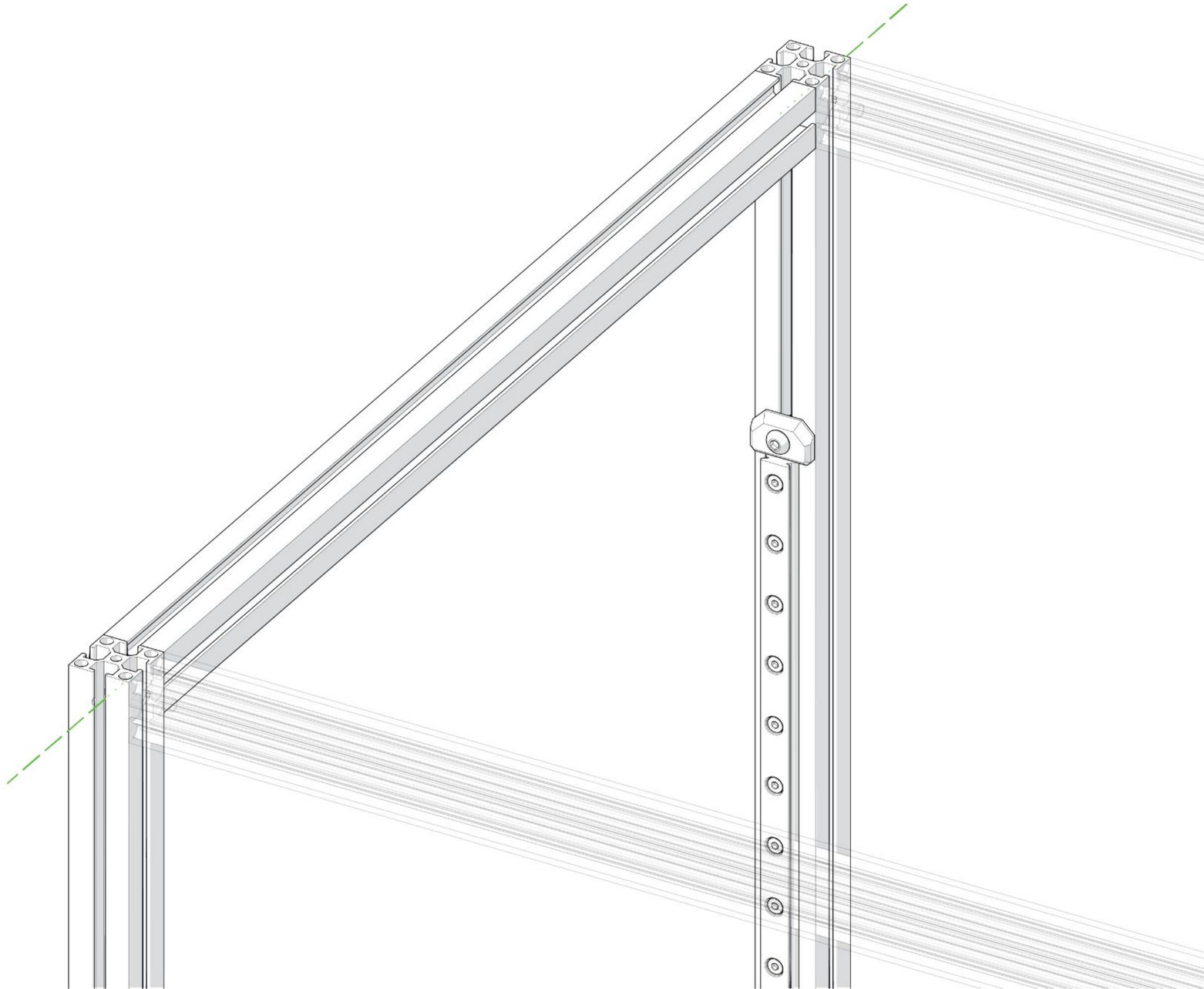
NOTE

Don’t forget the preloaded nuts.

E / FRAME

OUTER FRAME

Insert the hex driver through the blind holes to reach each BHCS and tighten.

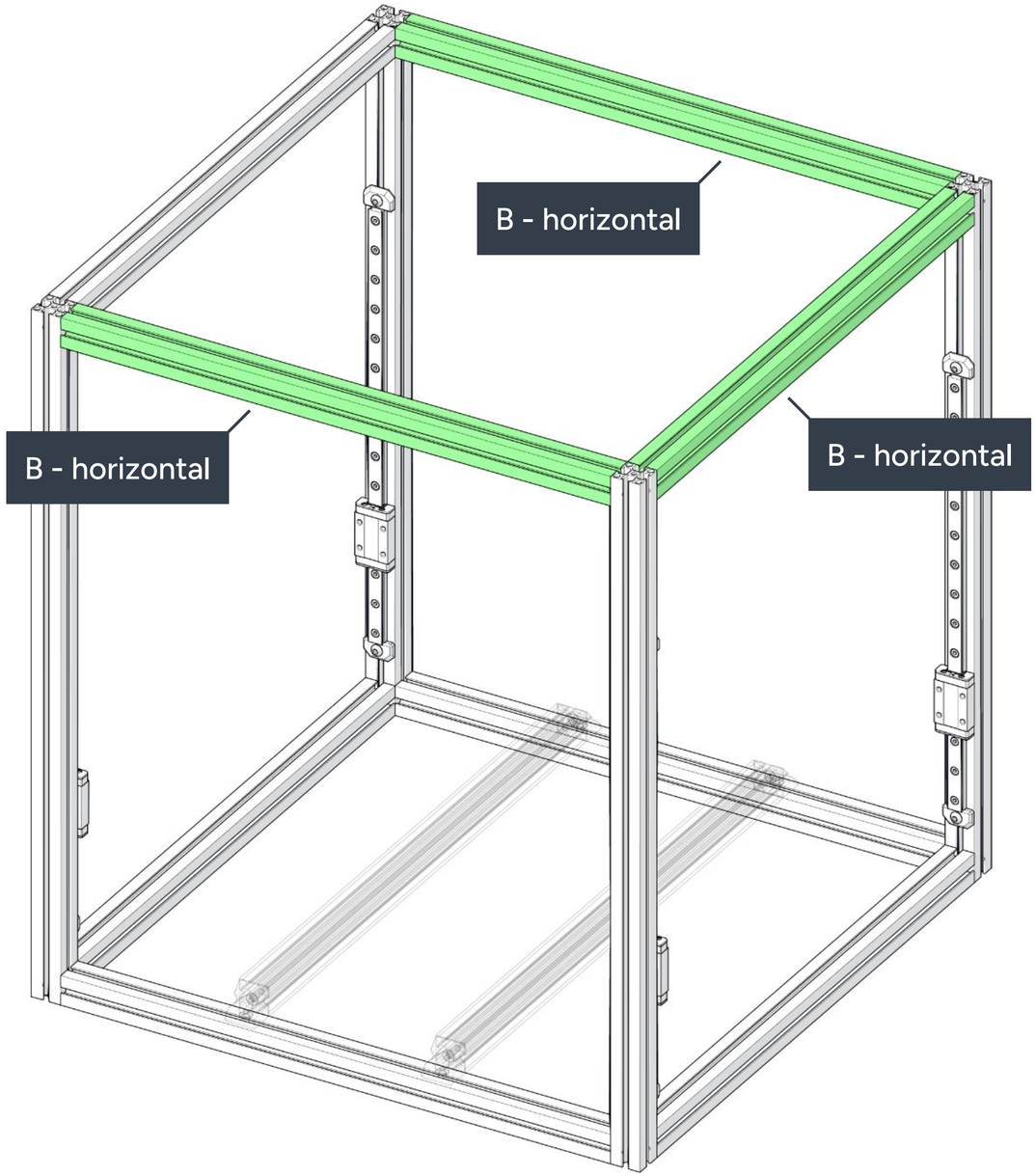


E / FRAME

OUTER FRAME

Repeat this process until the top frame of four B horizontal extrusions is assembled.

The following pages detail the preloaded nuts.



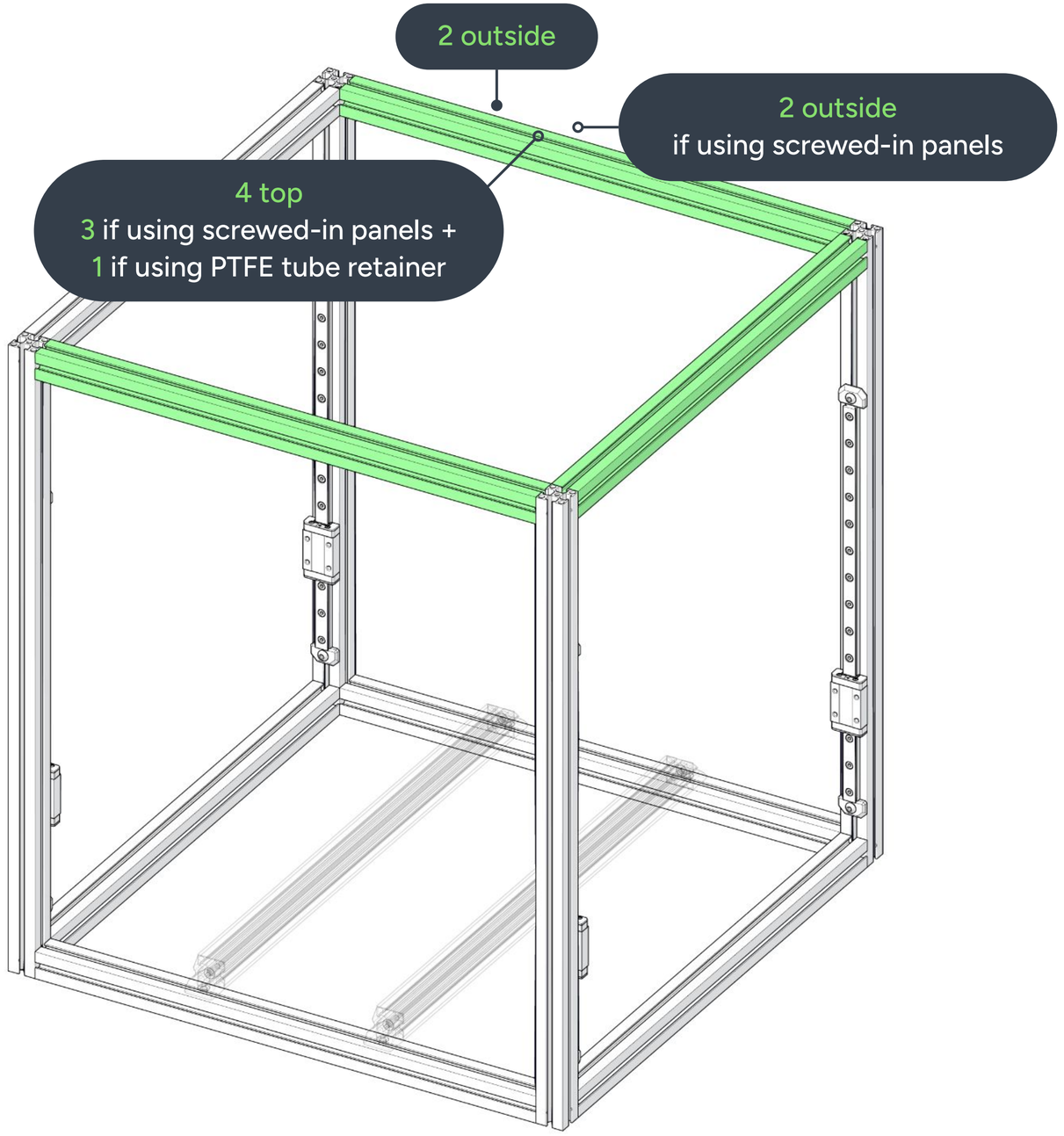
NOTE

Don’t forget the preloaded nuts.

E / FRAME

OUTER FRAME

Preloaded nuts for the rear
B horizontal extrusion.



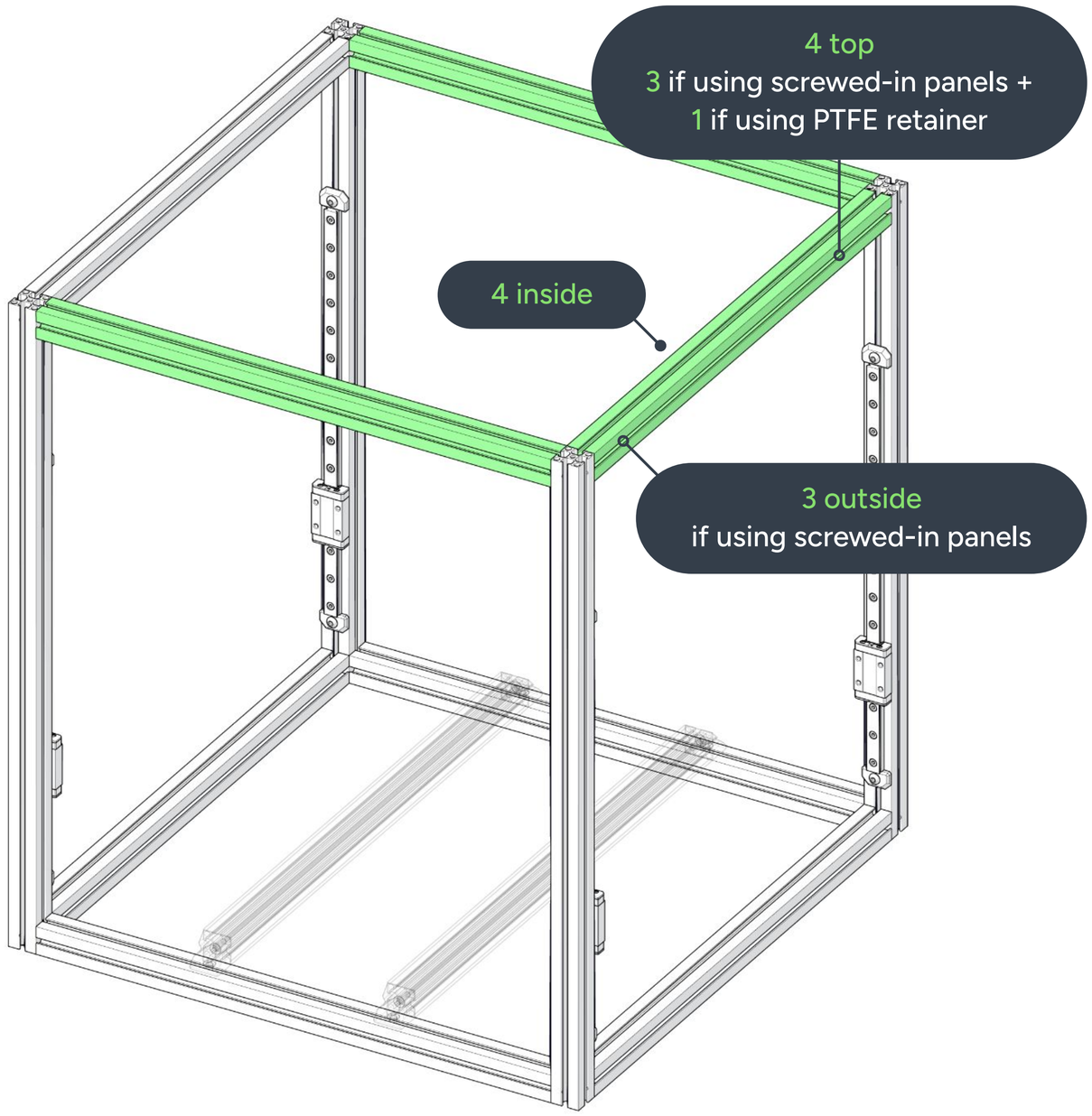
NOTE

Don’t forget the preloaded nuts.

E / FRAME

OUTER FRAME

Preloaded nuts for the right
B horizontal extrusion.



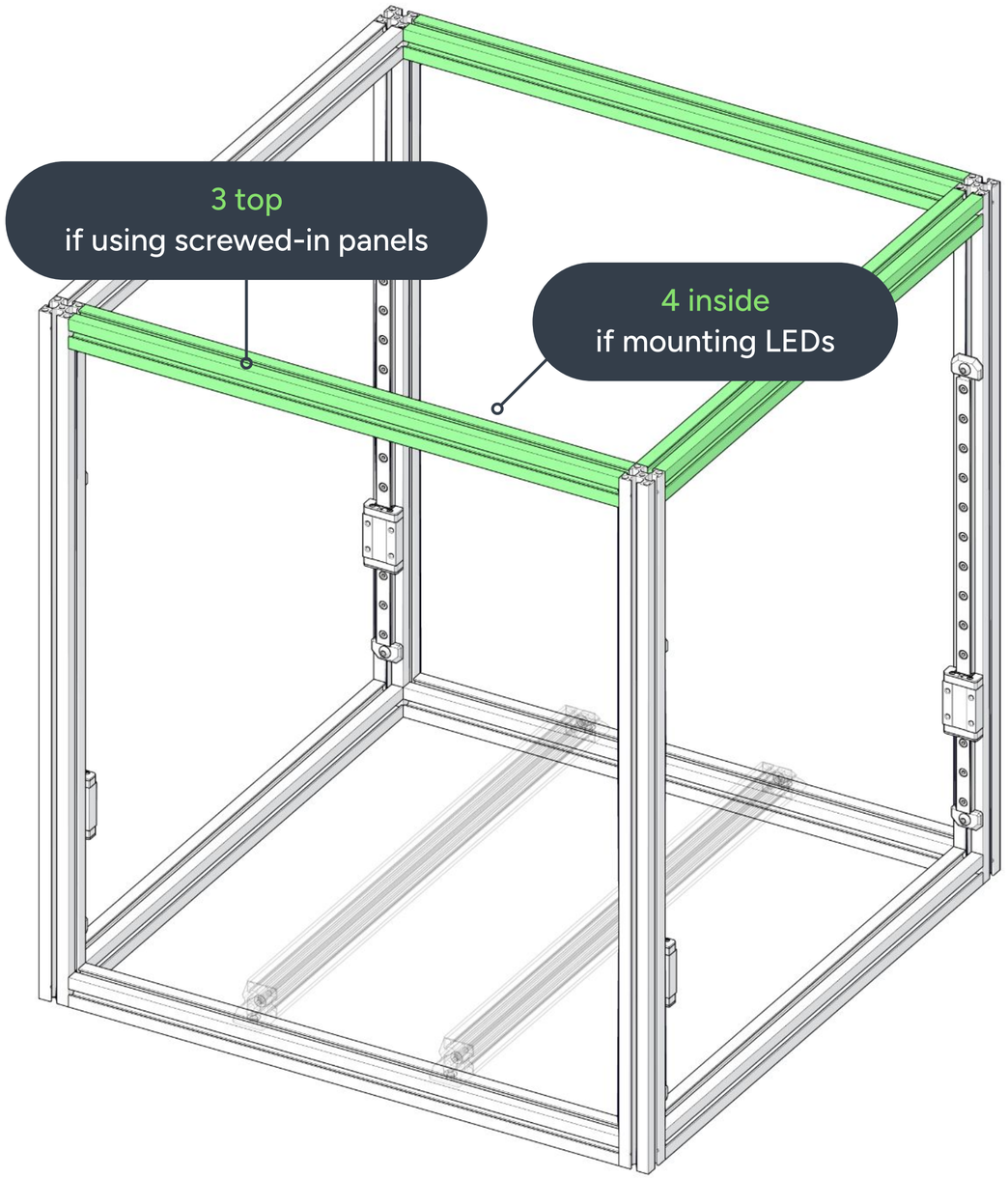
NOTE

Don’t forget the preloaded nuts.

E / FRAME

OUTER FRAME

Preloaded nuts for the front
B horizontal extrusion.



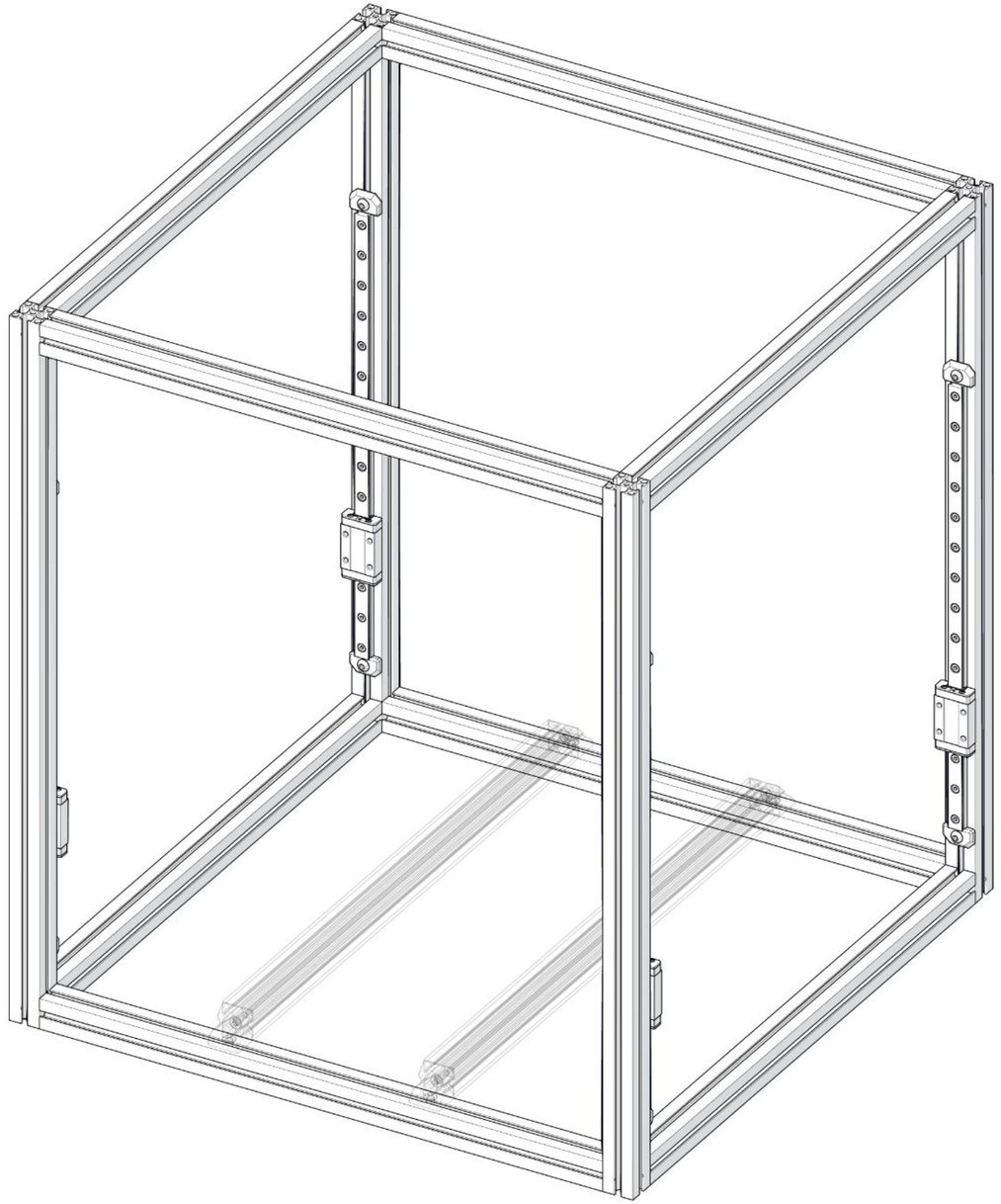
NOTE

Don’t forget the preloaded nuts.

E / FRAME

OUTER FRAME

The outer frame assembly is now complete.



NOTE

This is a good time to check the frame for square. Once additional parts are installed, correcting the frame becomes much more difficult.

A simple check is to measure the diagonals across the same face. If the measurements differ by less than 1 mm, the frame is sufficiently square.

CONCEPT

FRAME SQUARENESS

The frame defines the geometry of the entire printer. If it is not square, all motion components are forced out of alignment.

An unsquare frame can cause:

- Binding or rough linear rail movement
- Gantry tilt or racking
- Inconsistent belt tracking and tension
- Difficulty achieving accurate bed leveling

Target state:

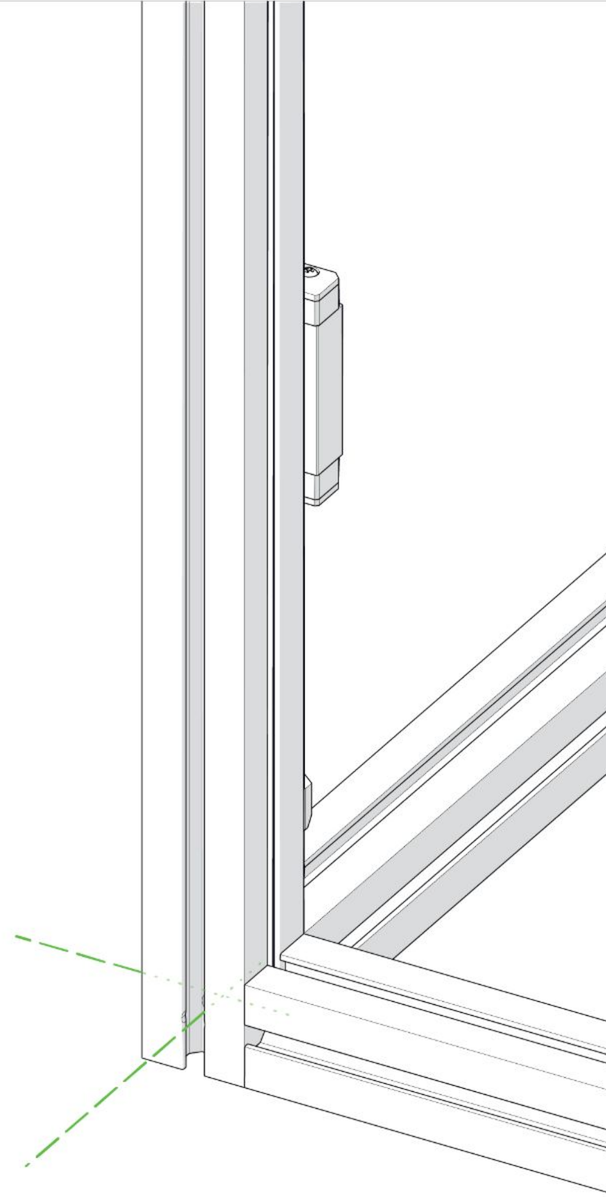
- Frame sits flat with no rocking
- Opposing diagonals on the same face match within ~1 mm

Common causes of misalignment:

- Tightening one corner fully before others
- Building on an uneven surface
- Blind joints pulling extrusions out of position during tightening

How to correct it:

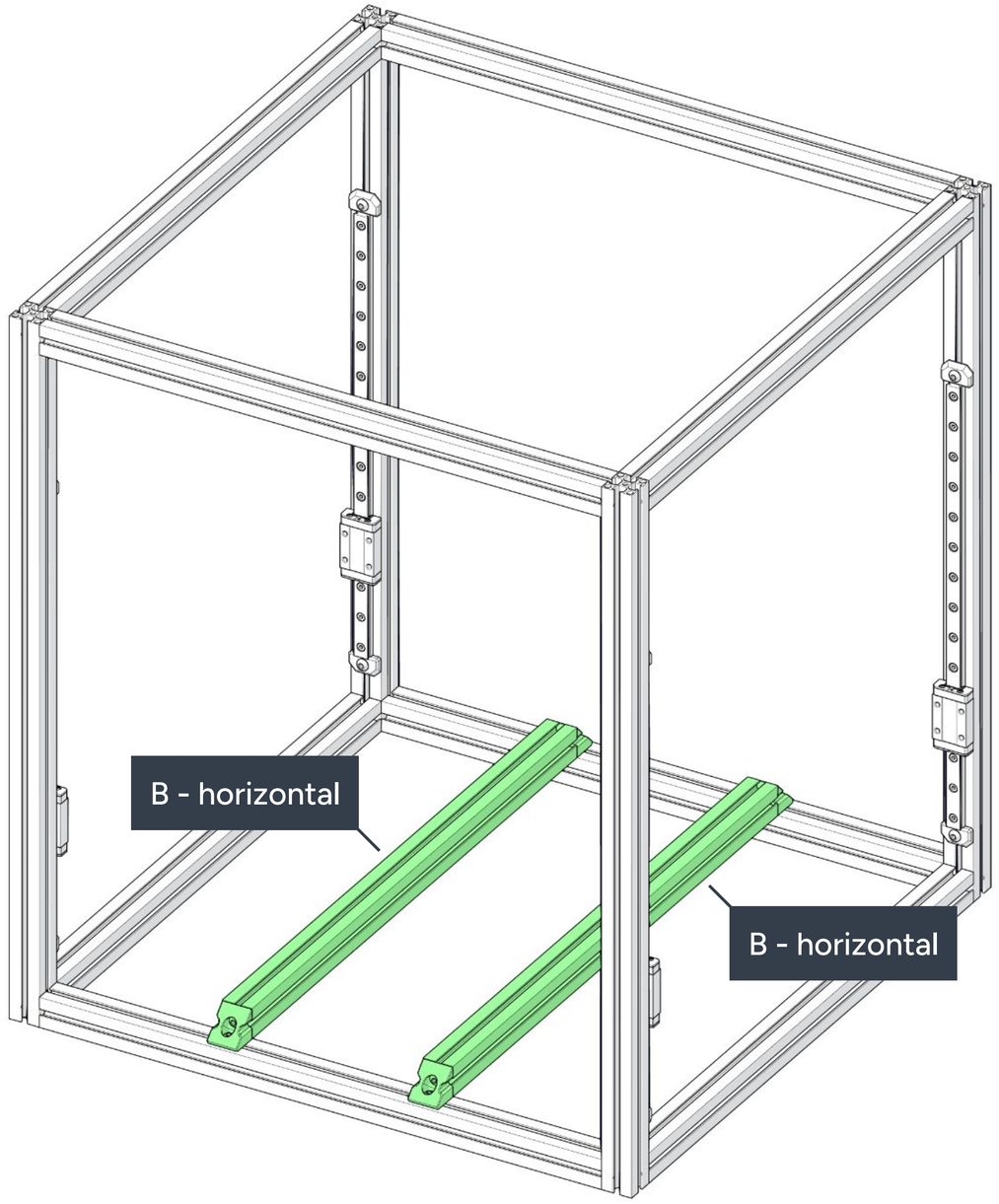
1. Loosen affected joints slightly
2. Re-seat the frame on a flat surface
3. Align corners and seat extrusions
4. Tighten gradually and evenly
5. Re-check diagonals after tightening



E / FRAME

BED EXTRUSIONS

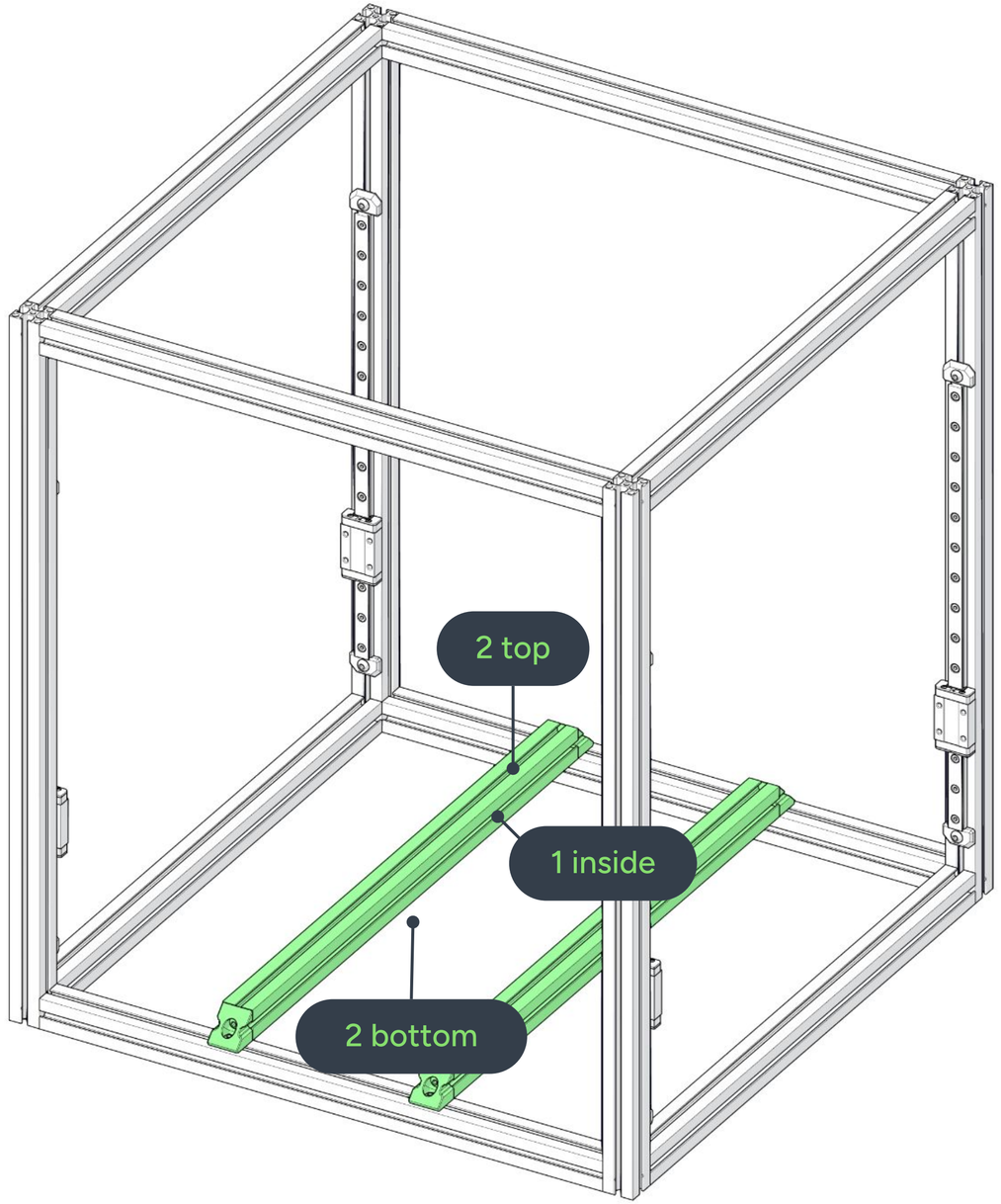
The final **two B horizontal extrusions** make up the bed mount.



E / FRAME

BED EXTRUSIONS

Preloaded nuts for the left bed extrusion.



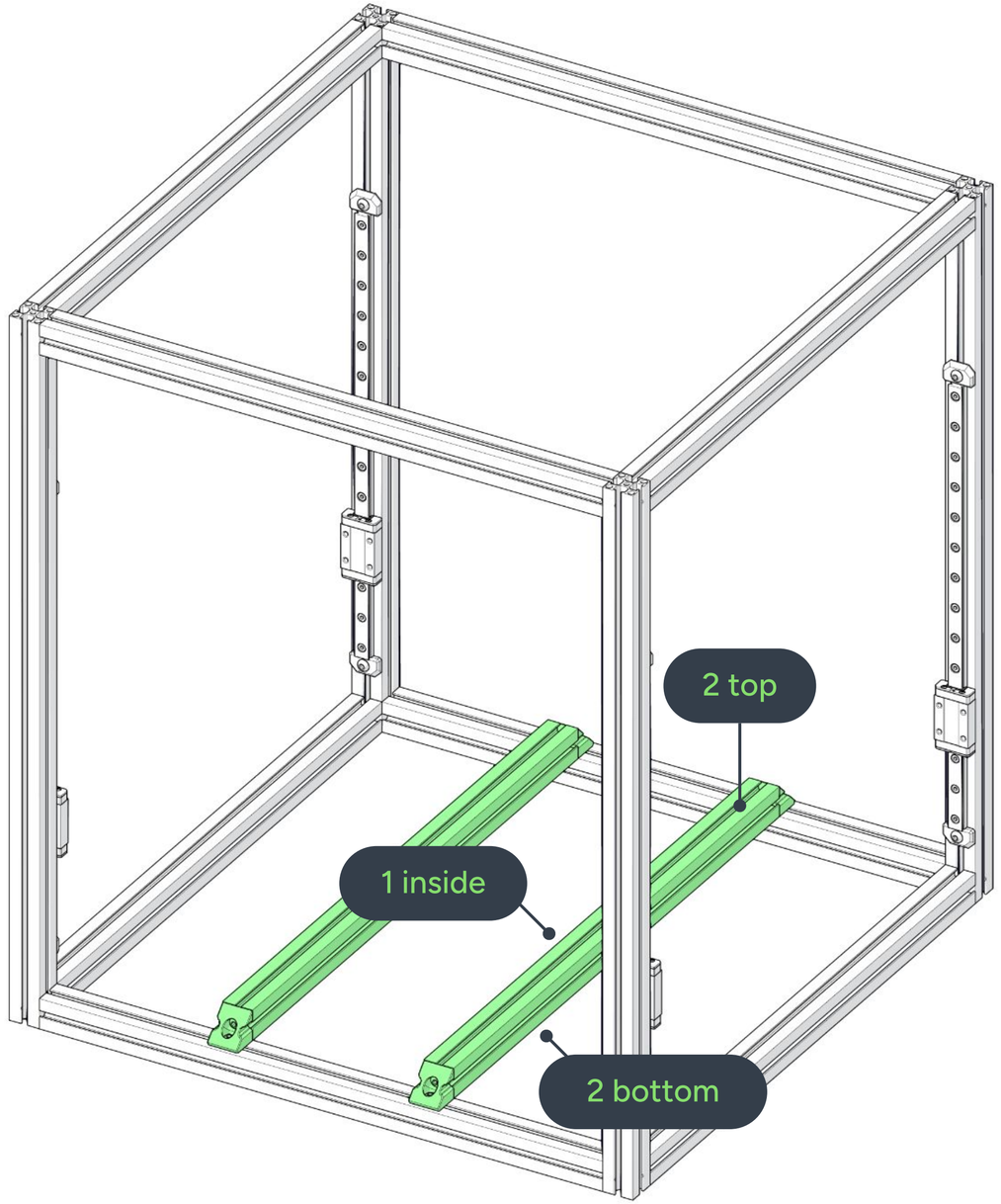
NOTE

Don’t forget the preloaded nuts.

E / FRAME

BED EXTRUSIONS

Preloaded nuts for the right bed extrusion.



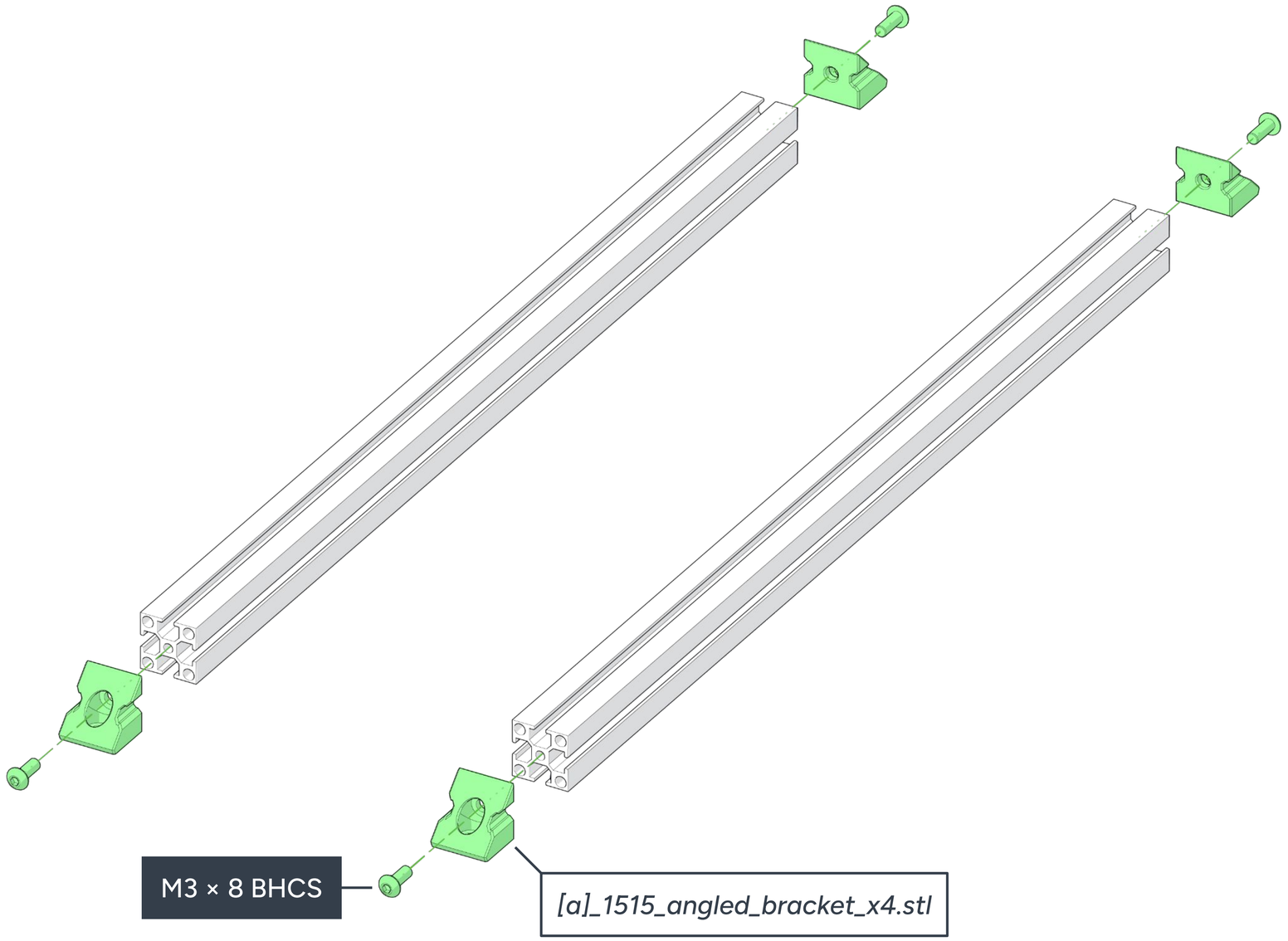
NOTE

Don’t forget the preloaded nuts.

E / FRAME

BED EXTRUSIONS

The two bed extrusions use printed corner brackets to attach to the outer frame. Secure them to the ends of the bed extrusions with **M3 × 8 BHCS**.



NOTE

Some kits include bed extrusions mounted with blind screws. **If so, the printed parts and fasteners on this page will not be needed.**

Don’t forget the preloaded nuts.

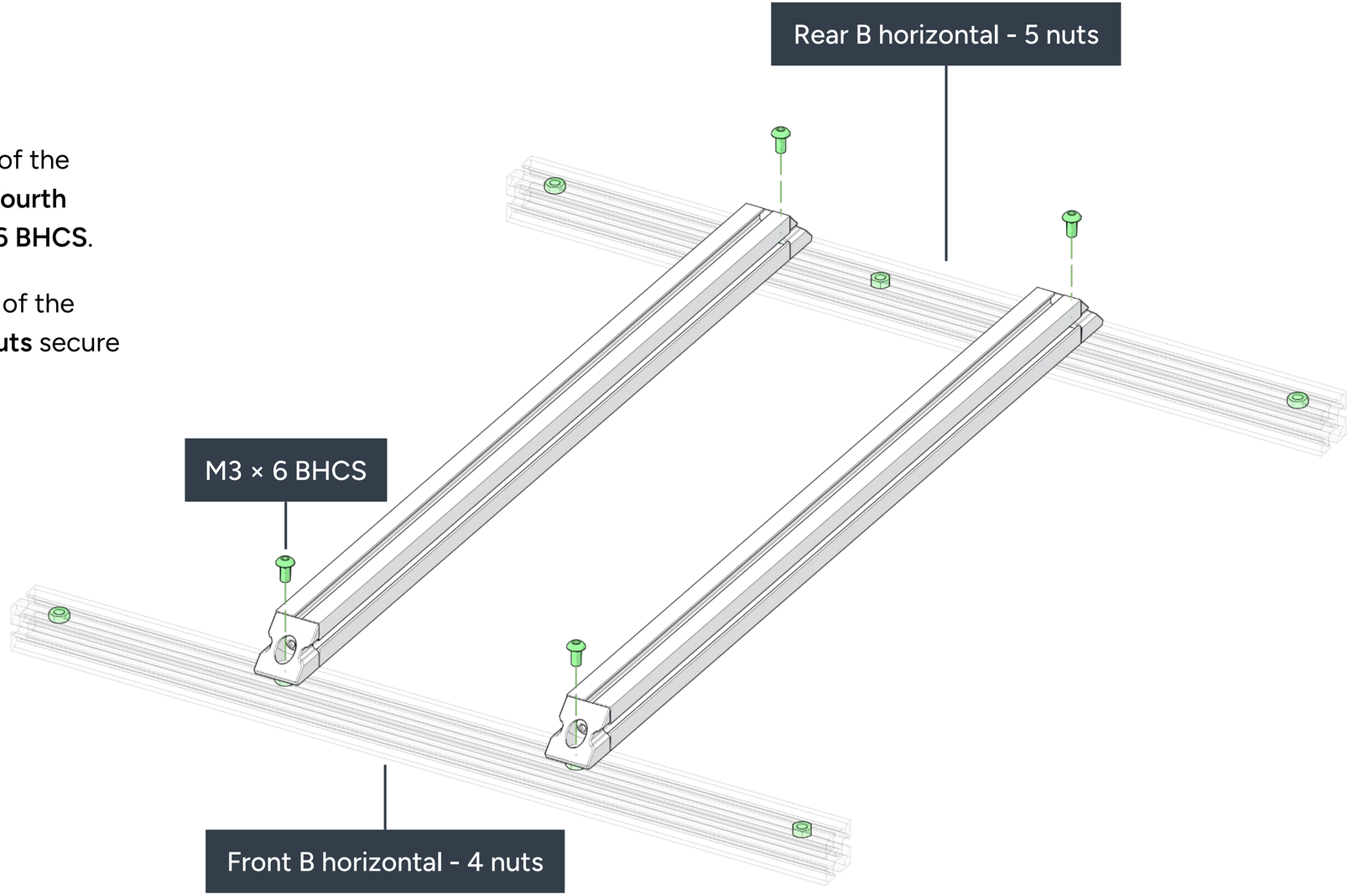
E / FRAME

BED EXTRUSIONS

There are five preloaded nuts in the top slot of the rear B horizontal extrusion. The **second and fourth nuts** secure the bed B extrusions with **M3 × 6 BHCS**.

There are four preloaded nuts in the top slot of the front B horizontal extrusion. The **two inner nuts** secure the bed B extrusions with M3 × 6 BHCS.

Do not tighten them down yet.



NOTE

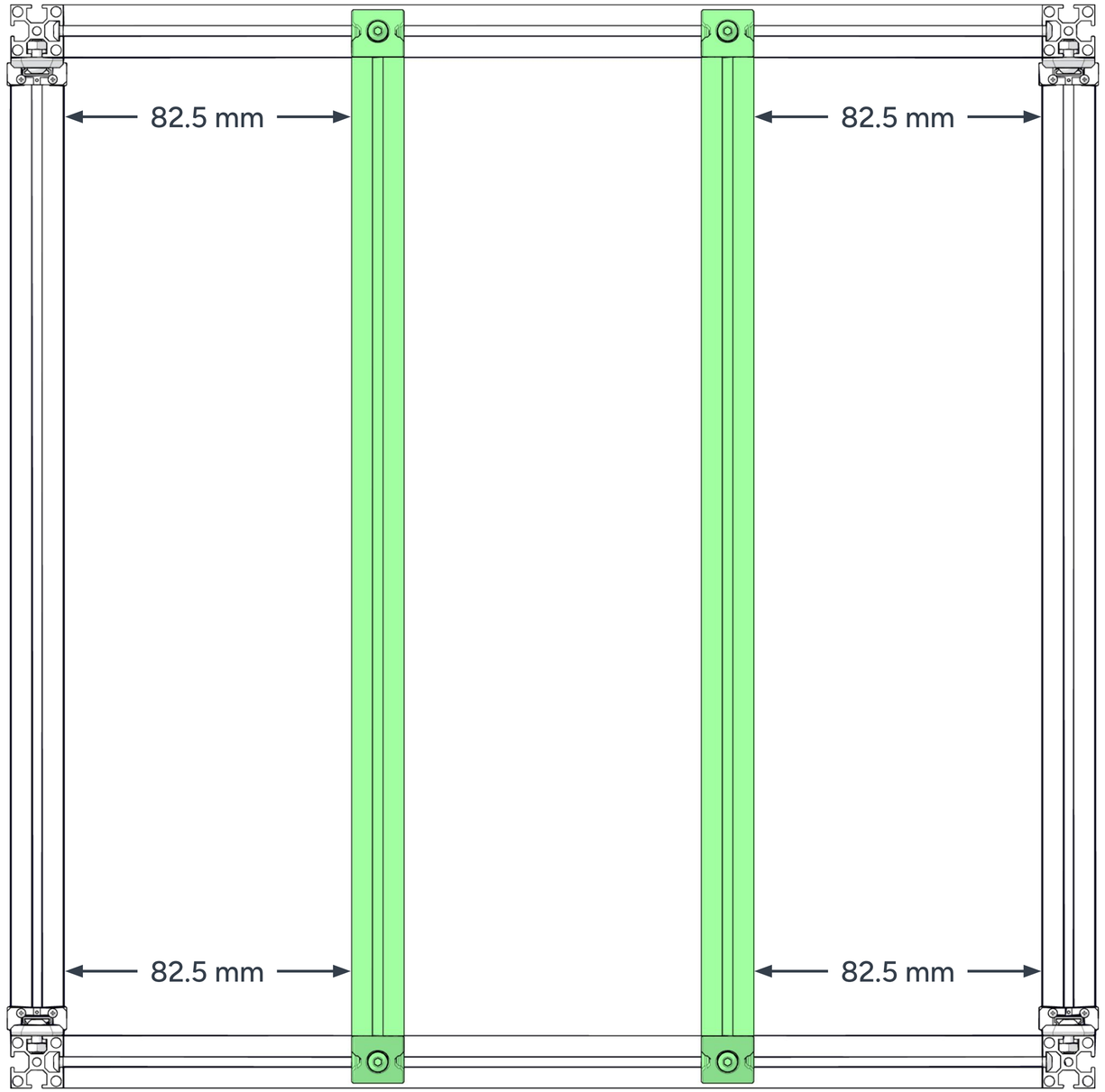
Some kits include bed extrusions mounted using blind screws. **If so, attach them to the same nuts using blind joints with M3 × 6 BHCS.**

E / FRAME

BED EXTRUSIONS

For proper placement, the bed extrusions need to sit inset 82.5 mm from the corner extrusions.

Once positioned correctly, tighten down the screws.



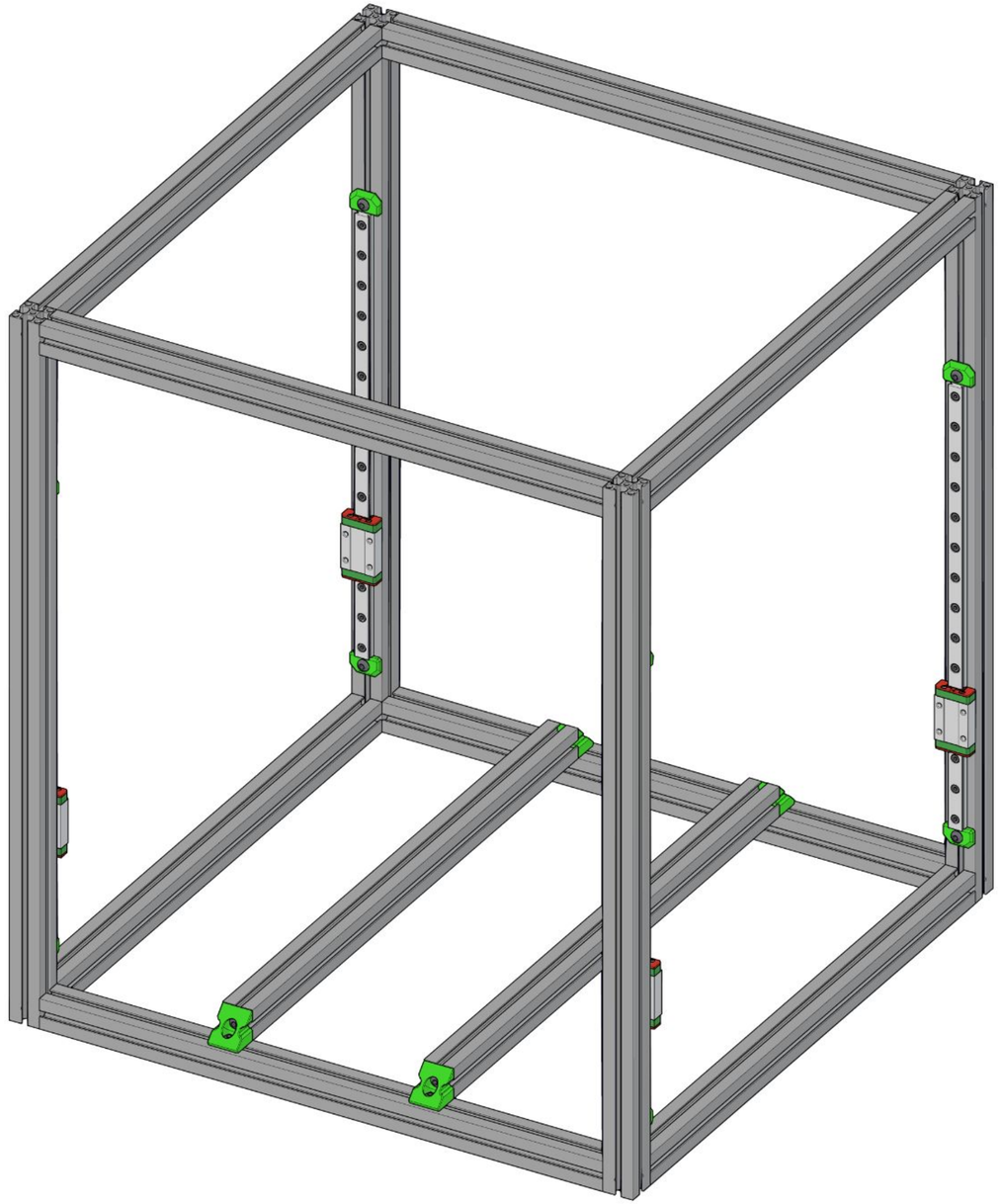
E / FRAME

BUILDER CHECK

The initial frame assembly is now complete.

CHECKLIST

- ▢ Frame sits flat on table
- ▢ Linear rails face inward
- ▢ Frame diagonals within 1 mm
- ▢ All blind joints tightened
- ▢ Linear rails are 36 mm from table top
- ▢ Bed extrusions inset 82.5 mm from corners



E / FRAME

DECK PANEL

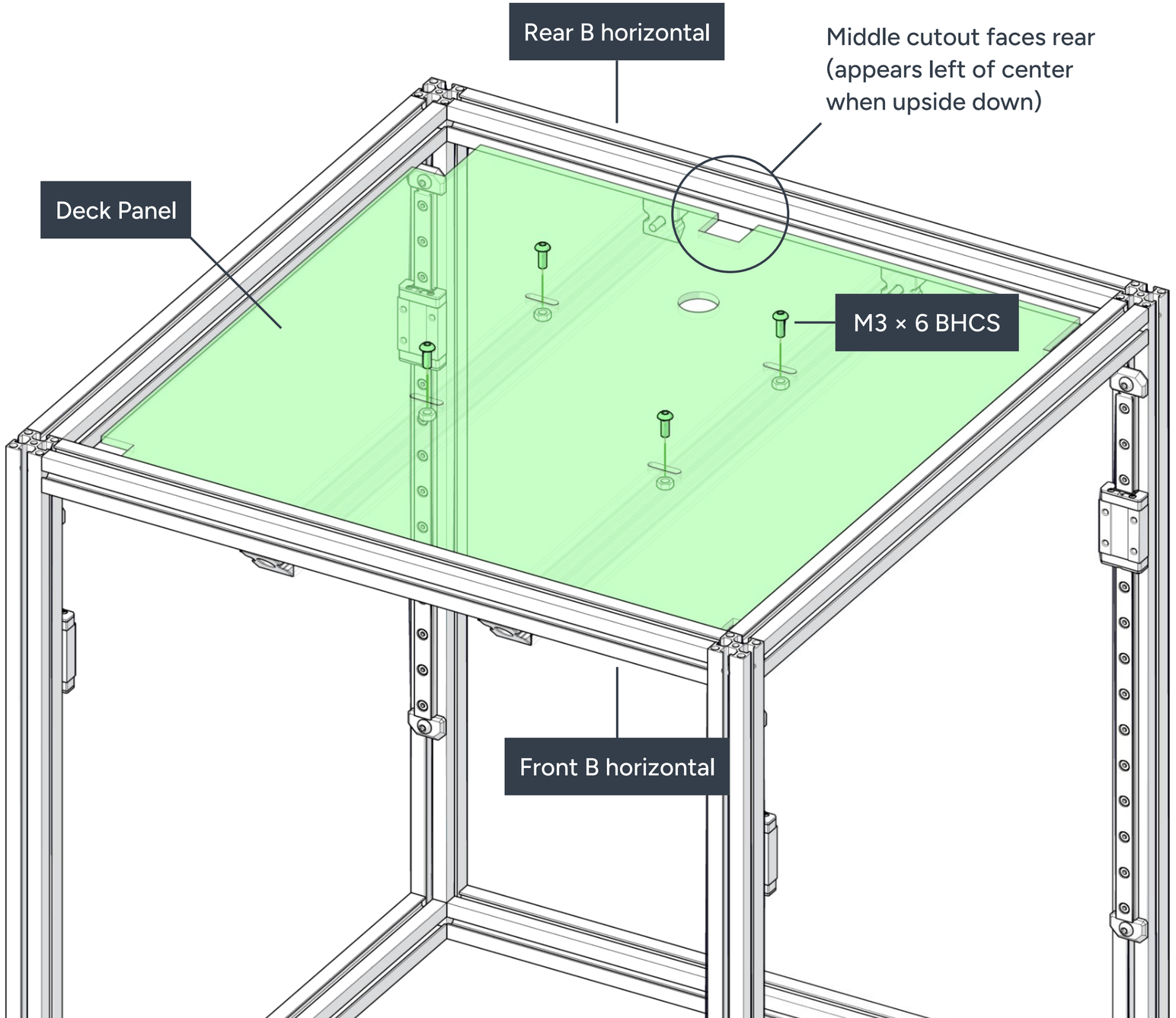
Carefully flip the printer upside down.

Each bed extrusion should have two preloaded M3 hex nuts in the bottom slots.

Place the **deck panel**, ensuring the cutouts are oriented correctly.

The middle cutout must face the rear of the printer.

Secure the deck panel with **four M3 × 6 BHCS** to the two bed extrusions' bottom preloaded M3 hex nuts.

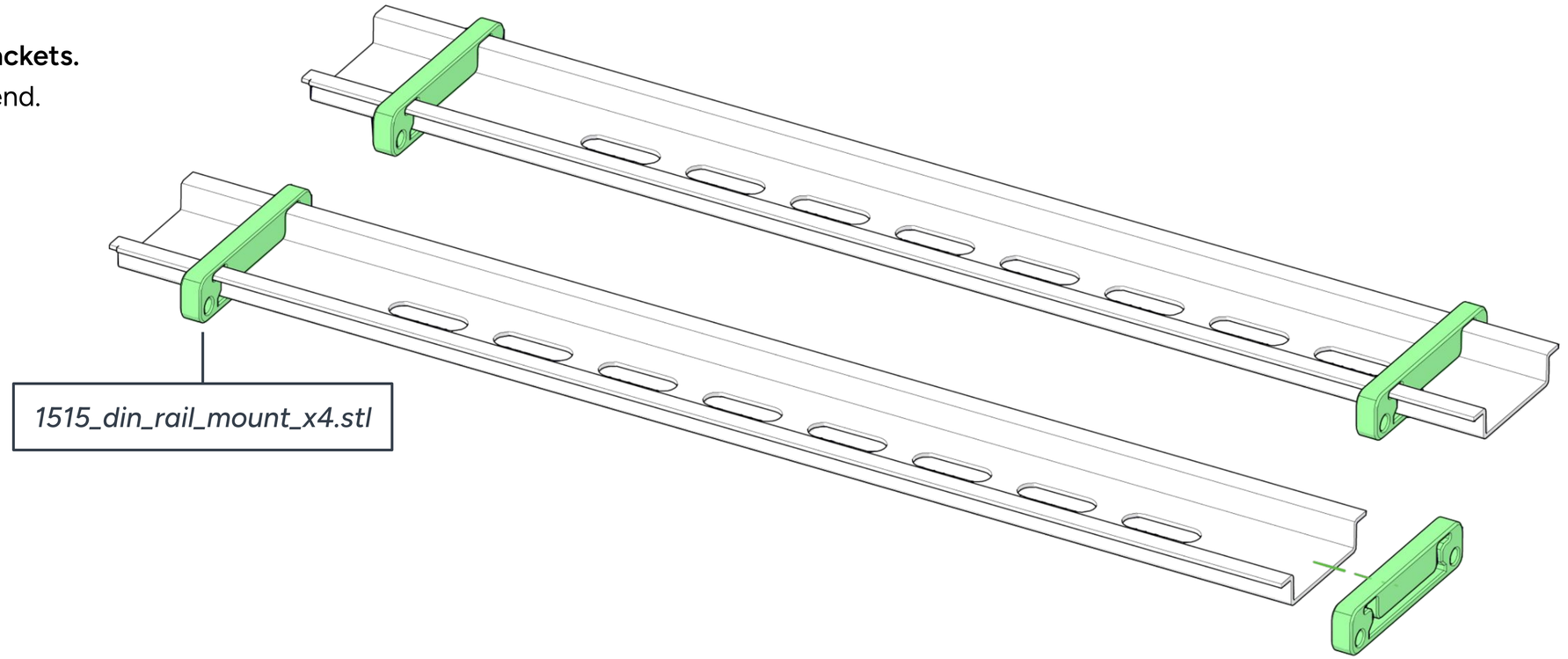


E / FRAME

DIN RAILS

These **DIN rails** are mounted below the deck panel and hold your electronics.

Each rail uses **two printed brackets.** Slide one bracket onto each end.

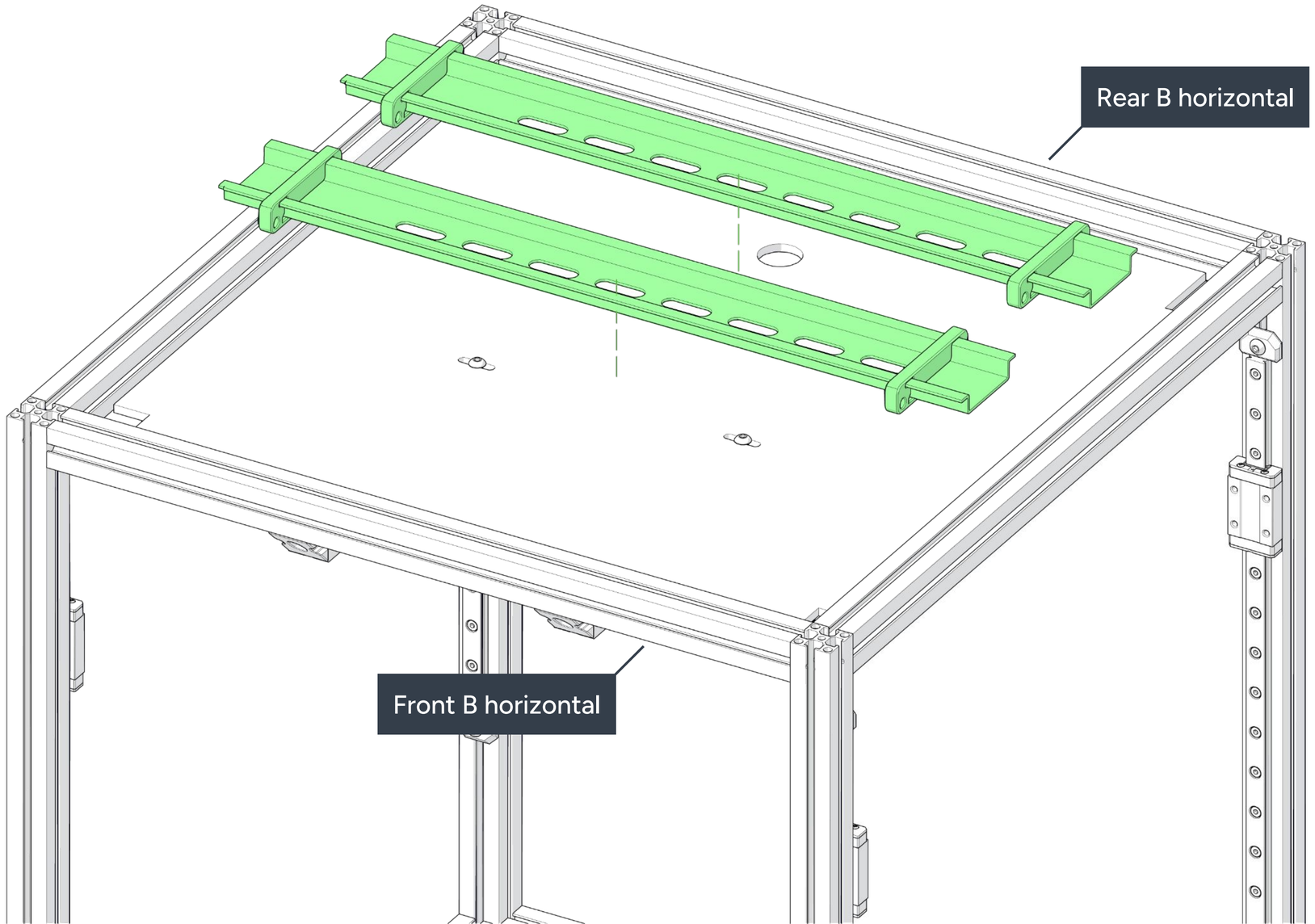


E / FRAME

DIN RAILS

Position the rails side-to-side within the frame.

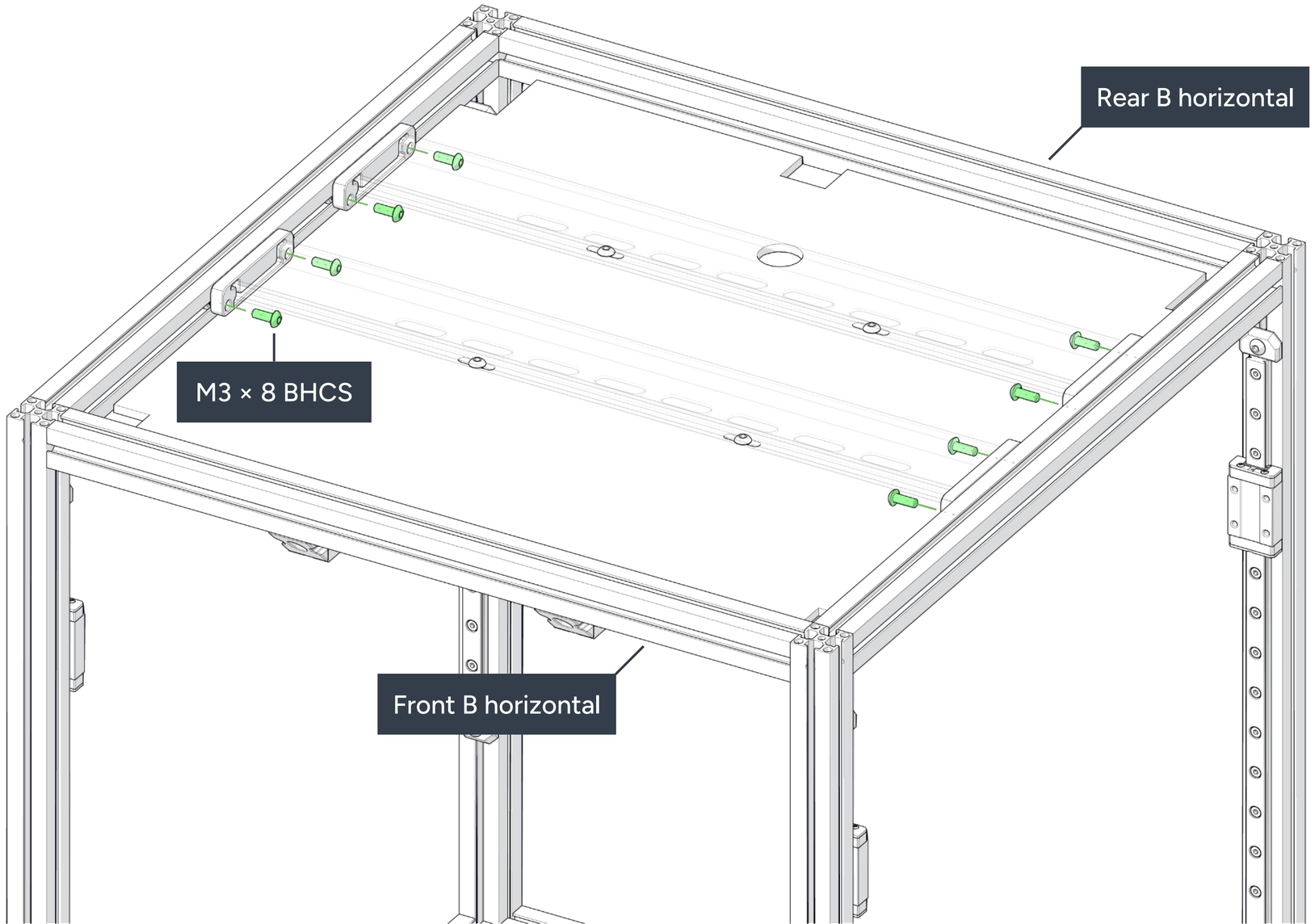
Final spacing depends on the electronics being installed.



E / FRAME

DIN RAILS

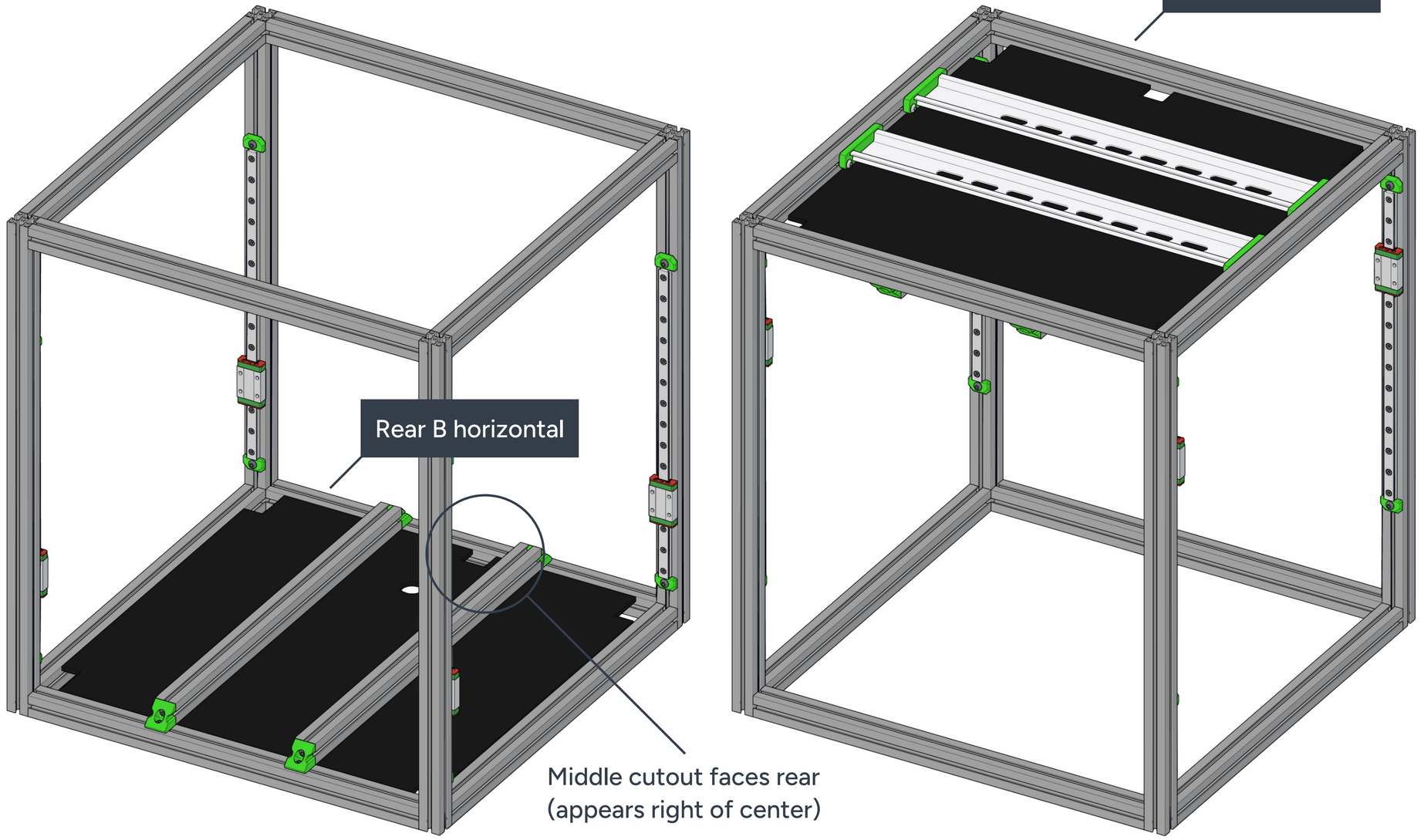
Slide the printed brackets against the extrusion and secure to the preloaded hex nuts with **M3 × 8 BHCS**.



E / FRAME

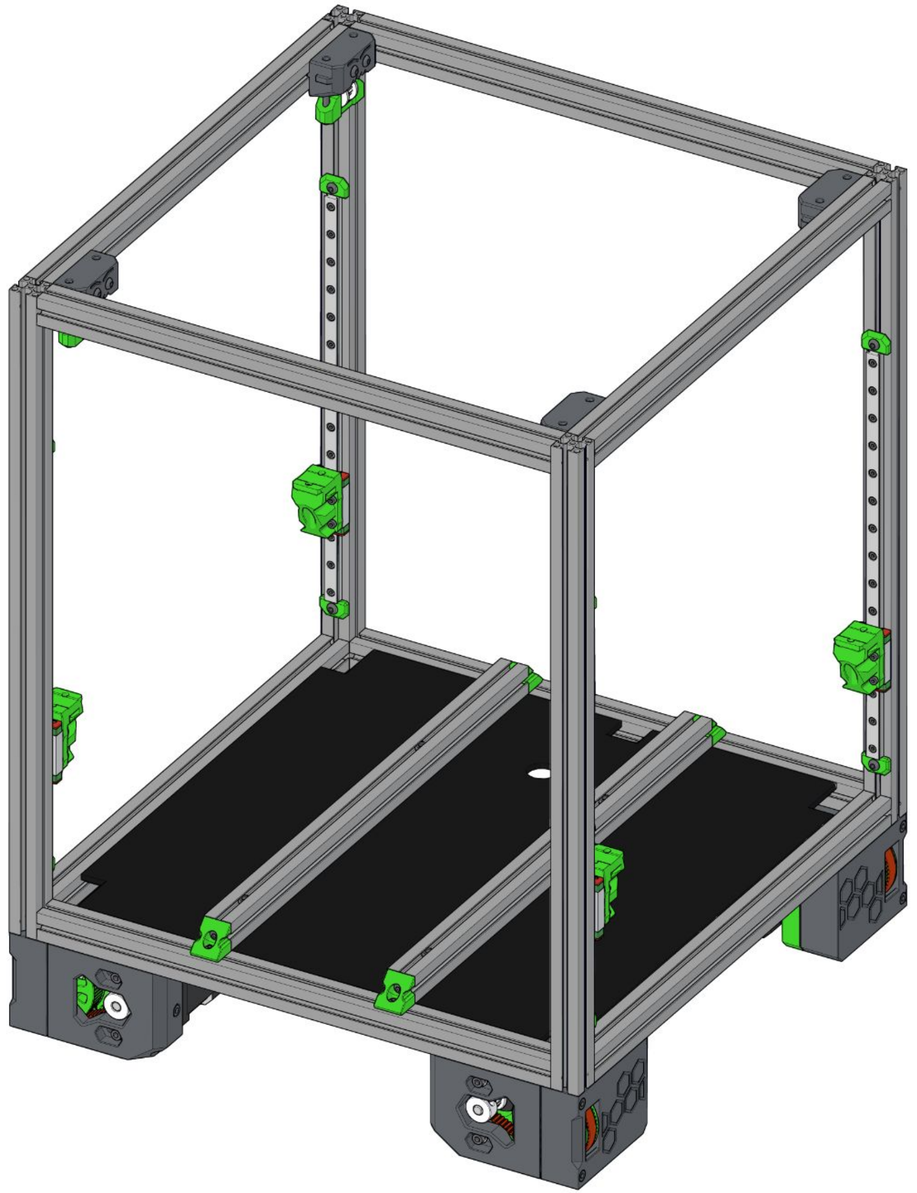
BUILDER CHECK

Your assembly should match these views.



F / Z-AXIS

The flying gantry is a defining feature of the printer. Four independent belted Z drives control each corner, allowing the gantry to automatically level itself relative to the rigidly mounted bed.



HARDWARE

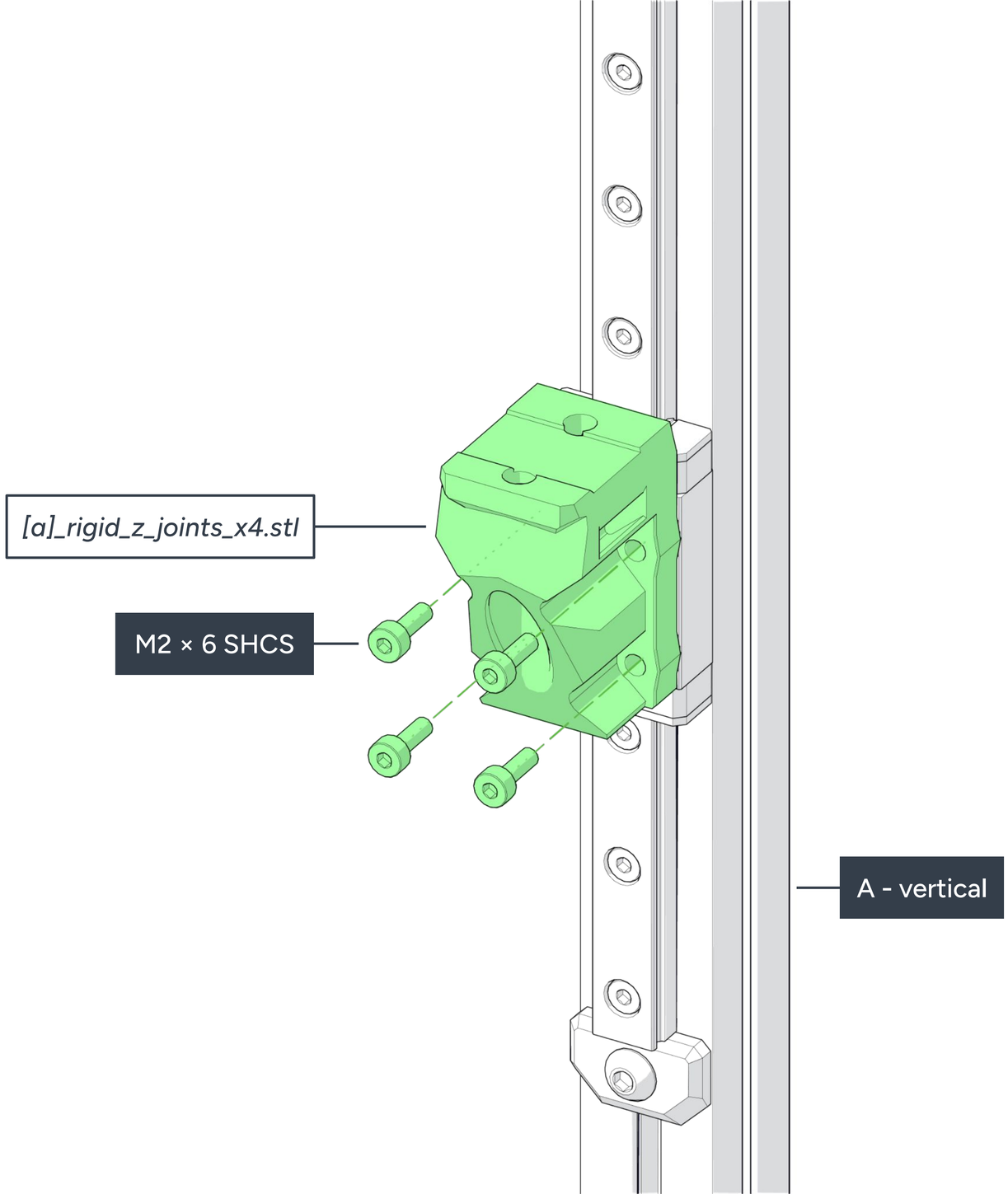
4	NEMA14 or NEMA17 motors
4	2GT 152 closed loop belts
8	2GT 16T pulleys
4	5 × 47 mm D shafts
8	625-2RS bearings
16	M3 × 6 set screws
12	M5 × 8 × 1 mm shims
32	M3 × 8 BHCS
16	M2 × 6 SHCS
12	M3 × 16 SHCS
8	M3 × 20 SHCS
8	M3 × 25 SHCS
12	M3 × 40 SHCS
8	M3 × 7 × 0.5 mm washers

F / Z-AXIS

Z JOINTS

Z joints are available in two variants: rigid-mounted or KGLM spherical bearings. This manual covers the rigid-mounted version.

Install ***[a]_rigid_z_joints_x4.stl*** onto the MGN7H carriages on the four A vertical assemblies using **four M2 × 6 SHCS** each.

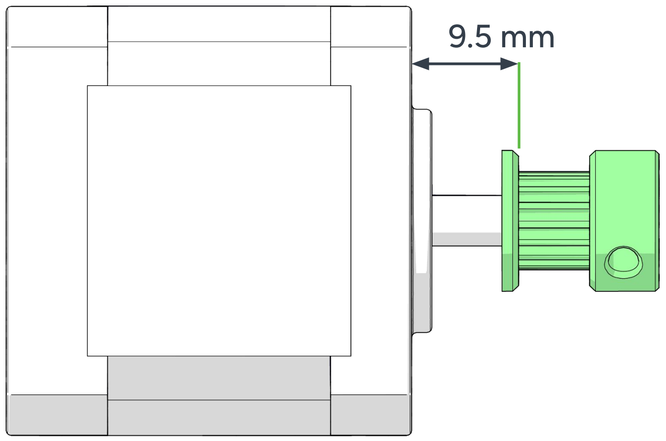


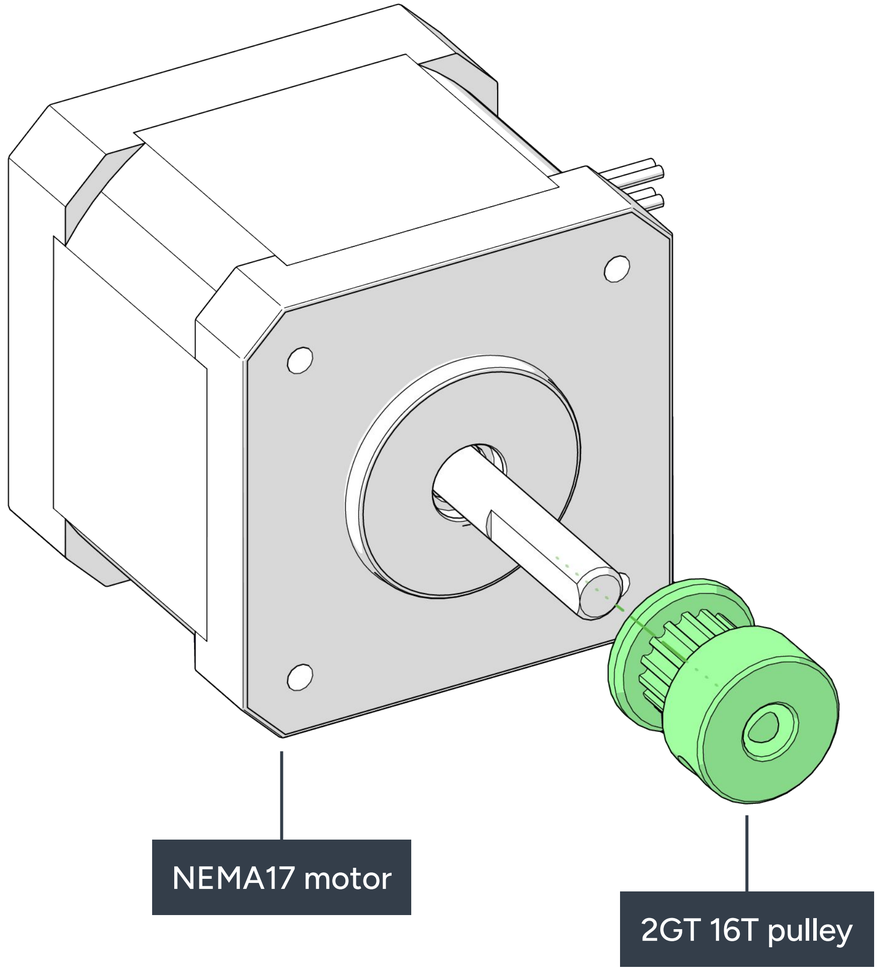
F / Z-AXIS

Z DRIVES - MOTOR ASSEMBLY

Each Z drive uses a **NEMA14 or NEMA17 motor**. Install a **2GT 16T pulley** onto the D shaft, positioned 9.5 mm from motor face to pulley teeth. Ensure the set screw aligns with the flat of the shaft. Apply threadlocker and tighten with a hex driver (no ball-end). This ensures correct engagement between pulley and belt.

Repeat for all four motors.





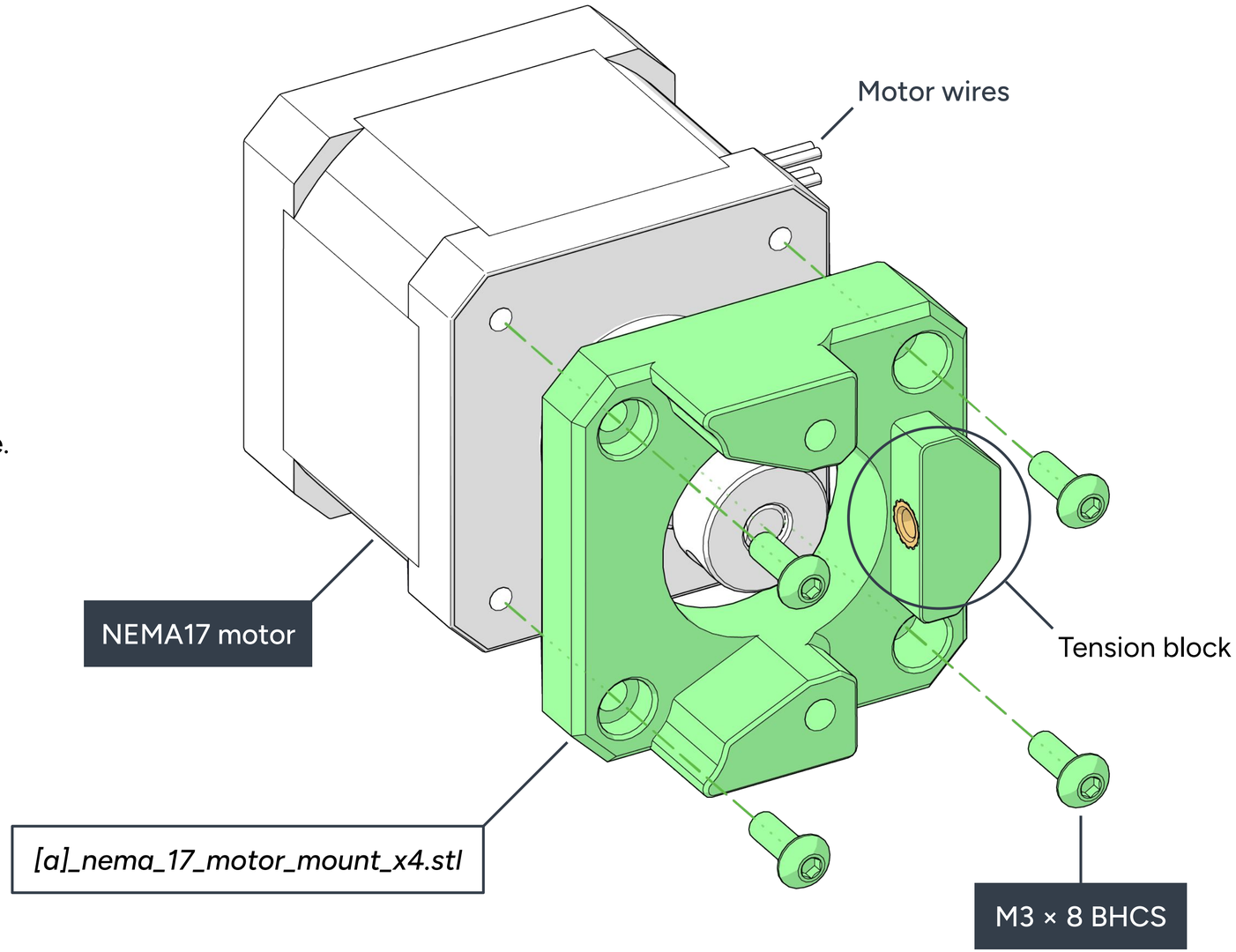
F / Z-AXIS

Z DRIVES - MOTOR ASSEMBLY

Attach ***[a]_nema_17_motor_mount_x4.stl*** to each motor with **four M3 × 8 BHCS**.

Orient the motor so the wires exit on the same side as the tension block.

You should have four motor assemblies when complete.

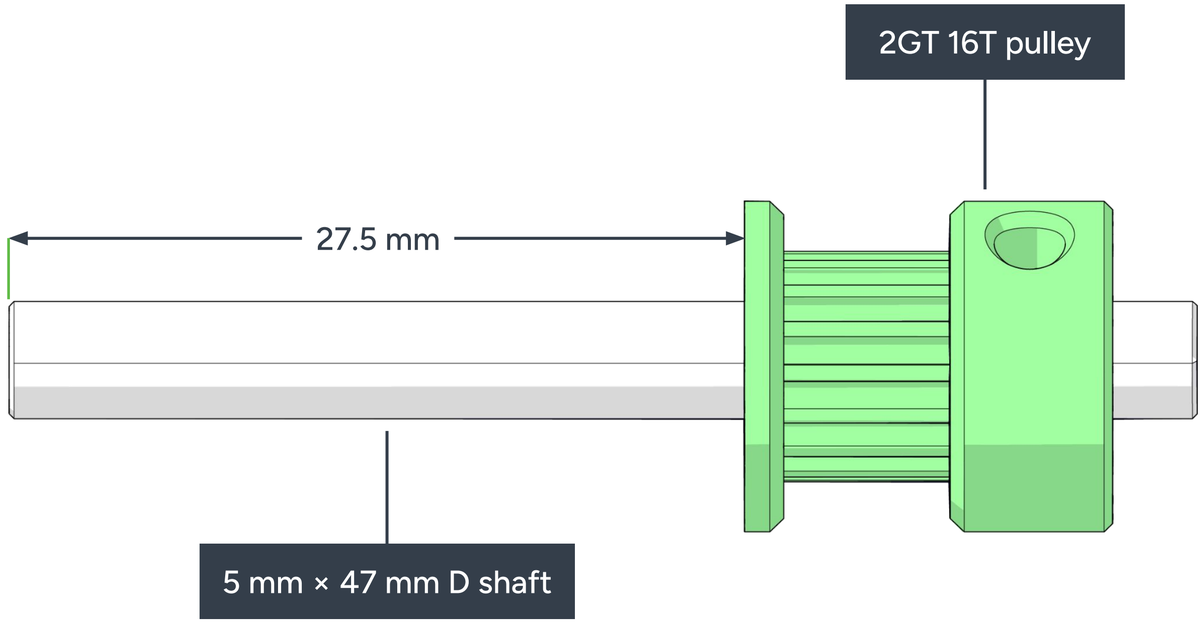


F / Z-AXIS

Z DRIVES - SHAFT ASSEMBLIES

Install a **2GT 16T pulley** onto a **5 × 47 mm D shaft** positioned 27.5 mm from the end to the pulley's flange. This 2GT 16T pulley stays flanged. Ensure the set screw is against the flat face of the D shaft. Do NOT use a ball ended driver to tighten.

This ensures correct position of the assembly in the Z drive.



NOTE

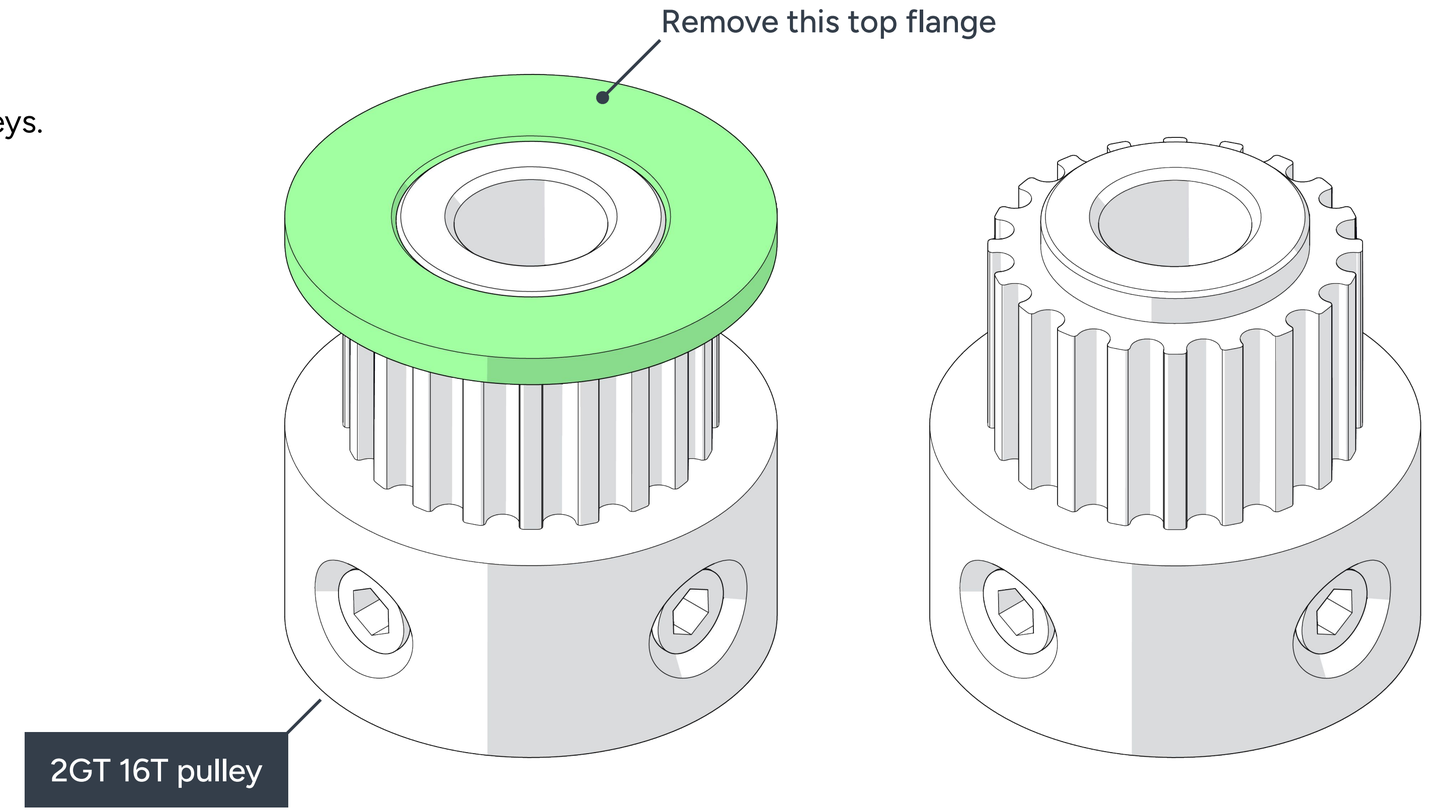
Apply threadlocker to all set screws. Loose set screws are a common cause of motion issues.

F / Z-AXIS

Z DRIVES - 2GT 16T PULLEYS

Each Z drive uses a **2GT 16T pulley**. Using pliers, carefully pull the top flange straight off four pulleys. Avoid damaging the teeth or bore.

You should have four deflanged 2GT 16T pulleys when complete.



NOTE

Some kits include pre-deflanged 2GT 16T pulleys. **If so, skip this step.**

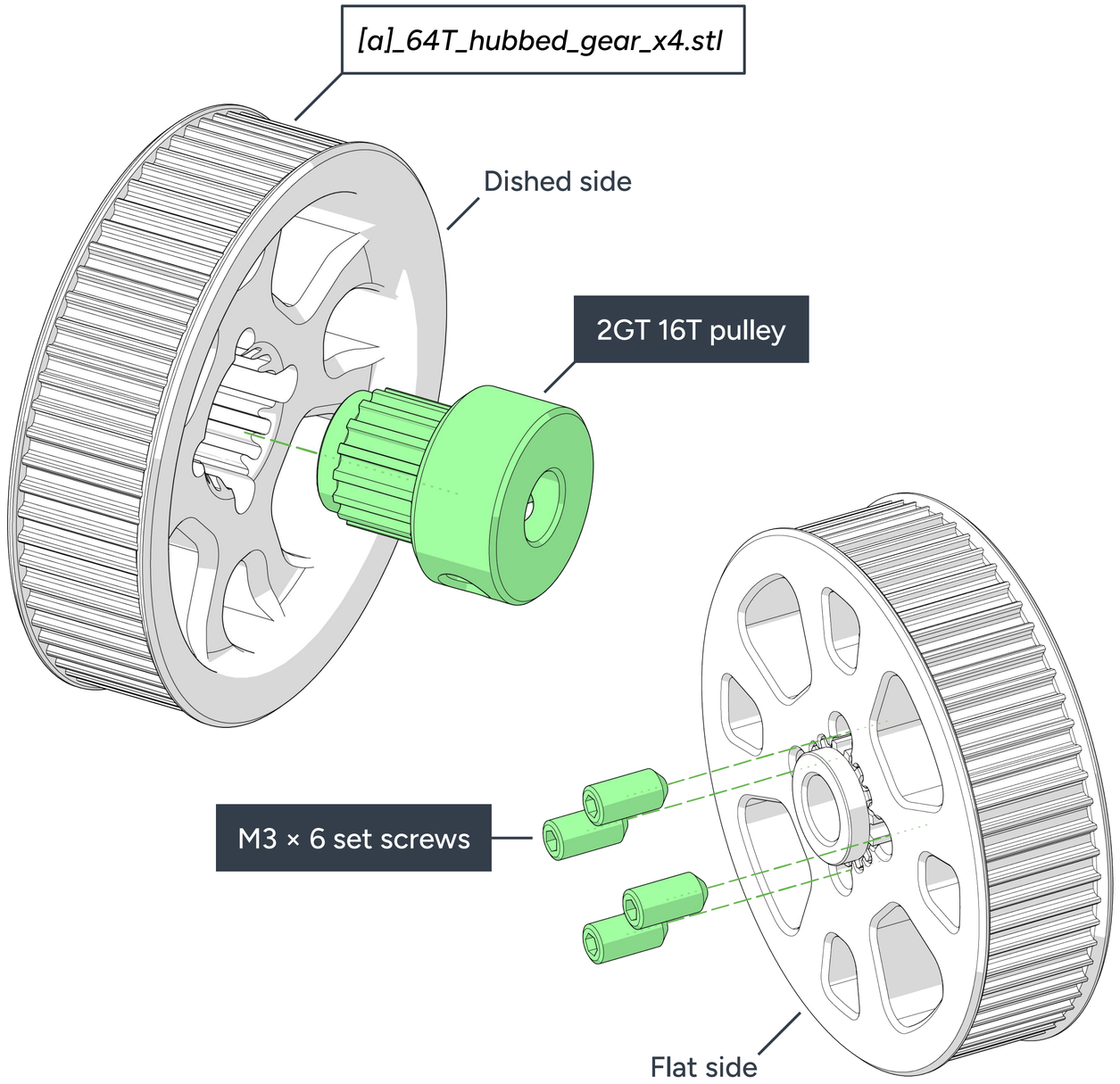
F / Z-AXIS

Z DRIVES - 64T HUBBED GEARS

Insert a deflanged 2GT 16T pulley into the dished side of a **64T hubbed gear**.

From the flat side of the hubbed gear, secure the pulley using **four M3 × 6 set screws**. Tighten gradually in an “X” pattern to keep the pulley centered. Stop when the set screws are flush with the printed face. Do not overtighten—this can skew the pulley.

You should have four 64T hubbed gear assemblies when complete.



NOTE

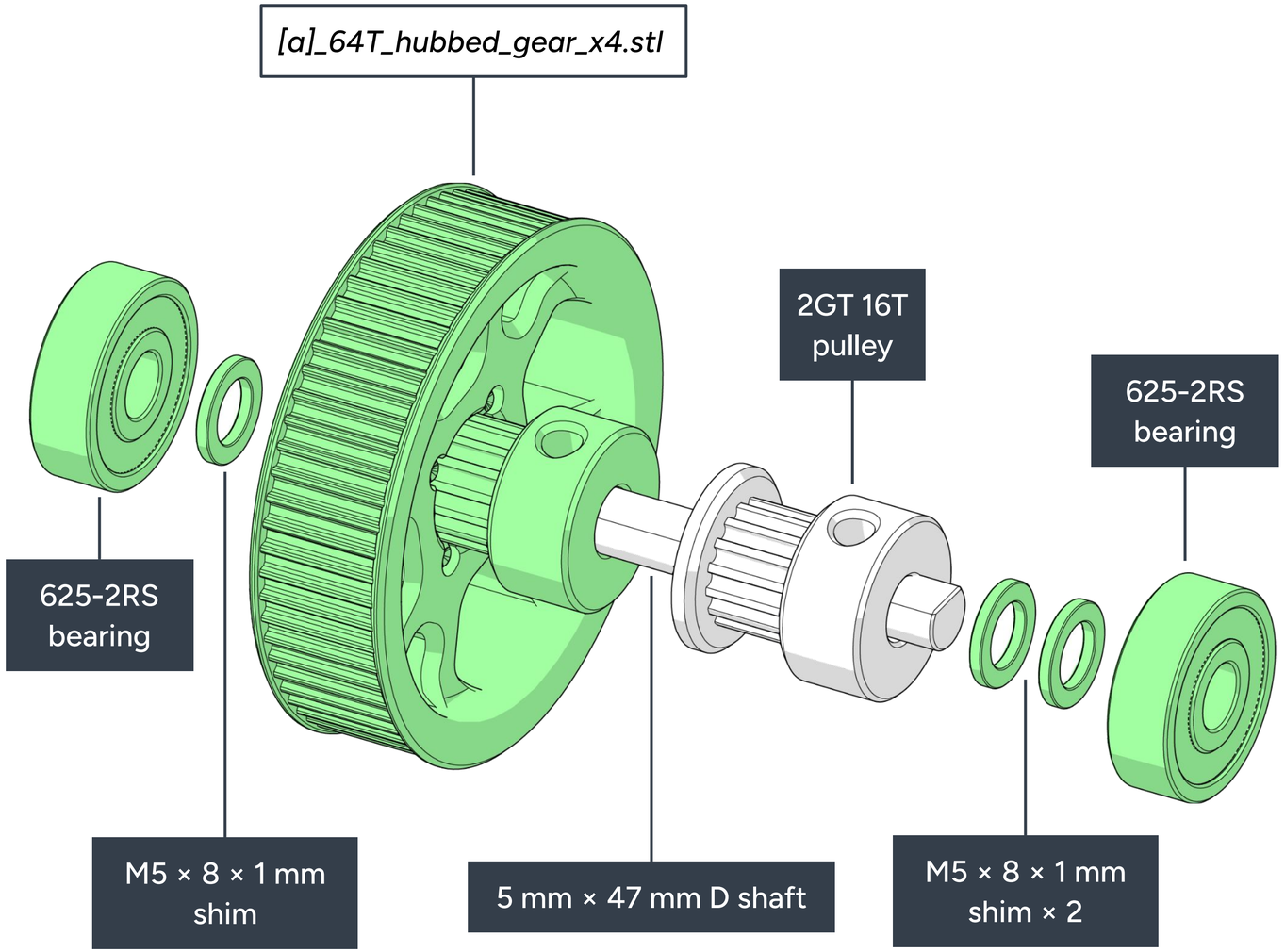
Some kits include machined 64T hubbed gears. **If so, skip this step.**

F / Z-AXIS

Z DRIVES - SHAFT ASSEMBLIES

Assemble the following component stack:

- **625-2RS bearing**
- **M5 × 8 × 1 mm shim**
- 64T hubbed gear assembly
- 2GT 16T pulley
- **M5 × 8 × 1 mm shim × 2**
- **625-2RS bearing**



NOTE

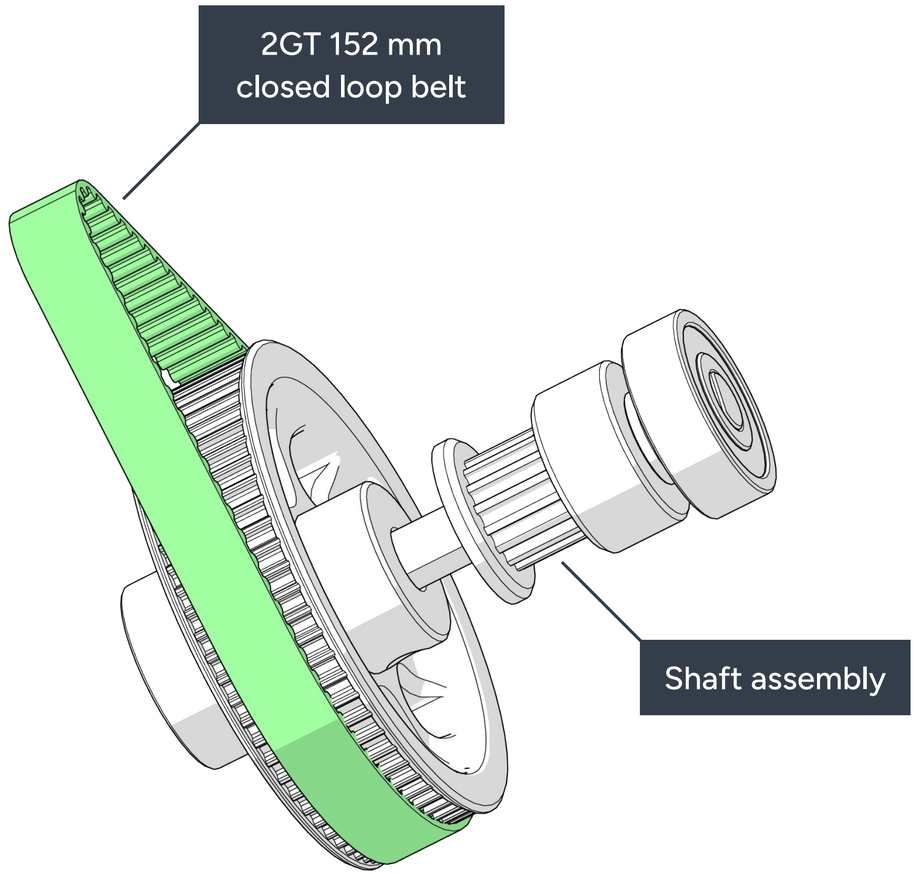
Apply threadlocker to all set screws. Loose set screws are a common cause of motion issues.

F / Z-AXIS

Z DRIVES

Loosely wrap a **2GT 152 mm closed loop belt** around the 64T hubbed gear. Ensure teeth are engaging.

You should have four shaft assemblies with loose belts when complete.

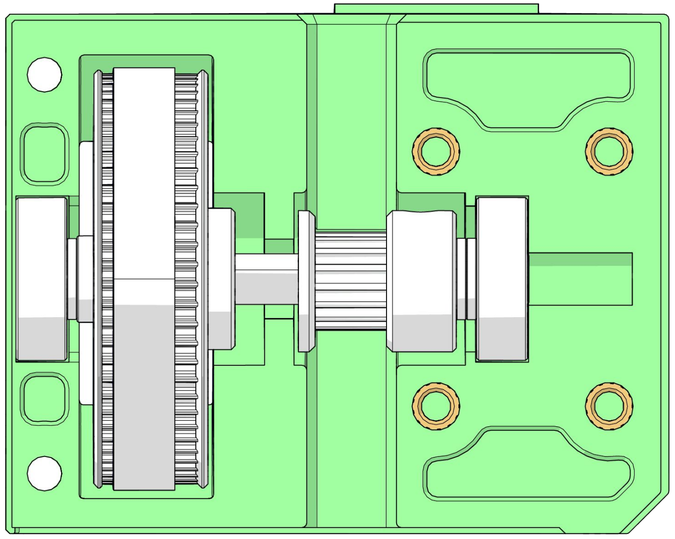


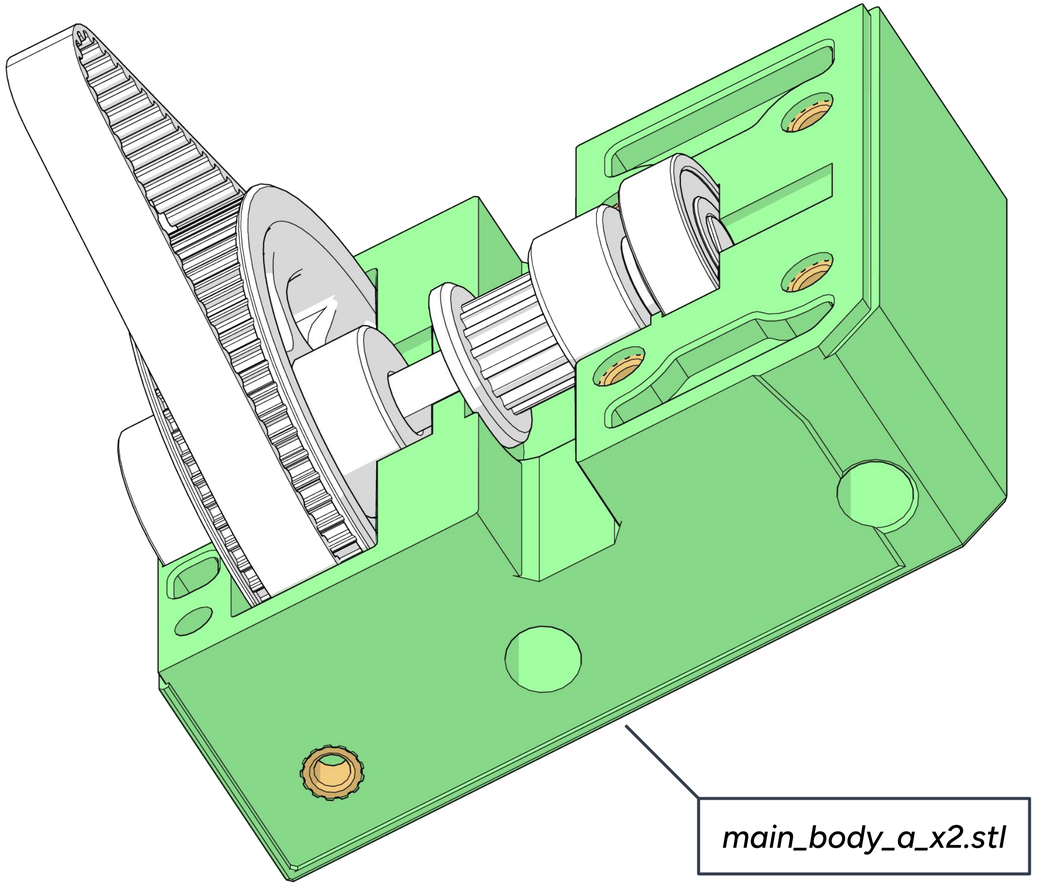
F / Z-AXIS

Z DRIVES

Of the four Z drives, you will build two ‘A’ versions and two mirrored ‘B’ versions.

Insert the shaft assembly with belt into ***main_body_a_x2.stl***. The shaft assembly should seat cleanly into the printed profile. Ensure the bearings are seating correctly and the exposed 2GT 16T pulley lines up with the belt channel.

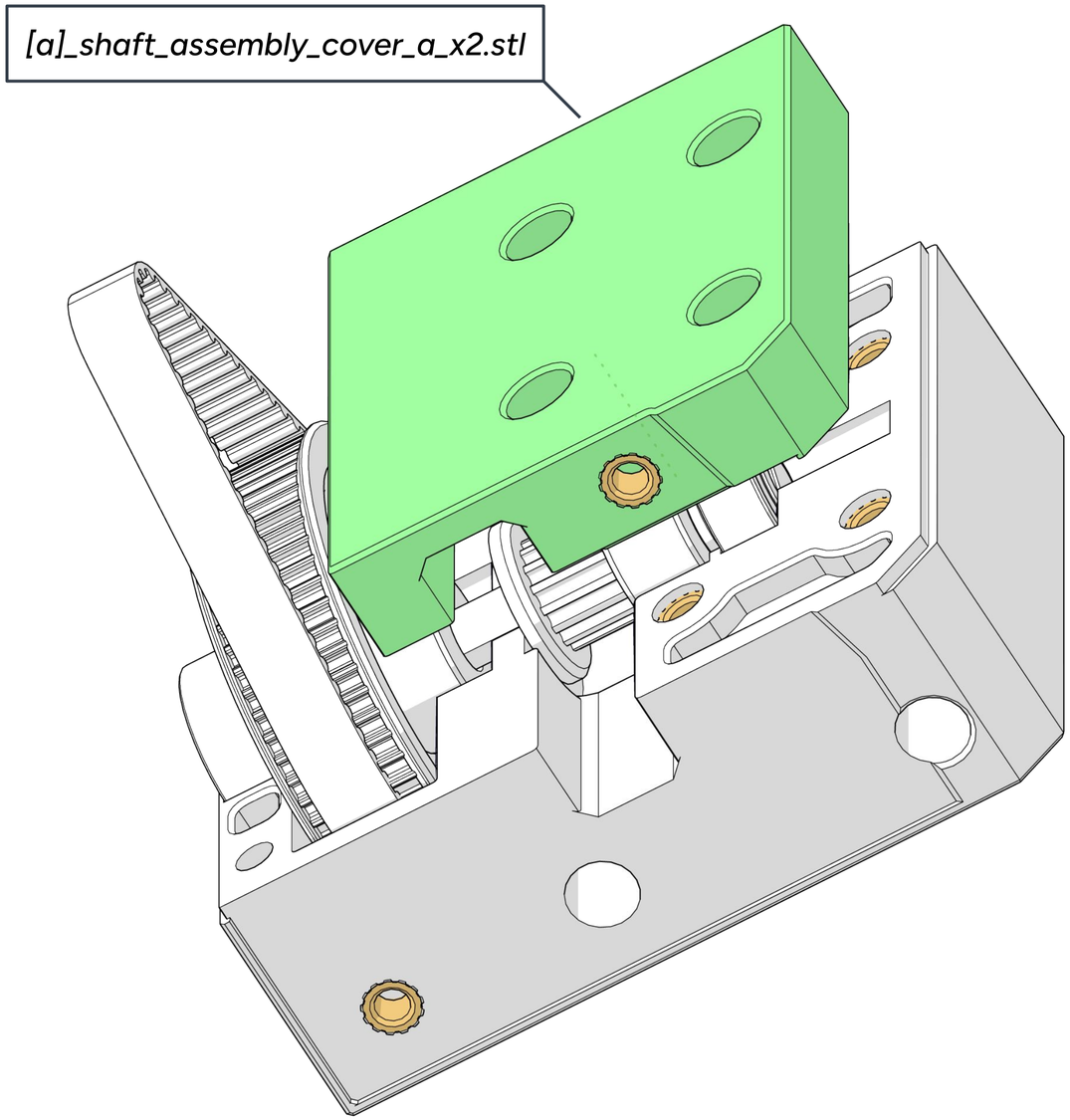




F / Z-AXIS

Z DRIVES

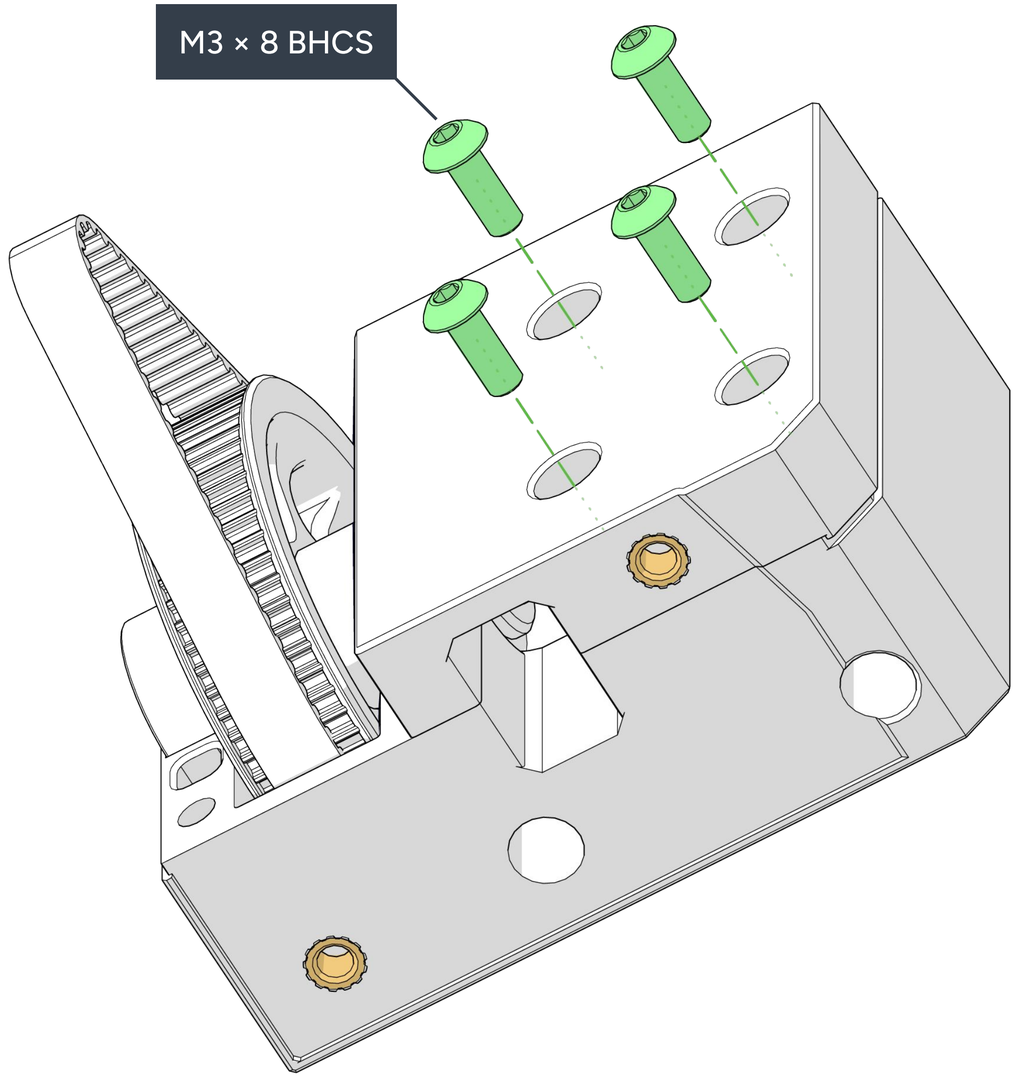
Install ***[a]_shaft_assembly_cover_a_x2.stl*** over the shaft assembly.



F / Z-AXIS

Z DRIVES

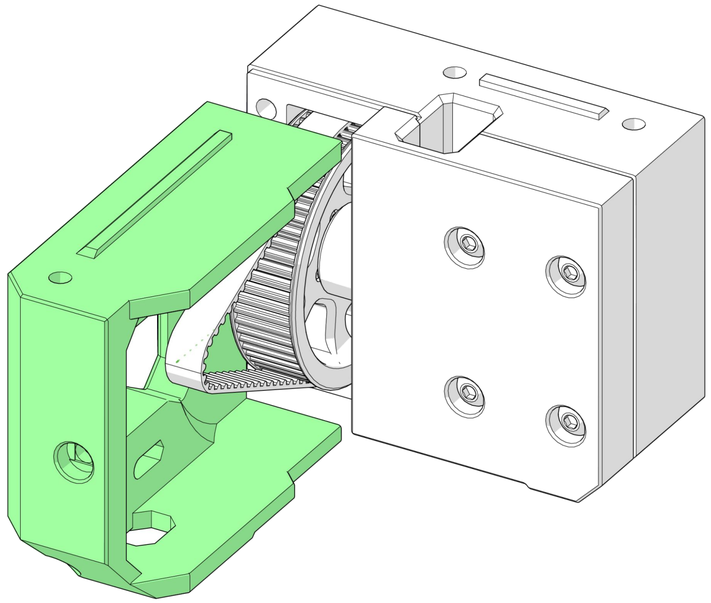
Secure the two printed parts with **four M3 × 8 BHCS** into the M3 heatset inserts. Tighten evenly.

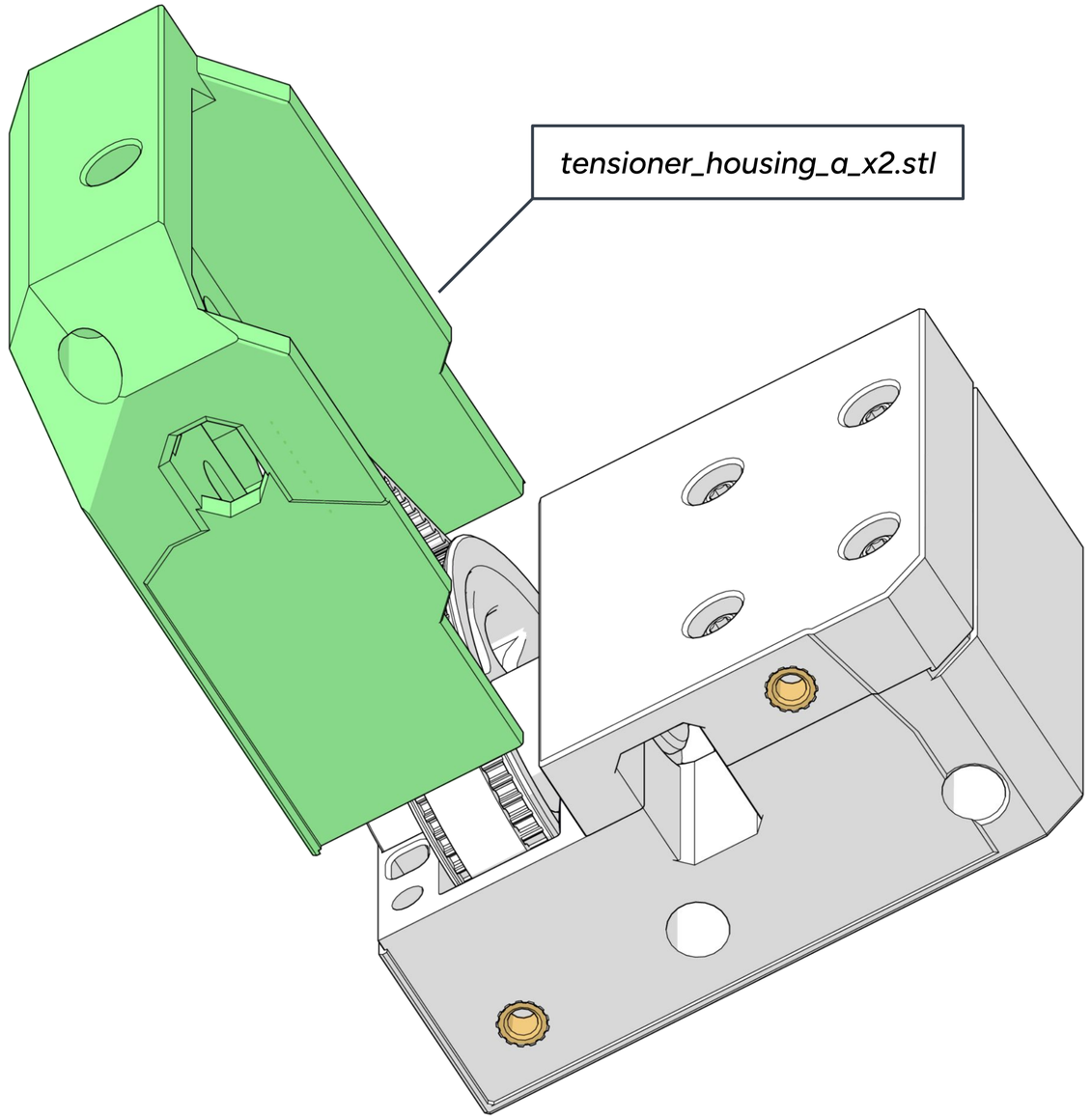


F / Z-AXIS

Z DRIVES

Position ***tensioner_housing_a_x2.stl*** over the hubbed gear and loose belt.

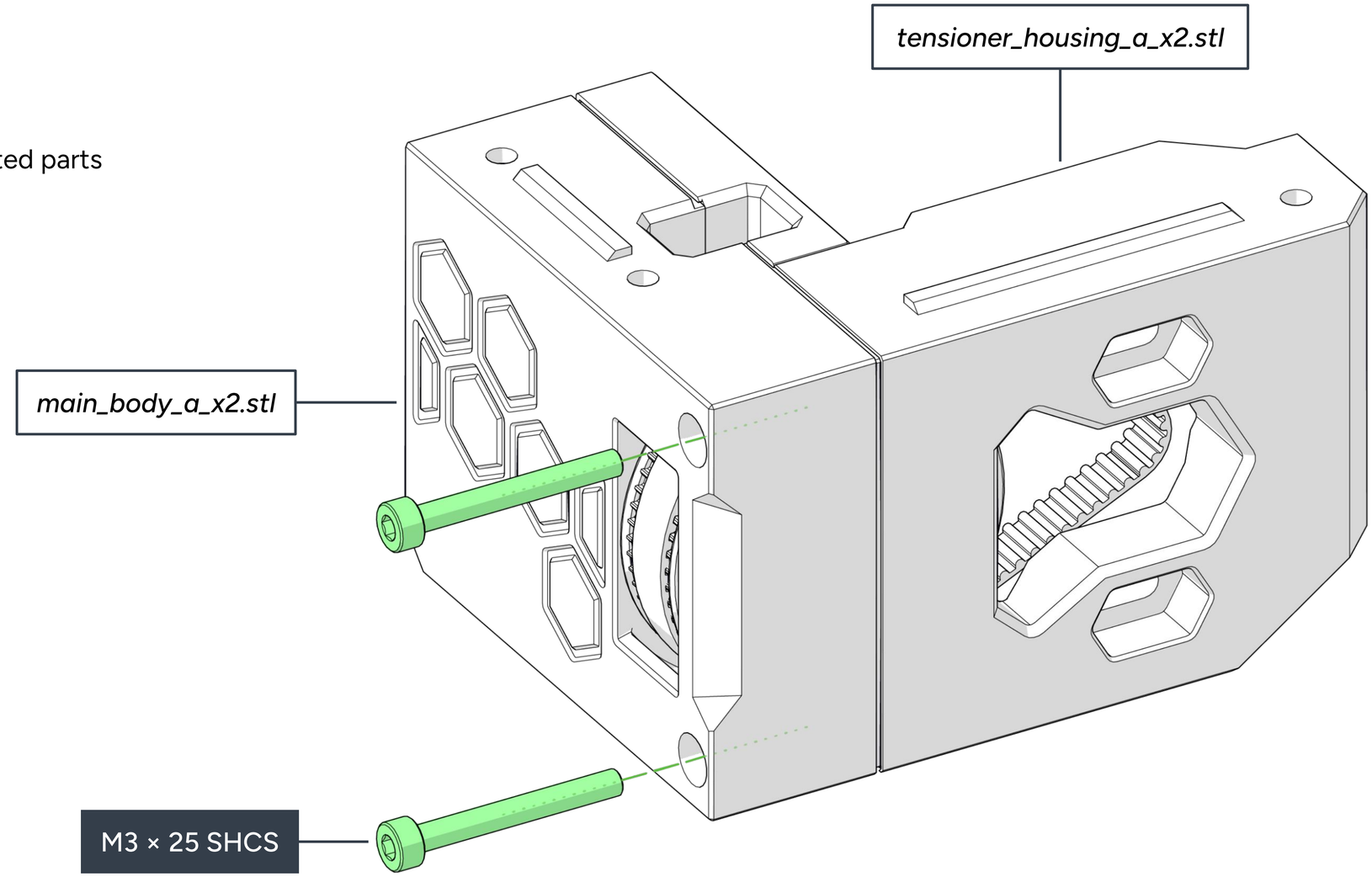




F / Z-AXIS

Z DRIVES

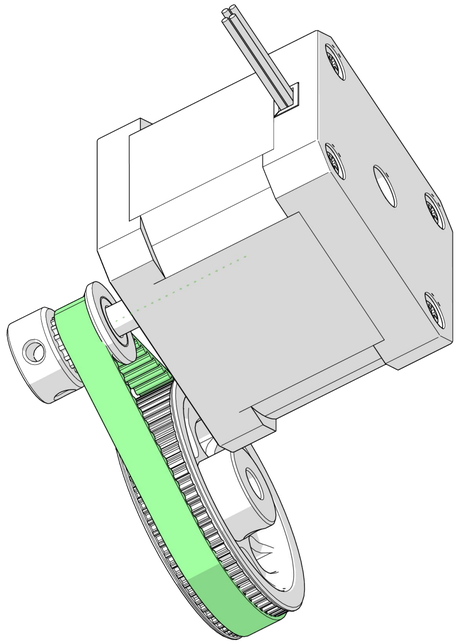
From the other side, join the two printed parts with **two M3 × 25 SHCS**.

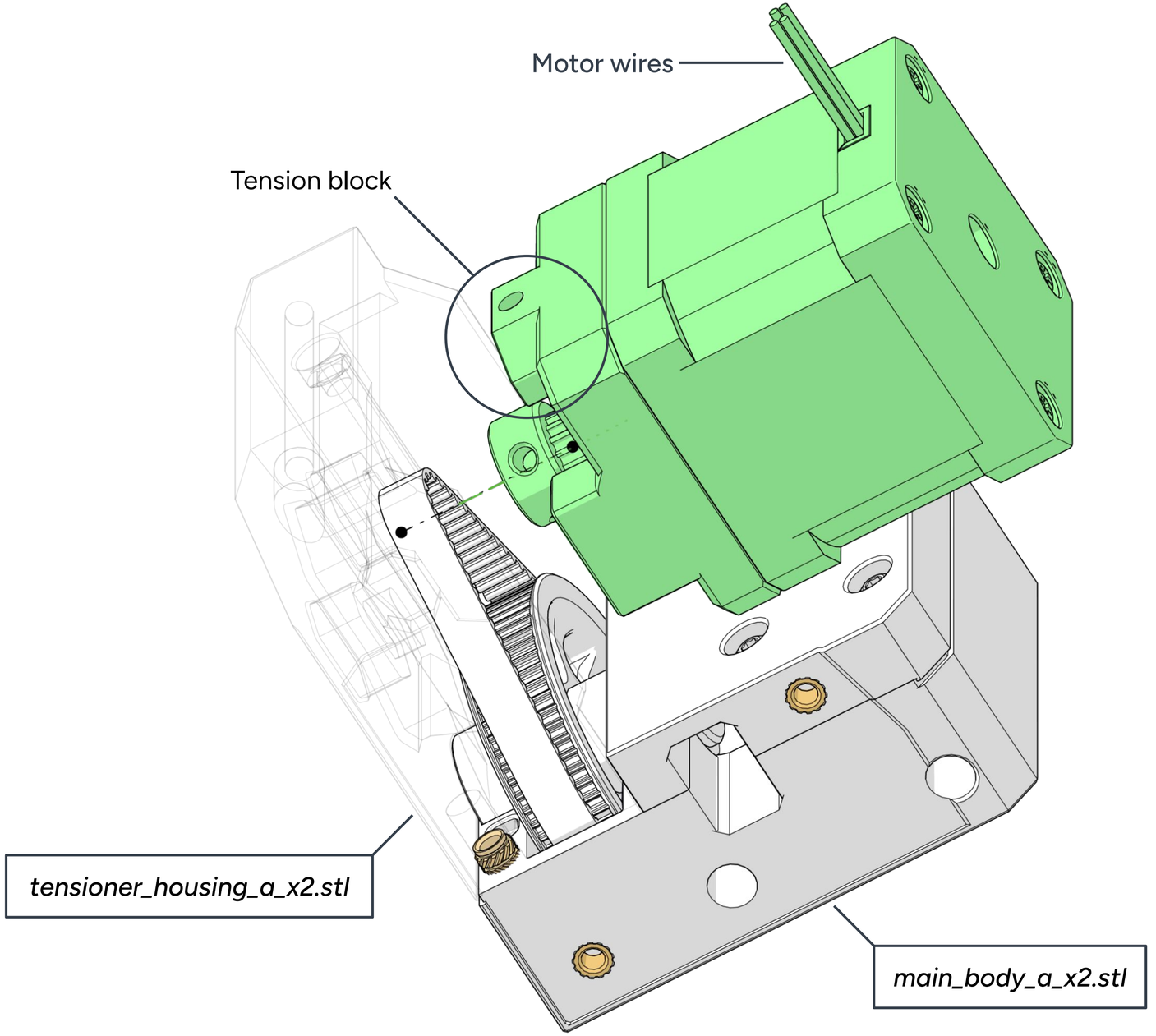


F / Z-AXIS

Z DRIVES

Insert a motor assembly so that the pulley engages with the loose belt. Note the direction of the motor wires and tension block.

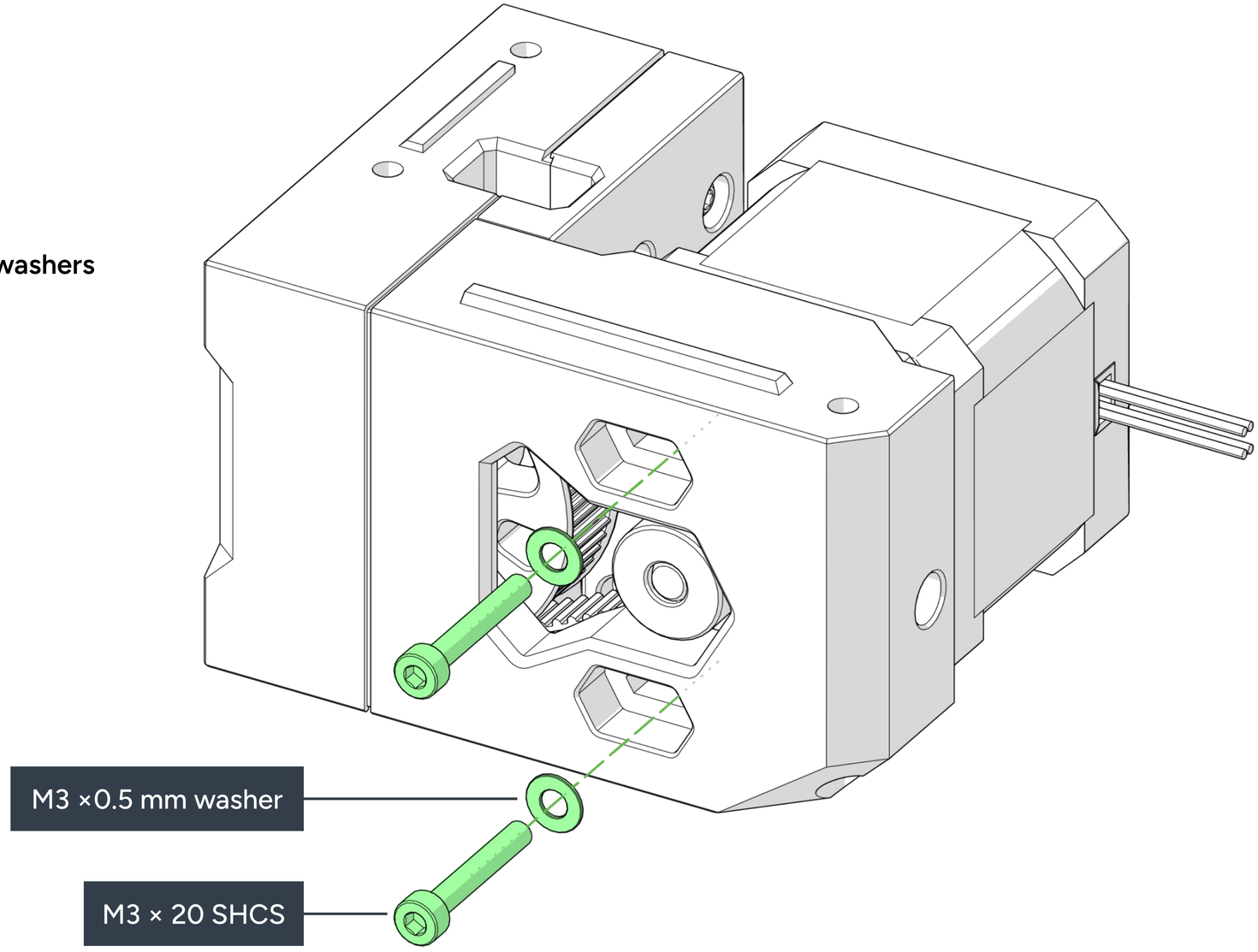




F / Z-AXIS

Z DRIVES

Secure the motor assembly with **two M3 × 0.5 mm washers** and **two M3 × 20 SHCS**. Do not tighten yet to allow alignment in the next step.



F / Z-AXIS

Z DRIVES - TENSION SCREW

Check that the 152 mm closed loop belt moves freely and the shaft assembly rotates smoothly. Install an **M3 × 16 SHCS** into the tension block.

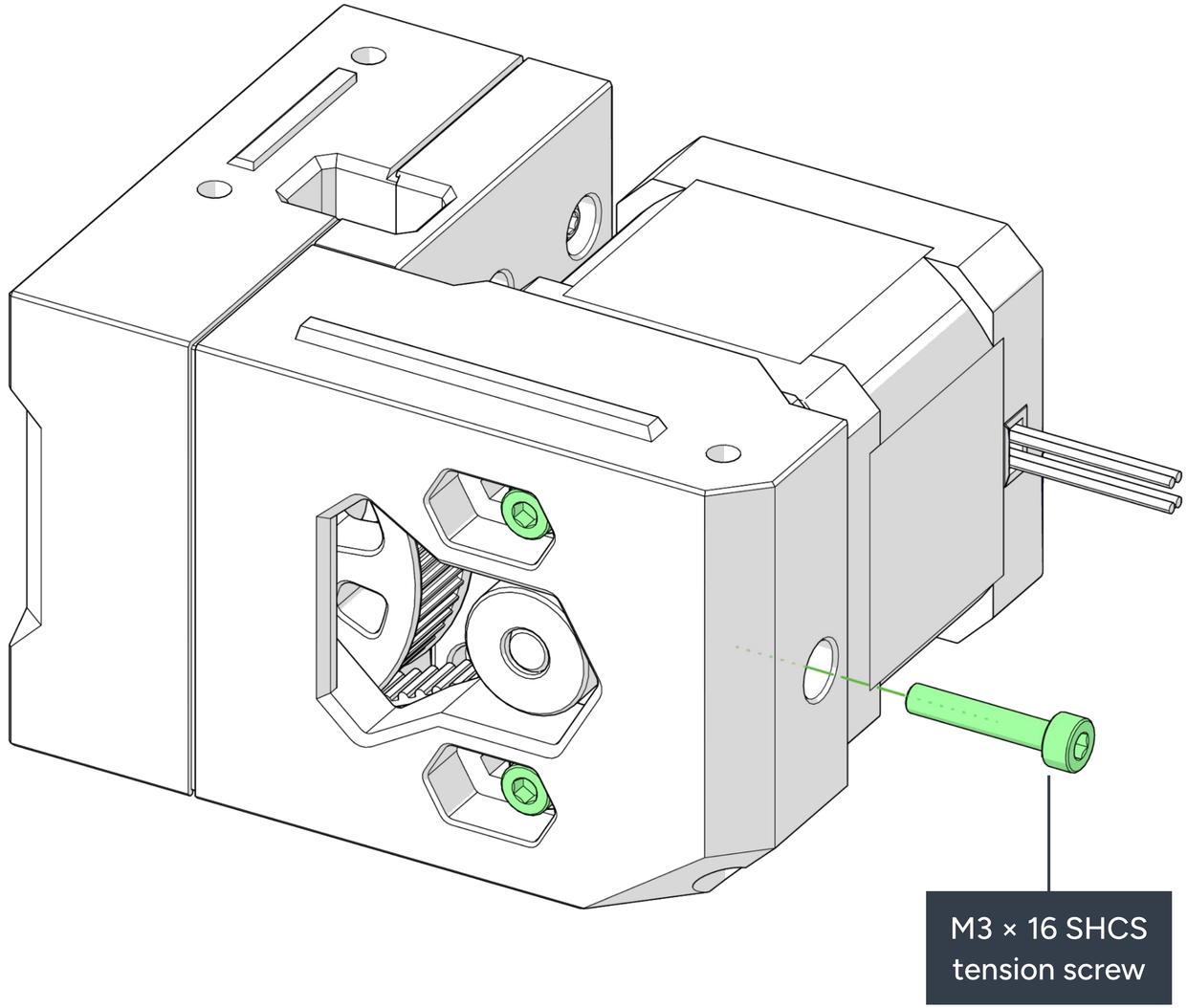
You should have four Z assemblies when complete.

TENSIONING THE Z MOTOR BELT

Loosen the **two mounting M3 × 20 SHCS** to allow the motor to slide side-to-side.

Slowly adjust the **M3 × 16 SHCS tension screw** until the desired tension is reached. Clockwise to tighten; counter-clockwise to loosen. Set tension just high enough for consistent pulley engagement without binding or slipping.

Tighten the **two mounting M3 × 20 SHCS** to lock the motor in place.

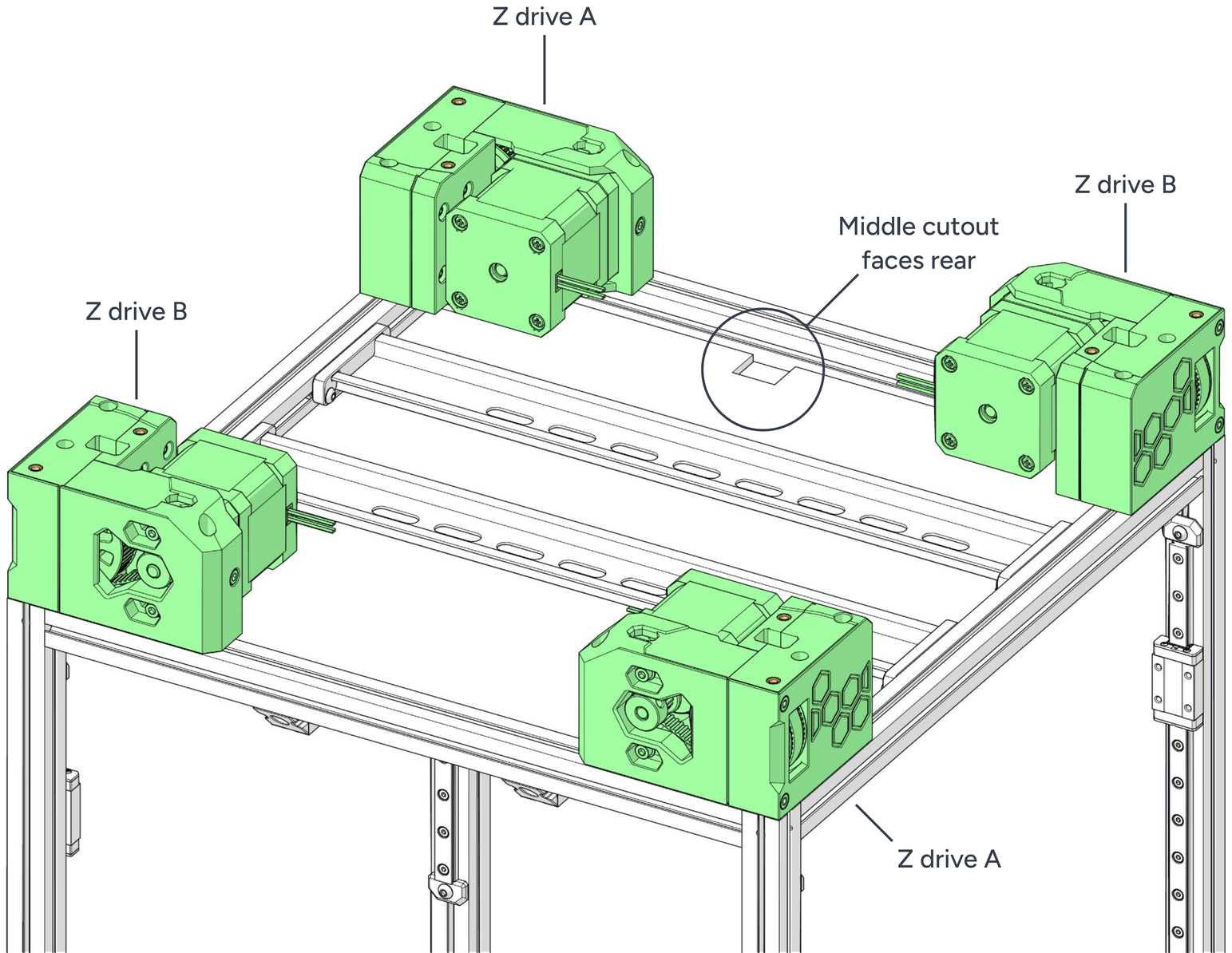


F / Z-AXIS

Z DRIVES - MOUNTING

You should have four assembled Z drives; two A assemblies and two mirrored B assemblies.

See the diagram for correct orientation. Use the deck panel cutout as a reference.



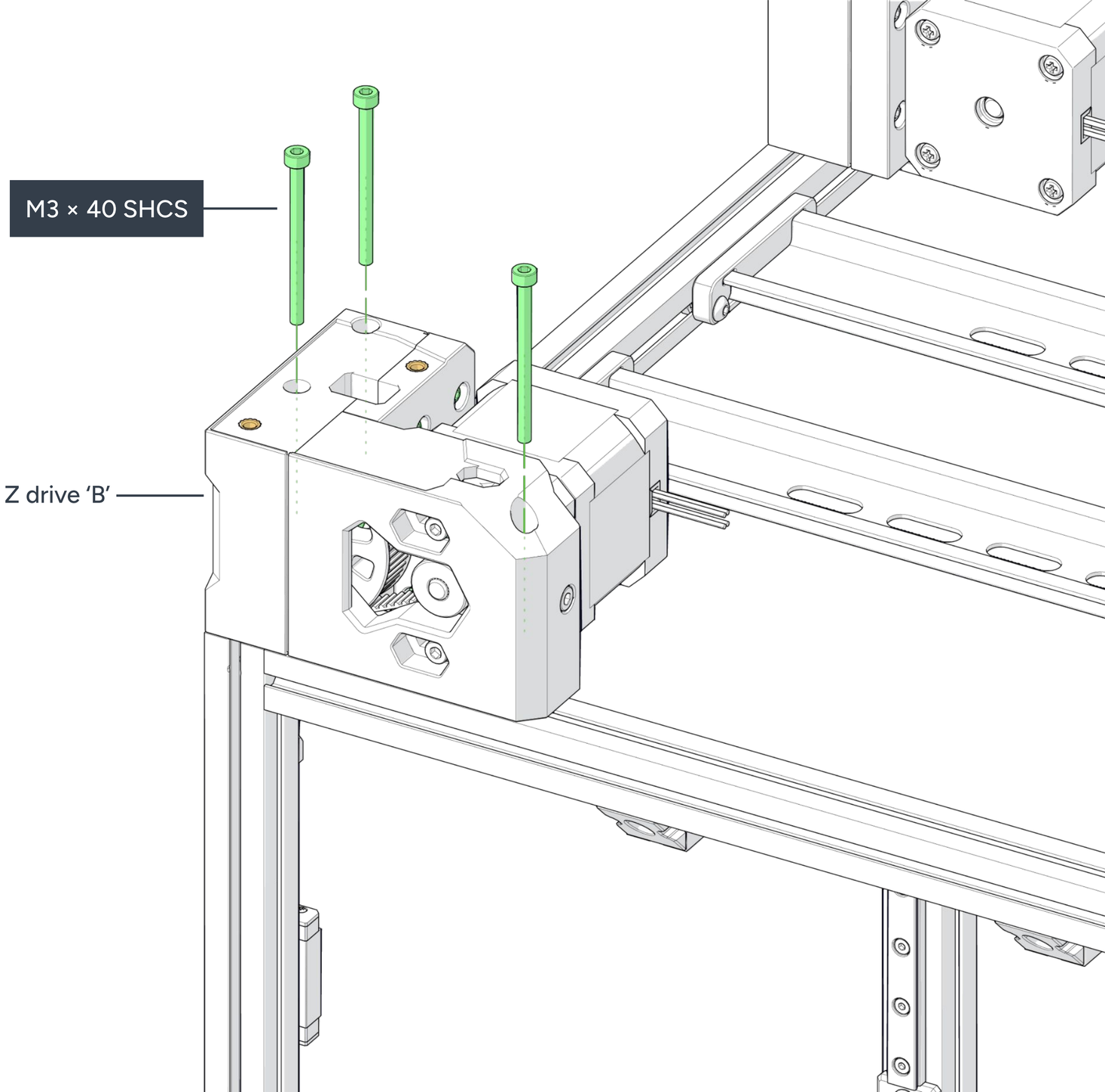
F / Z-AXIS

Z DRIVES - MOUNTING

Mount each Z drive to the underside of the frame using **three M3 × 40 SHCS** into preloaded M3 hex nuts.

The shaft assembly side receives two screws. The motor mount side receives one.

Repeat for all four Z drives.



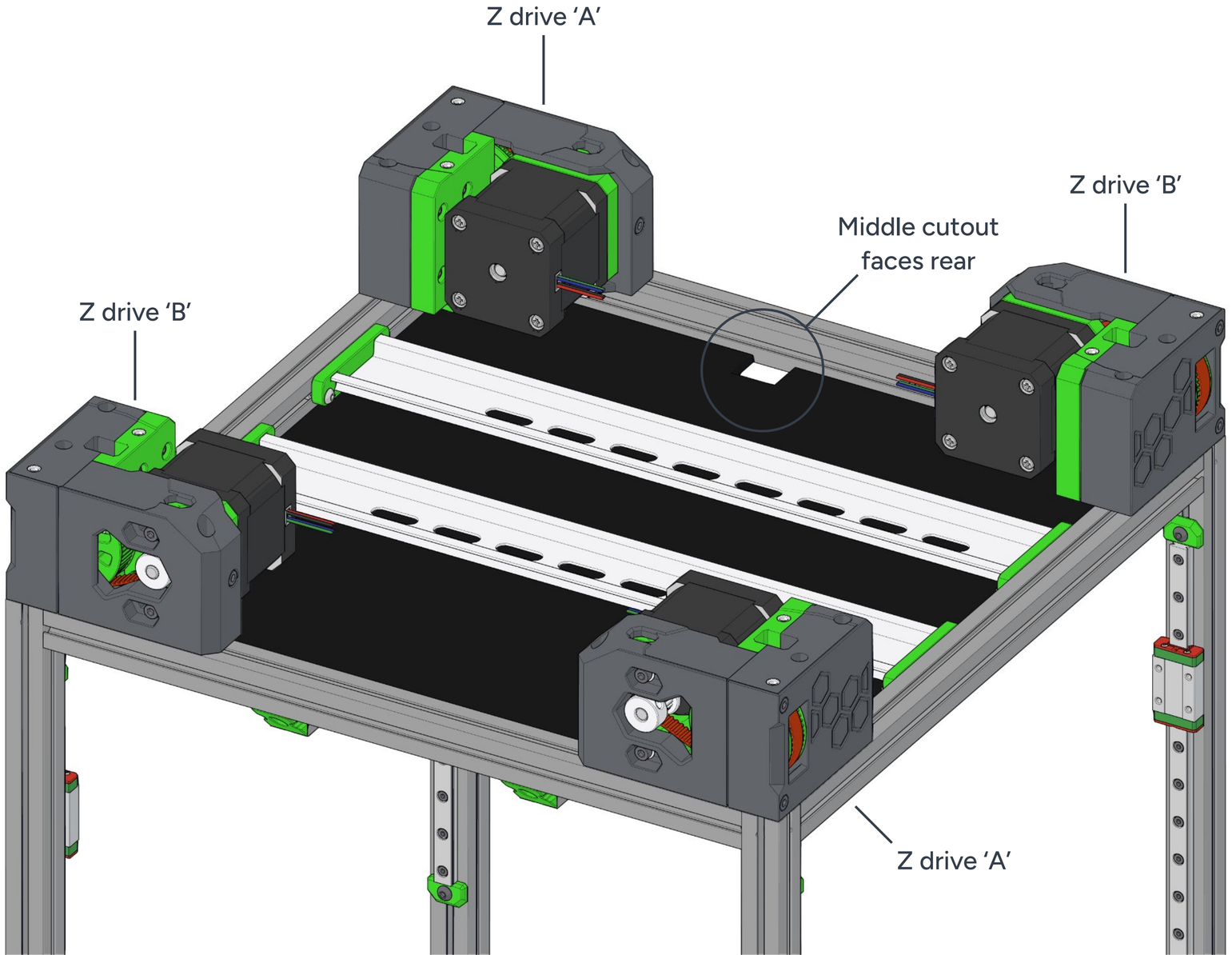
F / Z-AXIS

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

- ▢ Z drives are properly oriented
- ▢ Motor wires face the electronics chamber
- ▢ Motor mounting screws are tightened
- ▢ Z motor belts are properly tensioned



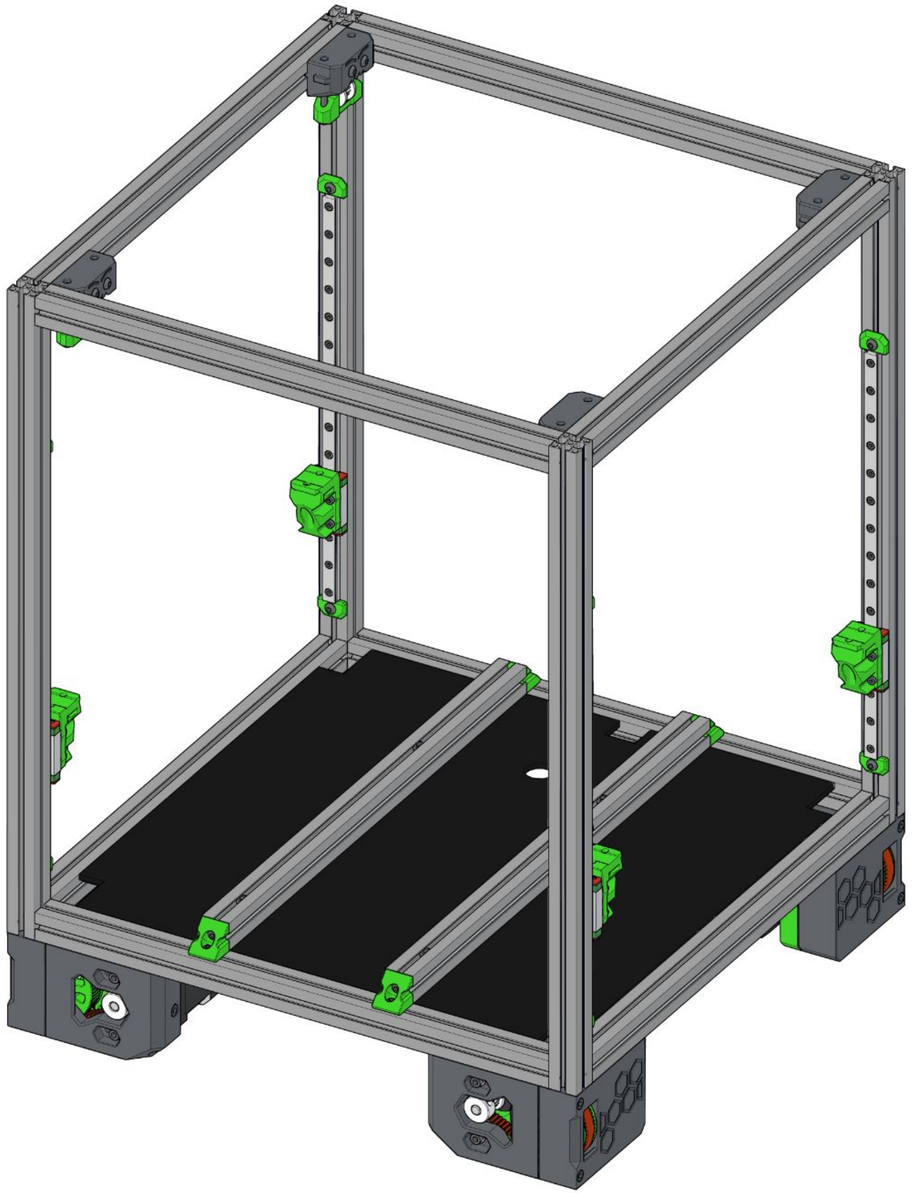
F / Z-AXIS

Z IDLERS

Next, we will install the Z idlers on the top frame, opposite each Z drive. The Z idlers redirect the belt and provide belt tension.

HARDWARE

4	2GT 16T idler
4	3 mm × 13 mm pin
8	M3 × 20 BHCS
8	M3 × 30 BHCS
8	M3 × 7× 0.5 mm shim
8	M3 hex nuts

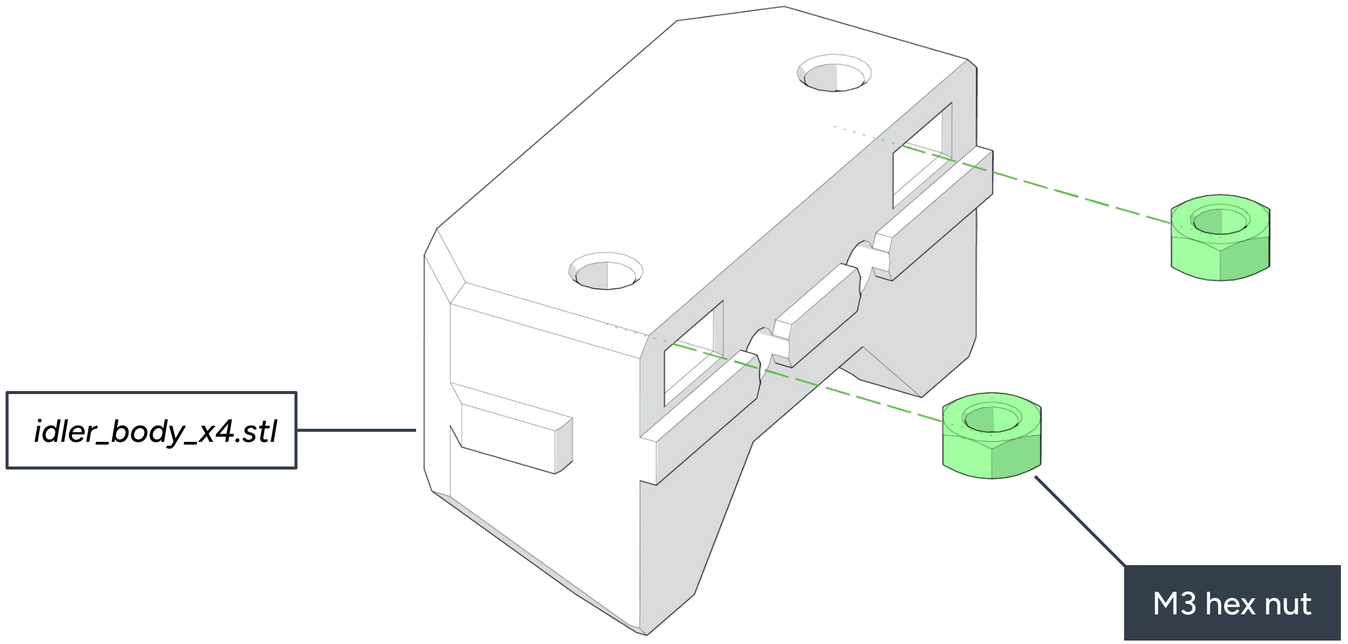


F / Z-AXIS

PREPPING Z IDLERS

Insert **two M3 hex nuts** into the slots of each **idler_body_x4.stl**.

You should have four idler bodies with two installed M3 hex nuts each.



NOTE

Z idler hardware varies by kit. The LDO kit uses a pinned idler assembly; some kits use a through-bolt (M3 × 13 SHCS). Match printed parts to hardware.

F / Z-AXIS

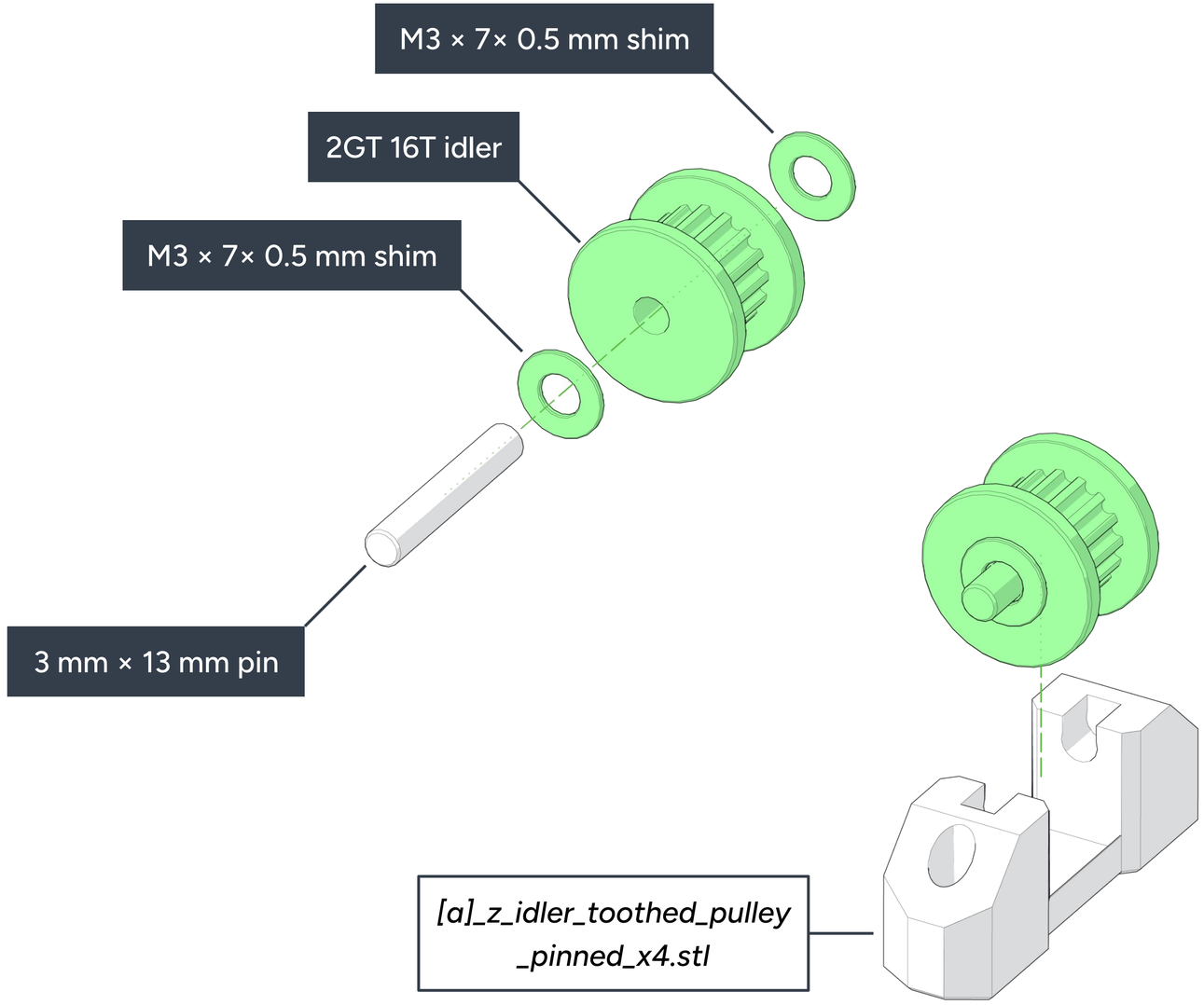
PREPPING Z IDLERS

Assemble the following component stack on a **3 mm × 13 mm pin**:

- **M3 × 7 × 0.5 mm shim**
- **2GT 16T idler**
- **M3 × 7 × 0.5 mm shim**

Insert the assembled stack into ***[a]_z_idler_toothed_pulley_pinned_x4.stl***. Ensure the idler spins freely.

Repeat for all four idler assemblies.

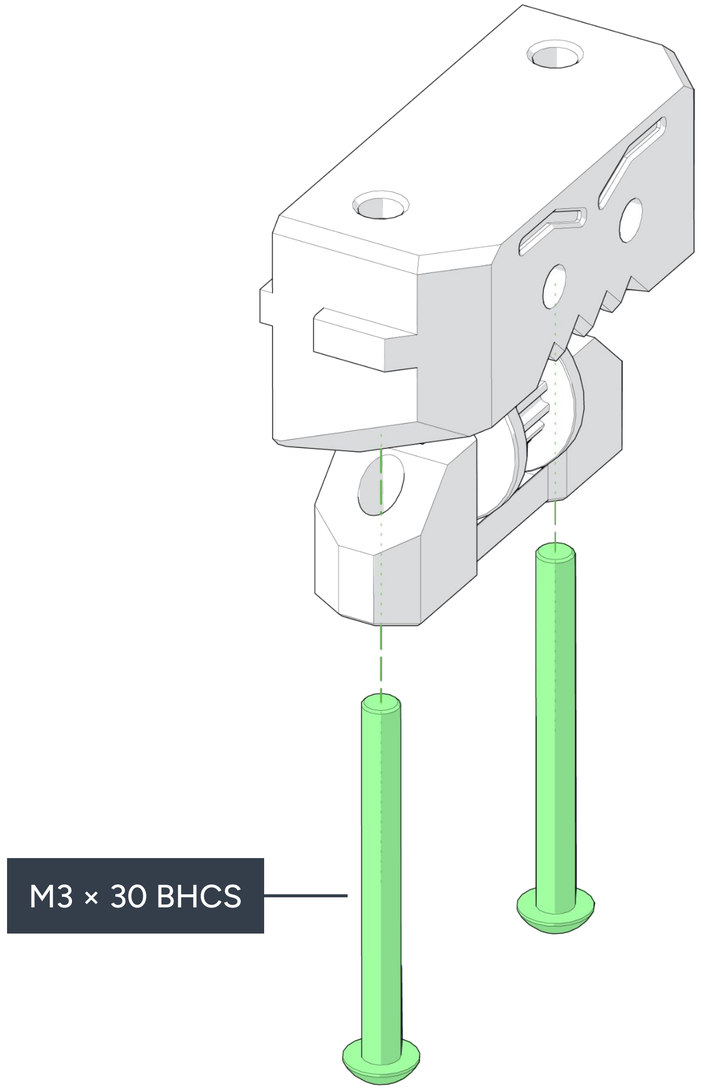


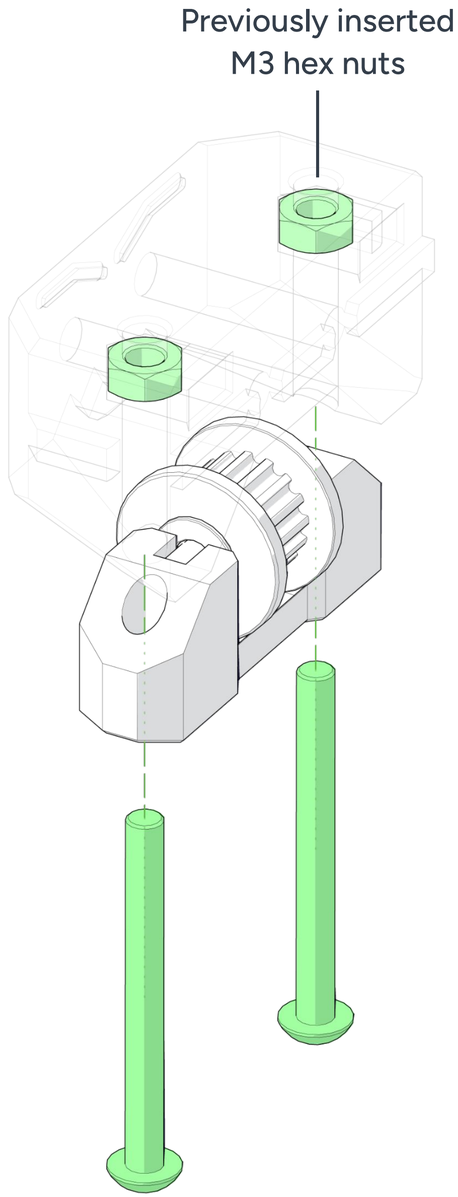
F / Z-AXIS

PREPPING Z IDLERS

Secure the idler assembly by installing **two M3 × 30 BHCS** from the bottom of the idler body into the previously inserted M3 hex nuts.

Repeat for all four idlers.



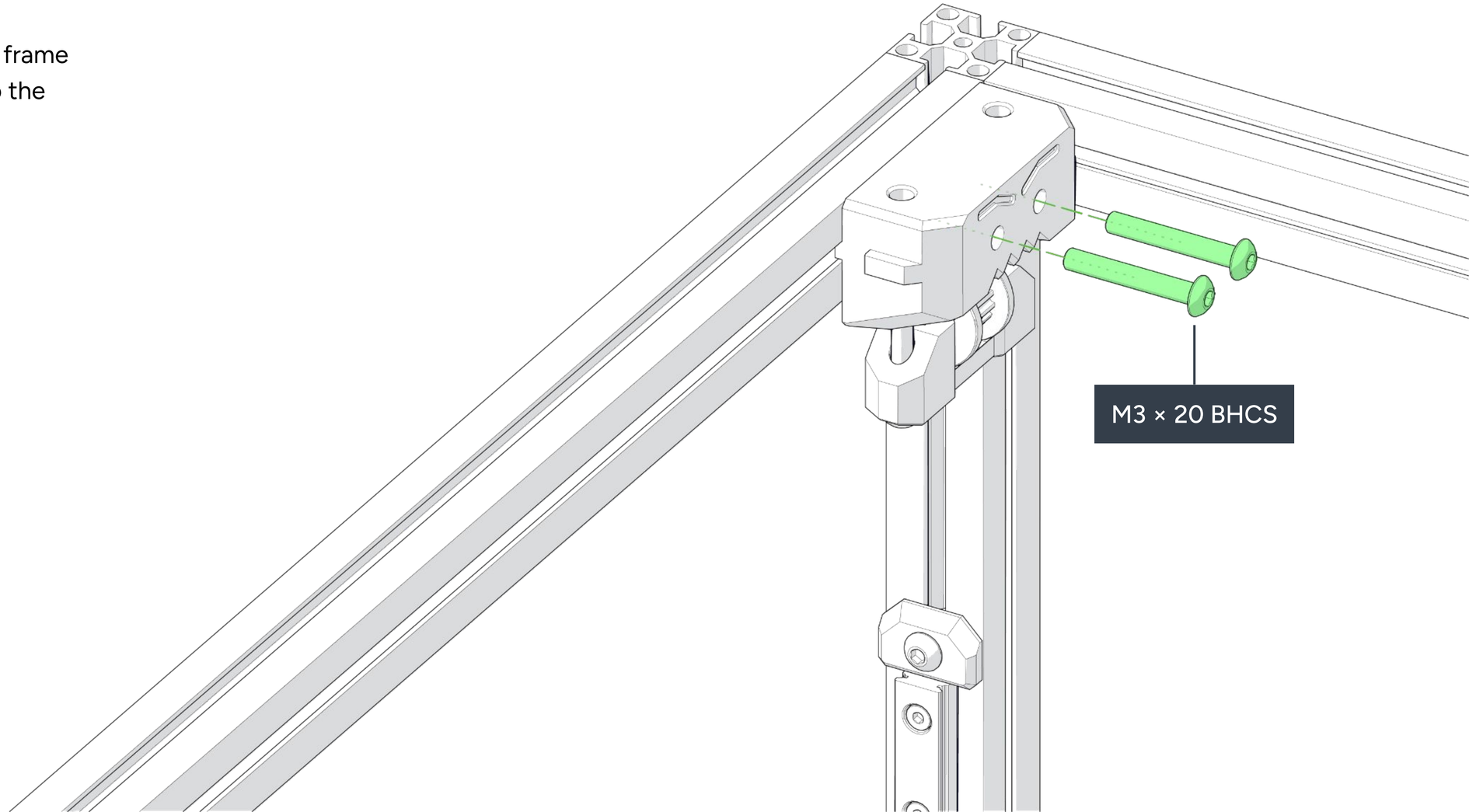


F / Z-AXIS

MOUNTING Z IDLERS

Mount each Z idler to the top frame using t**wo M3 × 20 BHCS** into the preloaded M3 hex nuts.

Repeat for all four idlers.



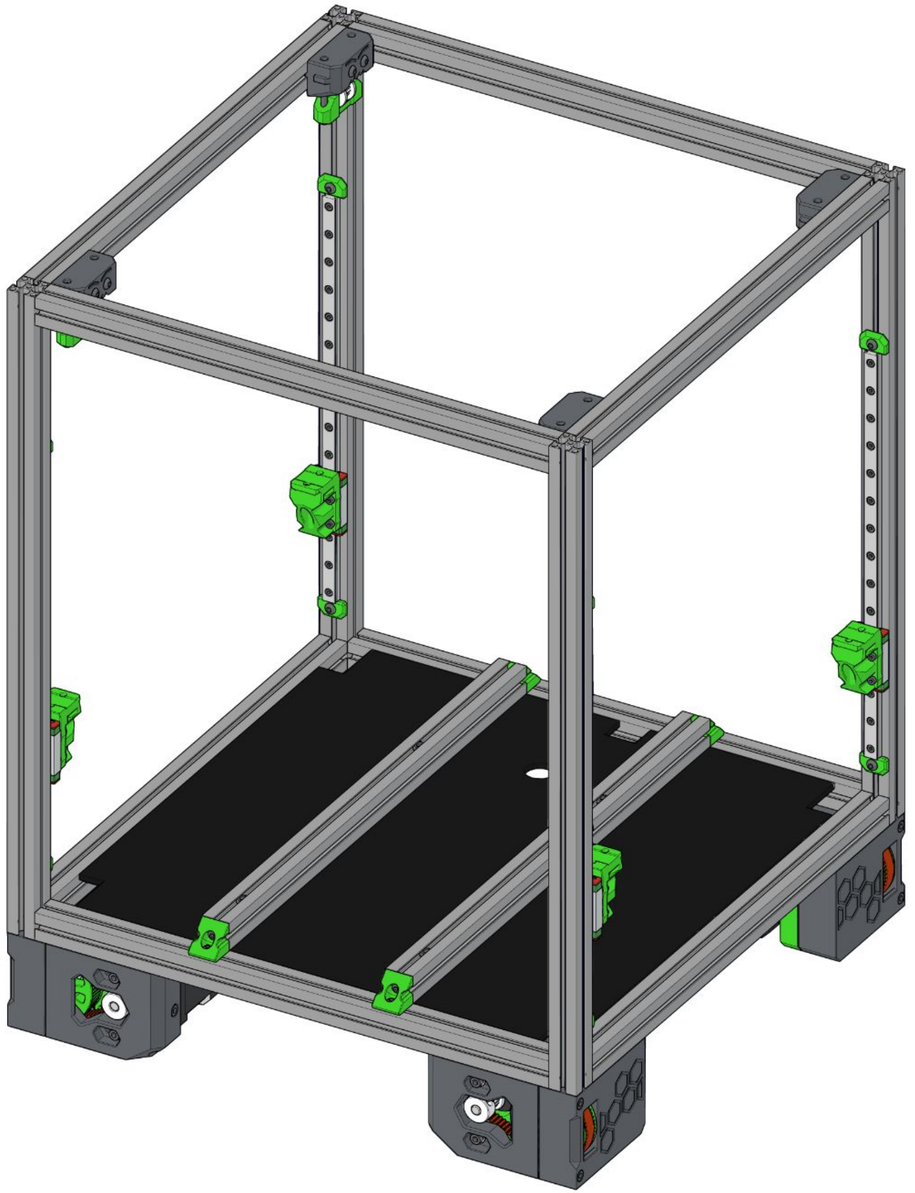
F / Z-AXIS

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

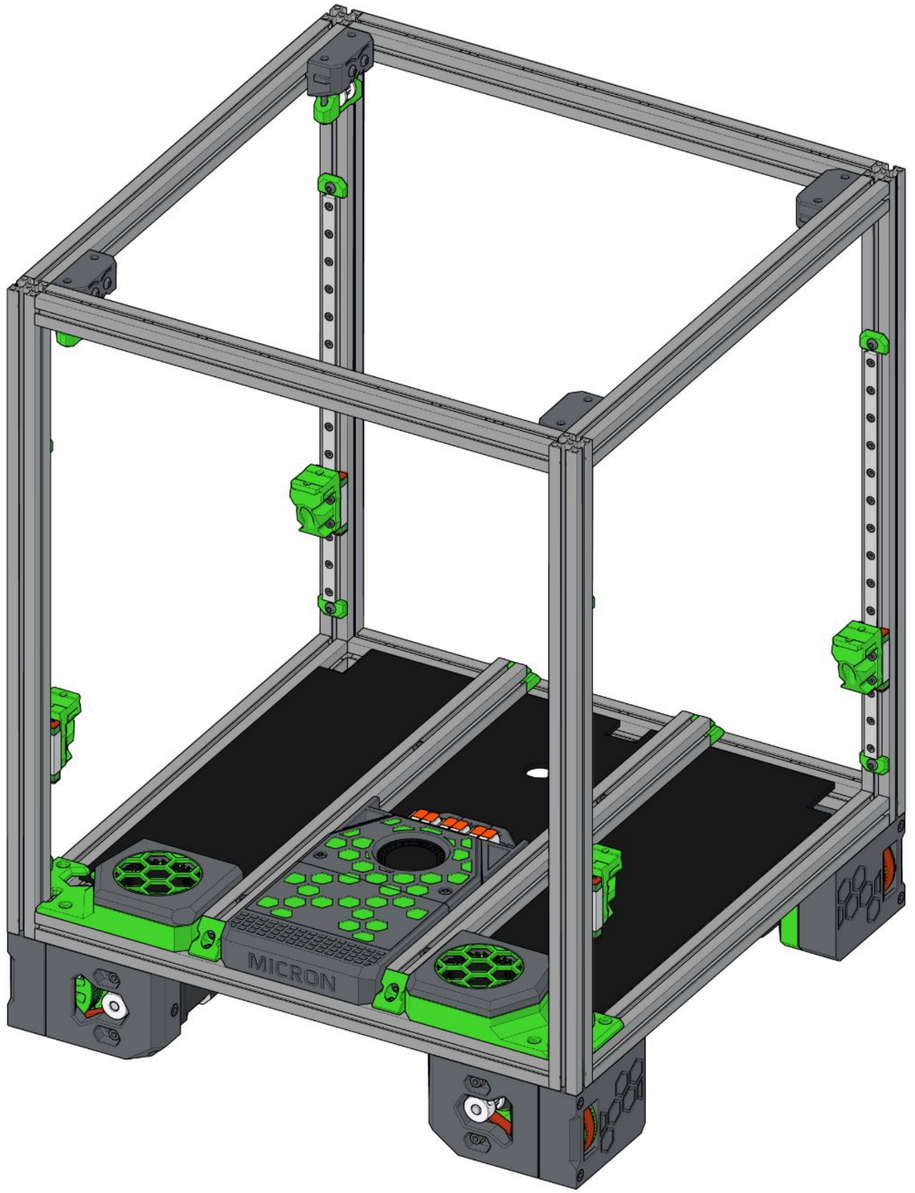
- ▢ Z idlers are properly oriented
- ▢ Z idler pulleys spin freely
- ▢ Z idlers are securely fastened to preloaded nuts



G/ FILTER ASSEMBLY

The chamber airflow system uses **three 5015 blower fans**. Two side fans circulate air around the bed to promote more even chamber temperatures, while the center plenum fan recirculates air through an activated-carbon basket to help reduce odors and some VOCs.

These fans recirculate air within the enclosure rather than exhausting it.



G/ FILTER ASSEMBLY

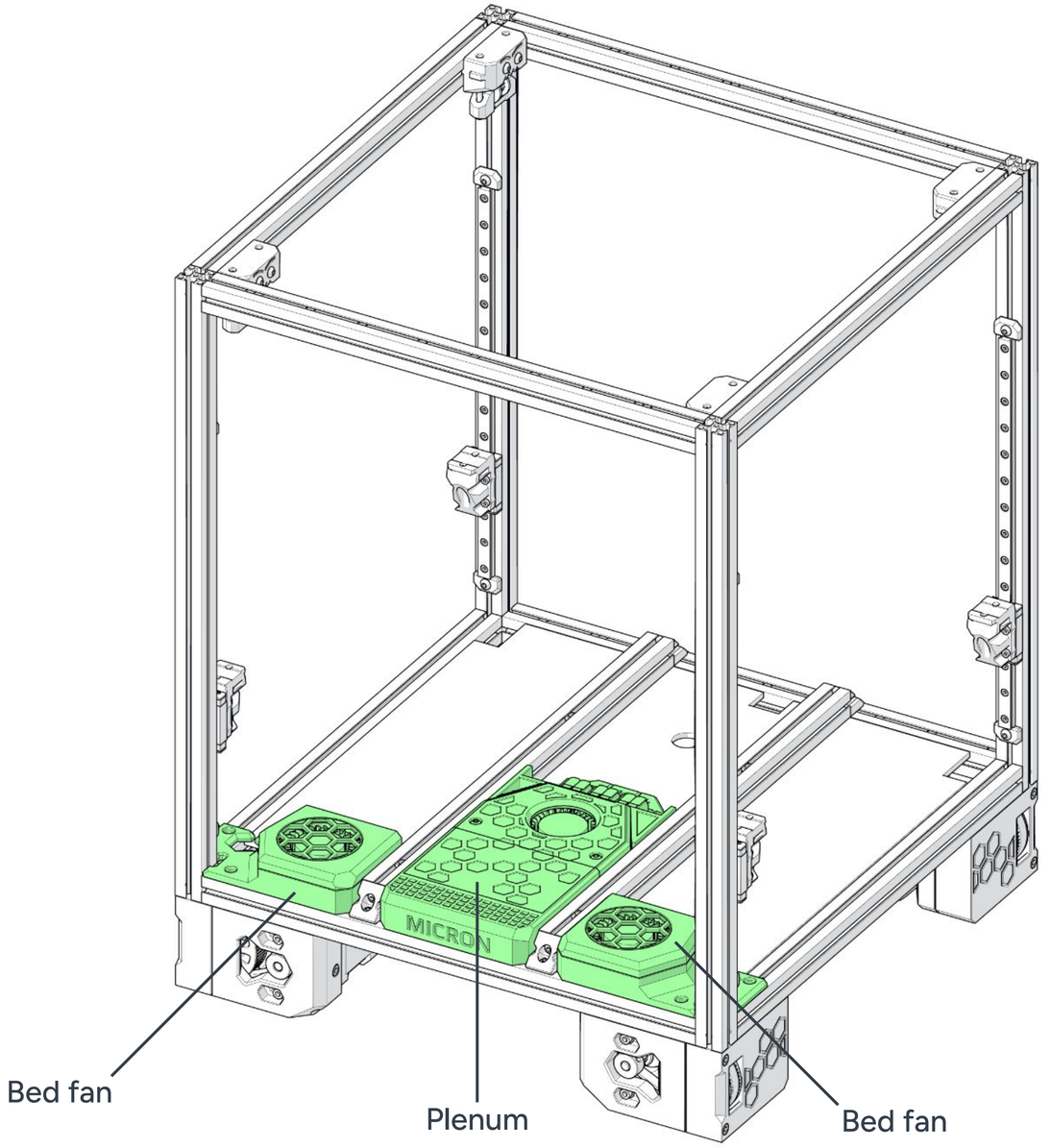
PLENUM & BED FANS

The center assembly is the filtered plenum. The two matching side assemblies are bed circulation fans.

Note their positions and orientation before assembly.

HARDWARE

4	6 × 3mm neodymium magnet
4	M3 heatset inserts
10	M3 × 8 BHCS
4	M3 × 12 BHCS
2	M3 hex nuts
3	5015 blower fans



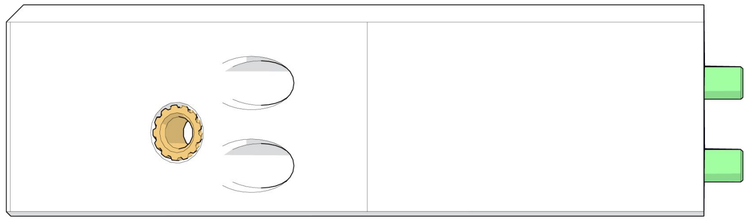
G/ FILTER ASSEMBLY

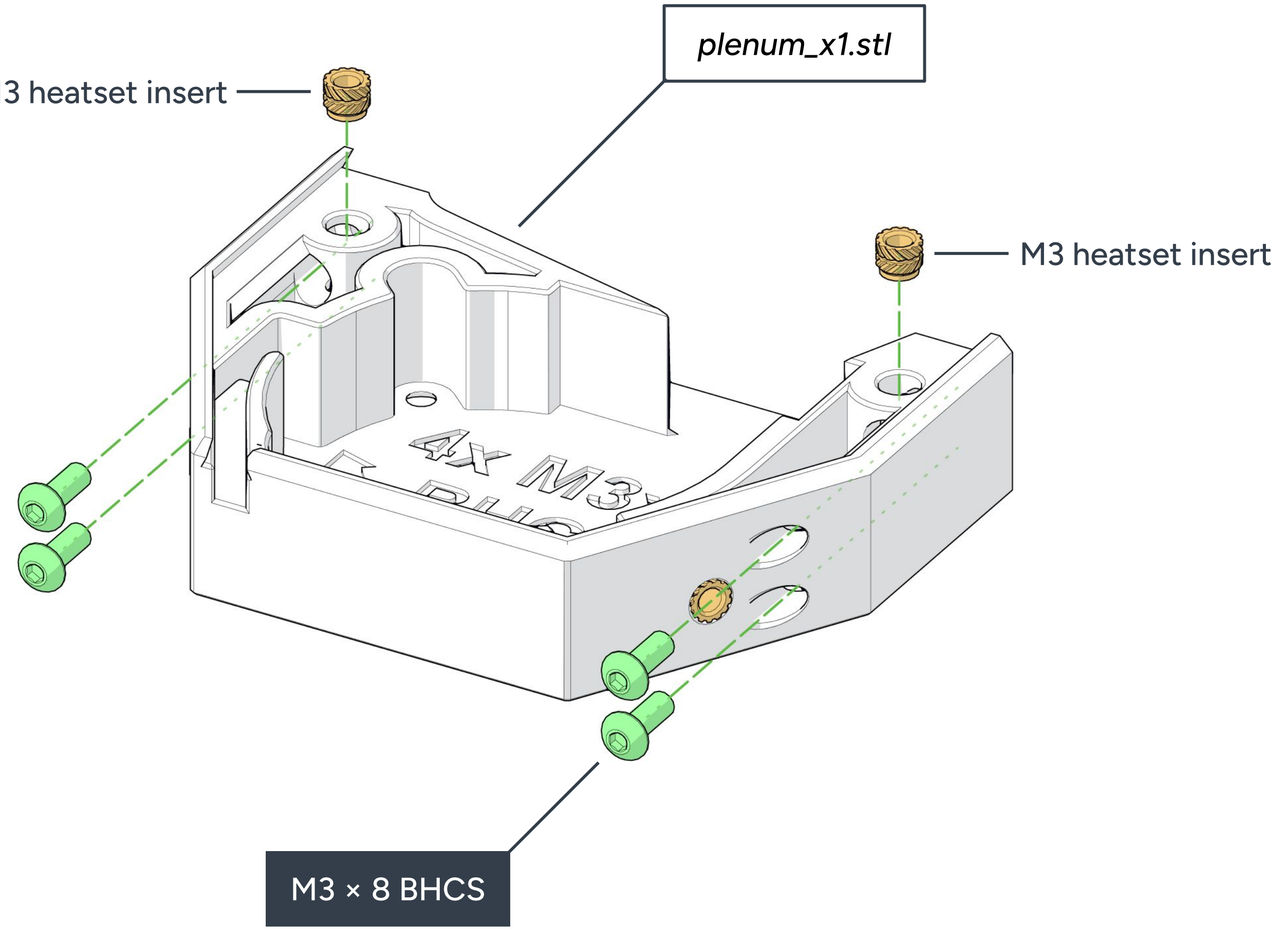
PLENUM PREP

Thread **four M3 × 8 BHCS** into ***plenum_x1.stl*** as shown. Use ferrous screws, as their ends serve as magnetic anchors for the cartridge.

Install the screws before the two indicated heatsets.

Once the screws are fully installed, install the **two M3 heatset inserts**.



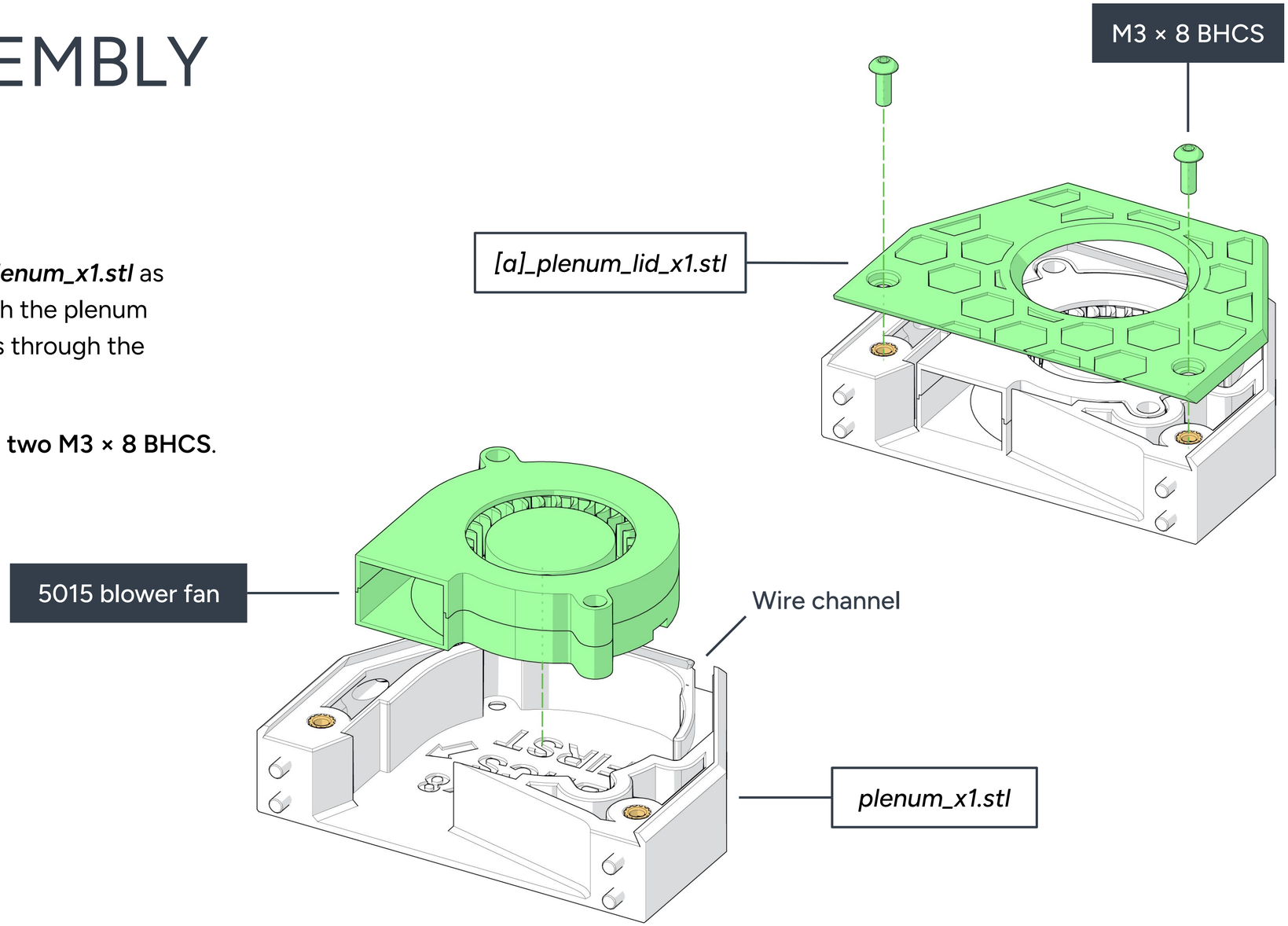


G/ FILTER ASSEMBLY

PLENUM PREP

Insert the **5015 blower fan** into ***plenum_x1.stl*** as shown, aligning the fan outlet with the plenum opening and routing the fan wires through the wire channel.

Secure ***[a]_plenum_lid_x1.stl*** with **two M3 × 8 BHCS**.

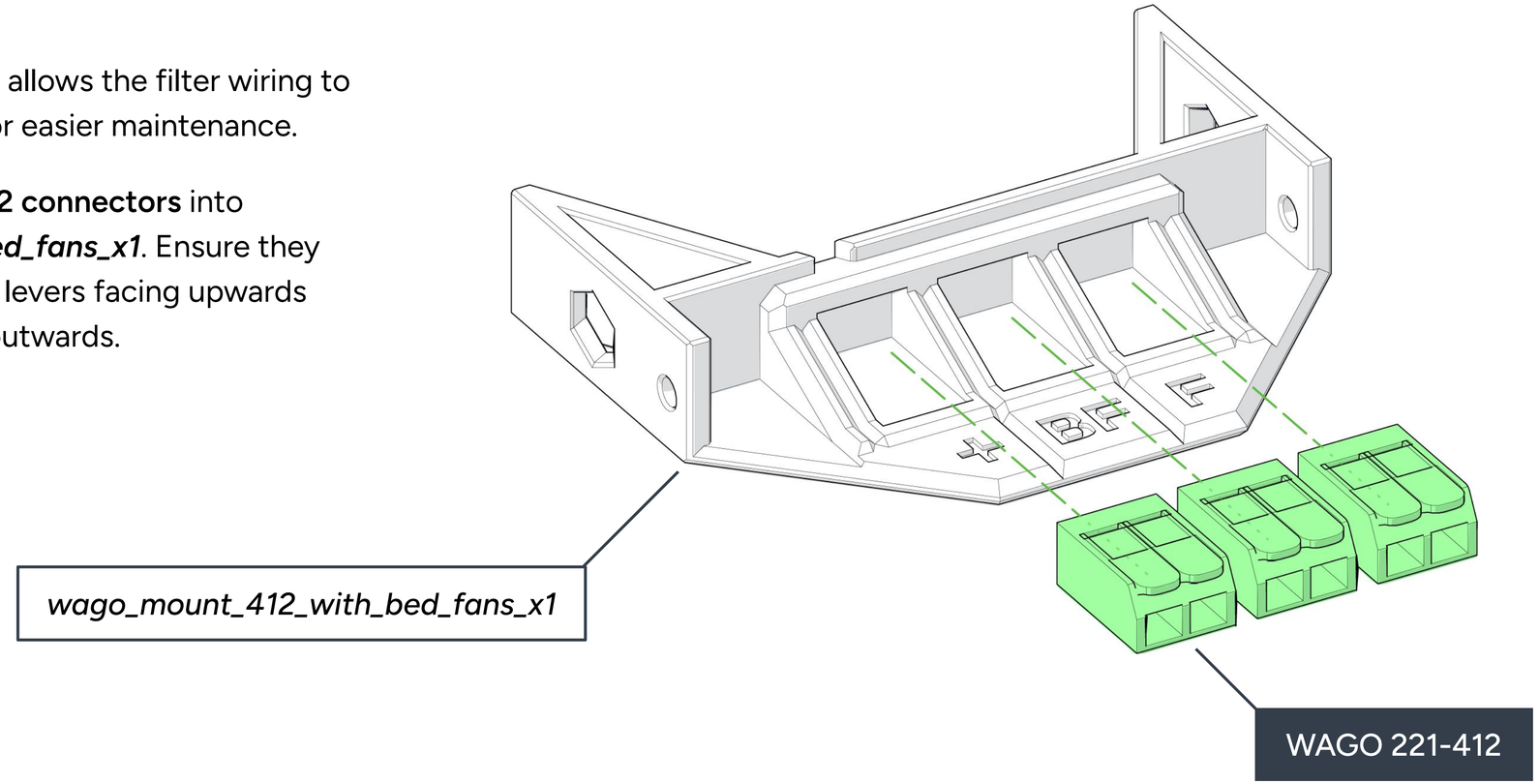


G/ FILTER ASSEMBLY

PLENUM PREP

The WAGO 221-412 mount allows the filter wiring to be quickly disconnected for easier maintenance.

Insert **three WAGO 221-412 connectors** into ***wago_mount_412_with_bed_fans_x1***. Ensure they click firmly into place, with levers facing upwards and wire openings facing outwards.

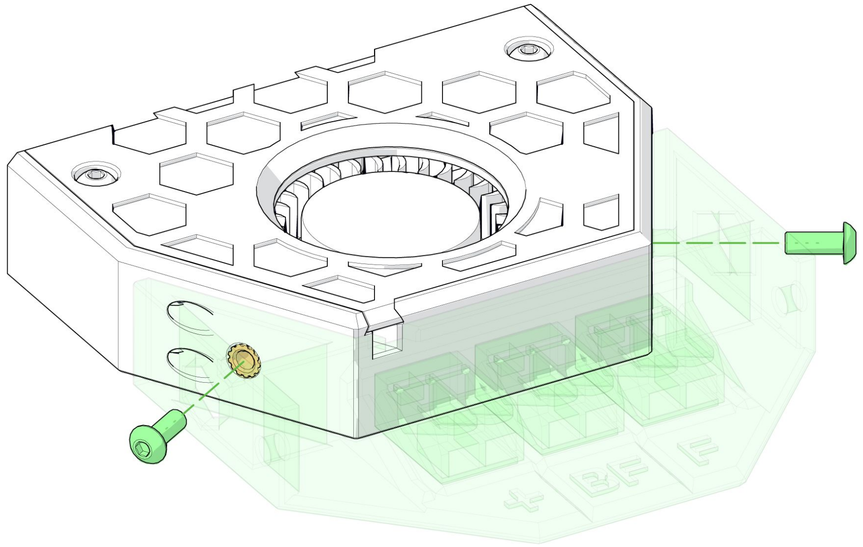


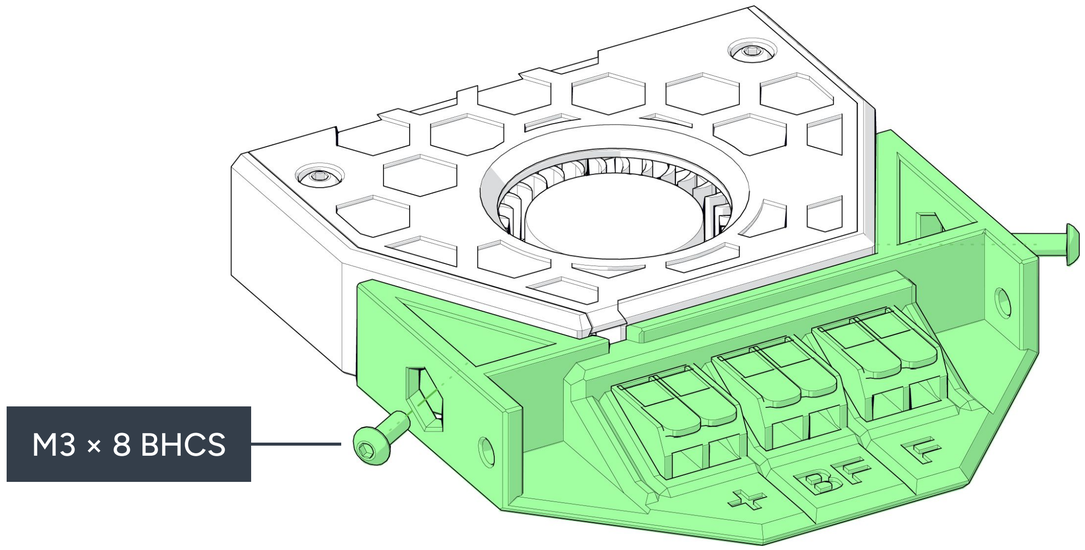
G/ FILTER ASSEMBLY

PLENUM PREP

Attach the WAGO mount assembly to the plenum with **two M3 × 8 BHCS** into the two M3 heatset inserts

Ensure fan wires exit without pinching.



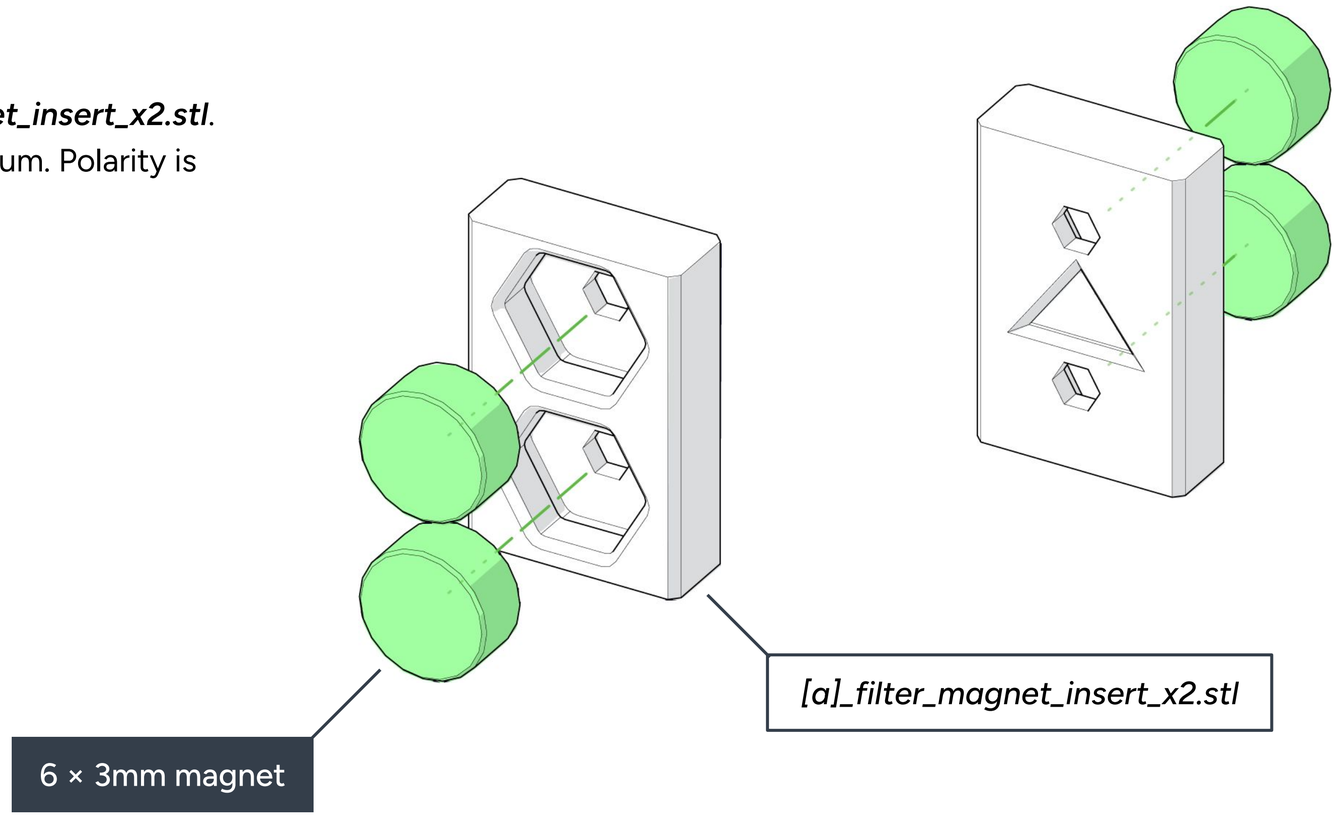


G/ FILTER ASSEMBLY

CARTRIDGE PREP

Insert **two 6 × 3mm magnets** into ***[a]_filter_magnet_insert_x2.stl***. The magnets secure the filter cartridge to the plenum. Polarity is not important.

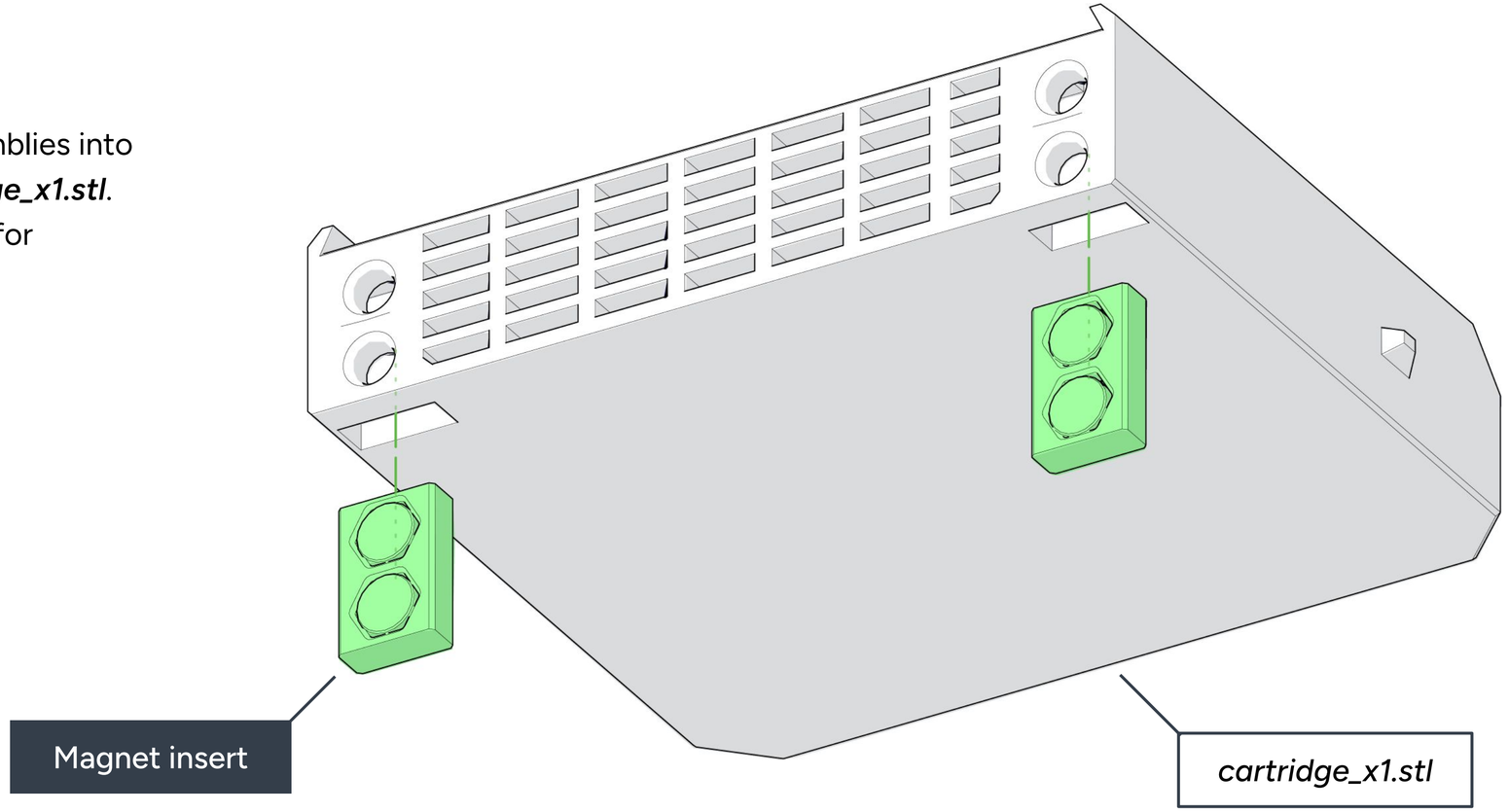
Prepare two identical assemblies.



G/ FILTER ASSEMBLY

CARTRIDGE PREP

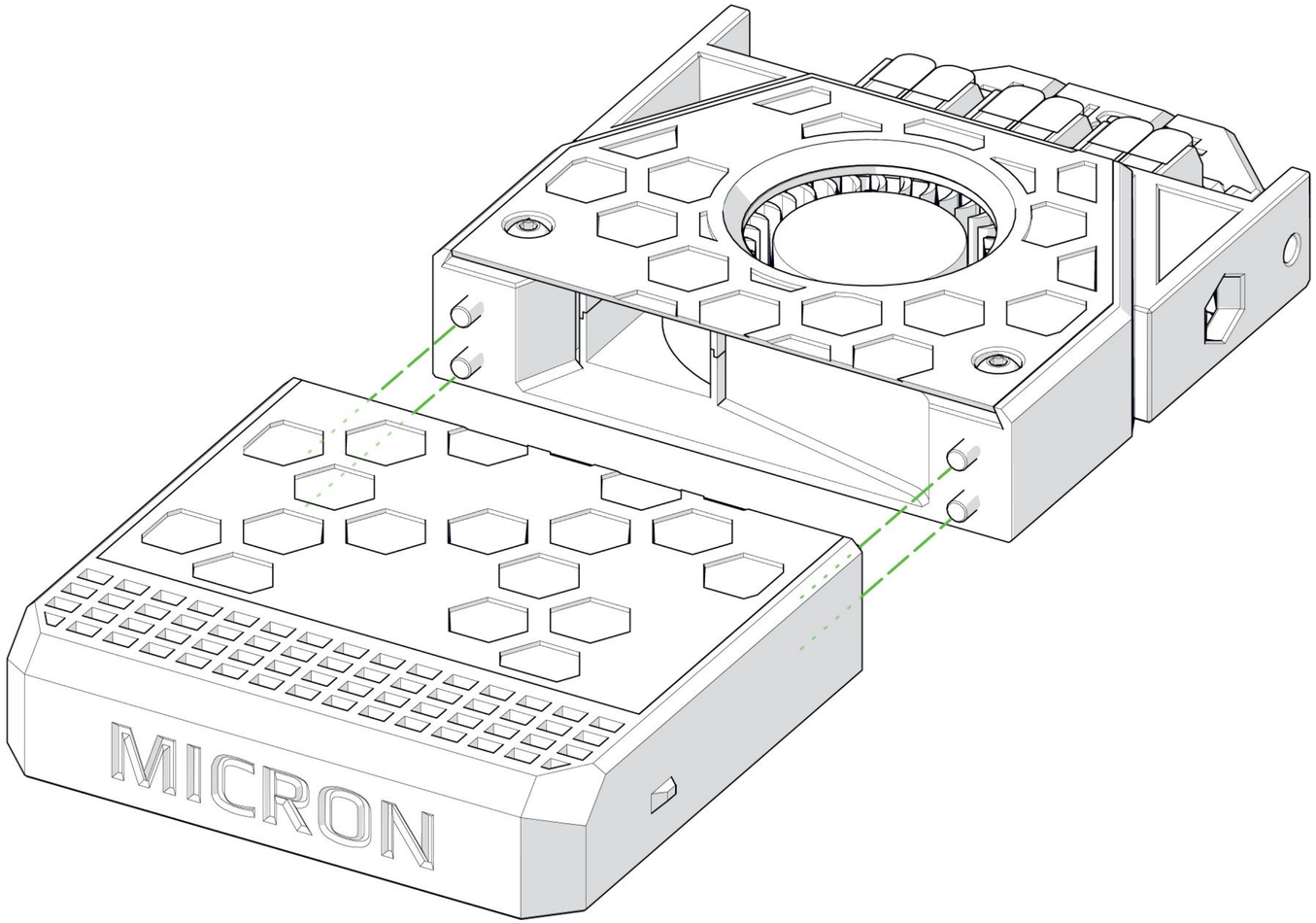
Insert the two magnet insert assemblies into the slots on the bottom of ***cartridge_x1.stl***. Ensure the magnets face outward for maximum holding force.



G/ FILTER ASSEMBLY

CARTRIDGE PREP

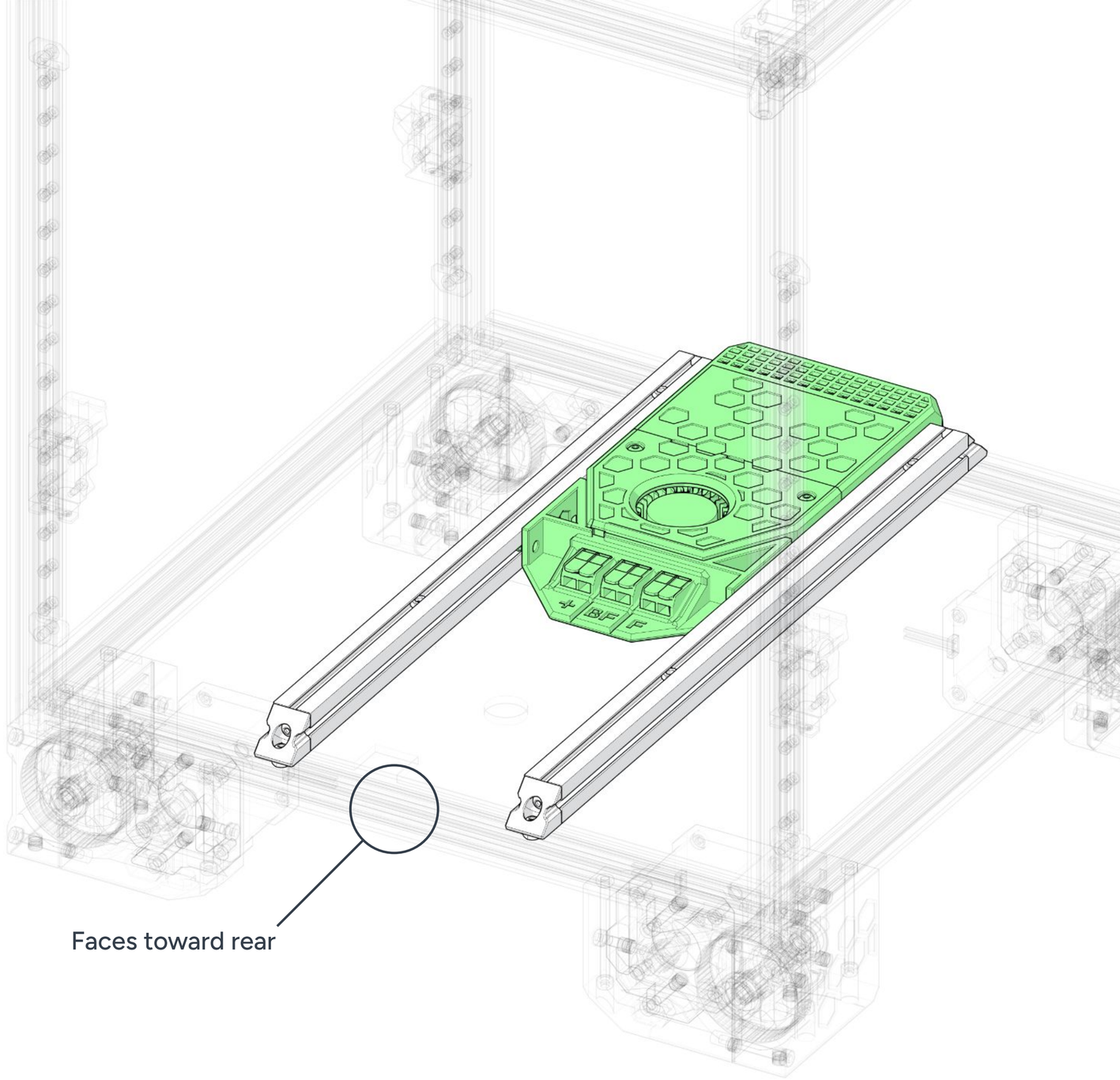
The cartridge attaches magnetically to the plenum. The four magnets align with the ends of the four M3 × 8 BHCS.



G/ FILTER ASSEMBLY

PLENUM MOUNTING

Fit the plenum assembly between the two bed extrusions as shown.



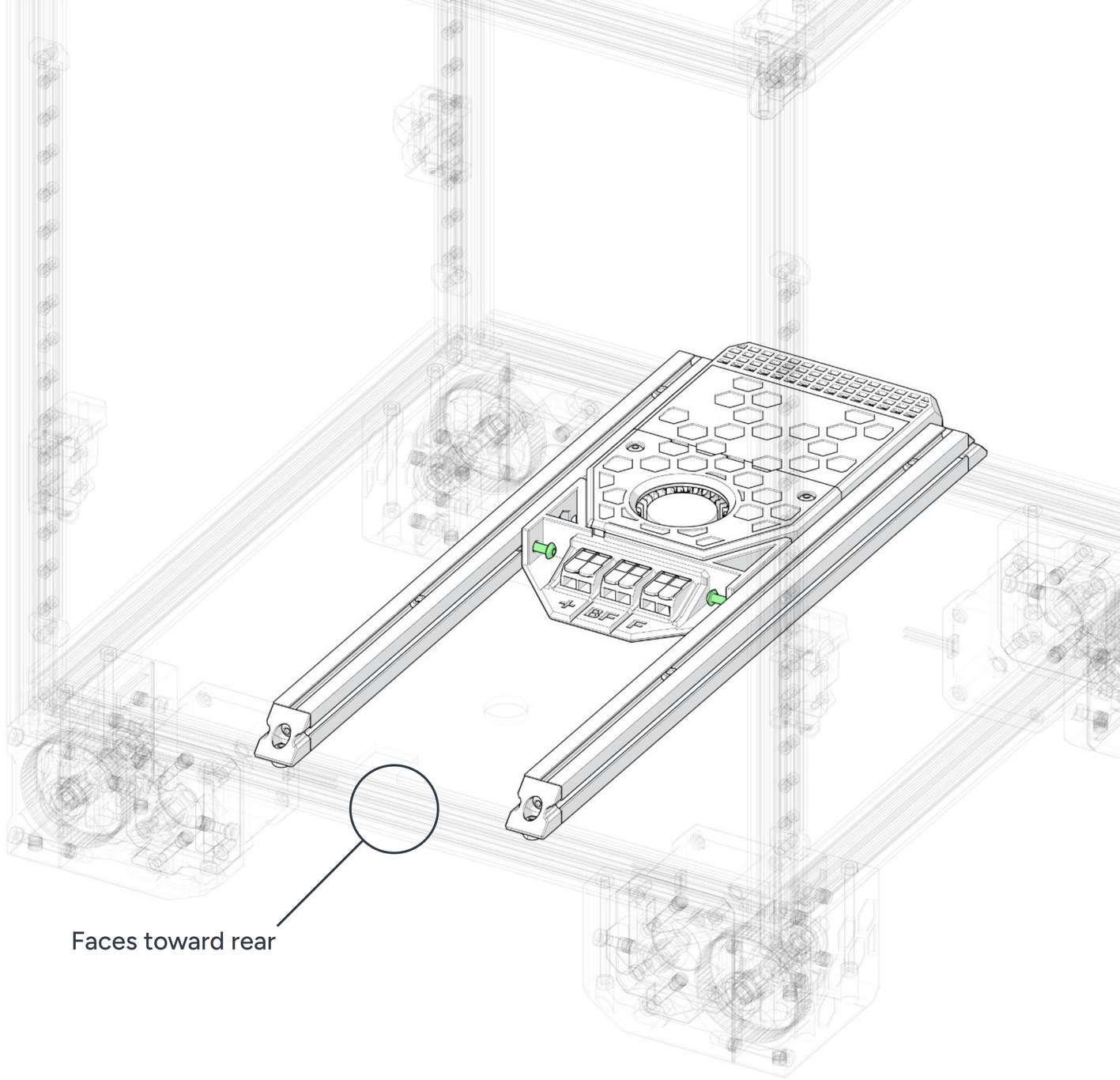
NOTE

This view is shown from the rear of the printer to better illustrate this step.

G/ FILTER ASSEMBLY

PLENUM MOUNTING

Secure the plenum assembly to the two preloaded hex nuts with **two M3 × 8 BHCS**.



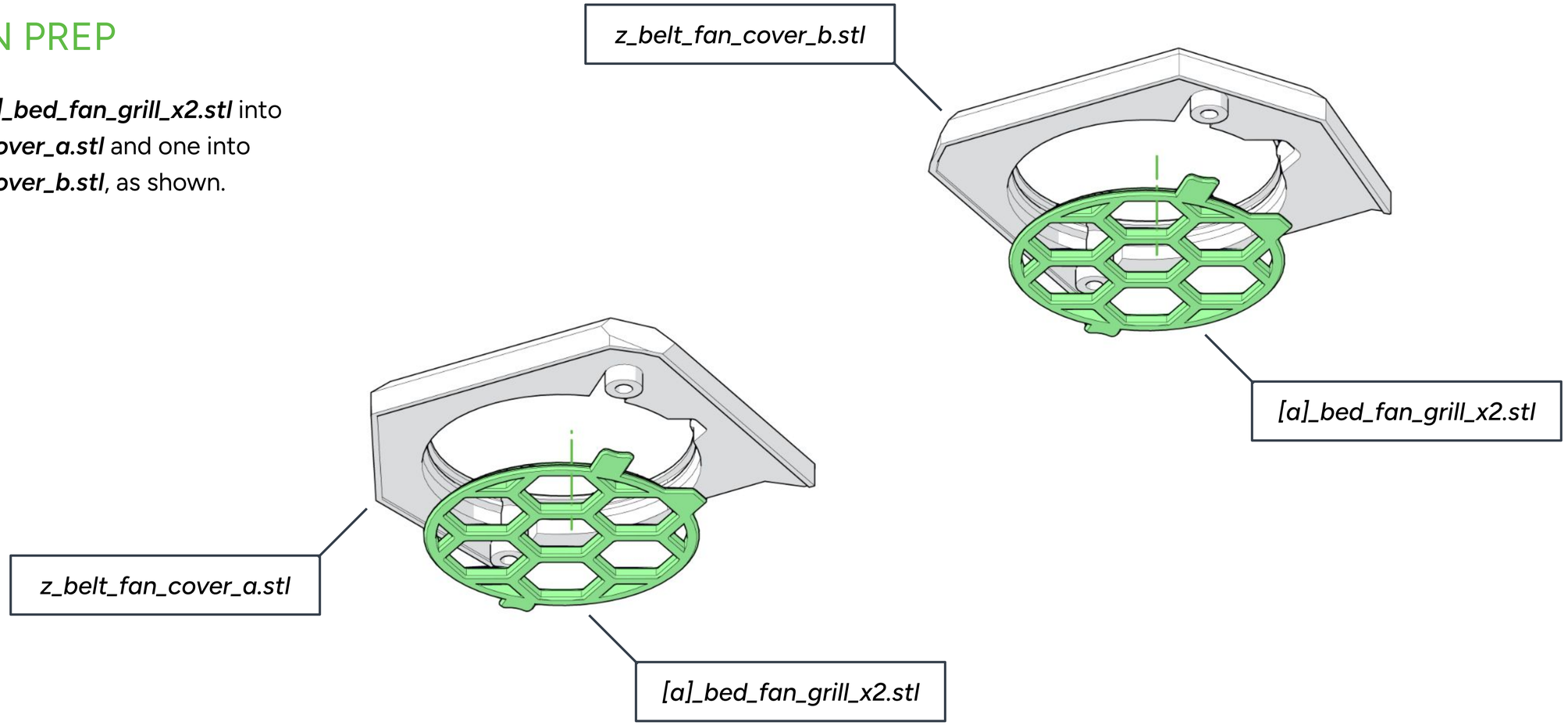
NOTE

This view is shown from the rear of the printer to better illustrate this step.

G/ FILTER ASSEMBLY

BED FAN PREP

Insert **one** ***[a]_bed_fan_grill_x2.stl*** into ***z_belt_fan_cover_a.stl*** and one into ***z_belt_fan_cover_b.stl***, as shown.

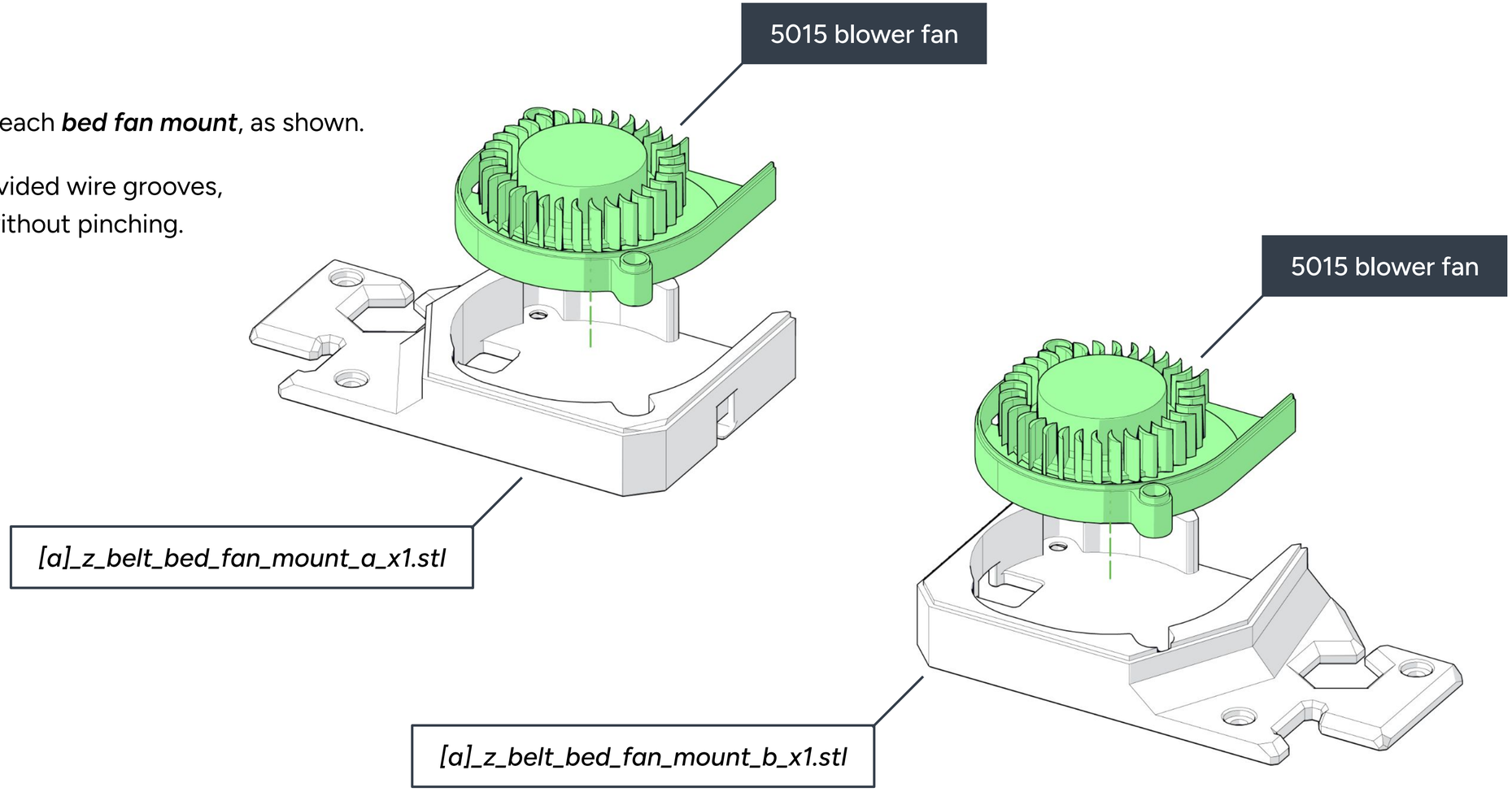


G/ FILTER ASSEMBLY

BED FAN PREP

Insert **one *5015 fan*** into each ***bed fan mount***, as shown.

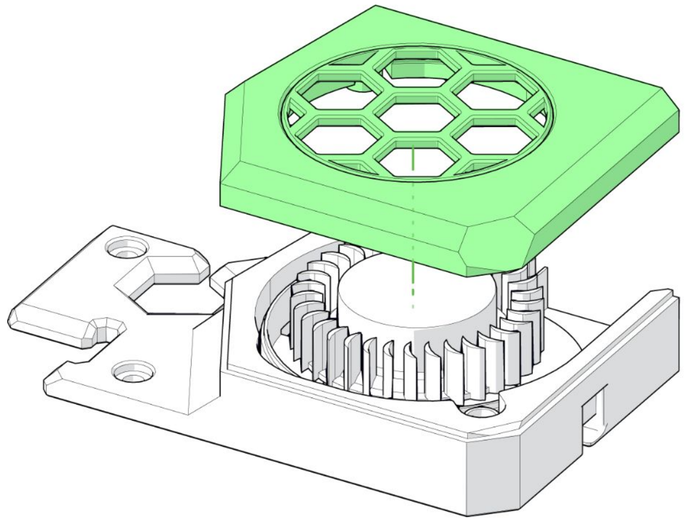
Route wires through provided wire grooves, ensuring fan wires exit without pinching.

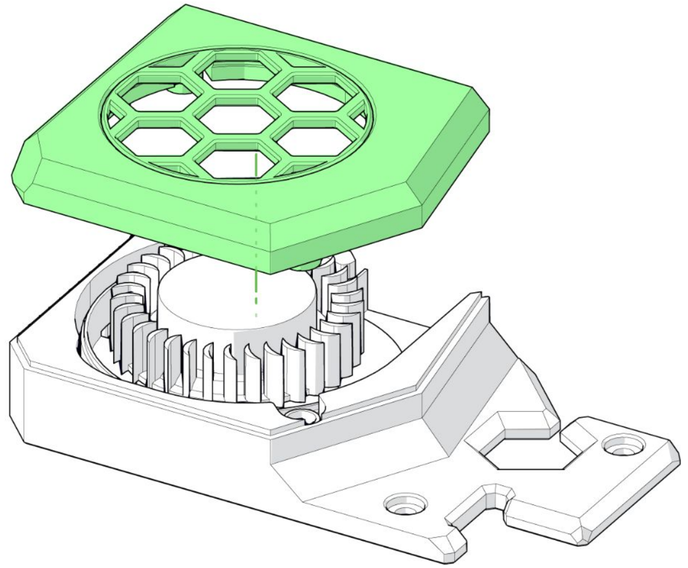


G/ FILTER ASSEMBLY

BED FAN PREP

Cap each bed fan assembly with the cover assemblies.

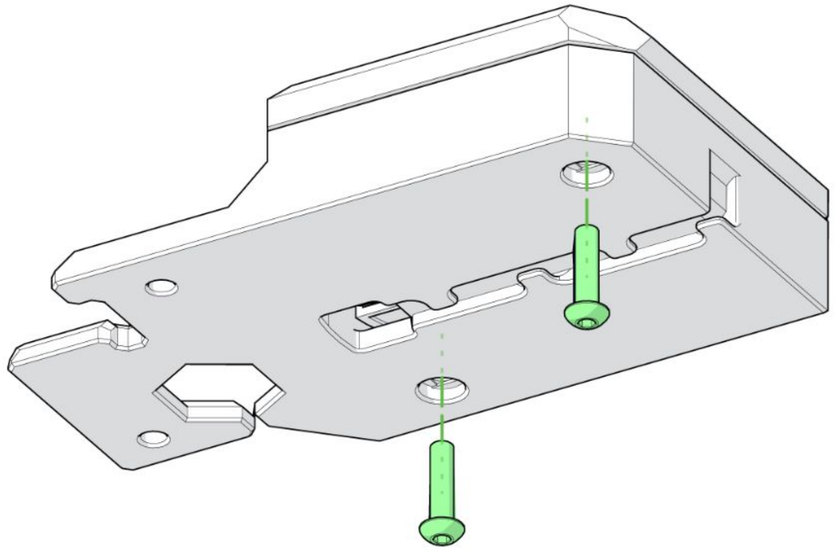


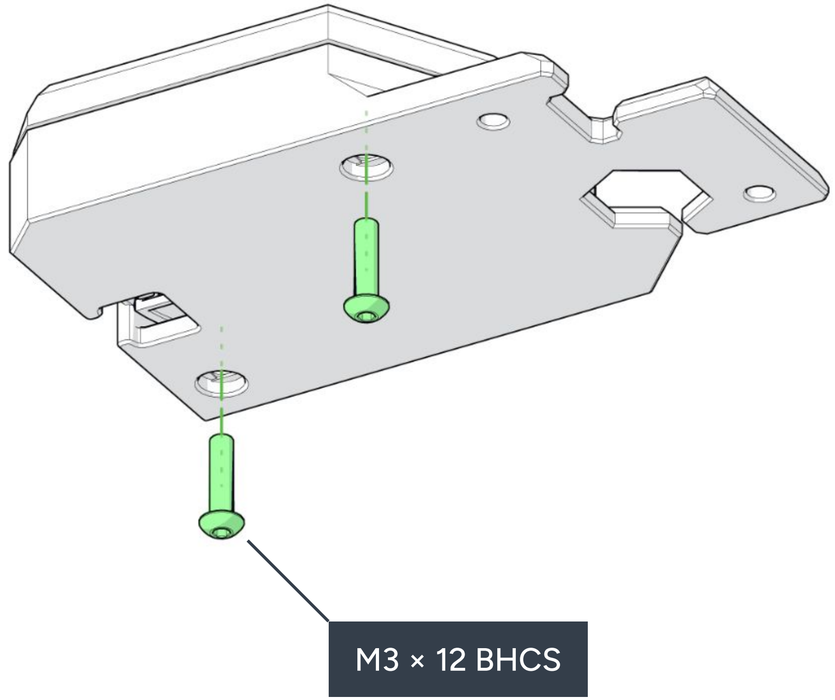


G/ FILTER ASSEMBLY

BED FAN PREP

Secure each bed fan assembly with **two M3 × 12 BHCS**. These screws are threaded directly into plastic.

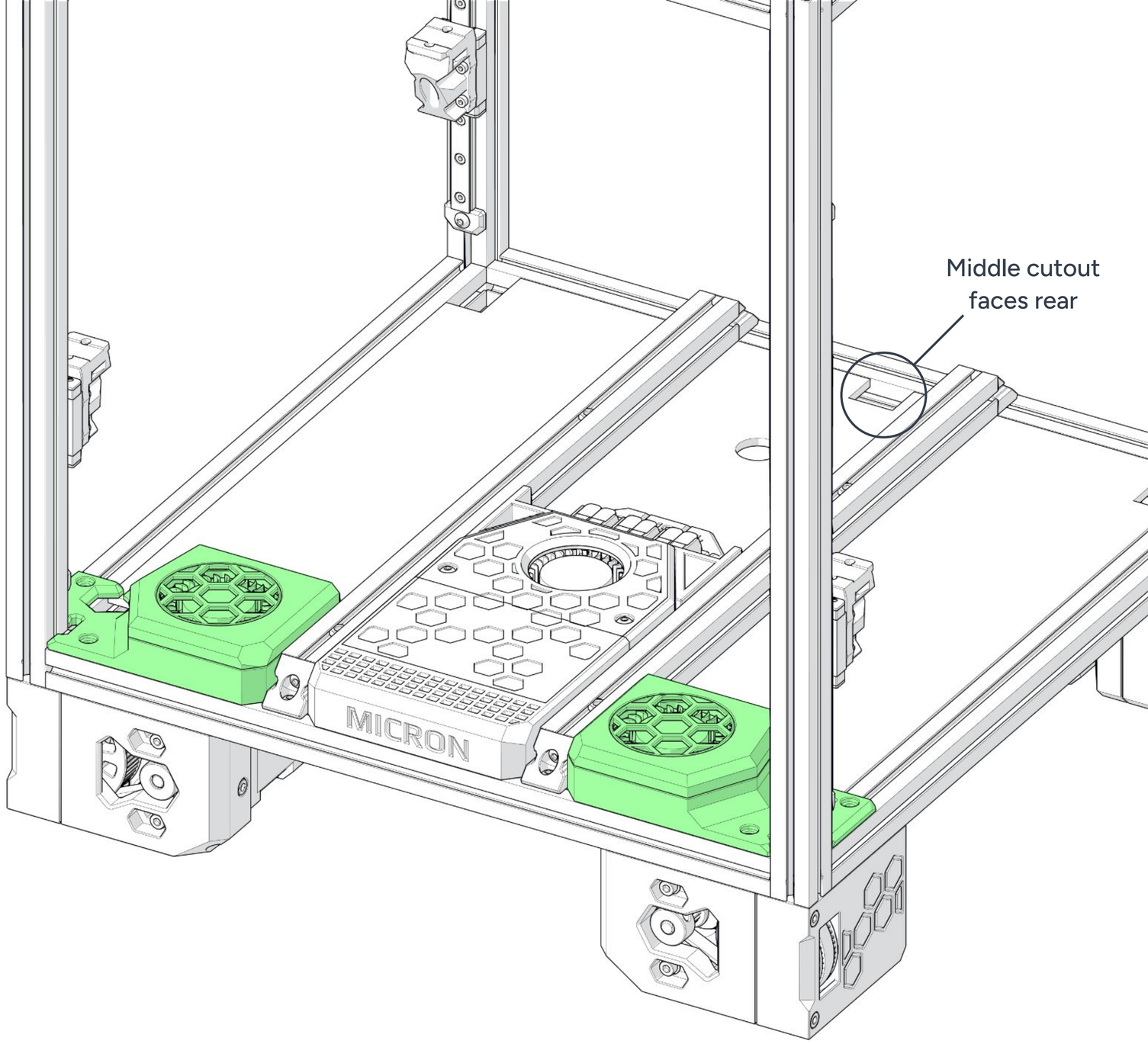




G/ FILTER ASSEMBLY

BED FAN MOUNTING

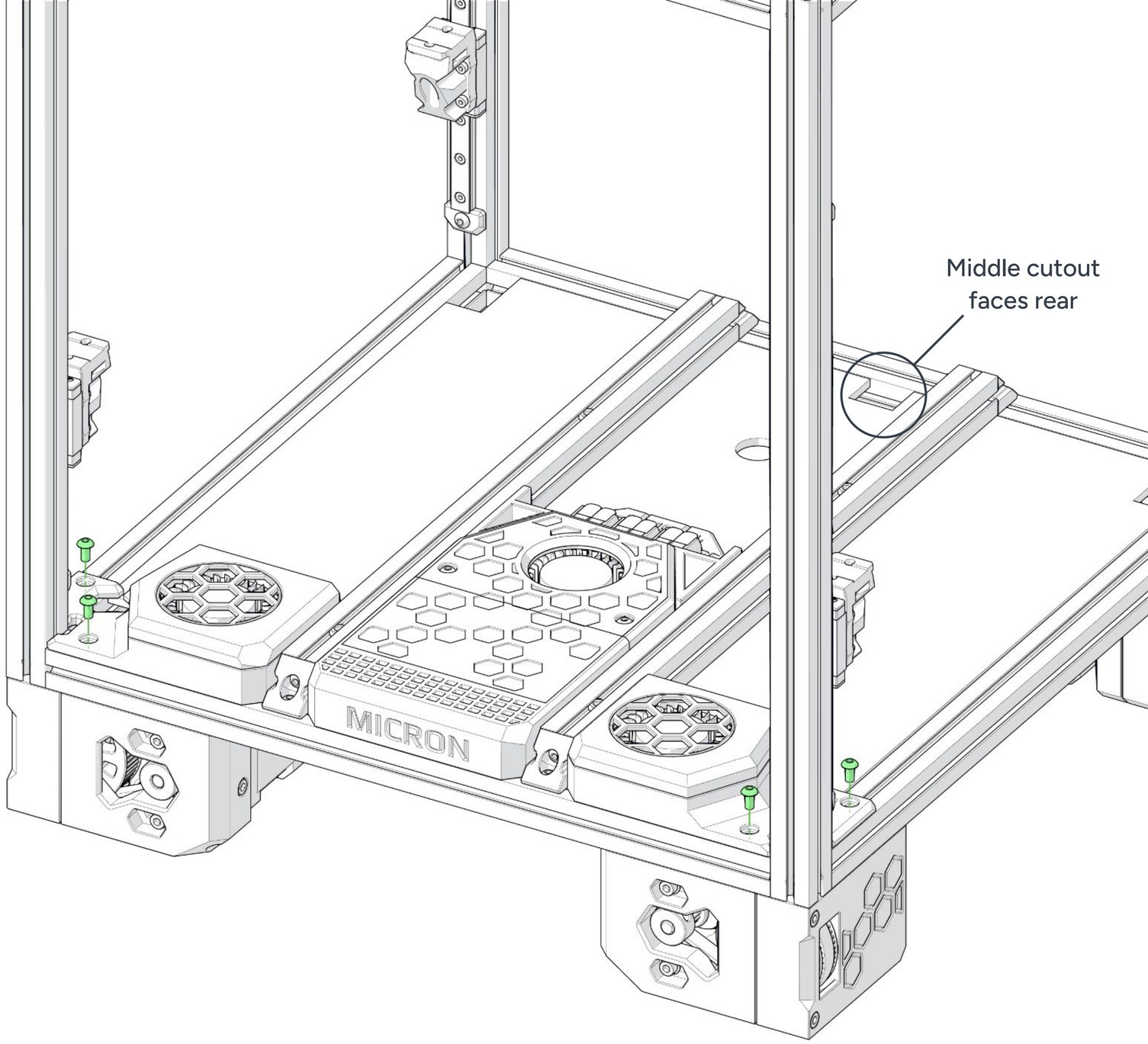
Fit the two bed fan assemblies as shown.



G/ FILTER ASSEMBLY

BED FAN MOUNTING

Secure the bed fan assemblies to preloaded hex nuts with **two M3 × 6 BHCS** each.



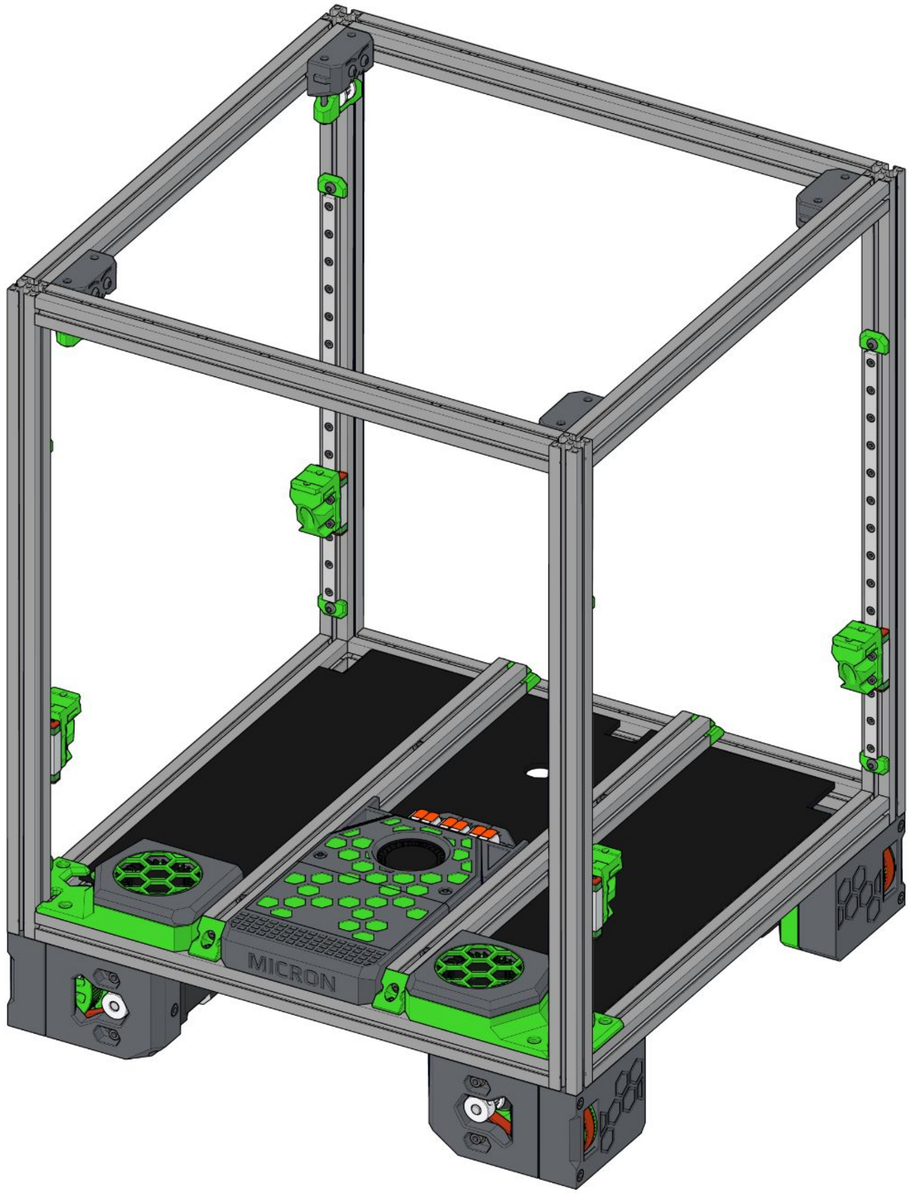
G/ FILTER ASSEMBLY

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

- ▢ Plenum is secured to the bed extrusions
- ▢ Cartridge attaches/detaches via magnets
- ▢ Fans are securely mounted with openings to match belt pass through cutouts



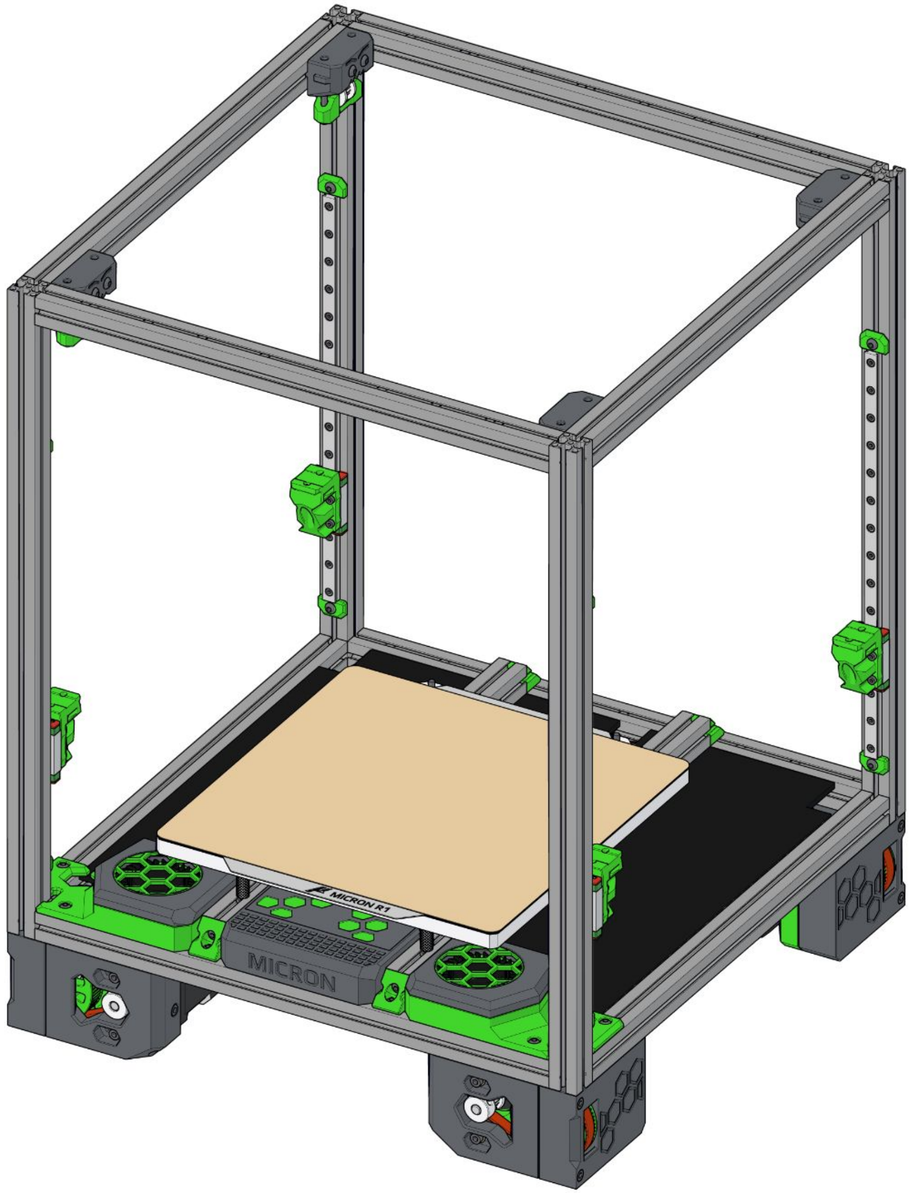
H/ BED ASSEMBLY

BED ASSEMBLY

The Micron+ bed assembly is mounted to aluminum extrusions fixed directly to the frame. This provides a rigid bed structure.

HARDWARE

1	Flex plate
1	Magnetic sheet
1	MIC6 aluminum bed
1	Bed heater
1	Thermal fuse
1	M3 × 6 BHCS
1	M3 × 8 BHCS
8	M3 × 16 SHCS
1	M3 washer
1	Ring terminal
4	M3 × 8 mm standoffs

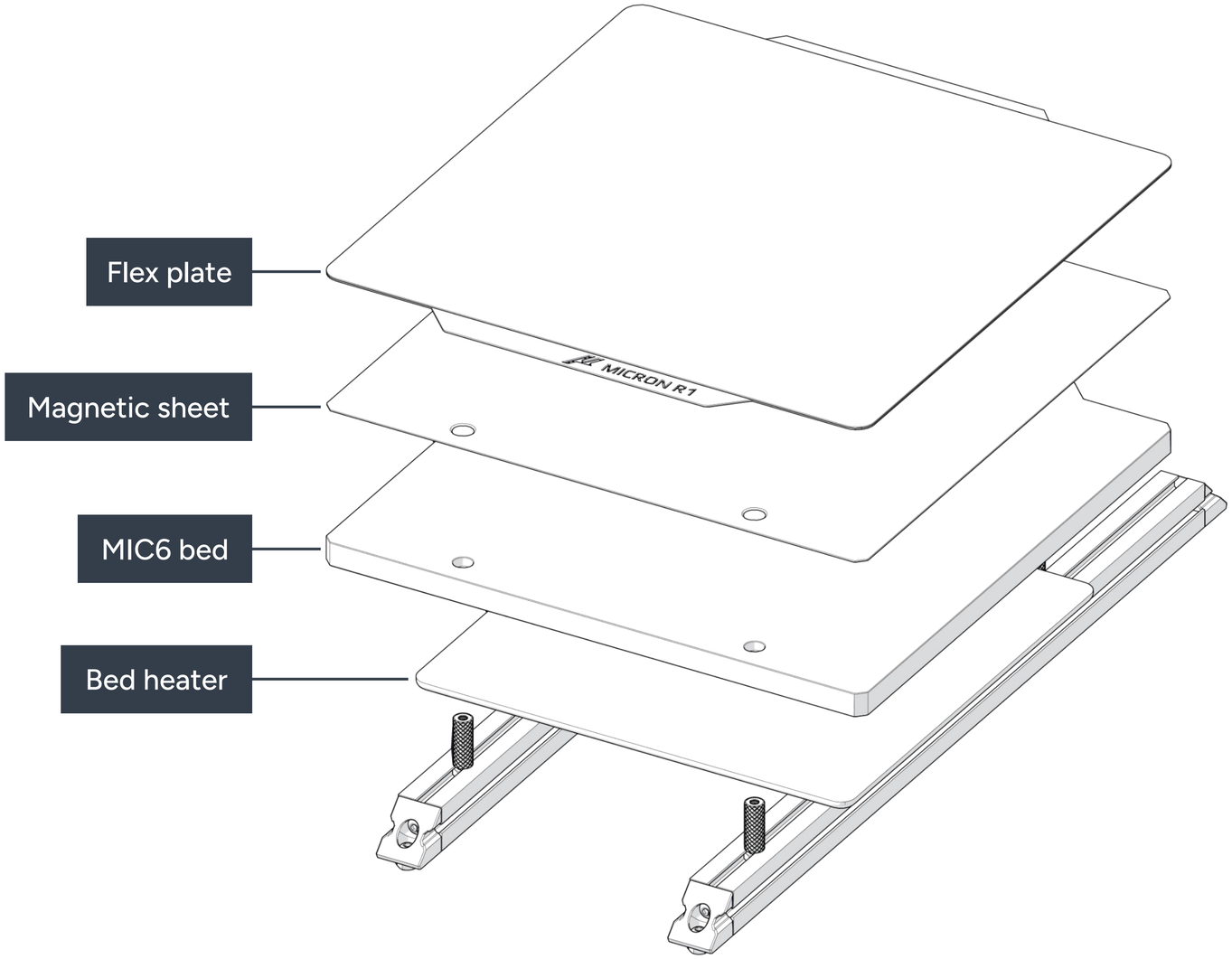


H/ BED ASSEMBLY

BED LAYERS

The bed assembly is comprised of the following stack:

- **Flex plate**
- **Magnetic sheet**
- **MIC6 bed**
- **Bed heater**



NOTE

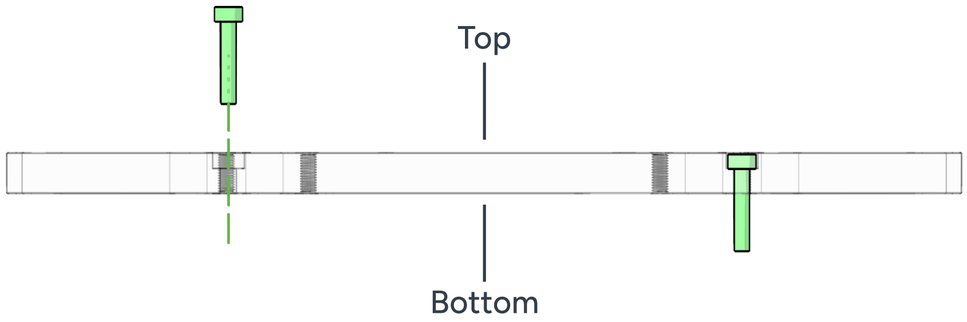
Included hardware will vary by kit.

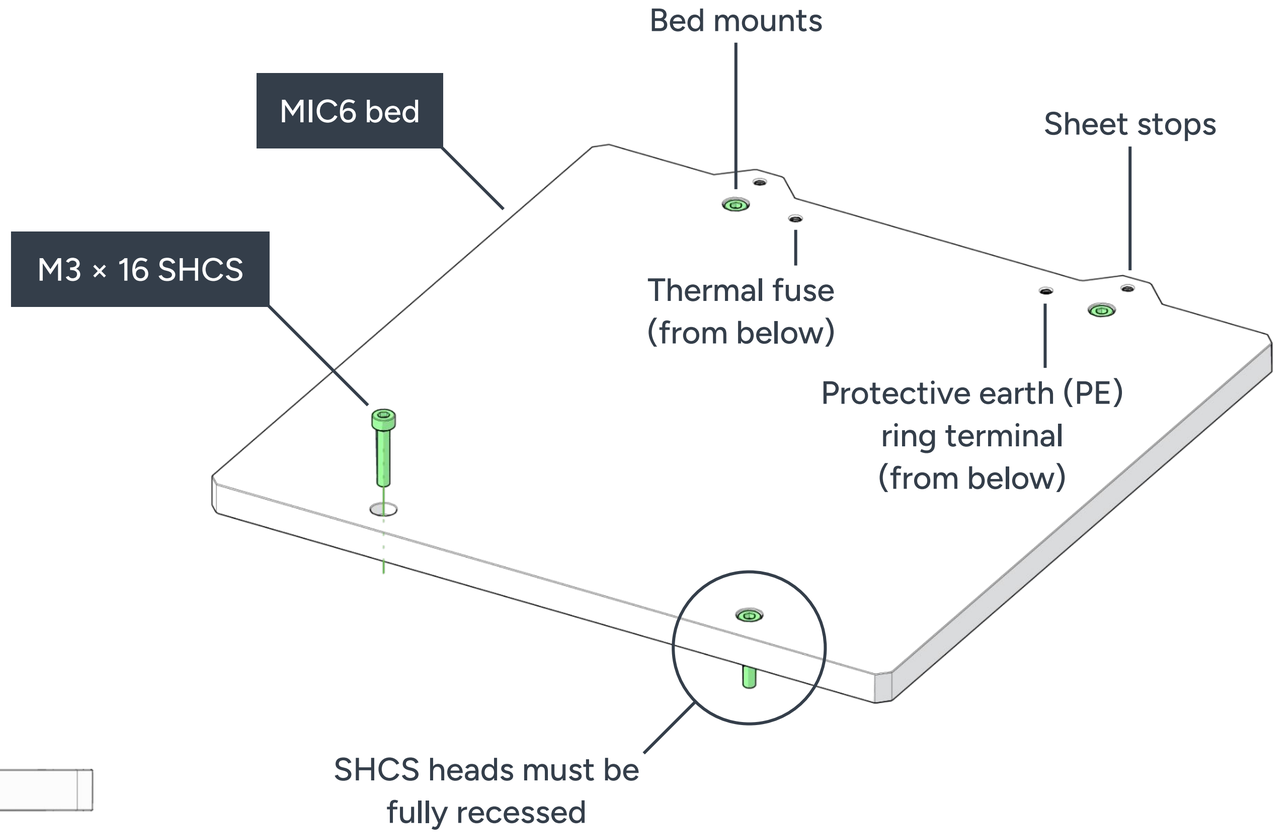
H/ BED ASSEMBLY

ORIENTING THE BED

The bed has a definitive top side with counterbores for recessed SHCS heads.

The diagram shows SHCS placement to help determine the top side. Do not install them yet.





H/ BED ASSEMBLY

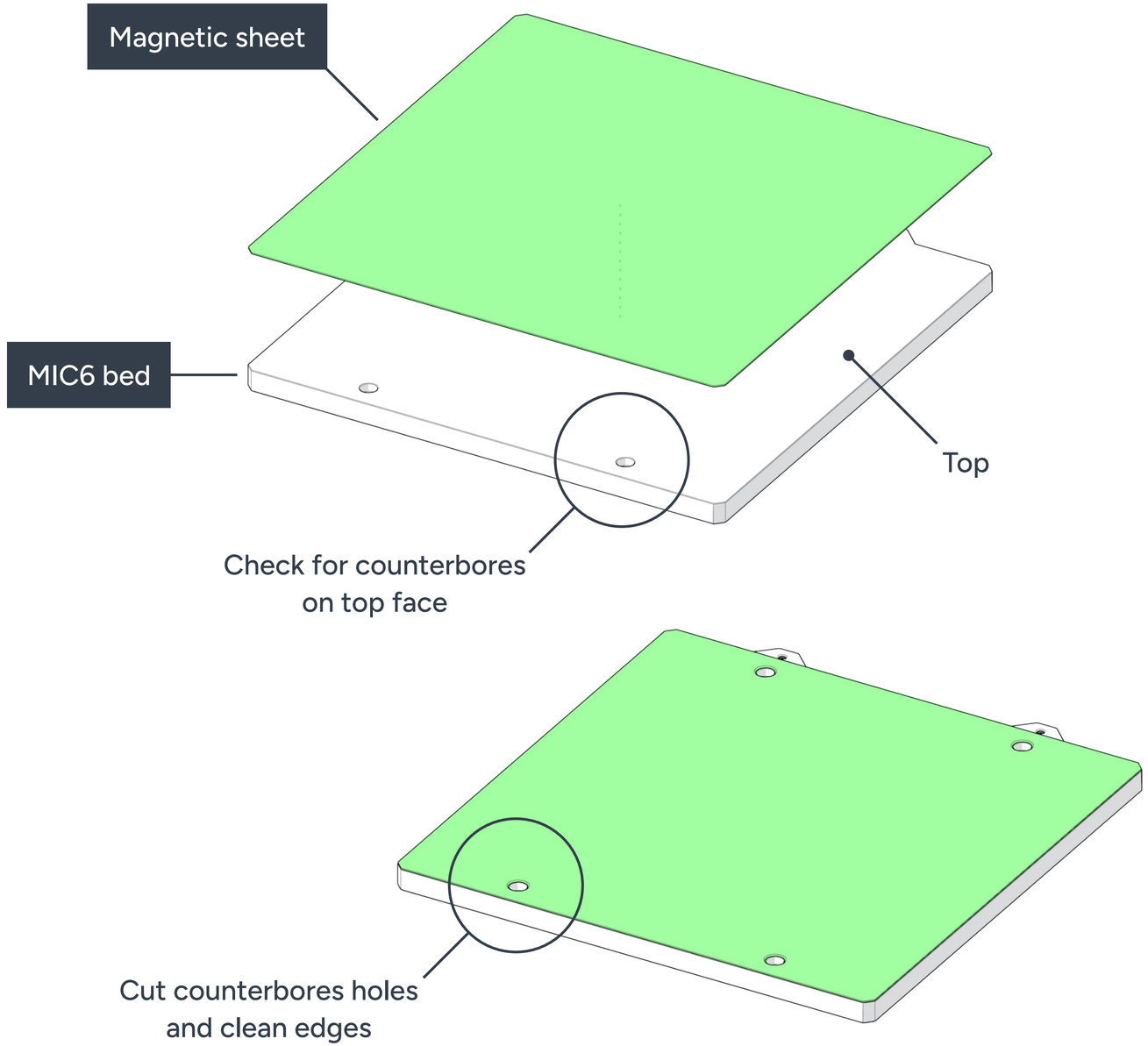
MAGNETIC SHEET INSTALL

Installing the magnetic sheet is straightforward with a bit of preparation. The peel-and-reveal method helps align the sheet while protecting the adhesive as it is applied.

Clean the bed surface with isopropyl alcohol.

Start by peeling a small section of liner along one edge, carefully align, then press firmly from the center outward. Gradually remove the liner, applying even pressure to avoid bubbles. Continue until the entire sheet is fully applied. Finally, cut the counterbore holes and clean the edges so they sit flush with the sheet surface.

Some magnetic sheets come oversized, reducing the need for precise alignment. Tack the sheet down using the same method, then trim the overhanging edges flush with the bed.



H/ BED ASSEMBLY

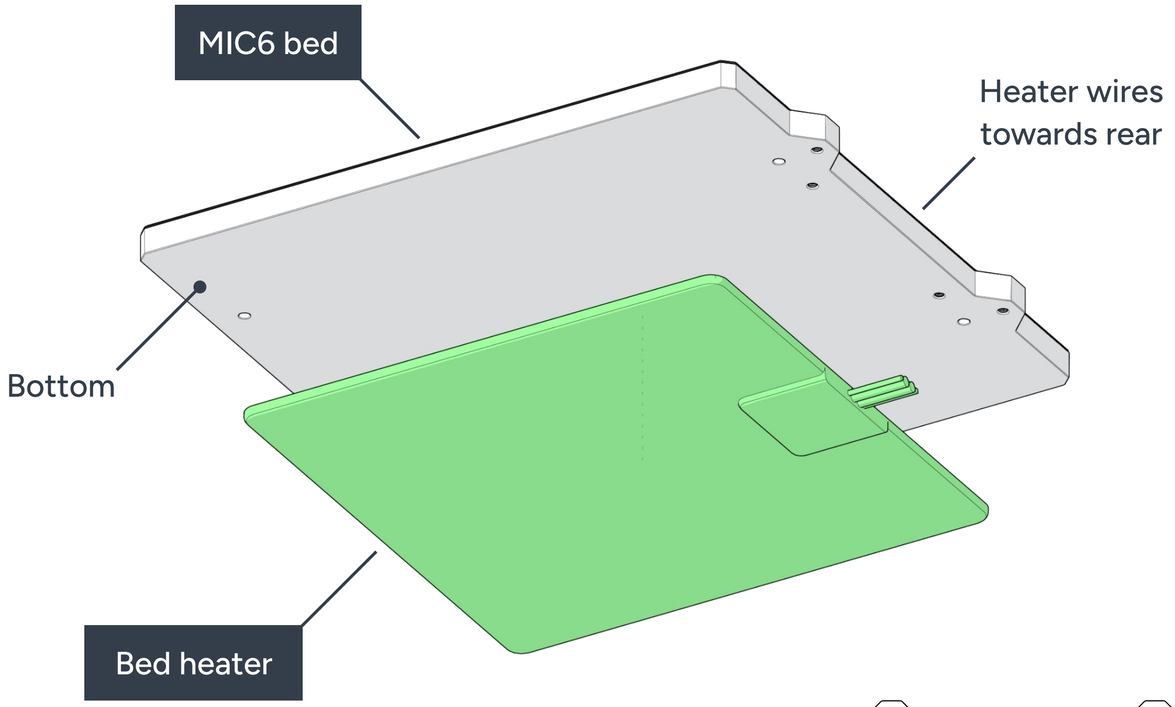
BED HEATER INSTALL

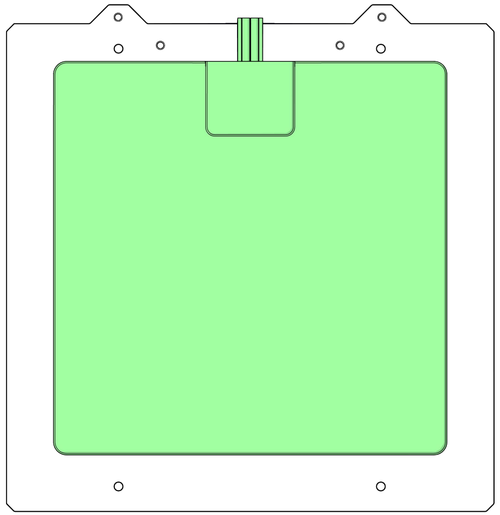
Install the magnetic sheet before the bed heater, as the heater wires will interfere.

Clean the bed surface with isopropyl alcohol.

Before installation, position the bed heater on the bed, check that the wires exit toward the rear, and trace its ideal location in pencil. This will provide alignment guidance and ensure even bed heating.

Install the bed heater using the same peel-and-reveal method described previously, taking care not to press on the heater wires.



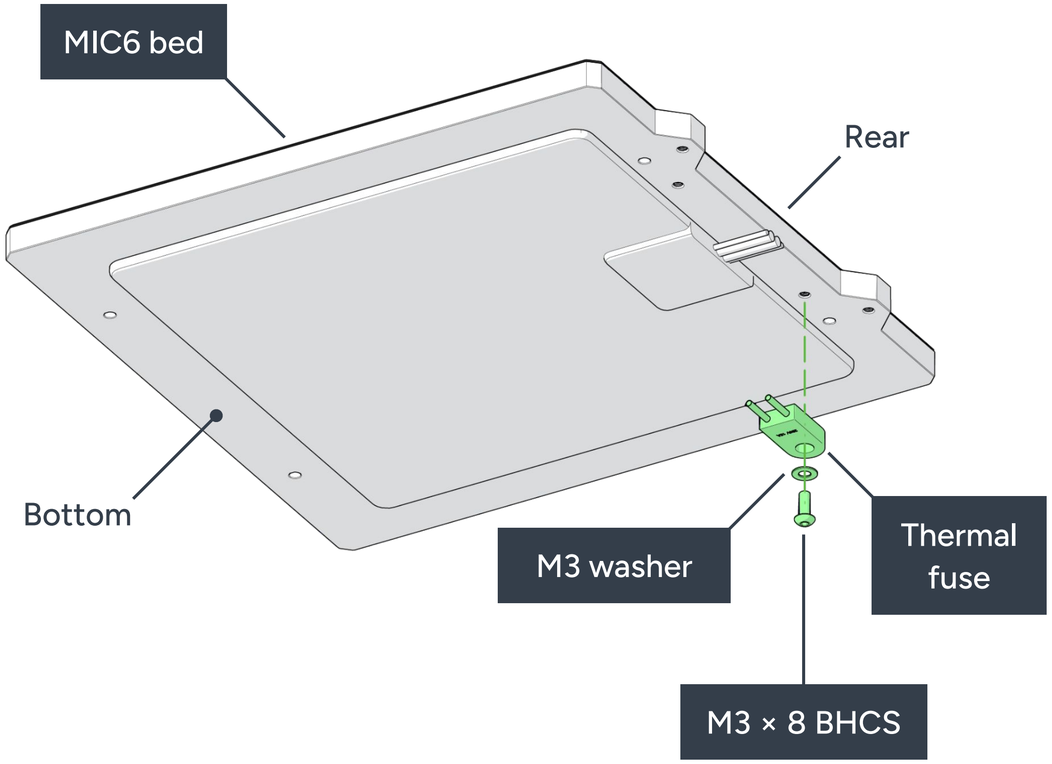


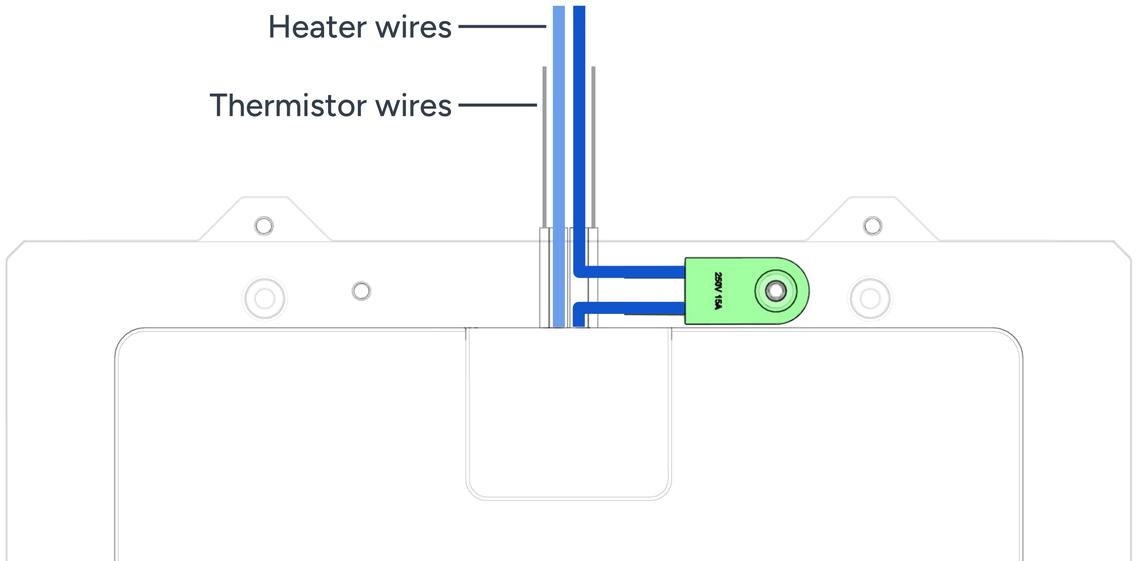
H/ BED ASSEMBLY

PREPPING THE BED ASSEMBLY

The **thermal fuse** provides a physical safety cutoff device in the event of heater runaway. Attach the thermal fuse to the underside of the bed plate using an **M3 washer** and **M3 × 8 BHCS**.

Connect the thermal fuse inline with the heater wires (not the thermistor wires). No polarity required.

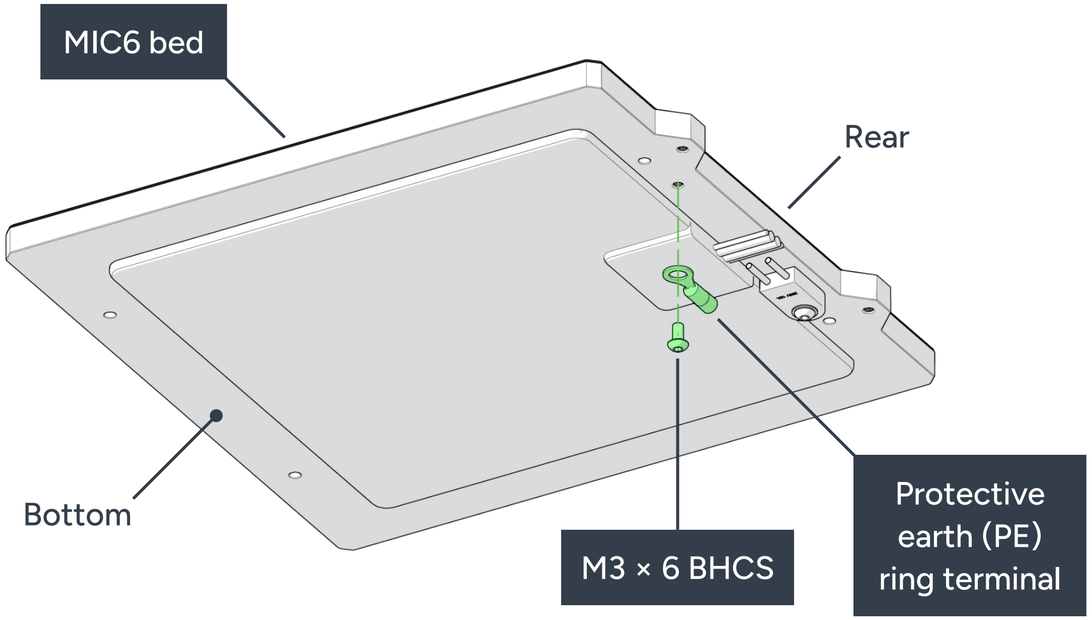


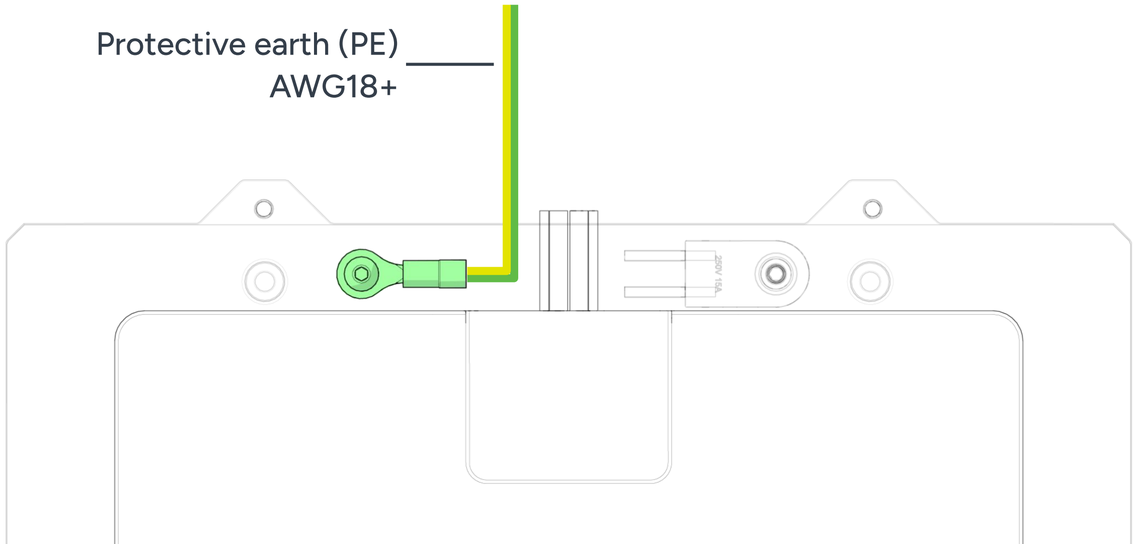


H/ BED ASSEMBLY

PREPPING THE BED ASSEMBLY

Attach the **protective earth (PE) ring terminal** to the underside of the bed plate using an **M3 × 6 BHCS**. Ensure the connection is secure and makes direct metal-to-metal contact.

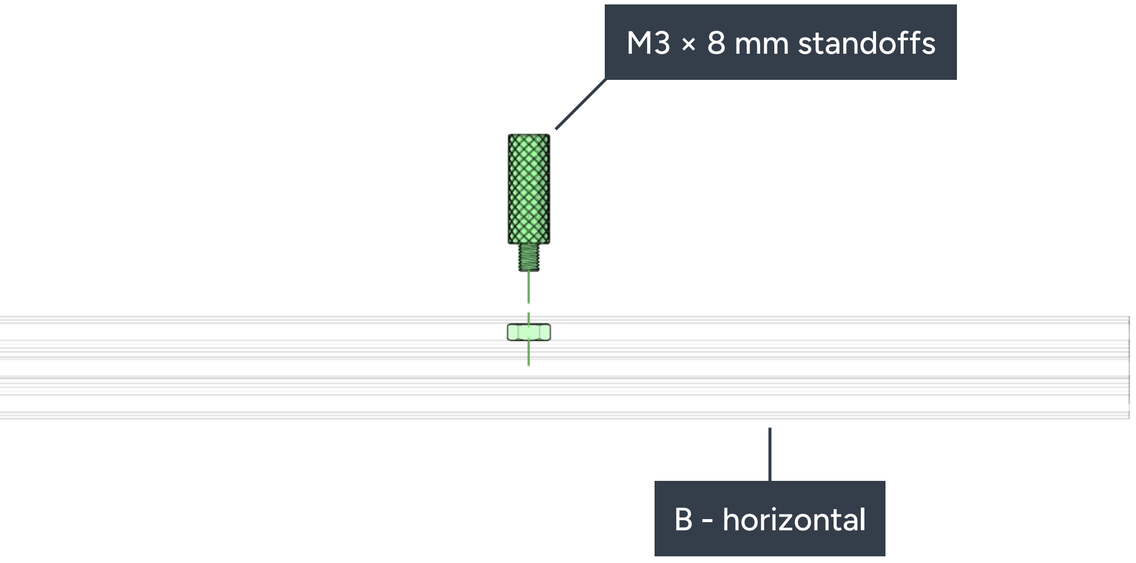


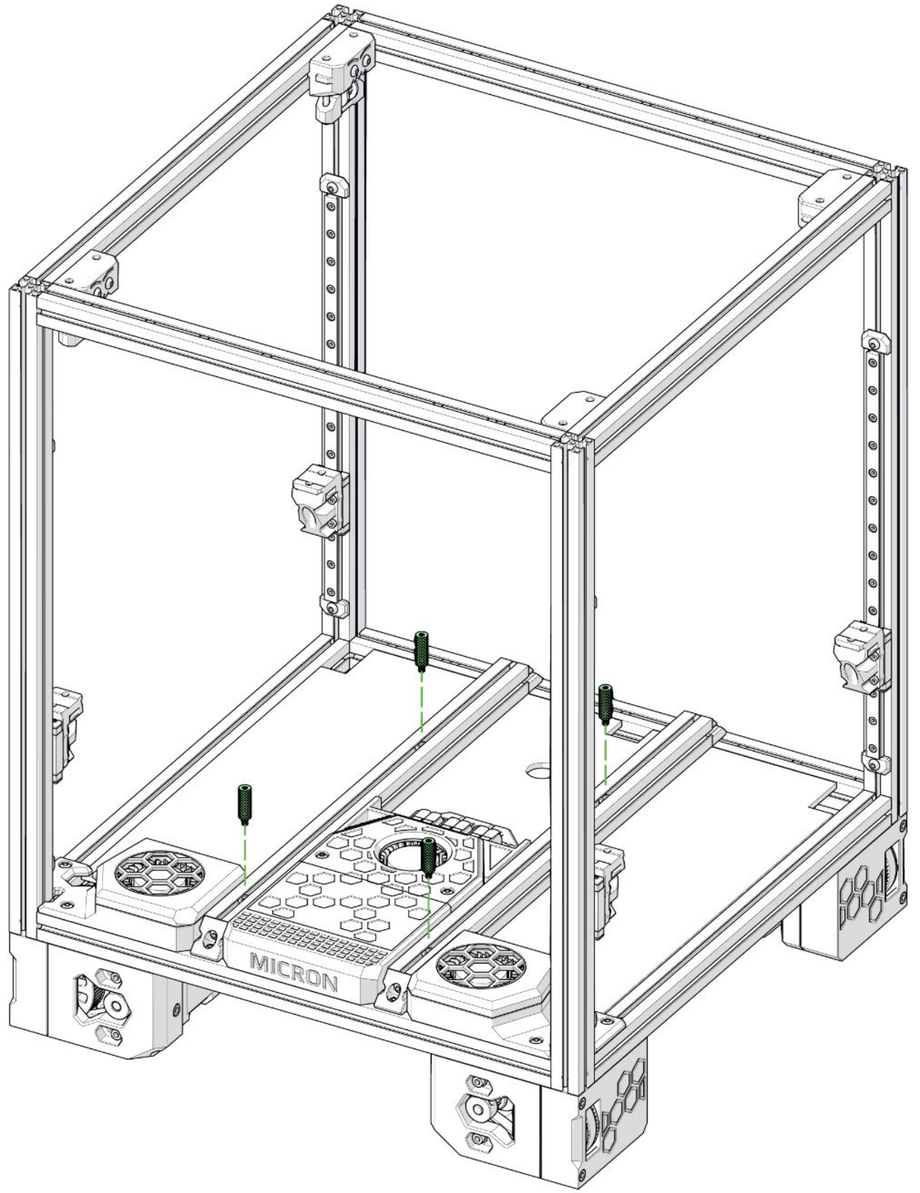


H/ BED ASSEMBLY

MOUNTING THE BED ASSEMBLY

The bed rests on **four M3 × 8 mm standoffs** fastened to preloaded hex nuts in the B horizontal bed extrusions.

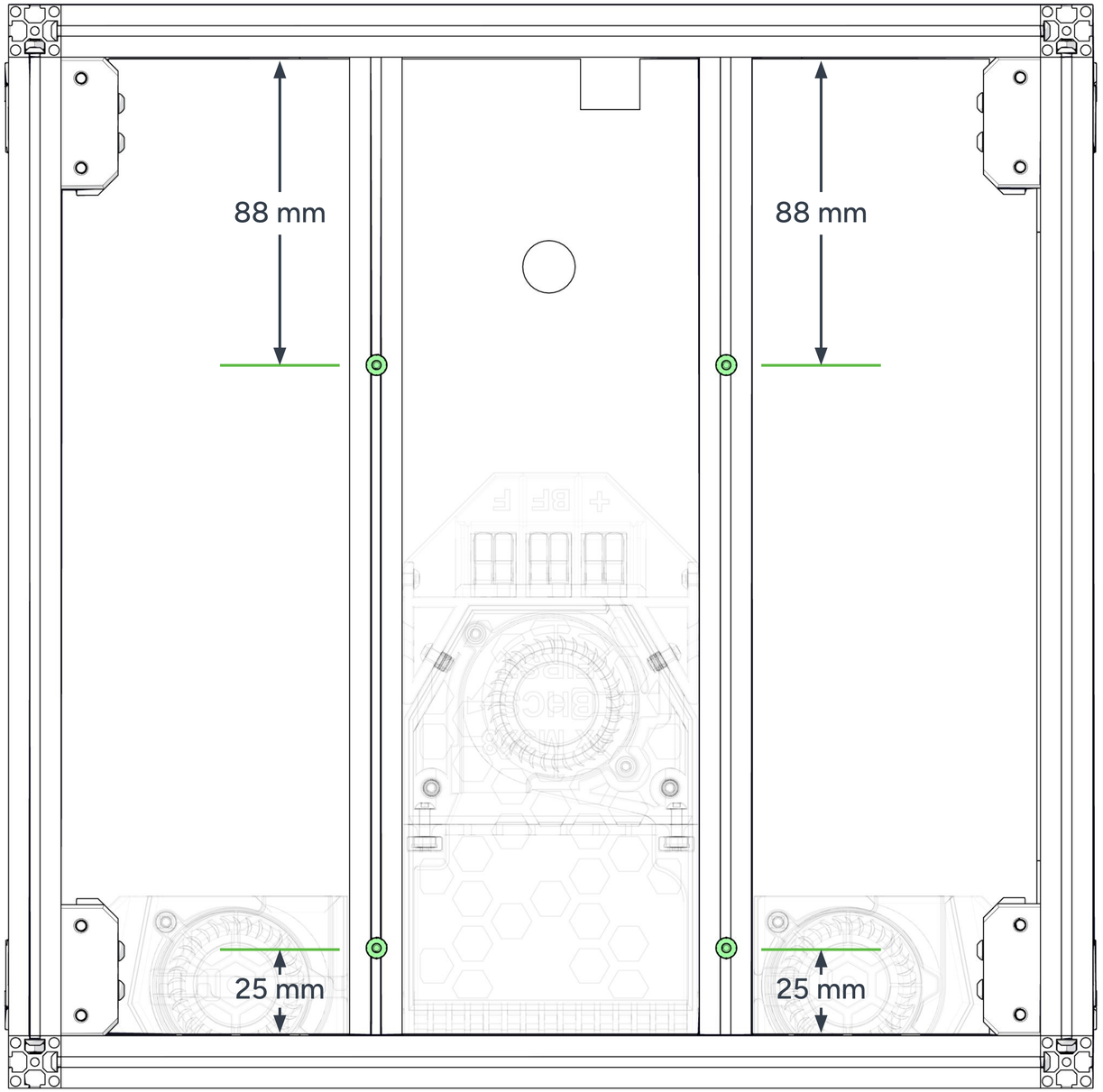




H/ BED ASSEMBLY

MOUNTING THE BED ASSEMBLY

Position the four M3 × 8 mm standoffs as shown and tighten them to the preloaded hex nuts.

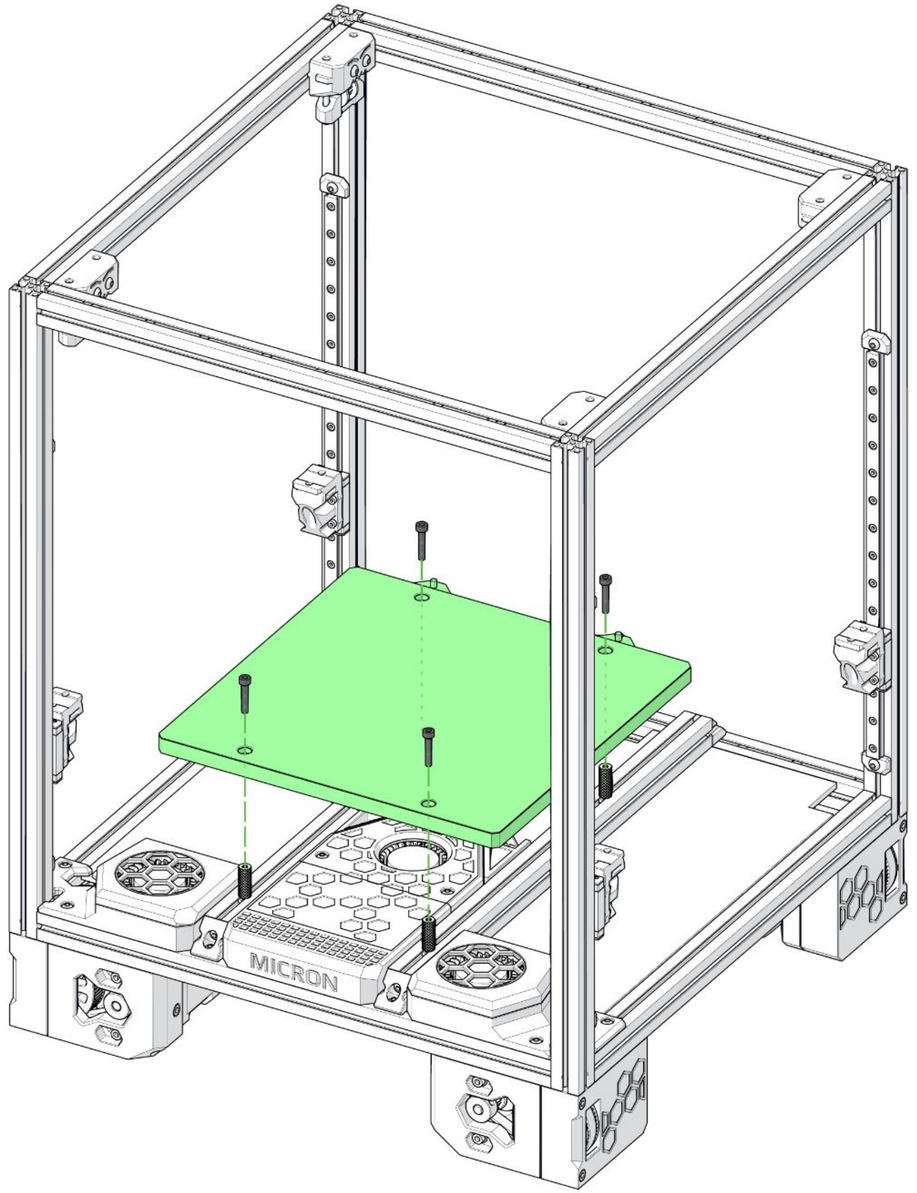


H/ BED ASSEMBLY

MOUNTING THE BED ASSEMBLY

Secure the bed over the four M3 × 8 mm standoffs with four M3 × 16 SHCS.

Tighten one screw fully to secure the bed. Leave slight play in the other three screws. This will allow for thermal expansion.



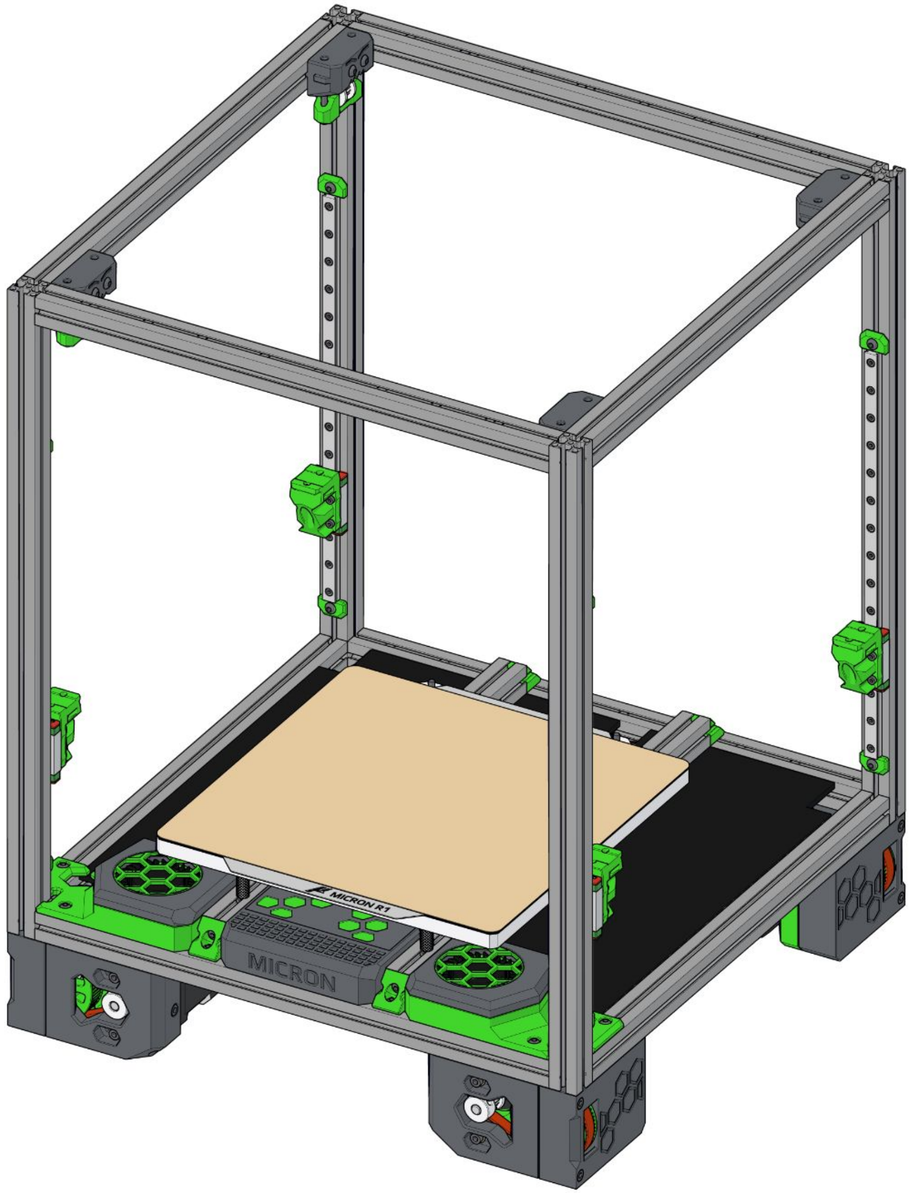
H/ BED ASSEMBLY

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

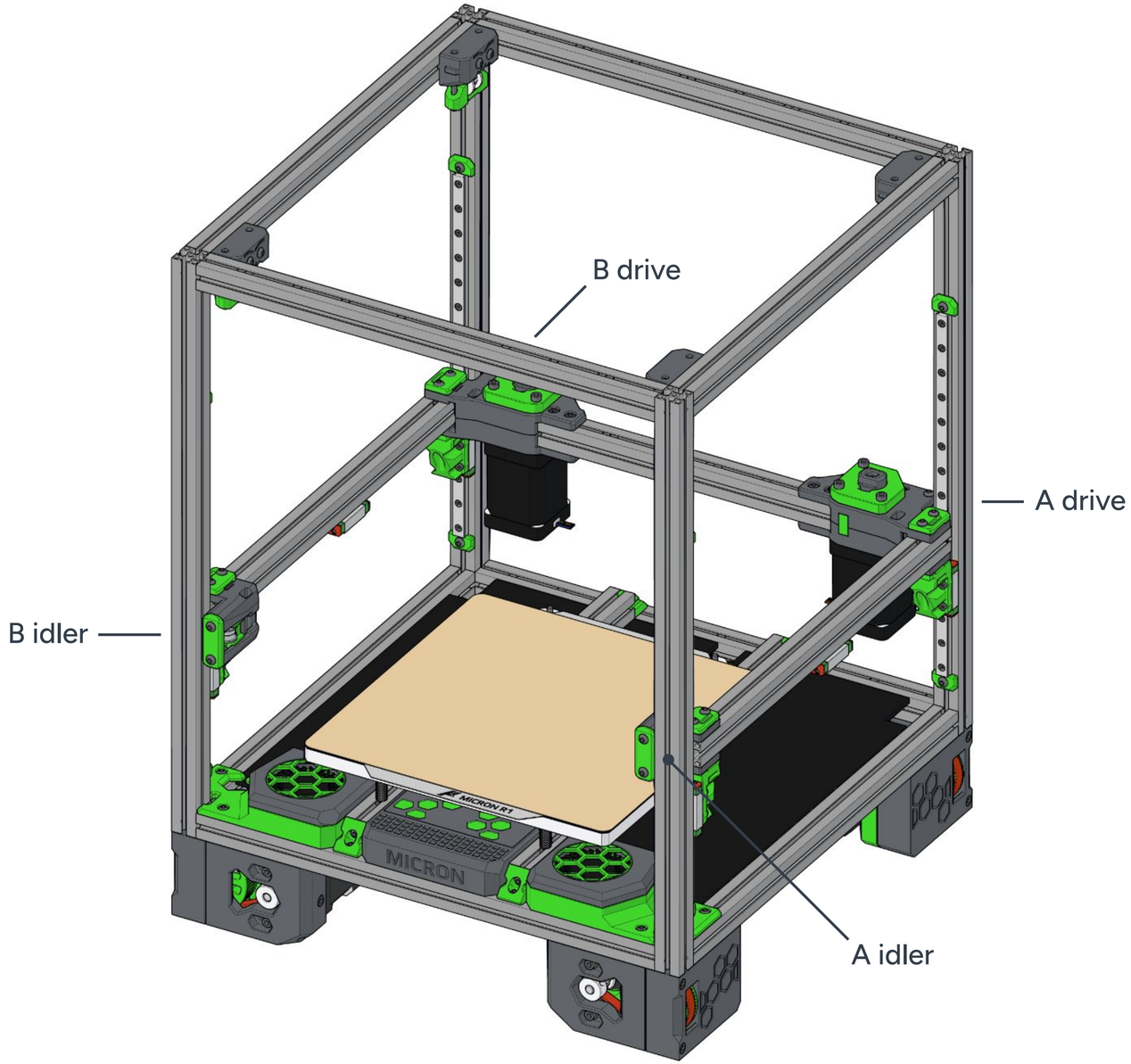
- ▢ Bed heater wires facing rear
- ▢ Thermal fuse connected inline with bed heater wires
- ▢ Protective earth ring terminal secured with direct metal-to-metal contact
- ▢ One bed mount screw fully tightened, three remaining loose



GANTRY ASSEMBLY

A/B DRIVES & FRONT IDLERS

The A/B drives and front idlers define the CoreXY motion system, translating belt movement into precise X/Y positioning of the gantry. Their alignment and belt paths directly affect accuracy, smoothness, and repeatability. Careful assembly here ensures consistent motion and reliable print performance.



NOTE

Included hardware varies by kit. The LDO kit uses a pinned idler assembly; some kits use a through-bolt (M3 × 12 SHCS). Match printed parts to hardware.

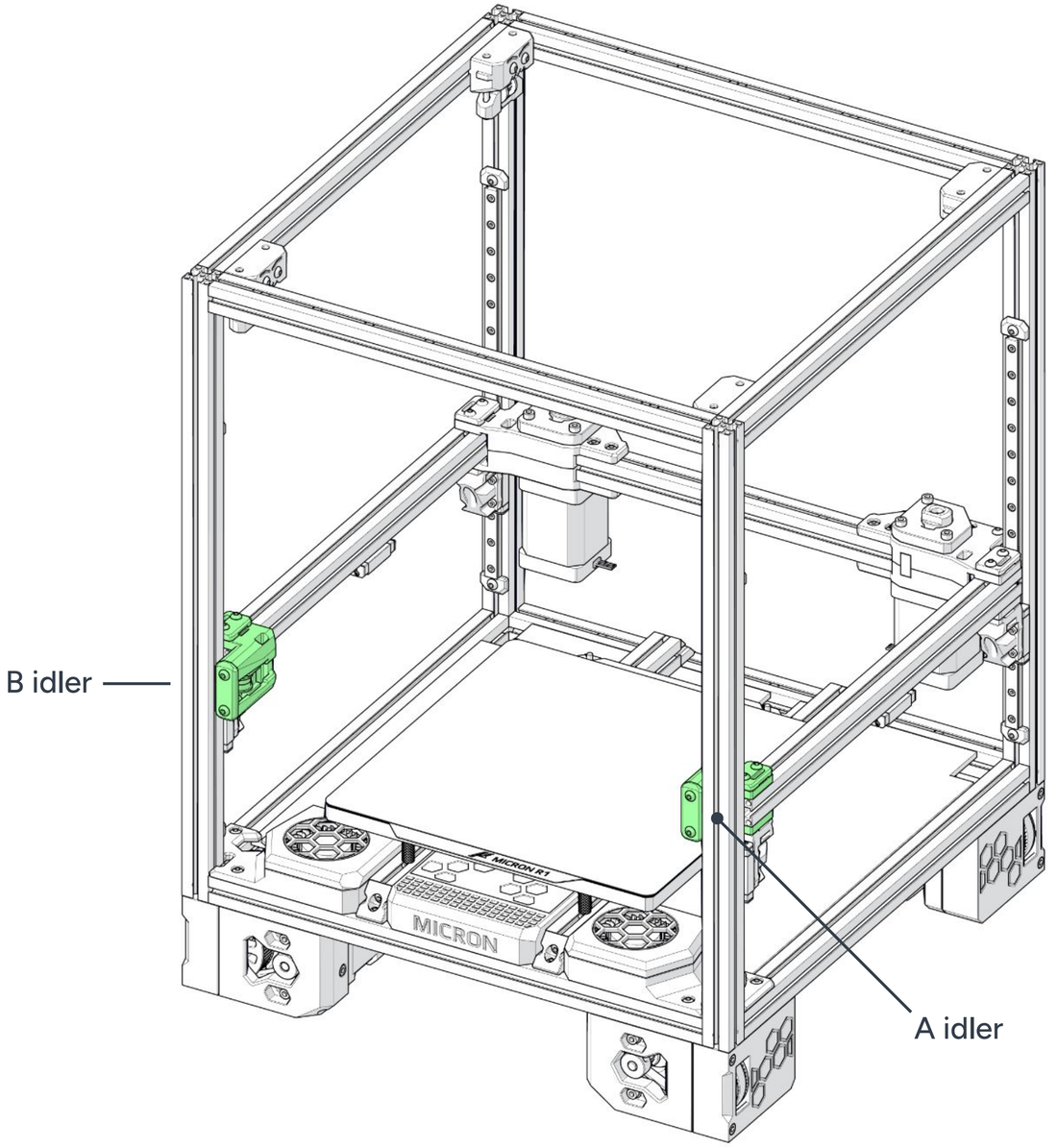
GANTRY ASSEMBLY

FRONT IDLERS

Front idlers vary by kit. This manual uses a pinned idler assembly.

HARDWARE

2	3 mm × 14 mm pin
4	F623-2RS bearing
4	M3 × 25 BHCS
4	M3 × 7 × 0.5 mm shim
8	M3 hex nuts



GANTRY ASSEMBLY

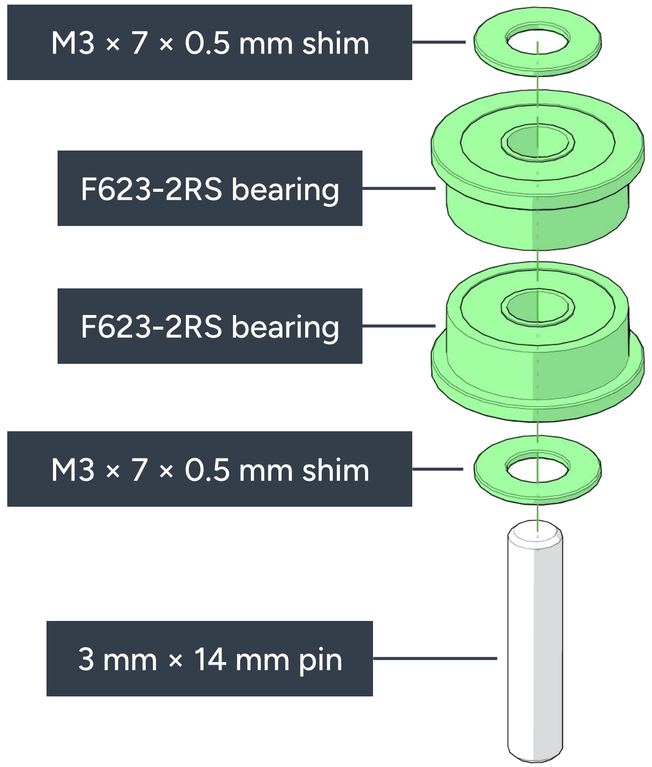
PREPPING FRONT IDLERS

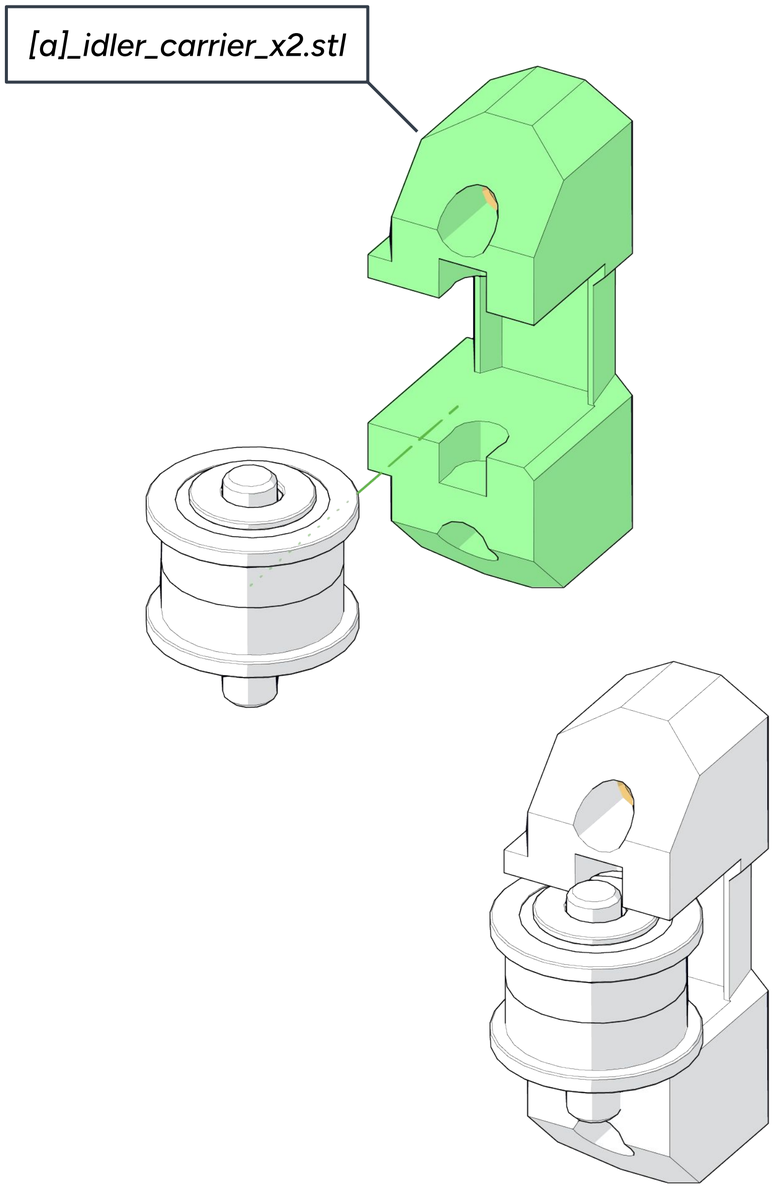
Assemble the following bearing stack on a **3 mm × 14 mm pin**:

- **M3 × 7 × 0.5 mm shim**
- **F623-2RS bearing (flange up)**
- **F623-2RS bearing (flange down)**
- **M3 × 7 × 0.5 mm shim**

Insert the assembled idler into ***[a]_idler_carrier_x2.stl***. Ensure the idler spins freely.

Repeat for the B idler assembly.



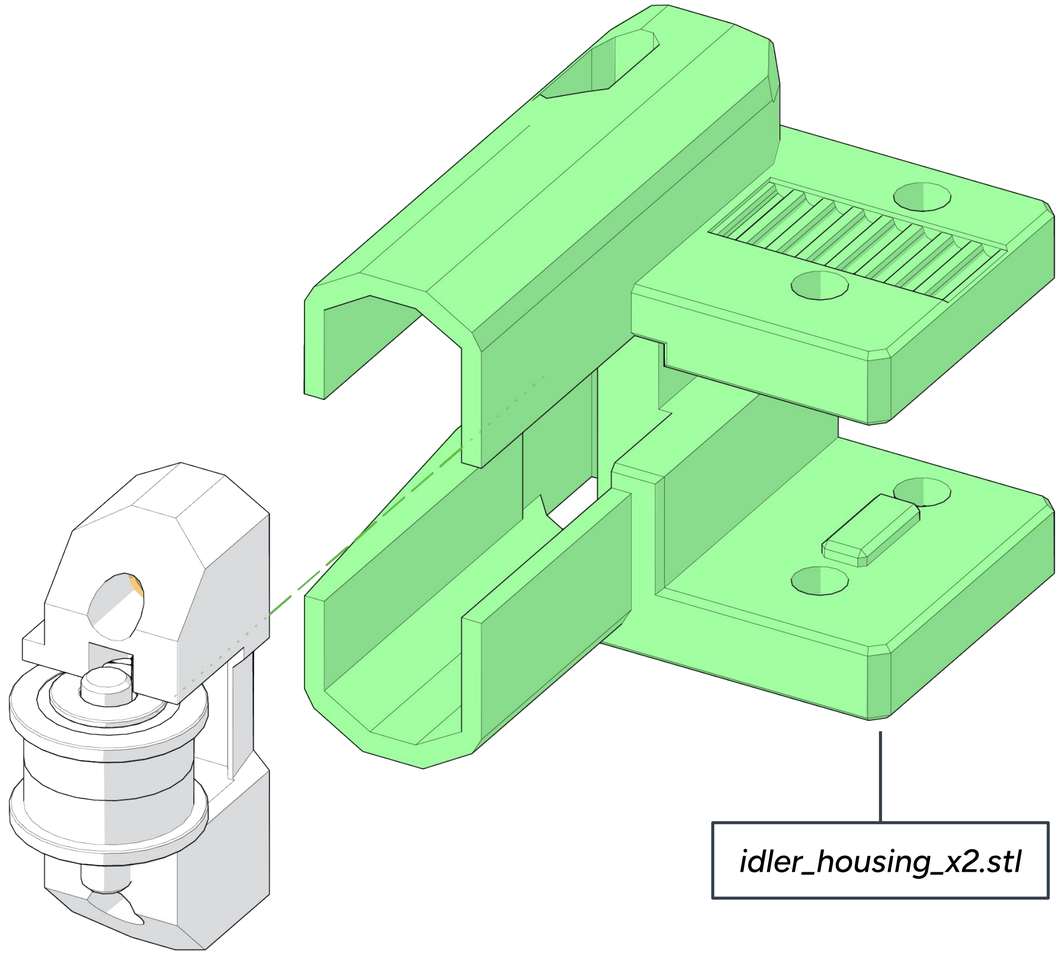


GANTRY ASSEMBLY

PREPPING FRONT IDLERS

Insert the assembled carrier into ***idler_housing_x2.stl***.

Repeat for the B idler assembly.



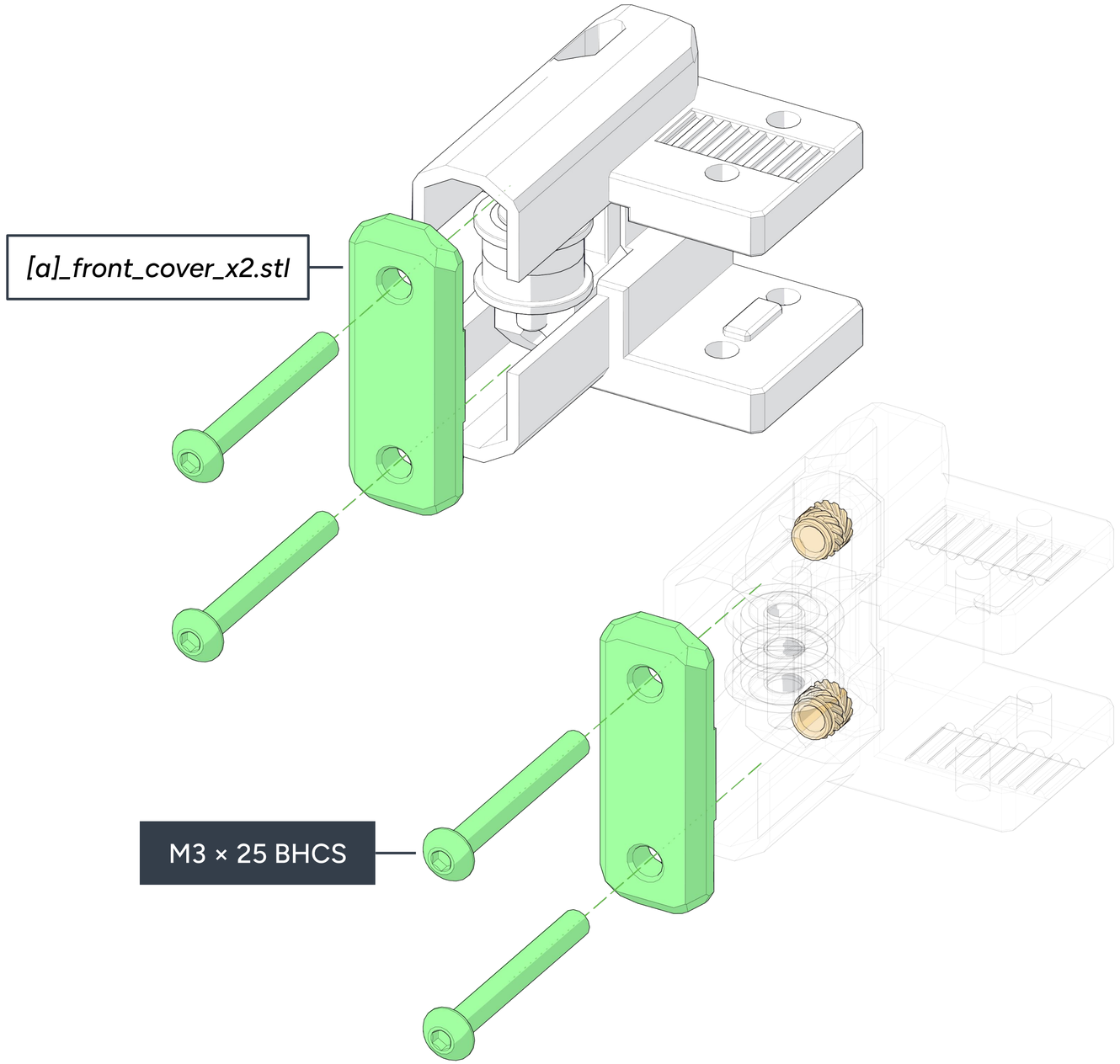
GANTRY ASSEMBLY

PREPPING FRONT IDLERS

Fit ***[a]_front_cover_x2.stl*** onto the assembly and secure with **two M3 × 25 BHCS** into the two headset inserts in the printed carrier.

Repeat for the B idler assembly.

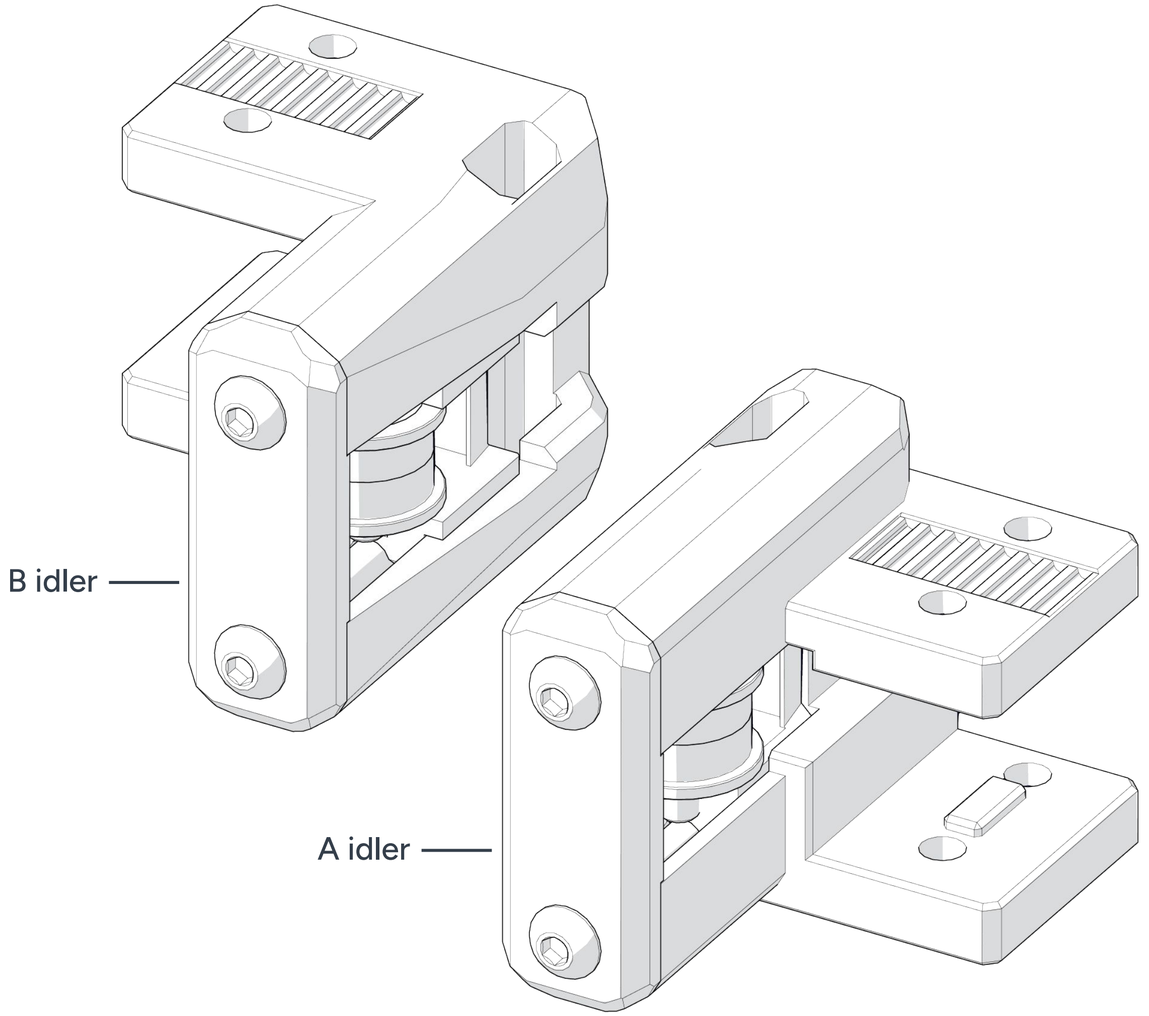
These screws tension the CoreXY belts by shifting the idler carrier position.



GANTRY ASSEMBLY

PREPPING FRONT IDLERS

You should now have assembled A and B front idlers.
Ensure both idlers spin freely before proceeding.



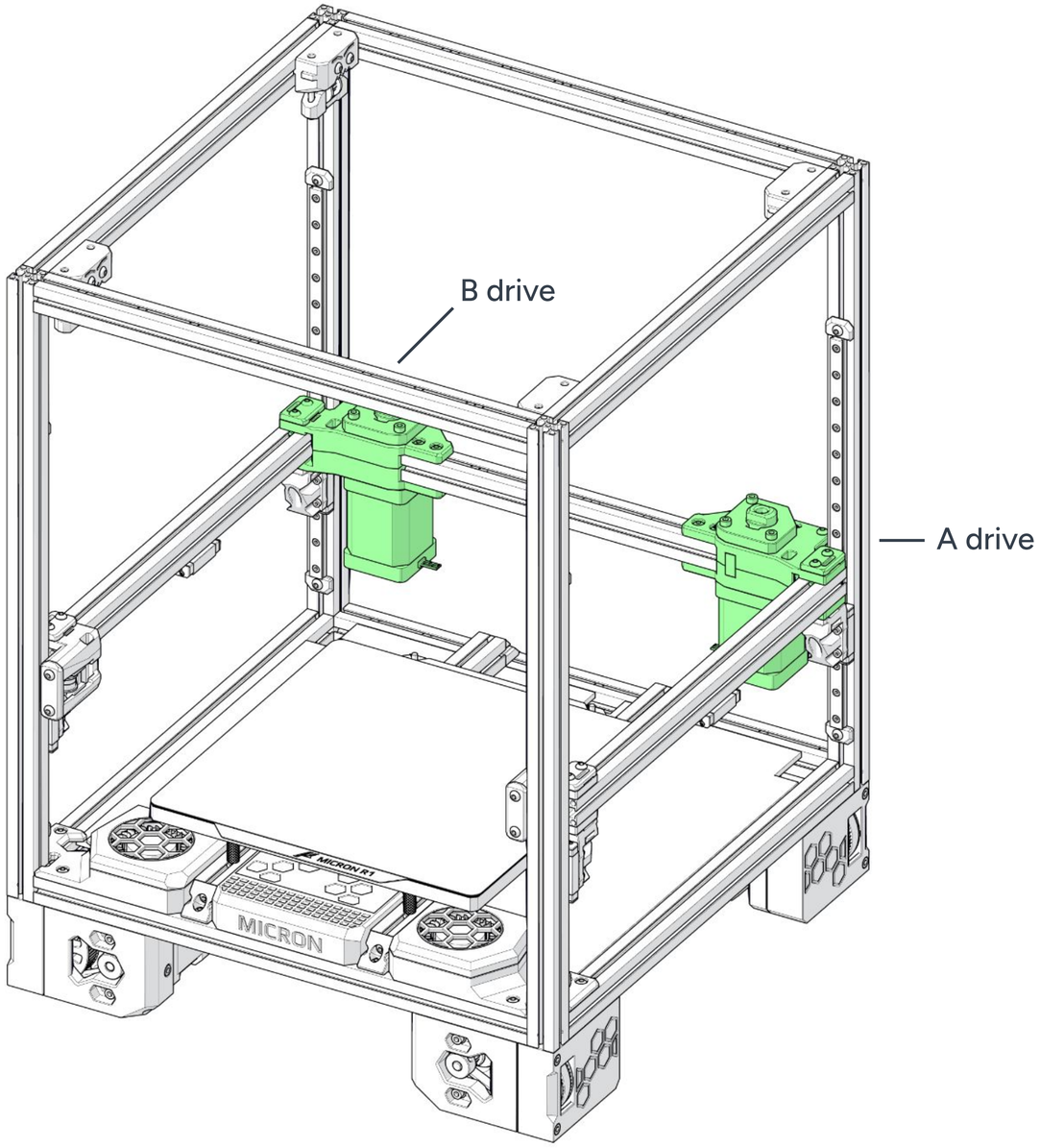
GANTRY ASSEMBLY

A/B DRIVES

Each A/B drive controls one belt loop, combining their motion to generate all X/Y movement in the CoreXY system.

HARDWARE

2	NEMA14 motors
2	2GT 16T pulley
14	F623-2RS bearing
2	M3 × 6 BHCS
6	M3 × 35 BHCS
12	M3 × 7 × 0.5 mm shim
2	3 mm × 25 mm pin

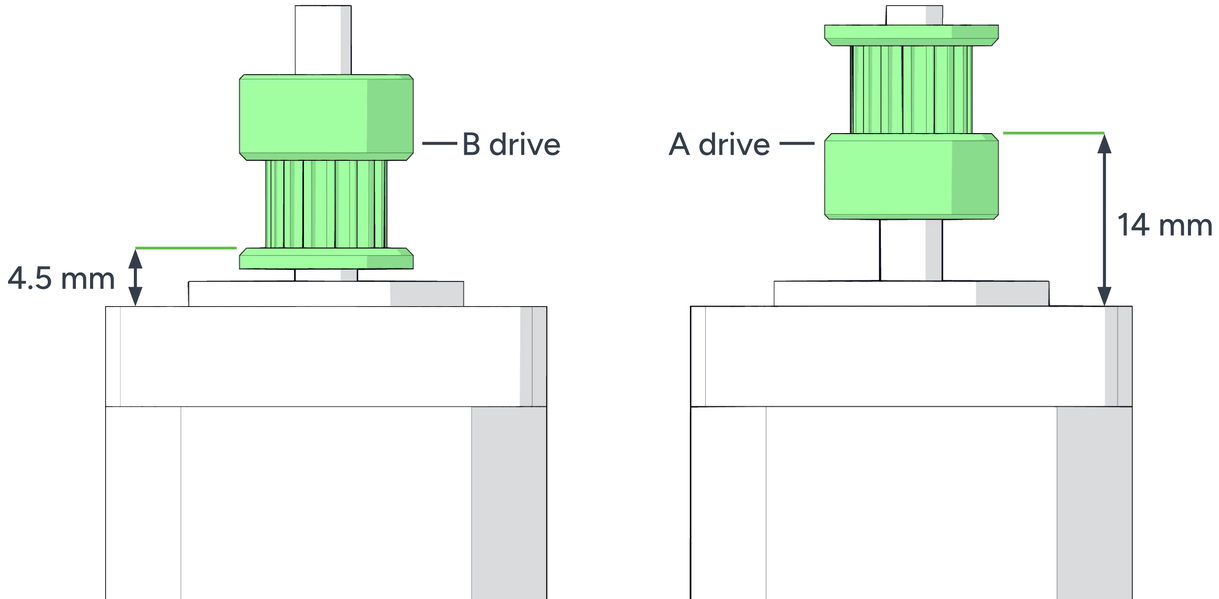


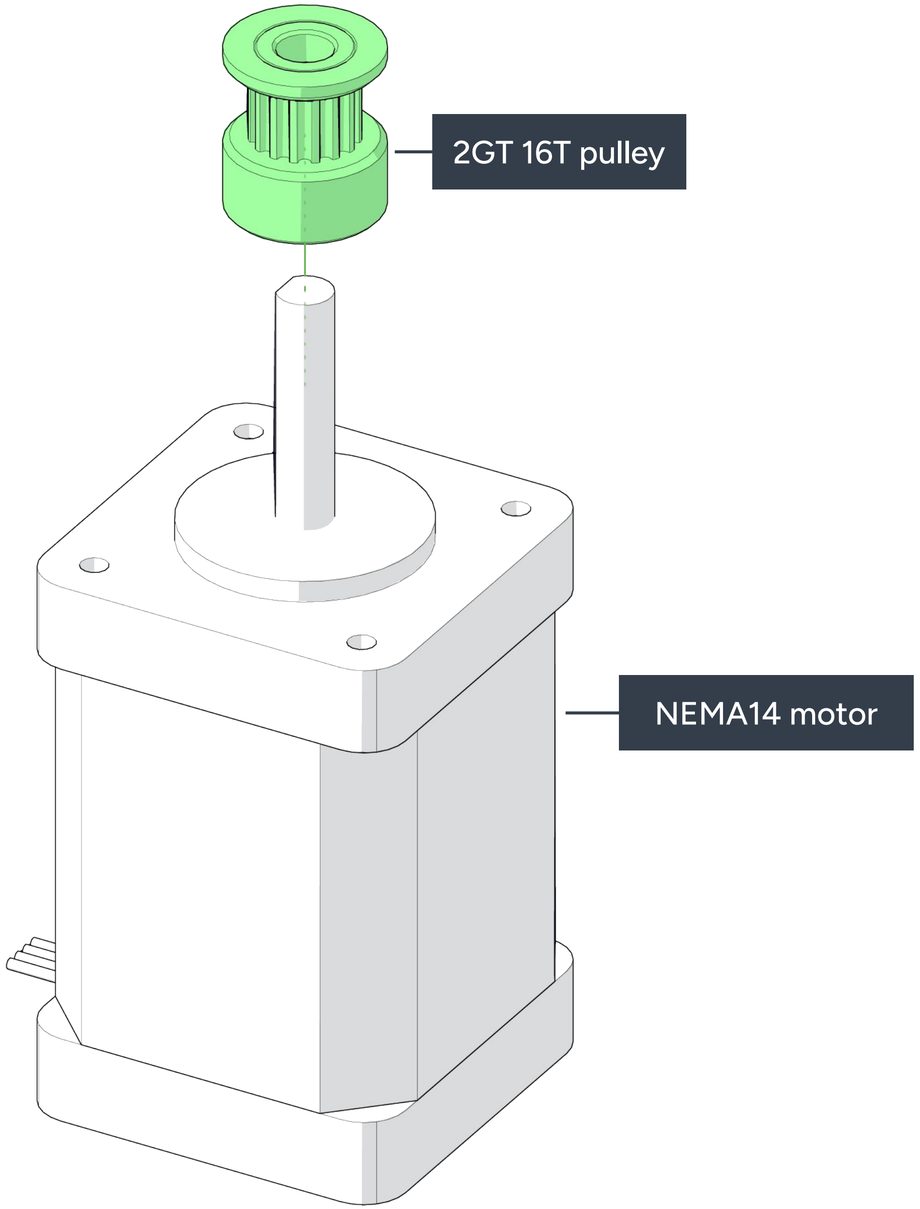
GANTRY ASSEMBLY

A/B DRIVES - MOTOR ASSEMBLIES

Install a **2GT 16T pulley** onto the D shaft of a **NEMA14 motor**, positioned 14 mm from motor face to pulley teeth. Ensure the set screw aligns with the flat of the shaft. Apply threadlocker and tighten with a hex driver (no ball-end). This ensures correct engagement between pulley and belt.

Repeat for B drive motor with pulley positioned at 4.5 mm.





GANTRY ASSEMBLY

PREPPING A DRIVE

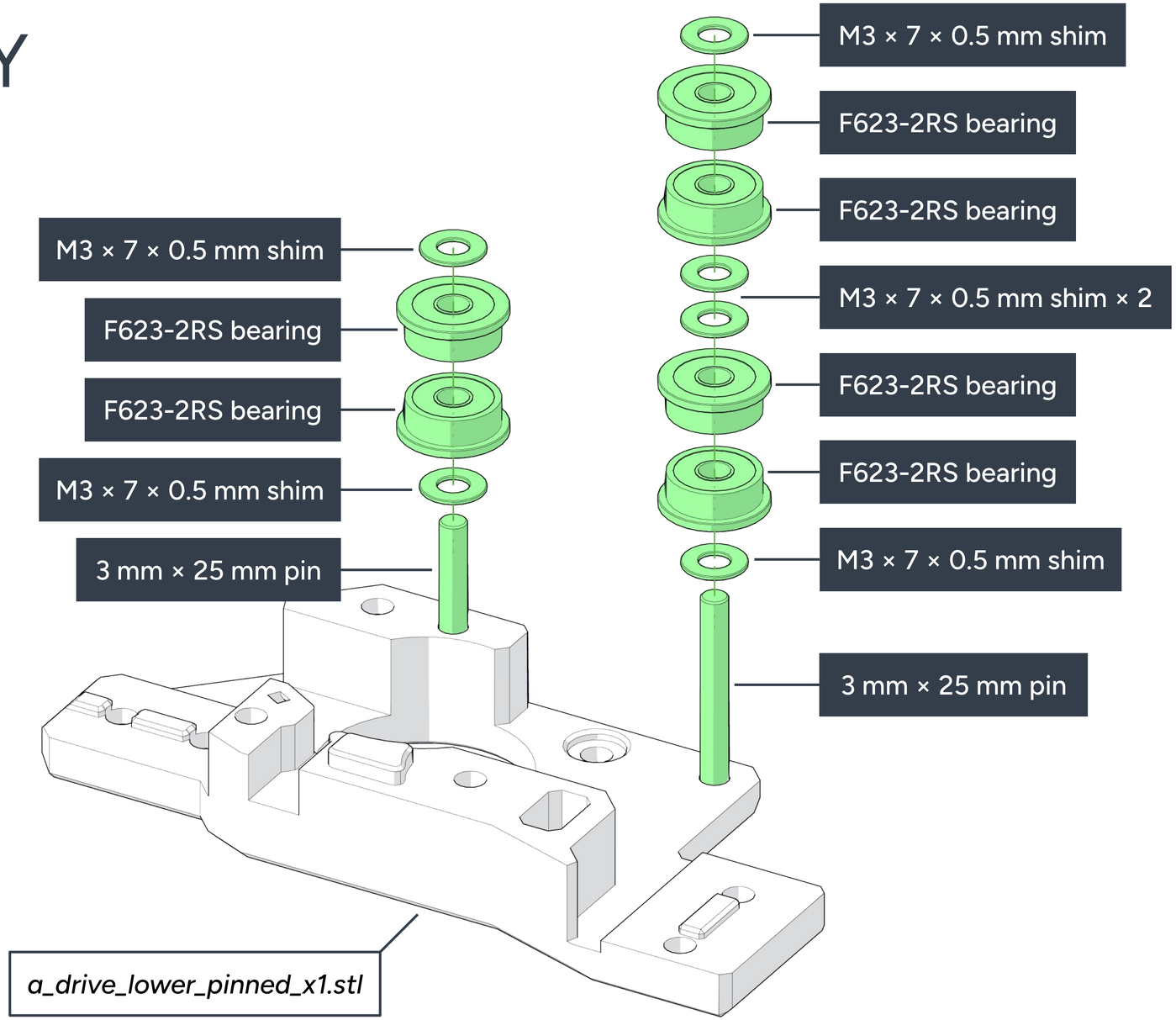
Assemble the following two bearing stacks on **two 3 mm × 25 mm pins** inserted in ***a_drive_lower_pinned_x1.stl*** as shown.

First stack:

- M3 × 7 × 0.5 mm shim
- F623-2RS bearing (flange up)
- F623-2RS bearing (flange down)
- M3 × 7 × 0.5 mm shim

Second stack:

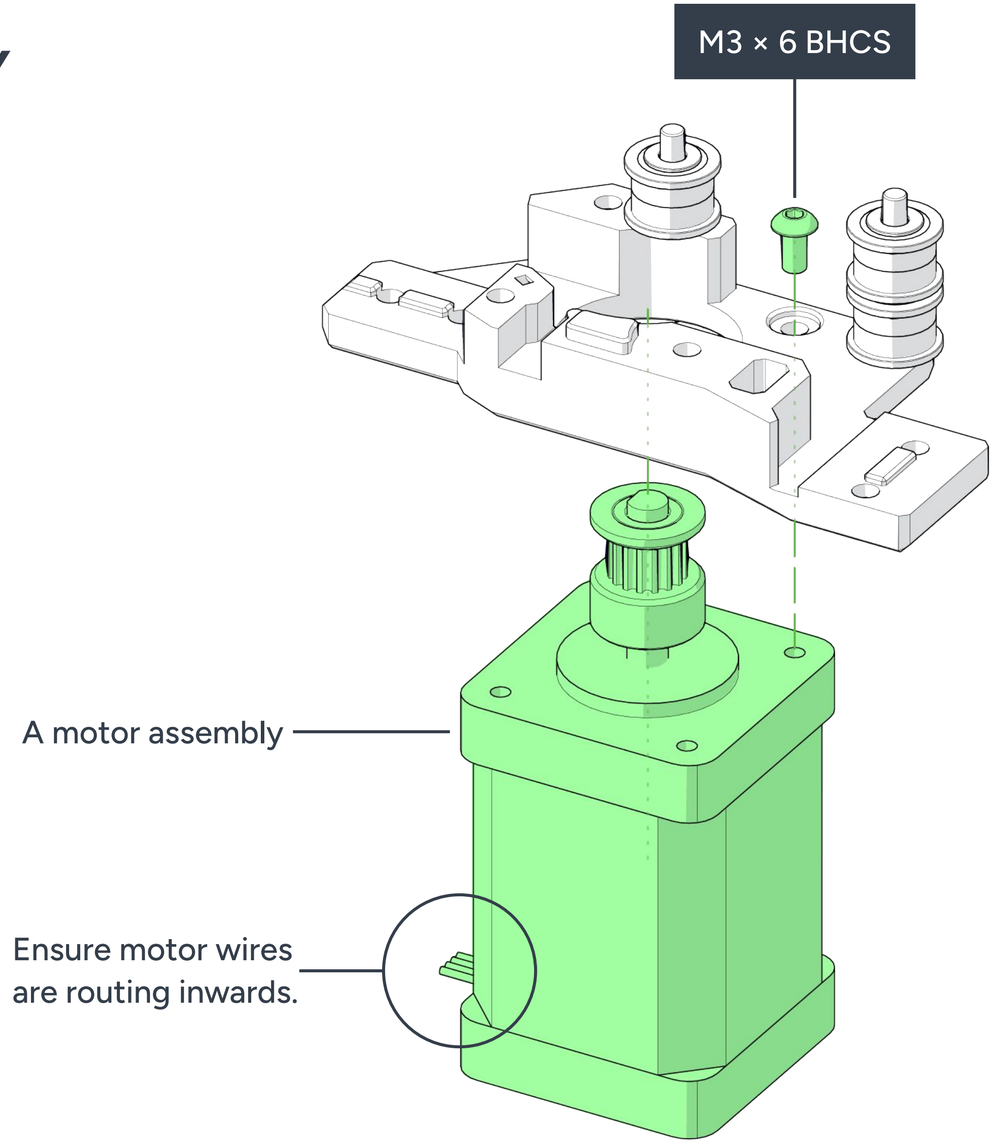
- M3 × 7 × 0.5 mm shim
- F623-2RS bearing (flange up)
- F623-2RS bearing (flange down)
- M3 × 7 × 0.5 mm shim
- M3 × 7 × 0.5 mm shim
- F623-2RS bearing (flange up)
- F623-2RS bearing (flange down)
- M3 × 7 × 0.5 mm shim



GANTRY ASSEMBLY

PREPPING A DRIVE

Insert the A drive motor assembly with 2GT 16T pulley from below and secure with an **M3 × 6 BHCS**. Ensure the motor wires' routing direction matches this view.



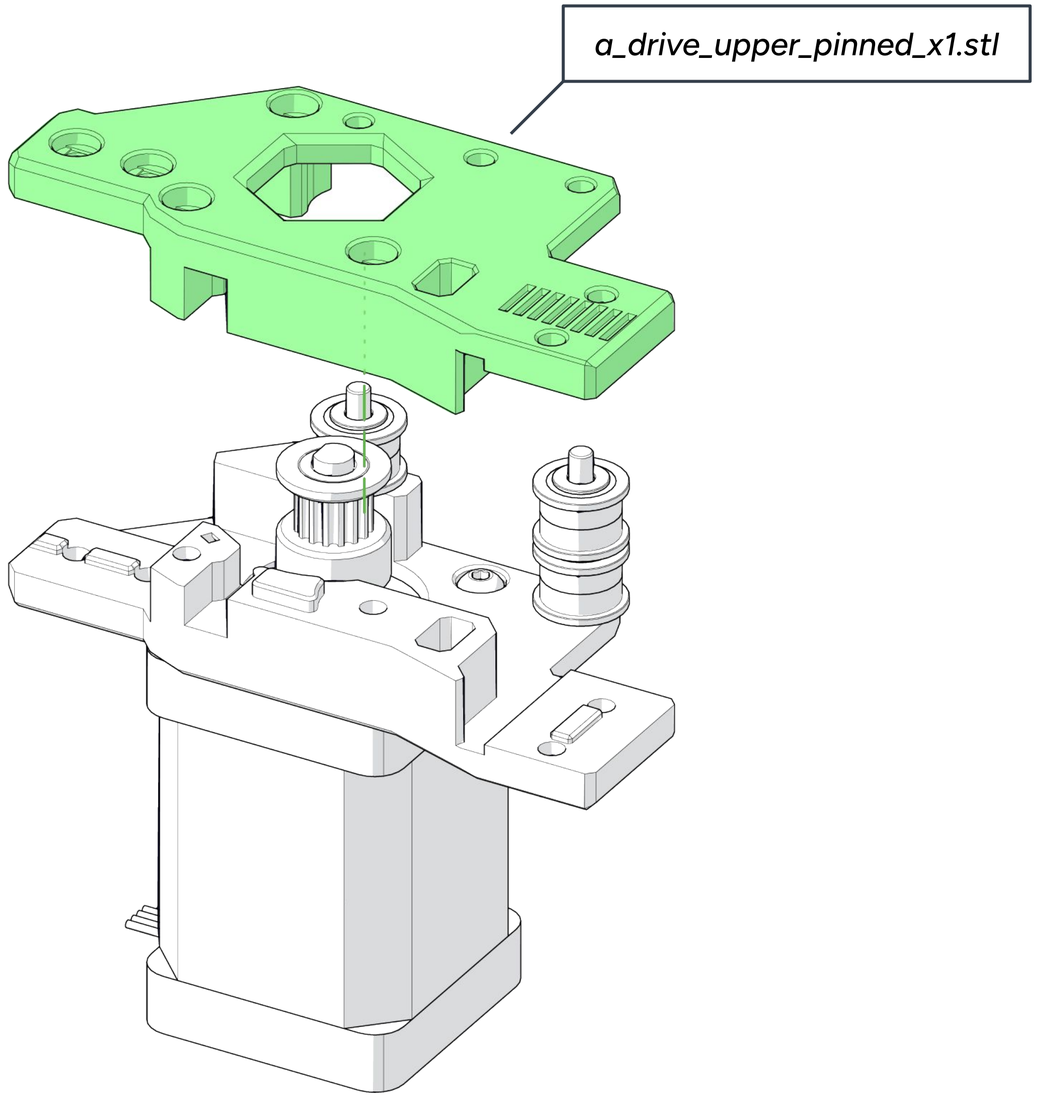
NOTE

The A and B drive motor assemblies differ in pulley position to align with the two belt paths. A drive pulley runs on the top 'lane' and B drive pulley runs on the bottom 'lane'.

GANTRY ASSEMBLY

PREPPING A DRIVE

Cover the assembly with ***a_drive_upper_pinned_x1.stl***. Ensure the pins are seated in their holes and the bearing stacks spin freely.

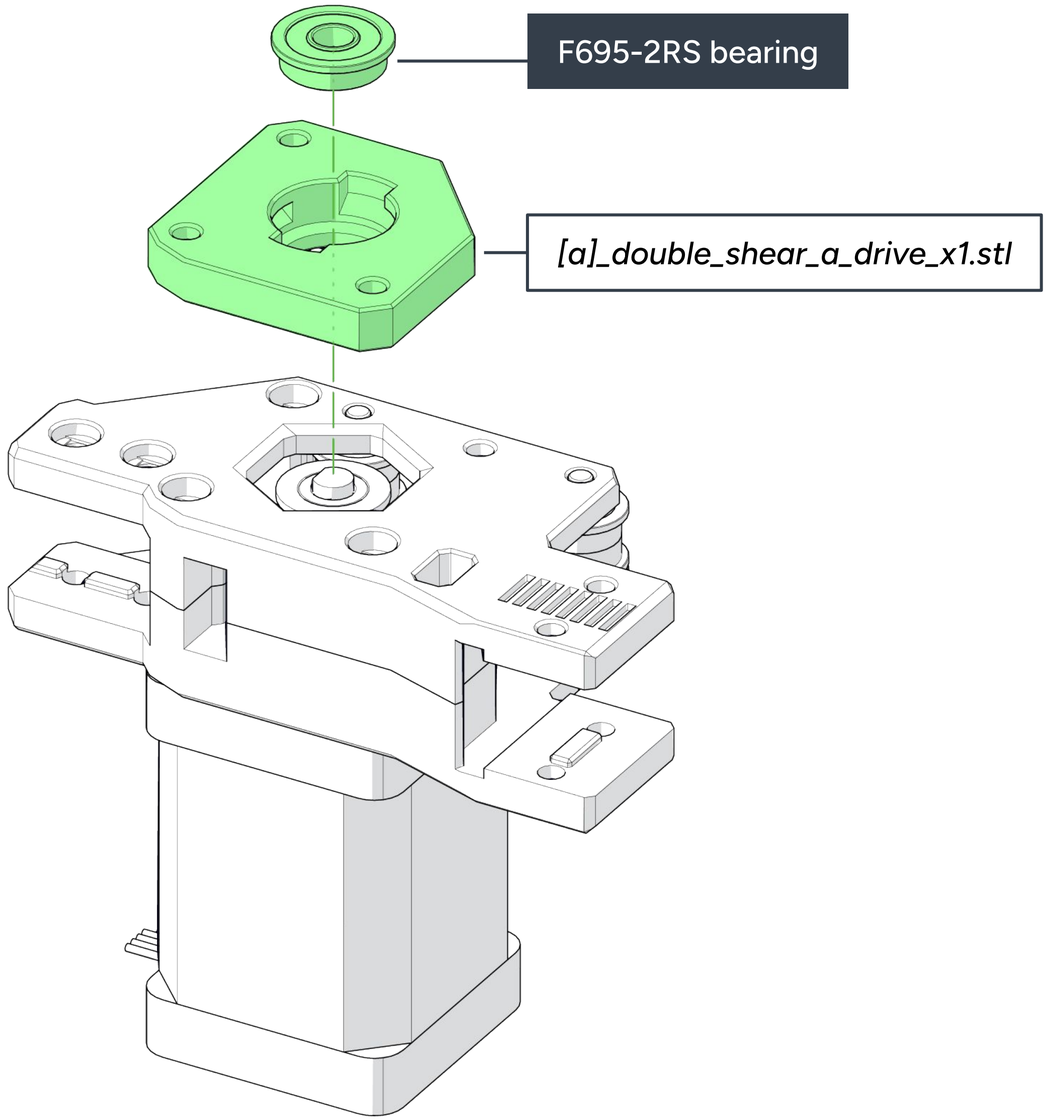


GANTRY ASSEMBLY

PREPPING A DRIVE

Add an **F695-2RS bearing** to the top of the pin, then cover the bearing with ***[a]_double_shear_a_drive_x1.stl***.

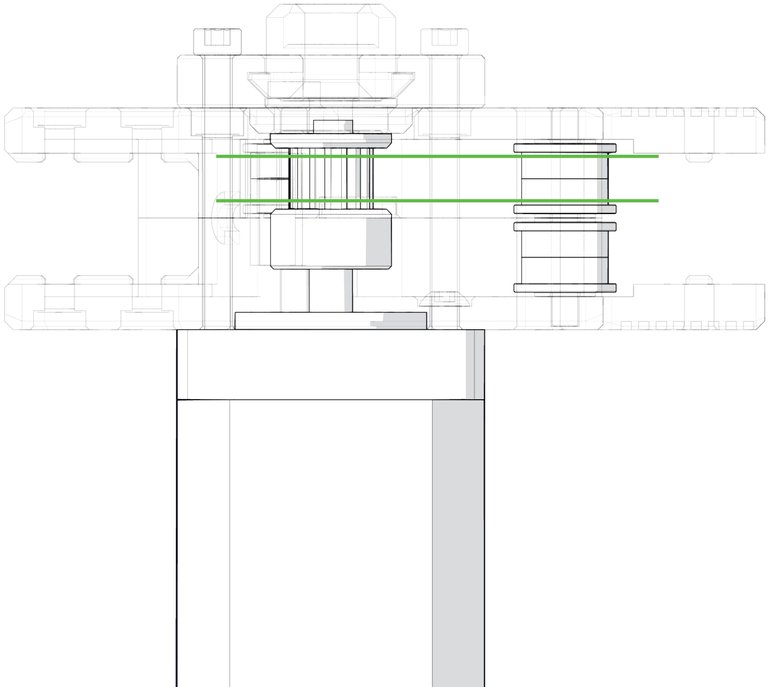
This bearing acts as the upper shaft support in the “double shear” setup.

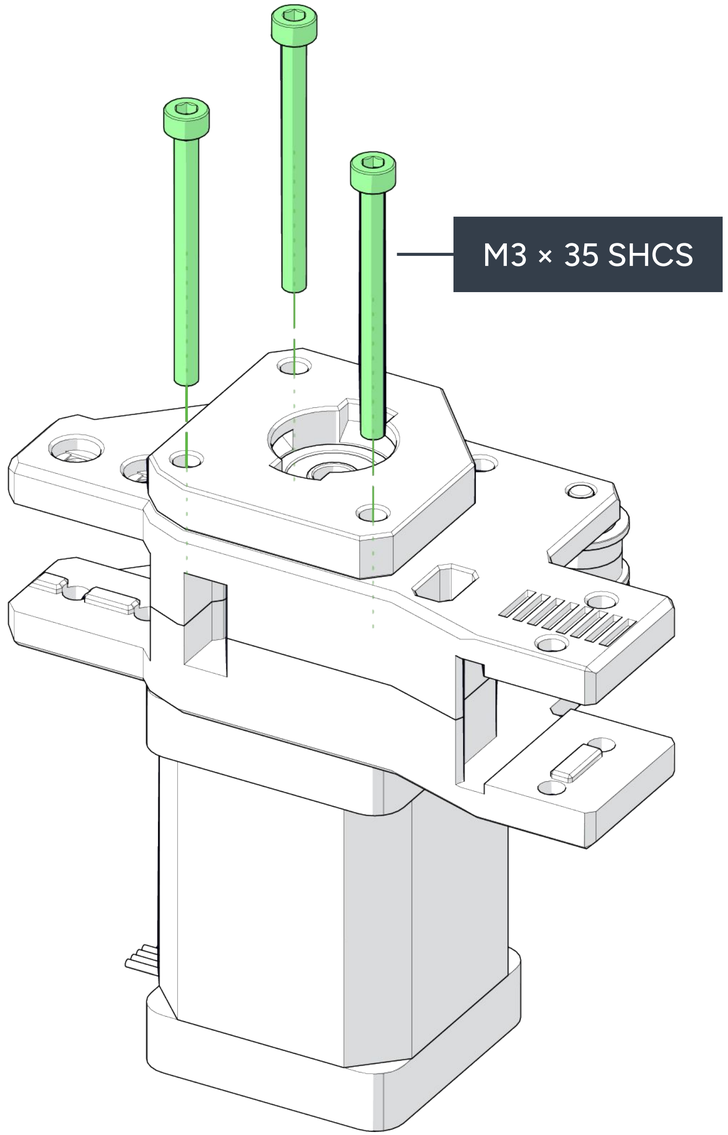


GANTRY ASSEMBLY

PREPPING A DRIVE

Secure the assembly with **three M3 × 35 SHCS** from the top through to the motor screw mounts. Ensure the exposed 2GT 16T pulley lines up with the top pair of bearings.

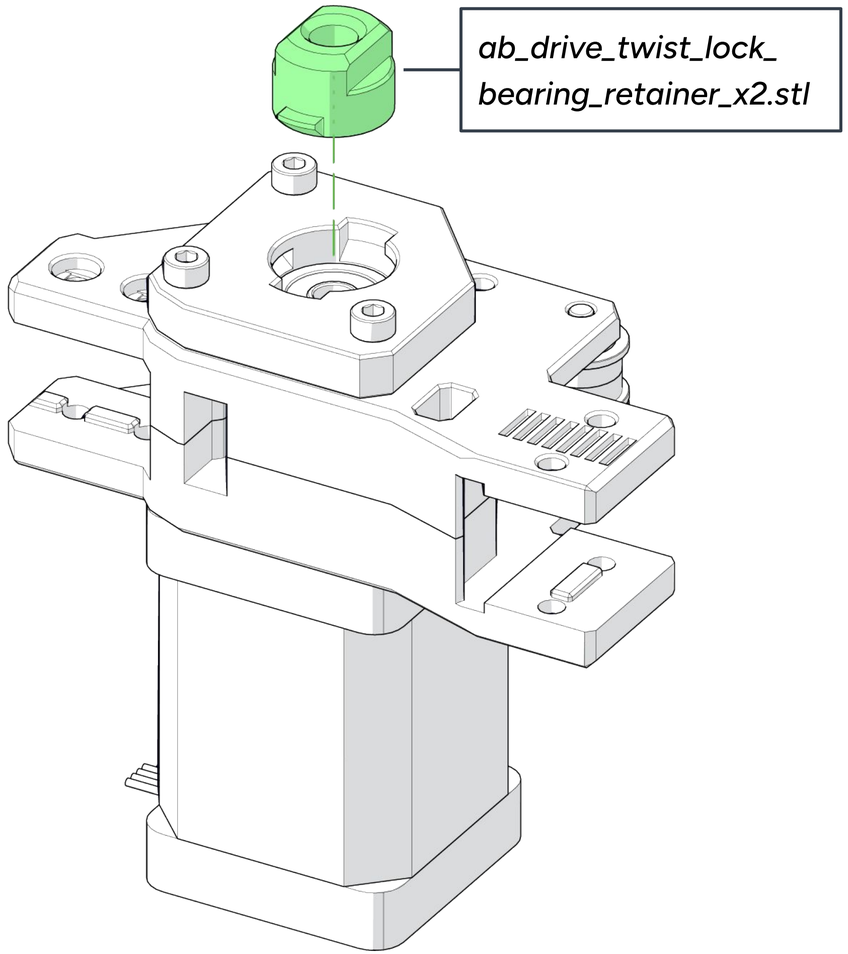




GANTRY ASSEMBLY

PREPPING A DRIVE

Drop in ***ab_drive_twist_lock_bearing_retainer_x2.stl***.



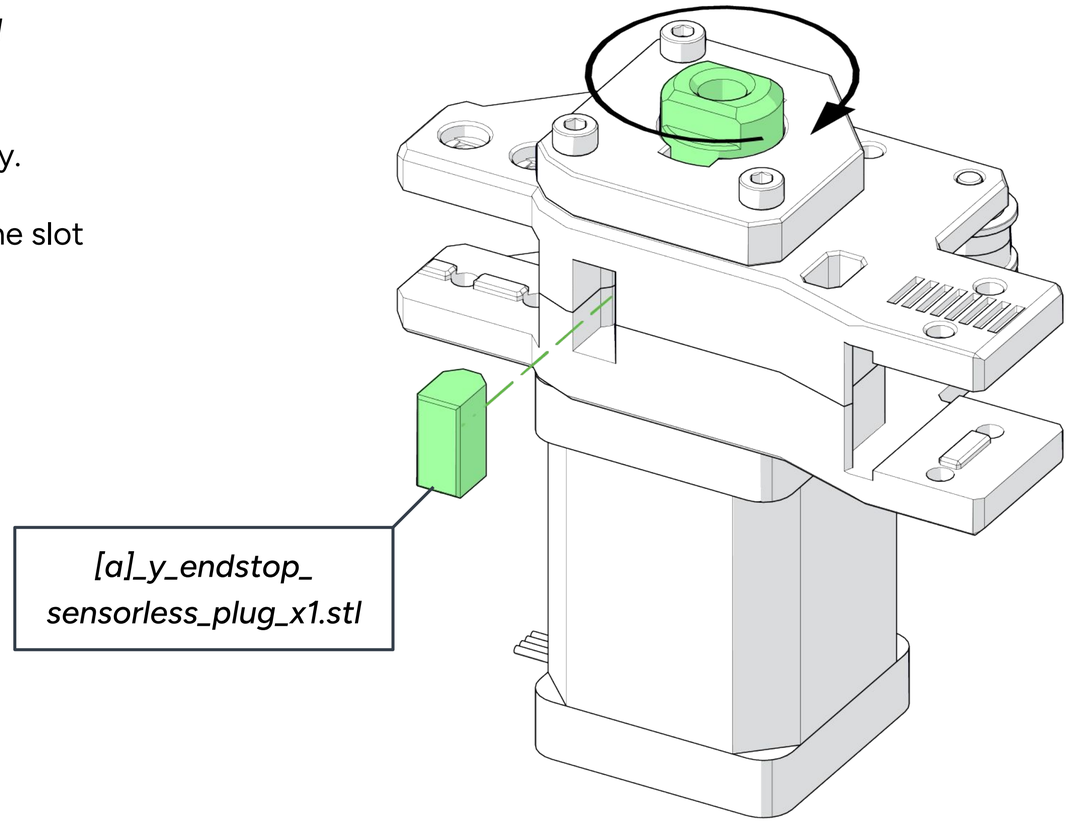
GANTRY ASSEMBLY

PREPPING A DRIVE

Rotate ***ab_drive_twist_lock_bearing_retainer_x2.stl*** clockwise to secure the assembly.

Ensure pins are seated and bearing stacks spin freely.

Insert ***[a]_y_endstop_sensorless_plug_x1.stl*** to fill the slot where the traditional Y limit switch would go.



NOTE

While the LDO kit uses sensorless homing and does not require X and Y limit switches, traditional limit switches can still be used.

GANTRY ASSEMBLY

PREPPING B DRIVE

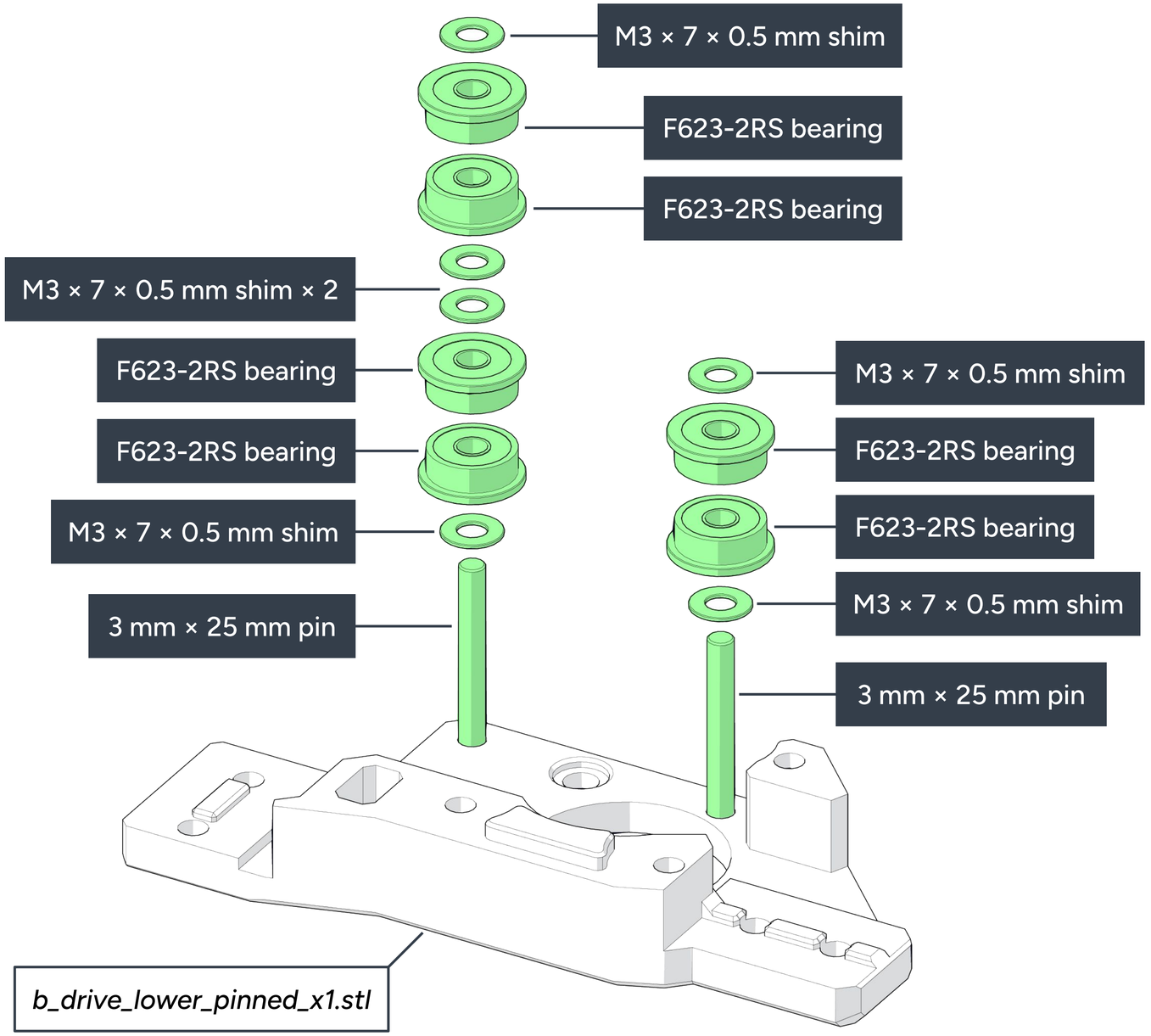
Assemble the following two bearing stacks on **two 3 mm × 25 mm pins** inserted in ***b_drive_lower_pinned_x1.stl*** as shown.

First stack:

- **M3 × 7 × 0.5 mm shim**
- **F623-2RS bearing (flange up)**
- **F623-2RS bearing (flange down)**
- **M3 × 7 × 0.5 mm shim**
- **M3 × 7 × 0.5 mm shim**
- **F623-2RS bearing (flange up)**
- **F623-2RS bearing (flange down)**
- **M3 × 7 × 0.5 mm shim**

Second stack:

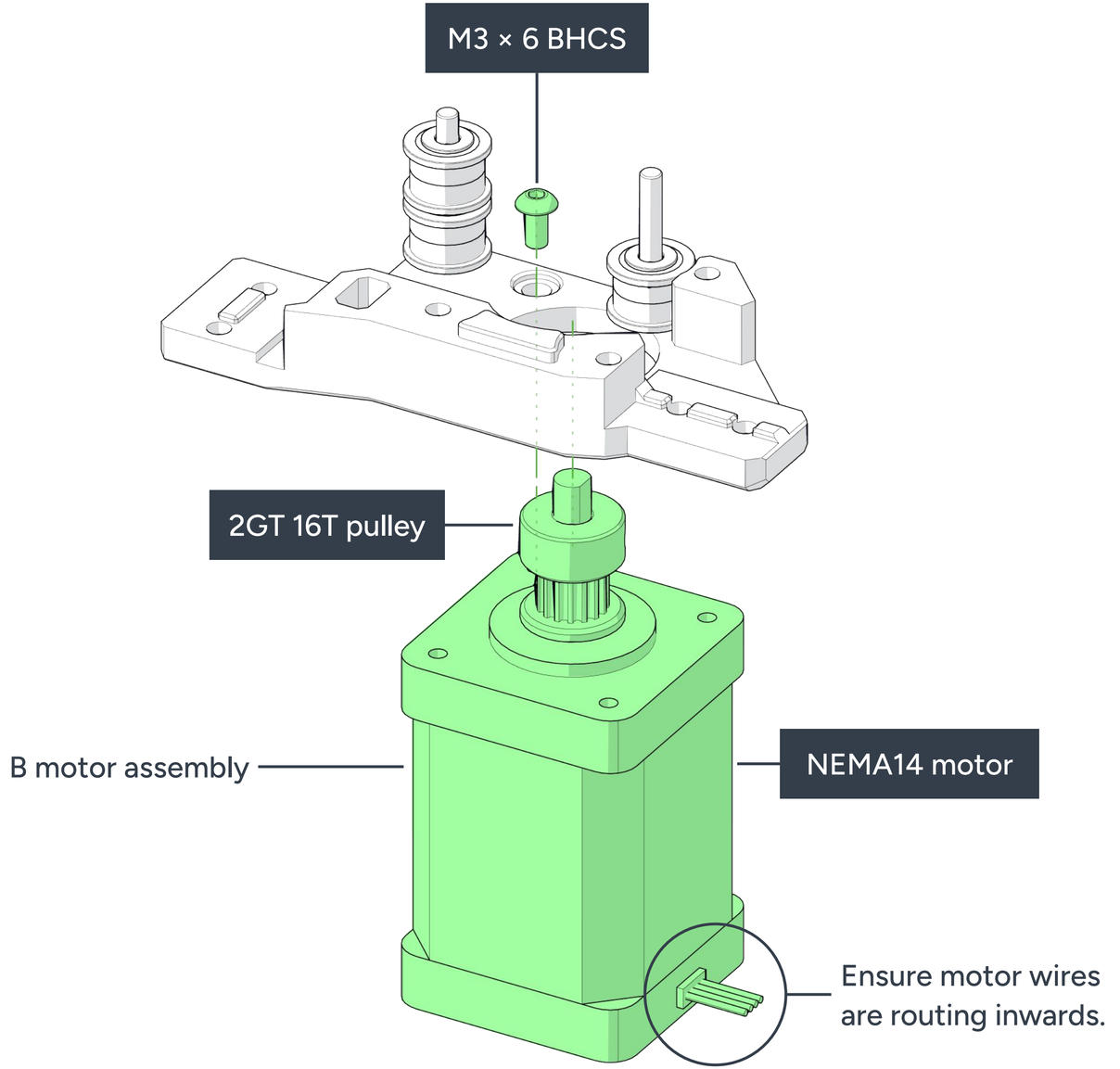
- **M3 × 7 × 0.5 mm shim**
- **F623-2RS bearing (flange up)**
- **F623-2RS bearing (flange down)**
- **M3 × 7 × 0.5 mm shim**



GANTRY ASSEMBLY

PREPPING B DRIVE

Insert the B drive motor assembly with 2GT 16T pulley from below and secure with an **M3 × 6 BHCS**. Ensure the motor wires' routing direction matches this view.



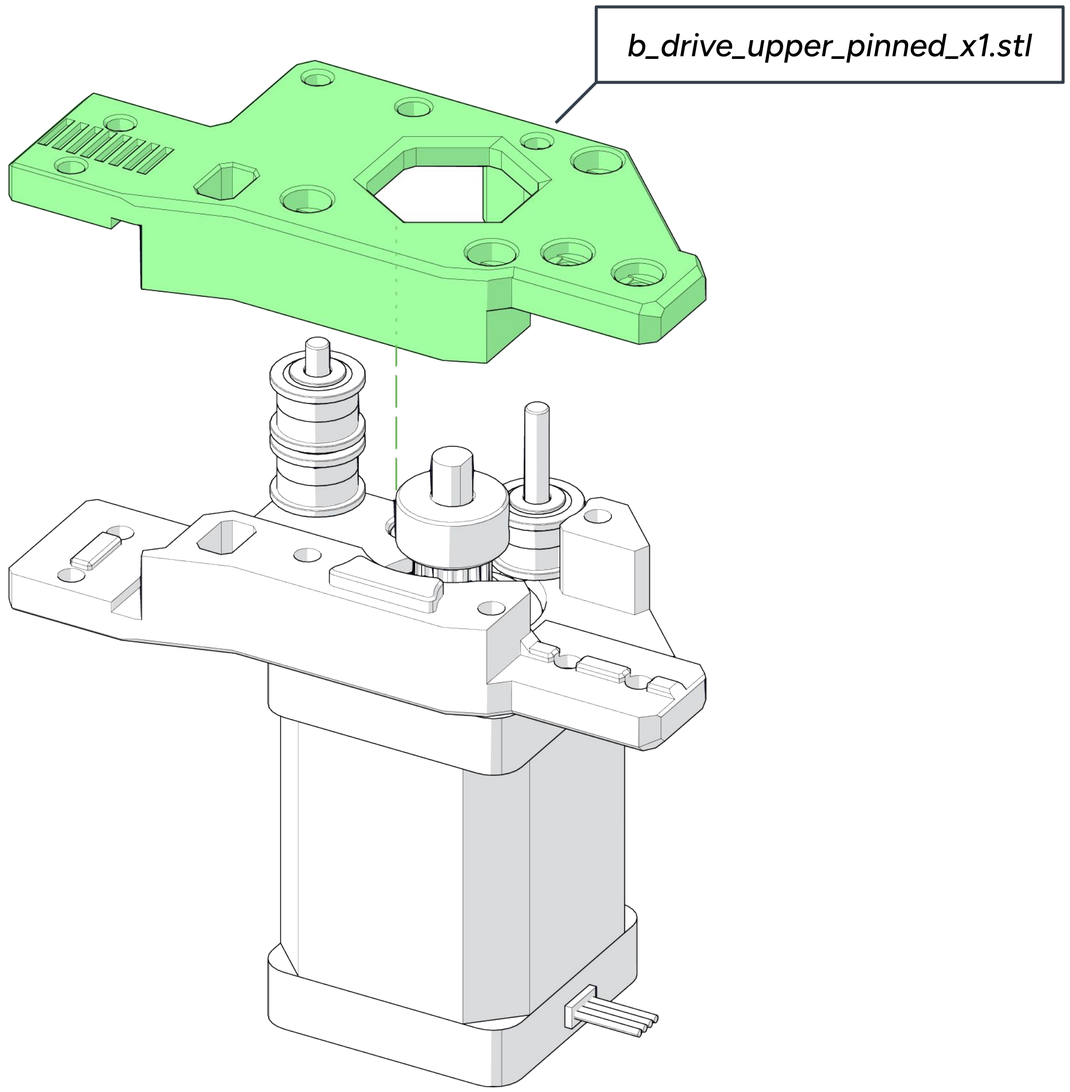
NOTE

The A and B drive motor assemblies differ in pulley position to align with the two belt paths. A drive pulley runs on the top 'lane' and B drive pulley runs on the bottom 'lane'.

GANTRY ASSEMBLY

PREPPING B DRIVE

Cover the assembly with ***b_drive_upper_pinned_x1.stl***. Ensure the pins are seated in their holes and the bearing stacks spin freely.

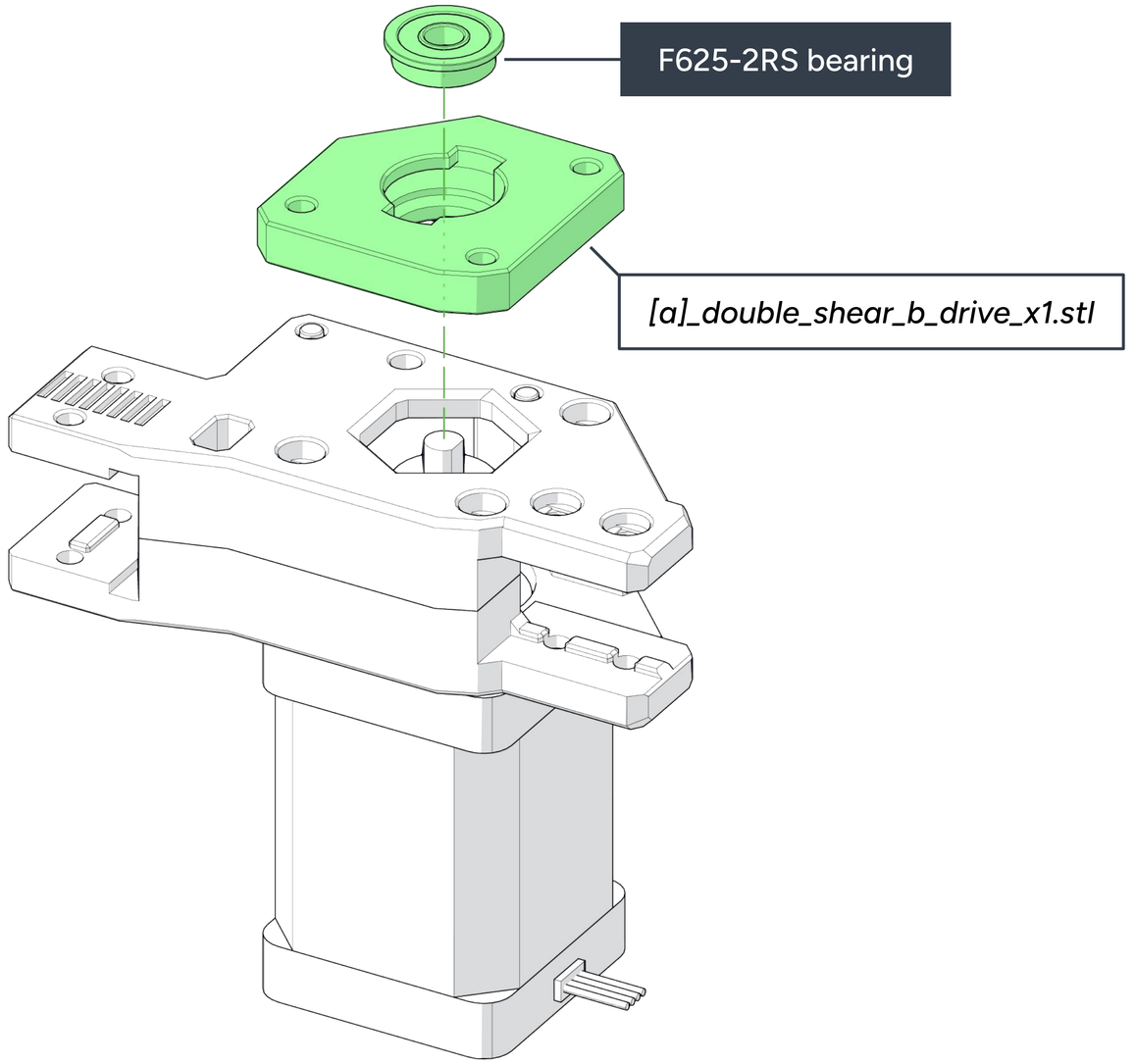


GANTRY ASSEMBLY

PREPPING B DRIVE

Add an **F625-2RS bearing** to the top of the pin, then cover the bearing with ***[a]_double_shear_b_drive_x1.stl***.

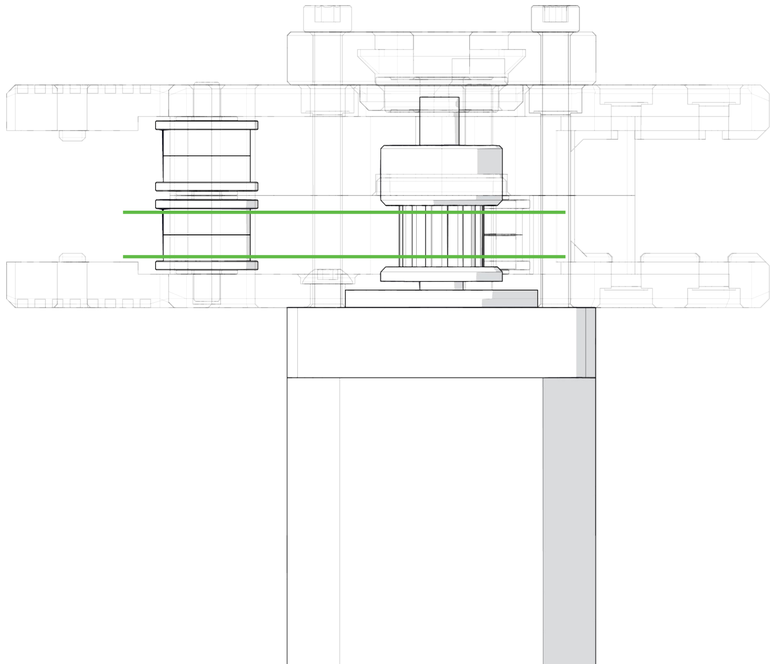
This bearing acts as the upper shaft support in the “double shear” setup.

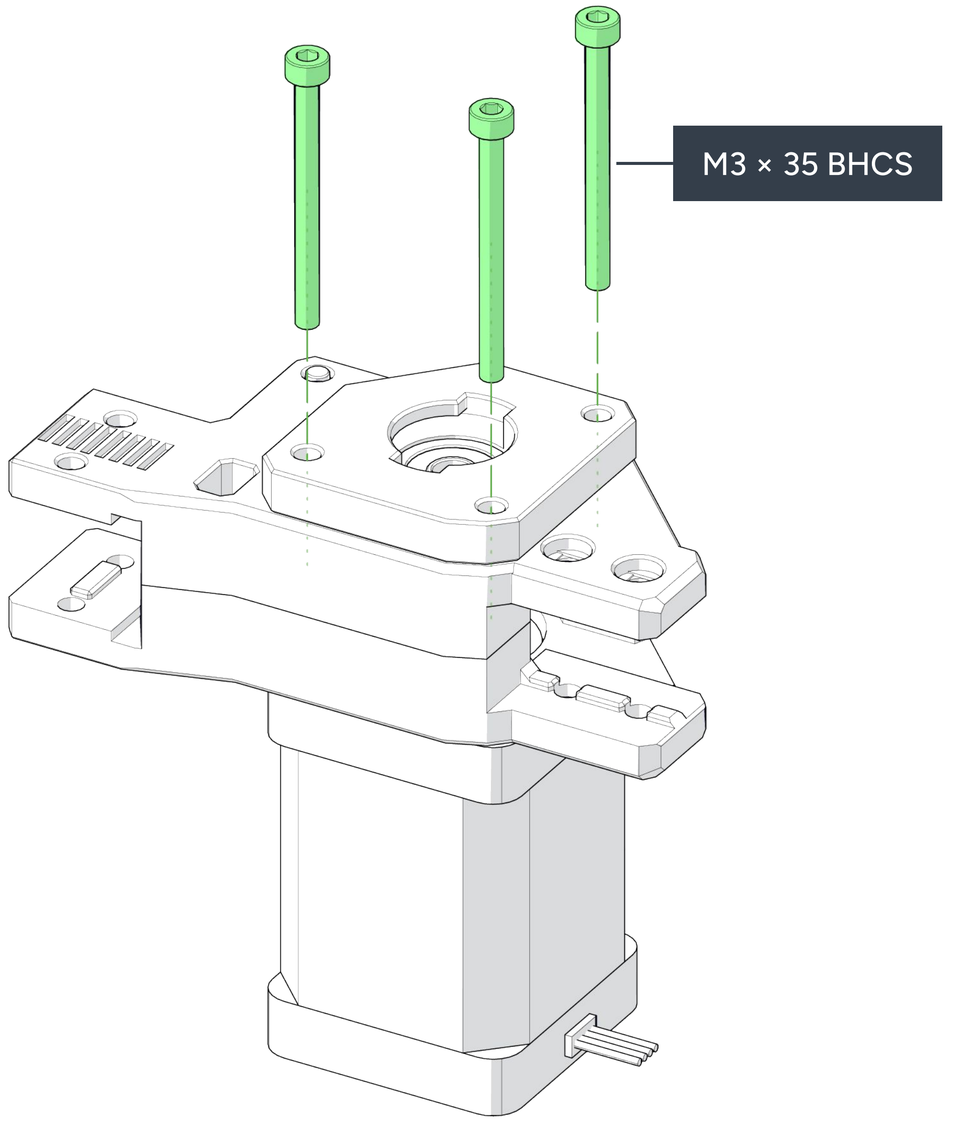


GANTRY ASSEMBLY

PREPPING B DRIVE

Secure the assembly with **three M3 × 35 BHCS** from the top through to the motor screw mounts. Ensure the exposed 2GT 16T pulley lines up with the bottom pair of bearings.

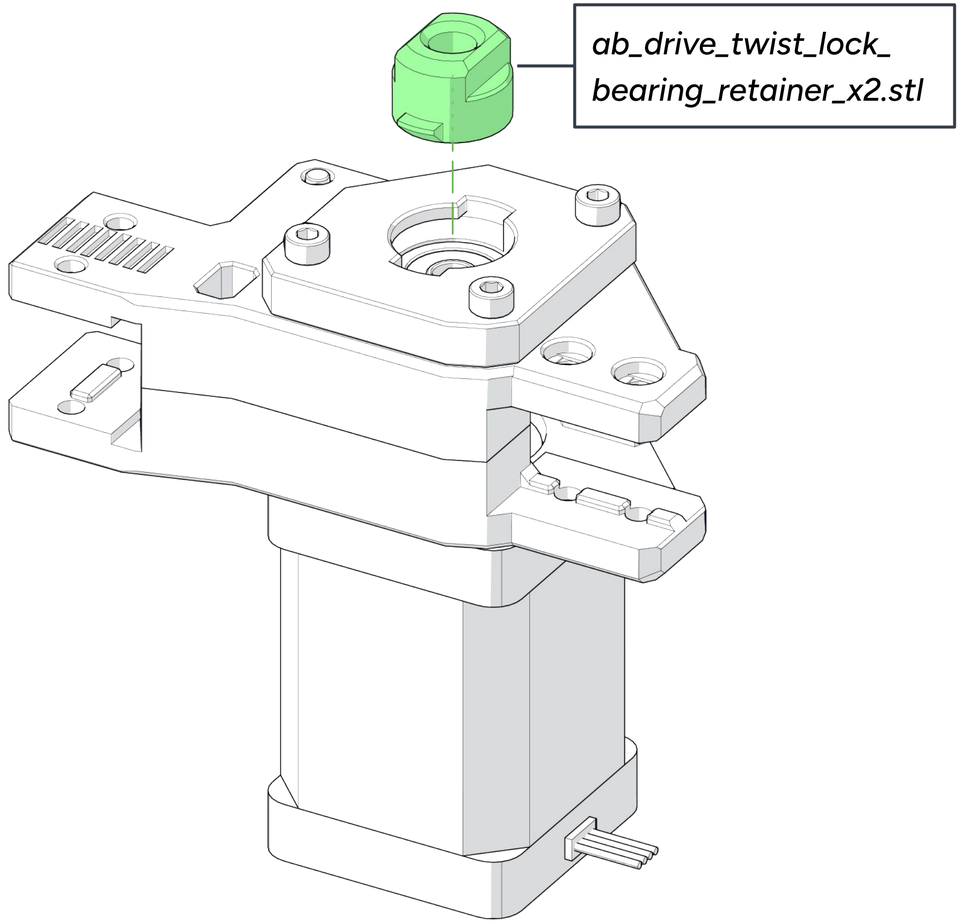




I / GANTRY ASSEMBLY

PREPPING B DRIVE

Drop in ***ab_drive_twist_lock_bearing_retainer_x2.stl***.

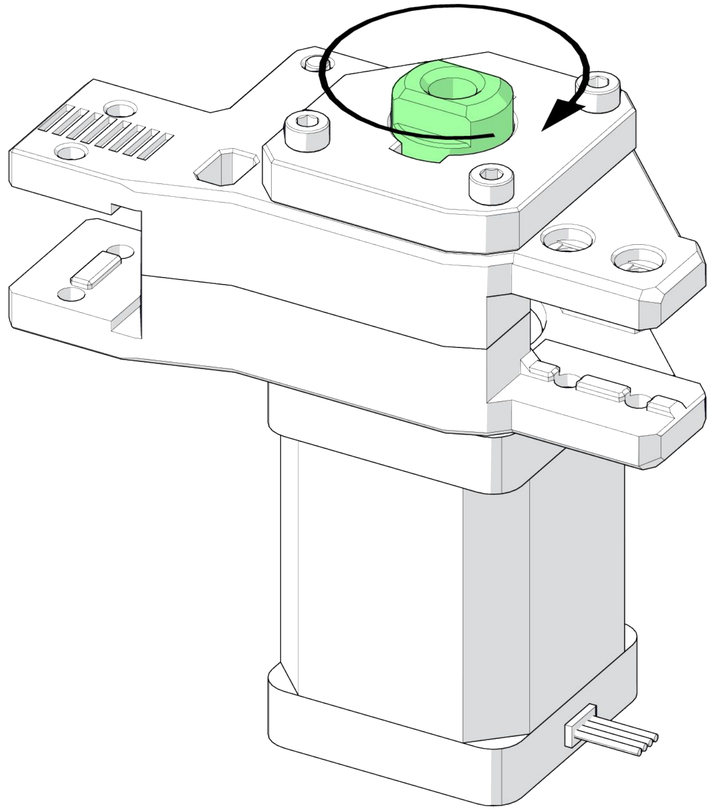


GANTRY ASSEMBLY

PREPPING B DRIVE

Rotate ***ab_drive_twist_lock_bearing_retainer_x2.stl*** clockwise to secure the assembly.

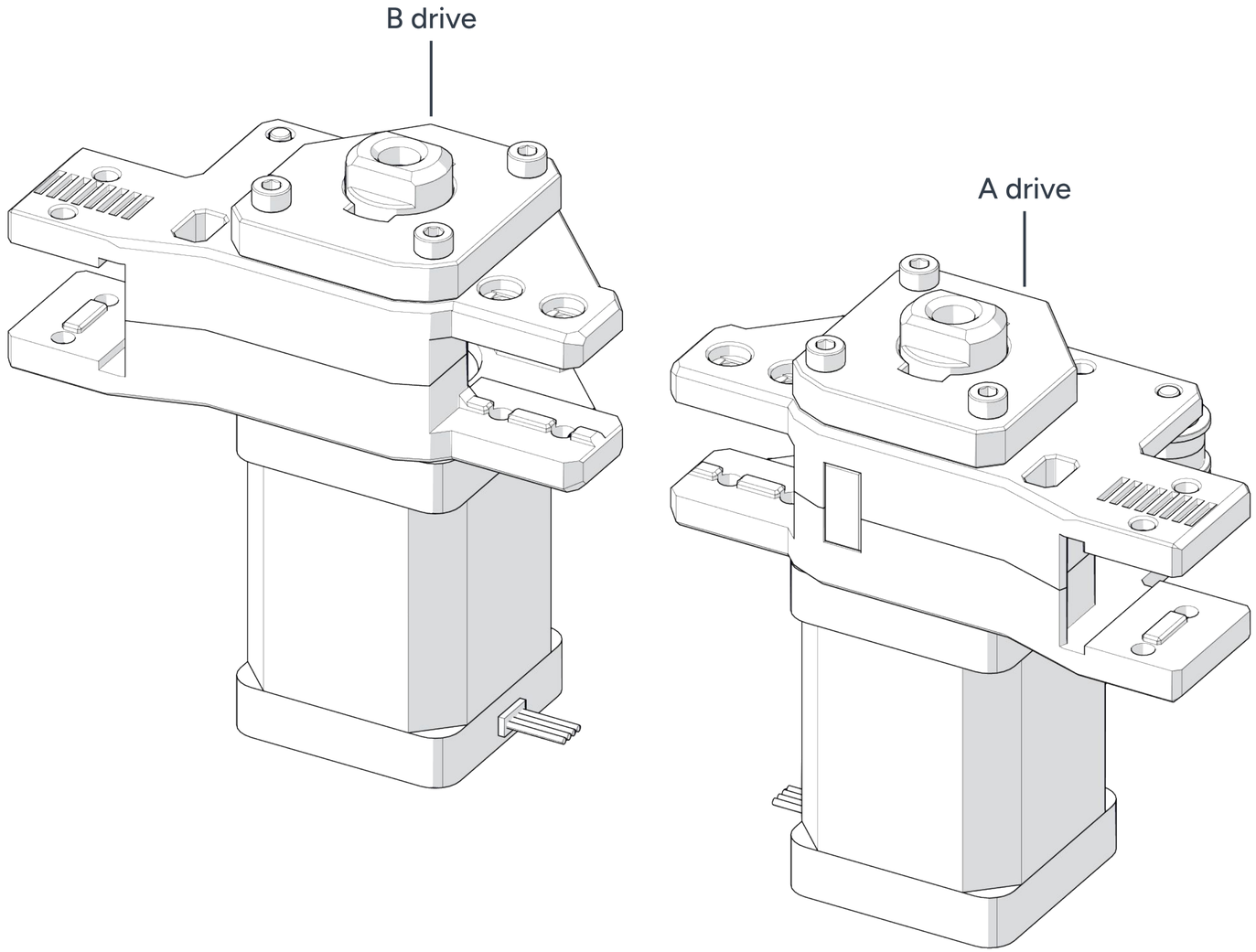
Ensure pins are seated and bearing stacks spin freely.



GANTRY ASSEMBLY

PREPPING A/B DRIVES

You should now have assembled A and B drives.
Ensure all bearings spin freely before proceeding.



GANTRY ASSEMBLY

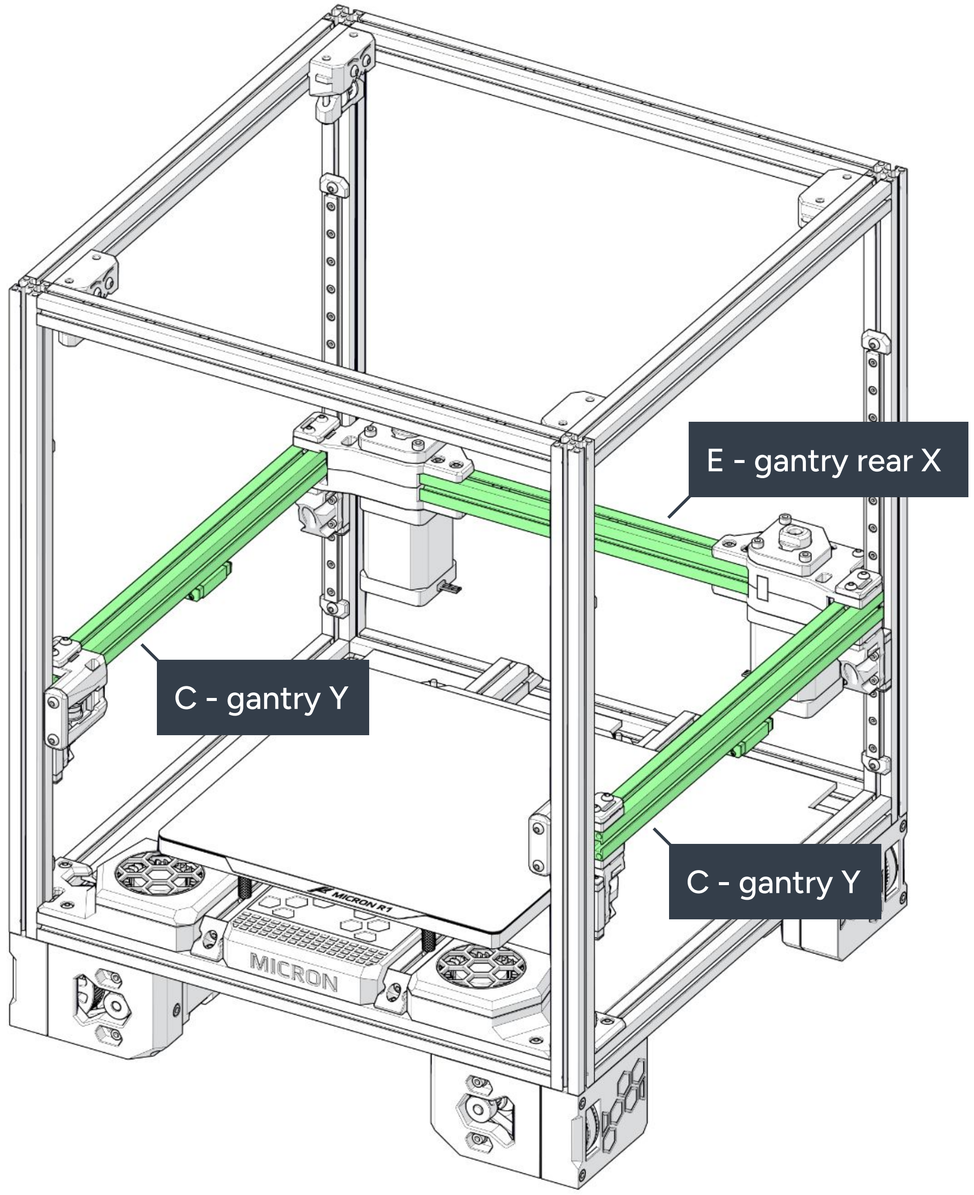
PREPPING GANTRY EXTRUSIONS

Two C extrusions form the gantry's Y arms. MGN7 linear rails mount to the underside using printed adapters. Preloaded nuts at the ends secure the front idlers and A/B drives.

The **E extrusion** sits at the rear of the gantry X. Preloaded nuts at the ends secure the A/B drives.

HARDWARE

2	1515 C extrusions (gantry Y)
1	1515 E extrusion (rear X)
2	MGN7H linear rails & carriages
30	M2 × 6 SHCS
30	M2 hex nuts
8	M3 × 8 BHCS
8	M3 × 12 BHCS
24	M3 hex nuts



GANTRY ASSEMBLY

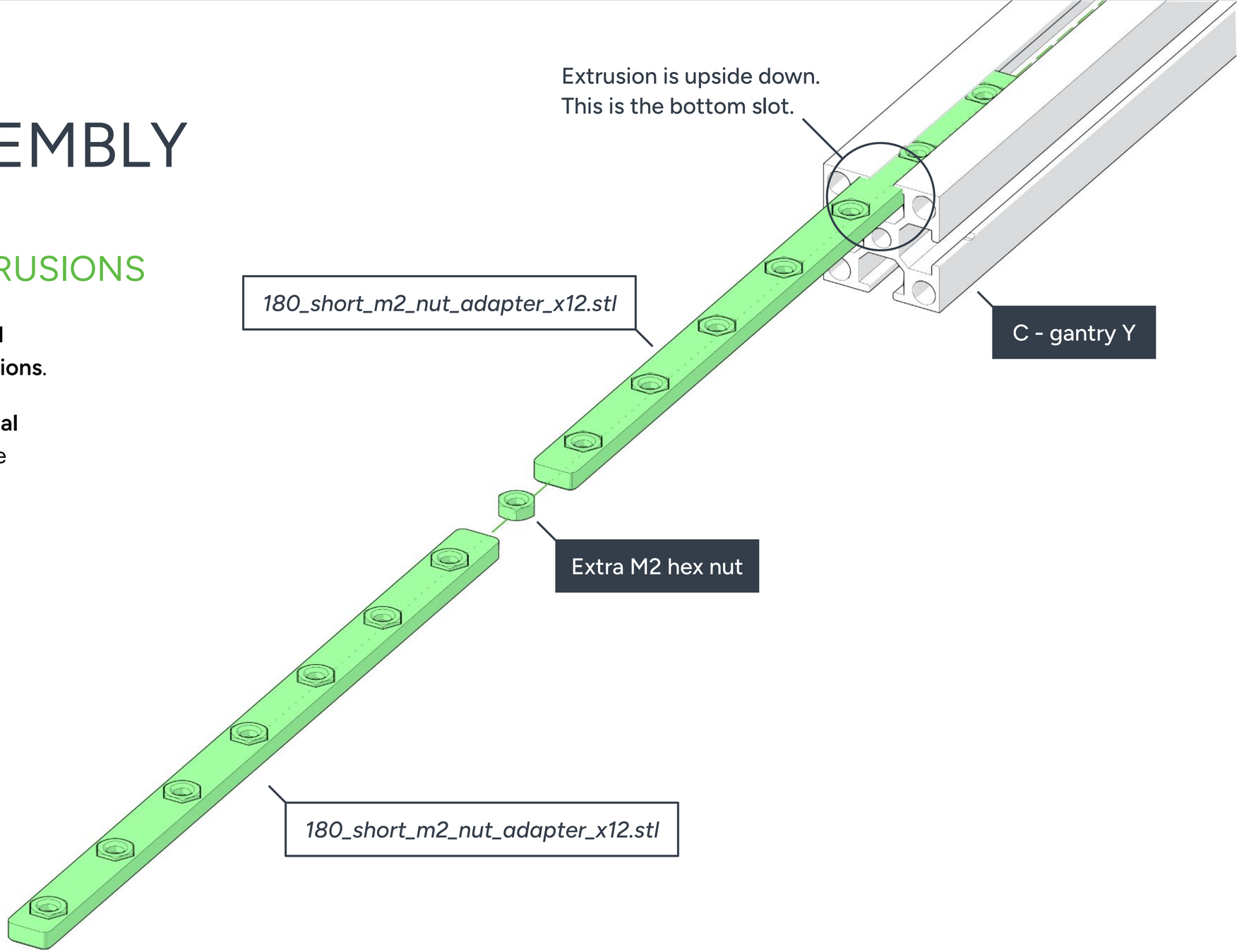
PREPPING GANTRY EXTRUSIONS

Y movement is handled by **two MGN7H rails** mounted to **two C gantry Y extrusions**.

Two printed adapters and **one additional M2 hex nut** are required to populate the 15 mounting holes of an MGN7H rail.

Slide them into the bottom of the C gantry Y extrusion. Ensure the nuts are facing outwards.

Repeat for both C gantry Y extrusions.

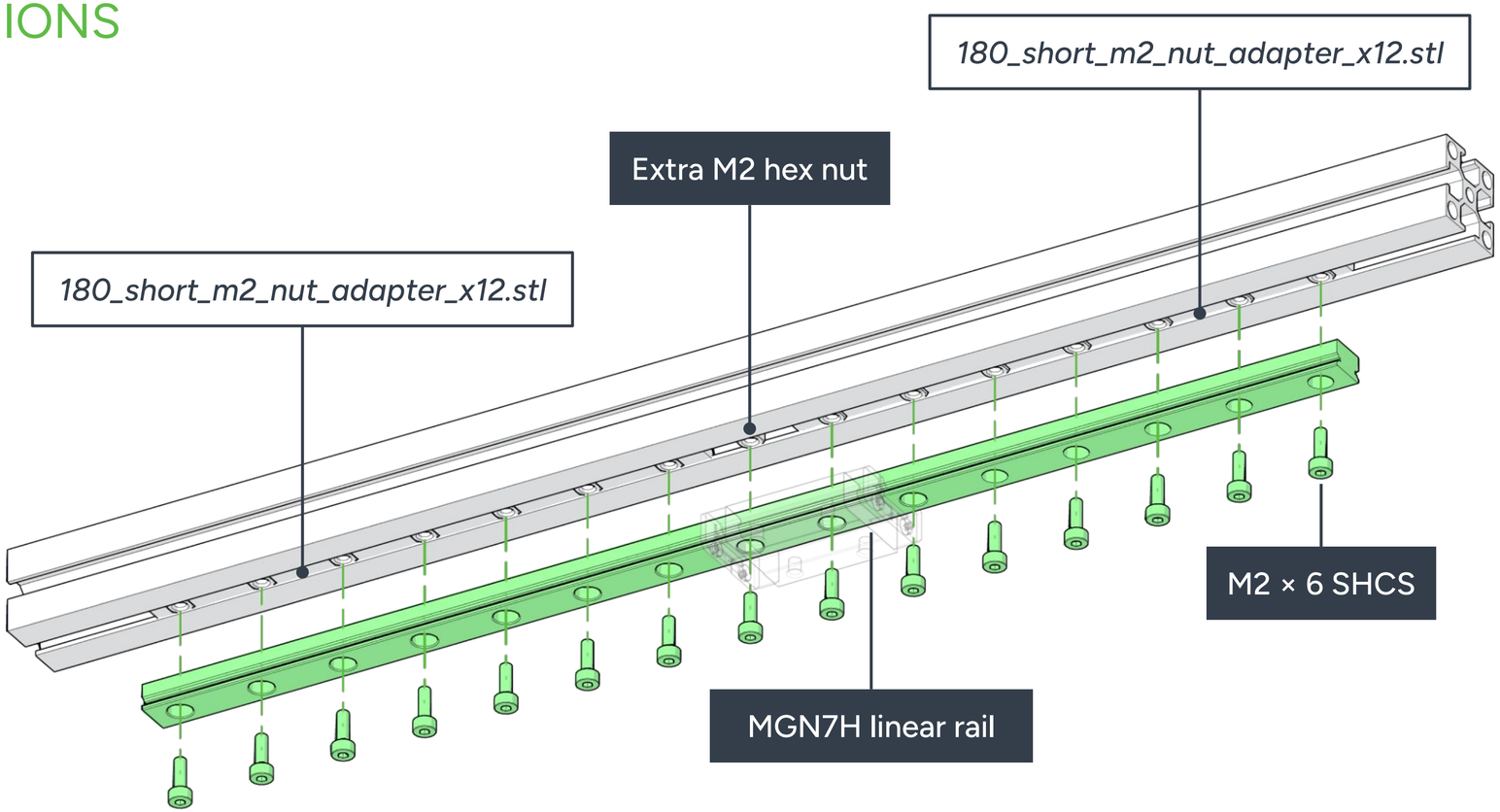


GANTRY ASSEMBLY

PREPPING GANTRY EXTRUSIONS

Line up the **MGN7H rail** over the nuts and lightly secure it with **15 M2 × 6 SHCS**. Do not tighten them down yet.

Repeat for both C gantry Y extrusions.



NOTE

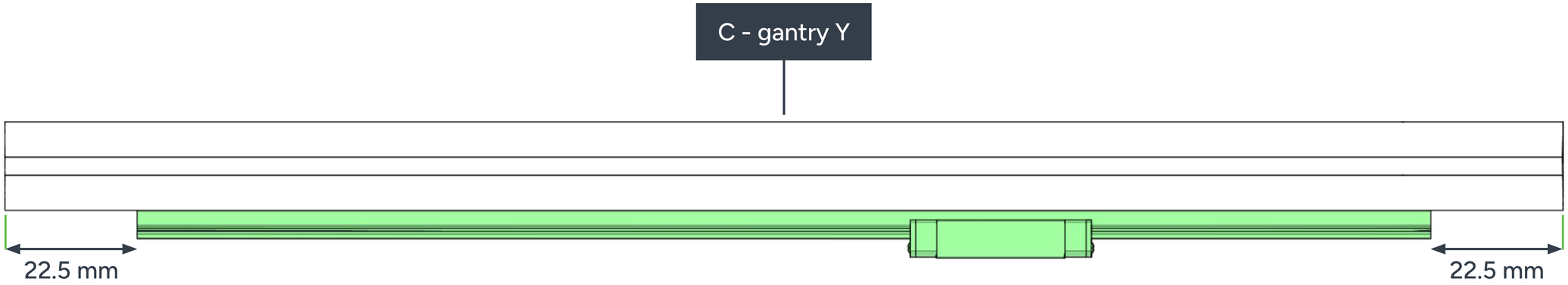
The linear rail is pictured without the carriage for illustrative purposes. Mind the carriage and do not let it fall off.

GANTRY ASSEMBLY

PREPPING GANTRY EXTRUSIONS

Position the rail 22.5 mm from the ends of the extrusion, then tighten down the 15 M2 × 6 SHCS.

Repeat for both C gantry Y extrusions.



GANTRY ASSEMBLY

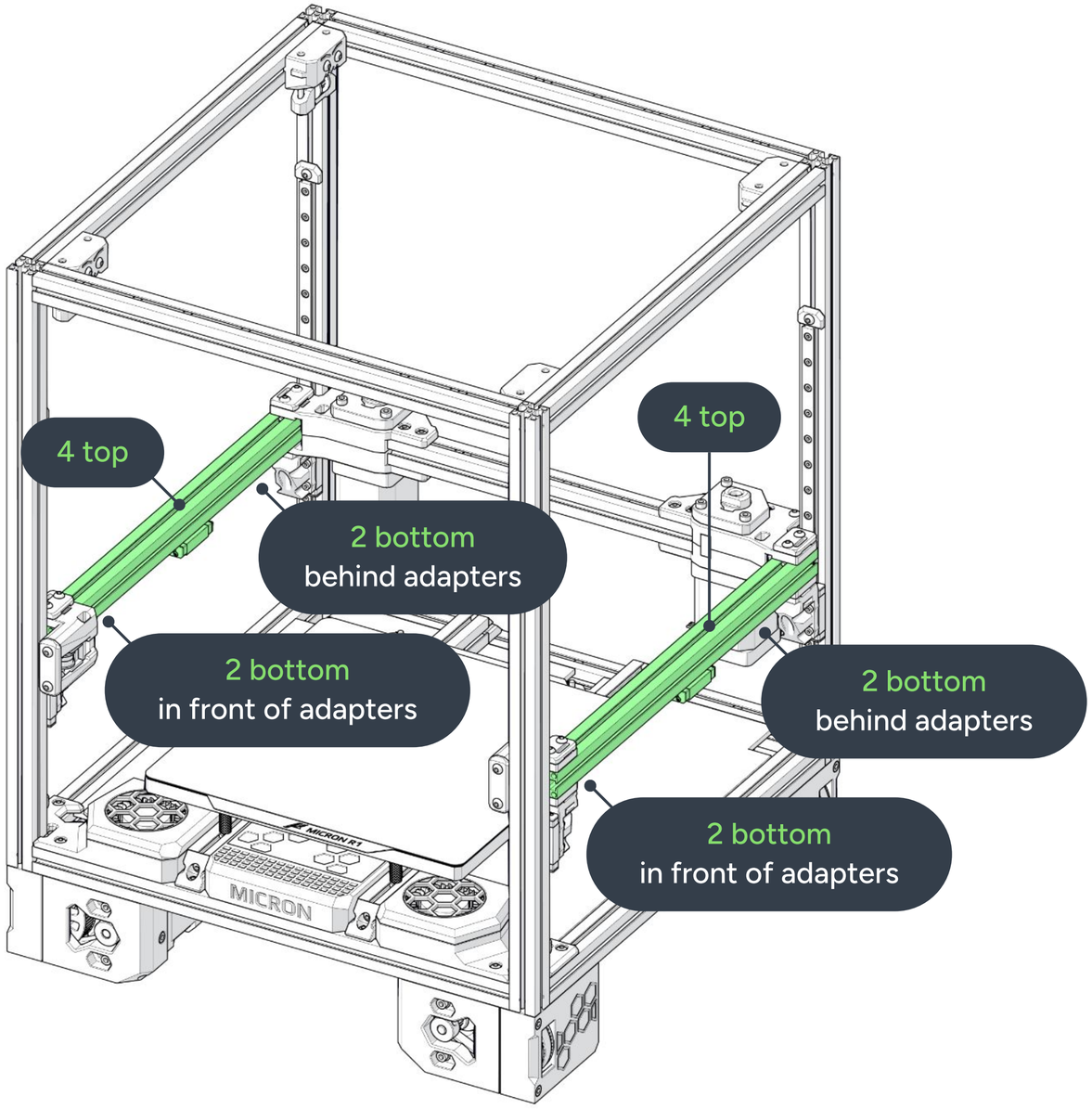
PREPPING GANTRY EXTRUSIONS

Preloaded nuts for the **two C gantry Y extrusions**.

Top slots: **4 M3 hex nuts**

Bottom slots: **4 M3 hex nuts**

- 2 in front of the adapter mounts
- 2 behind the adapter mounts



NOTE

Don’t forget the preloaded nuts.

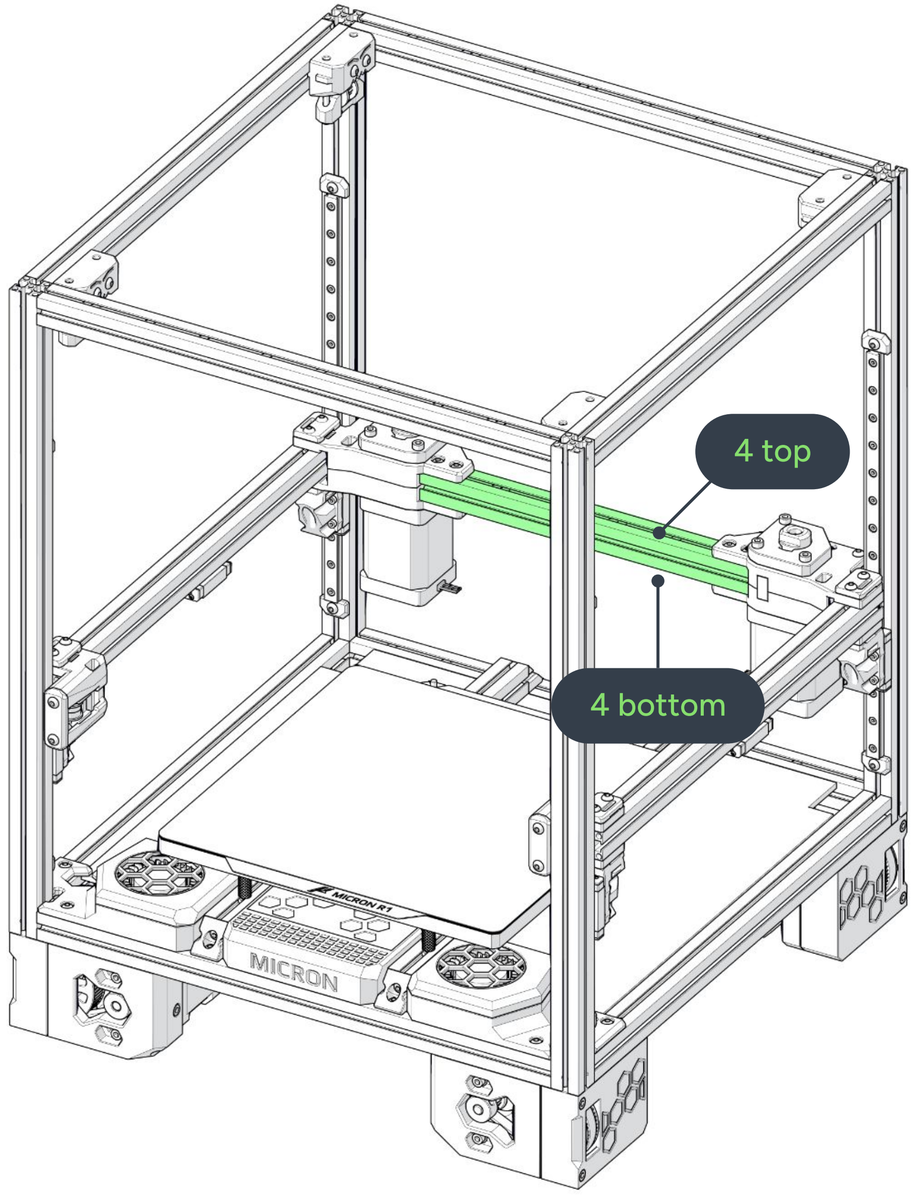
GANTRY ASSEMBLY

PREPPING GANTRY EXTRUSIONS

Preloaded nuts for the **E gantry rear X extrusion**.

Top slot: **4 M3 hex nuts**

Bottom slot: **4 M3 hex nuts**



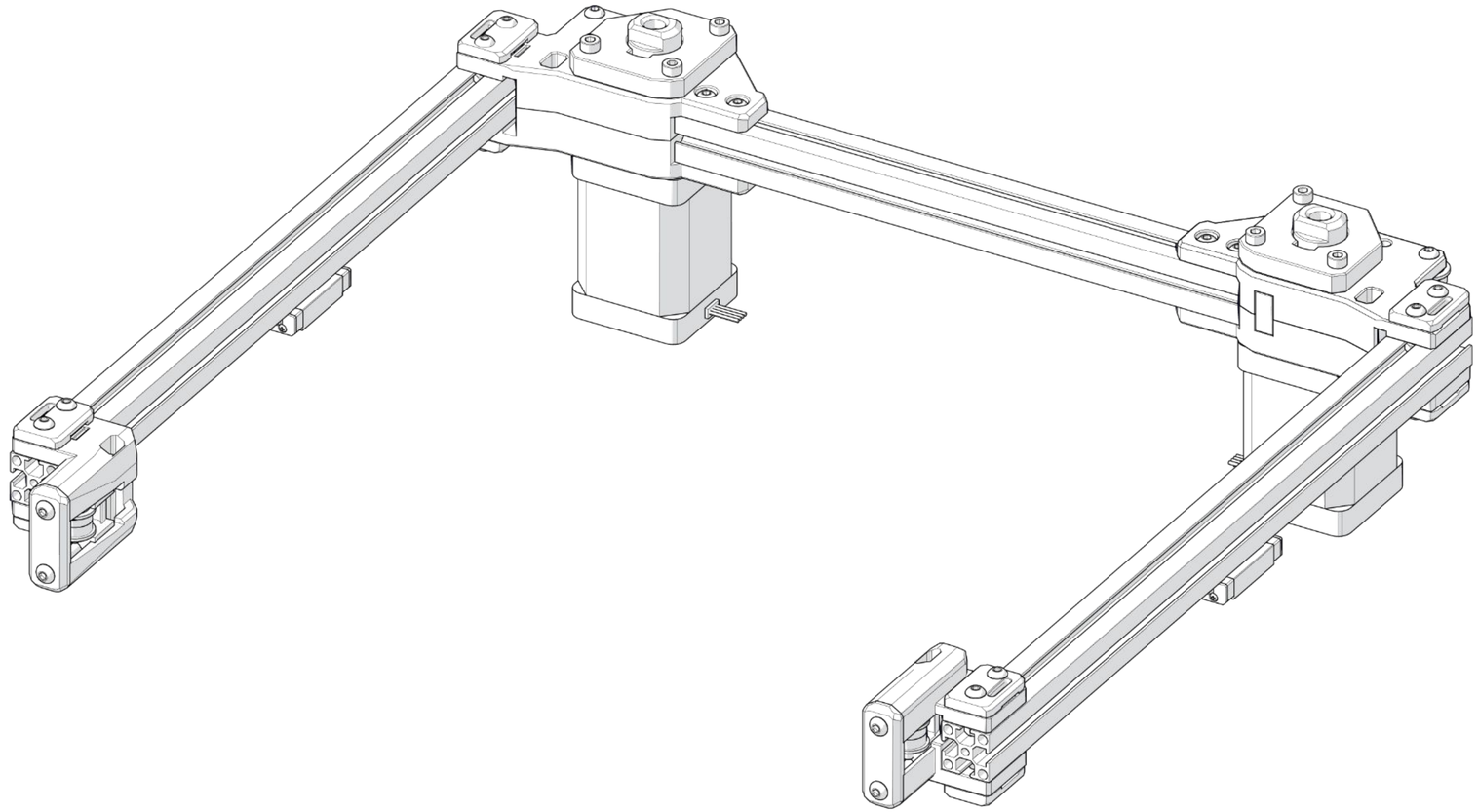
NOTE

Don’t forget the preloaded nuts.

GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

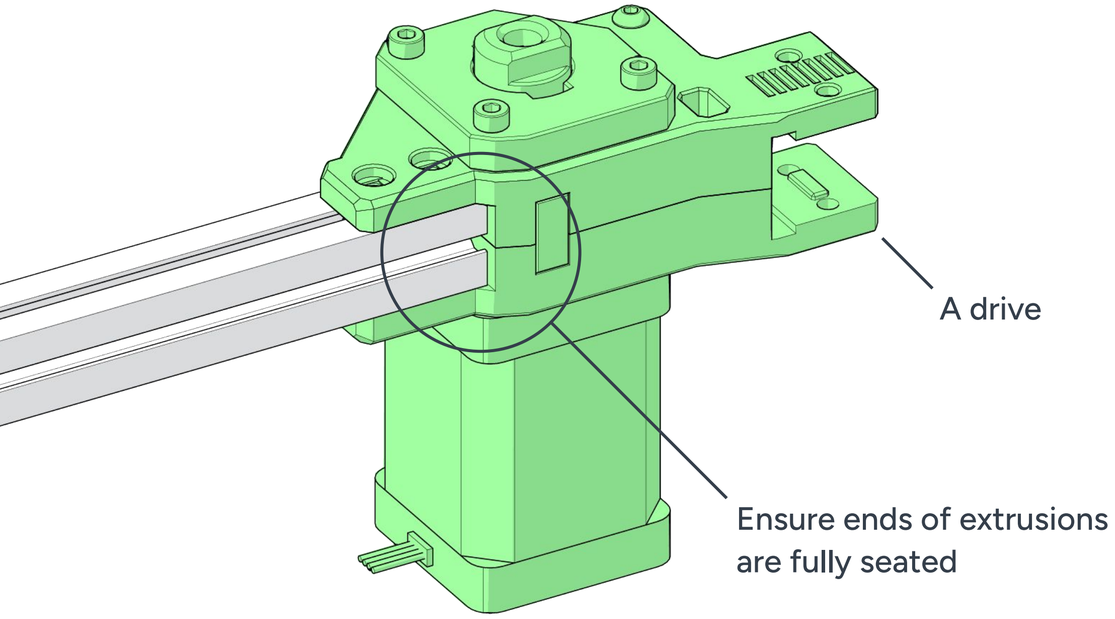
The front idlers and A/B drives mount to the gantry extrusions, defining the X/Y plane.

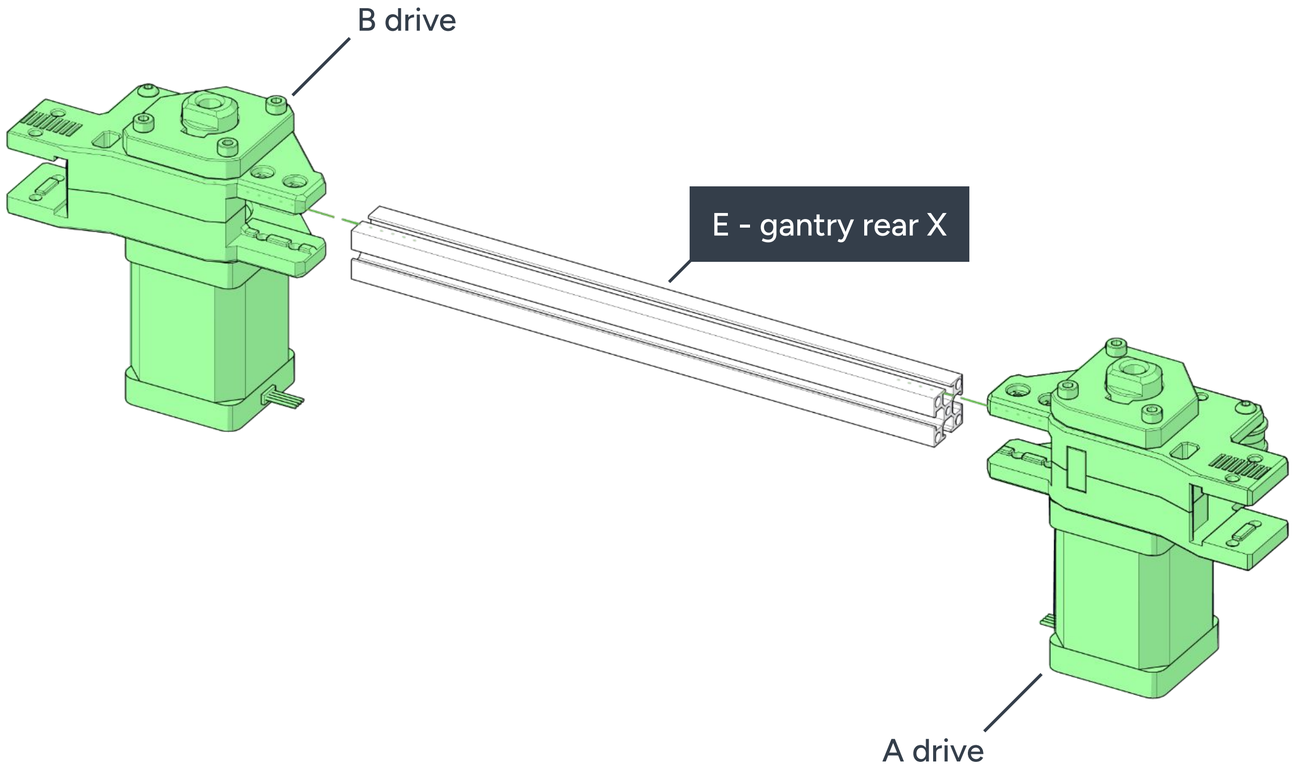


GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

Place the A/B drives on the ends of **E gantry rear X extrusion**. Ensure the ends of the extrusion are fully seated in the drives.

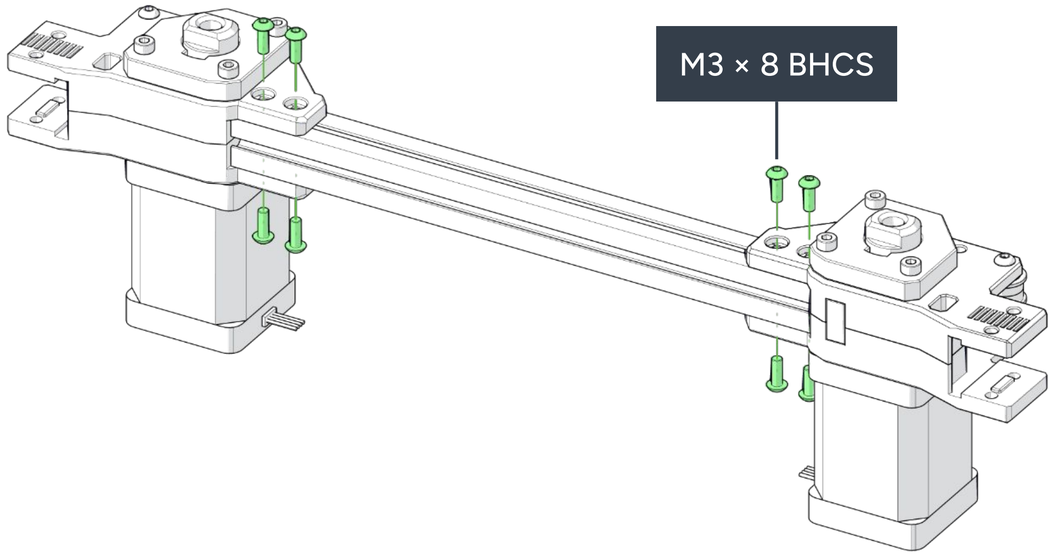




GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

Secure the A/B drives with **8 M3 × 8 BHCS** to the preloaded M3 hex nuts.

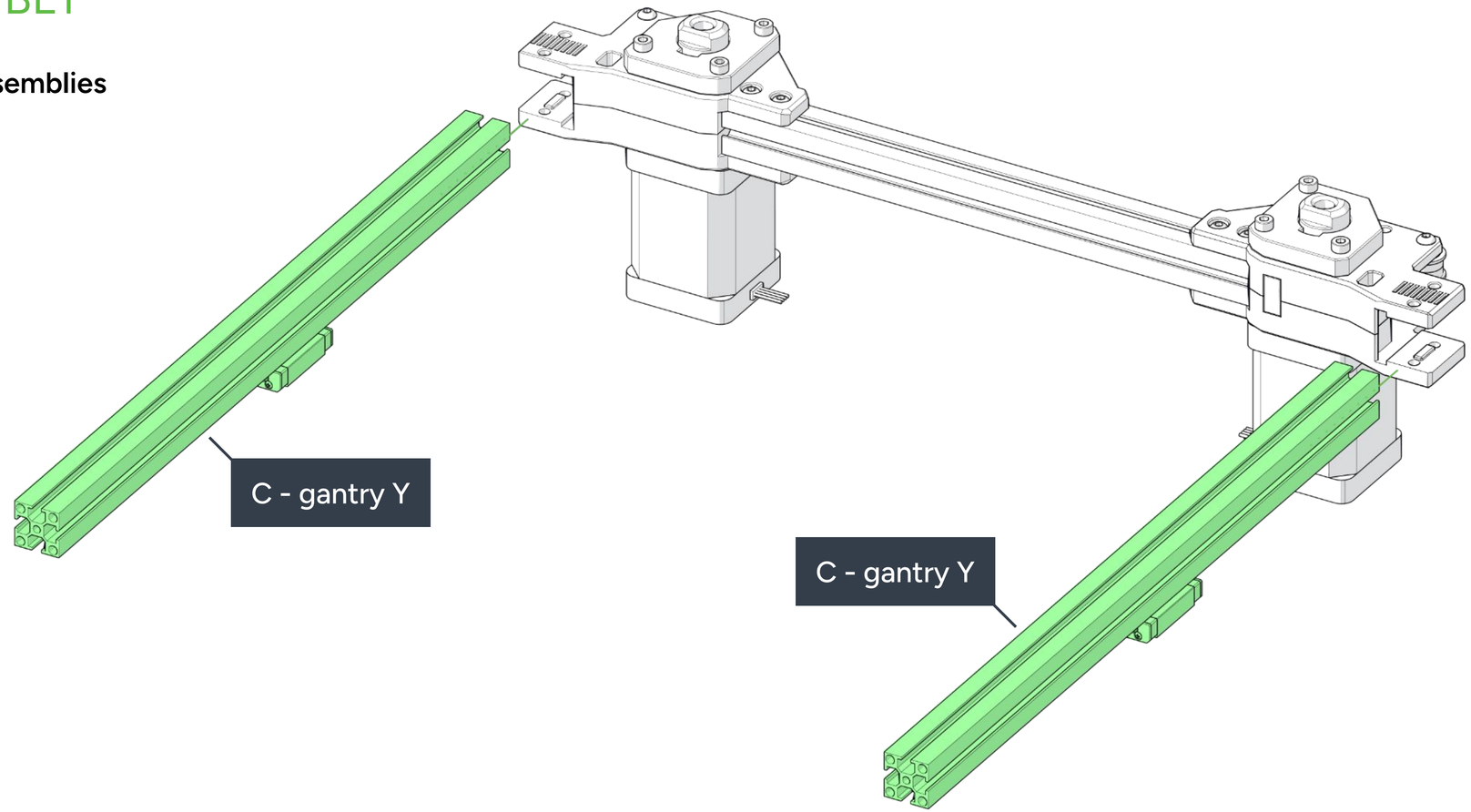


GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

Insert the **two C gantry Y extrusion assemblies** into the ends of the A/B drives.

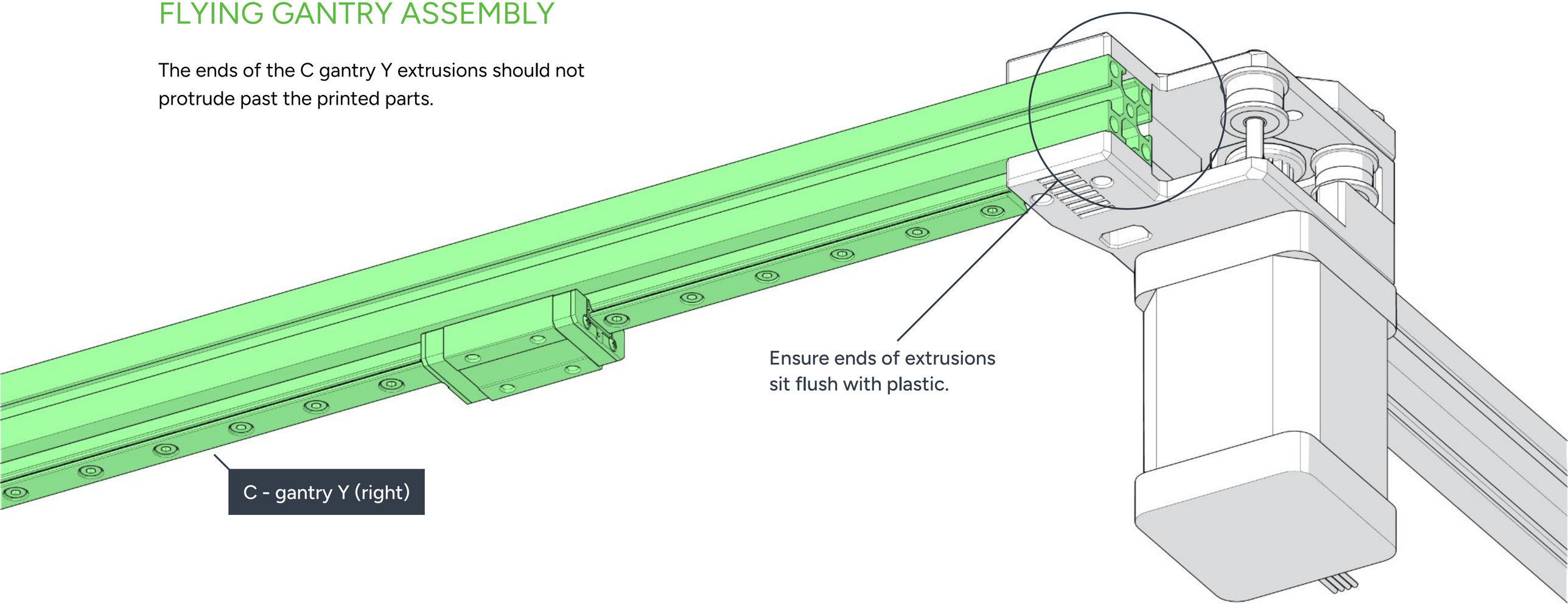
Rails are on the bottom slots.



GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

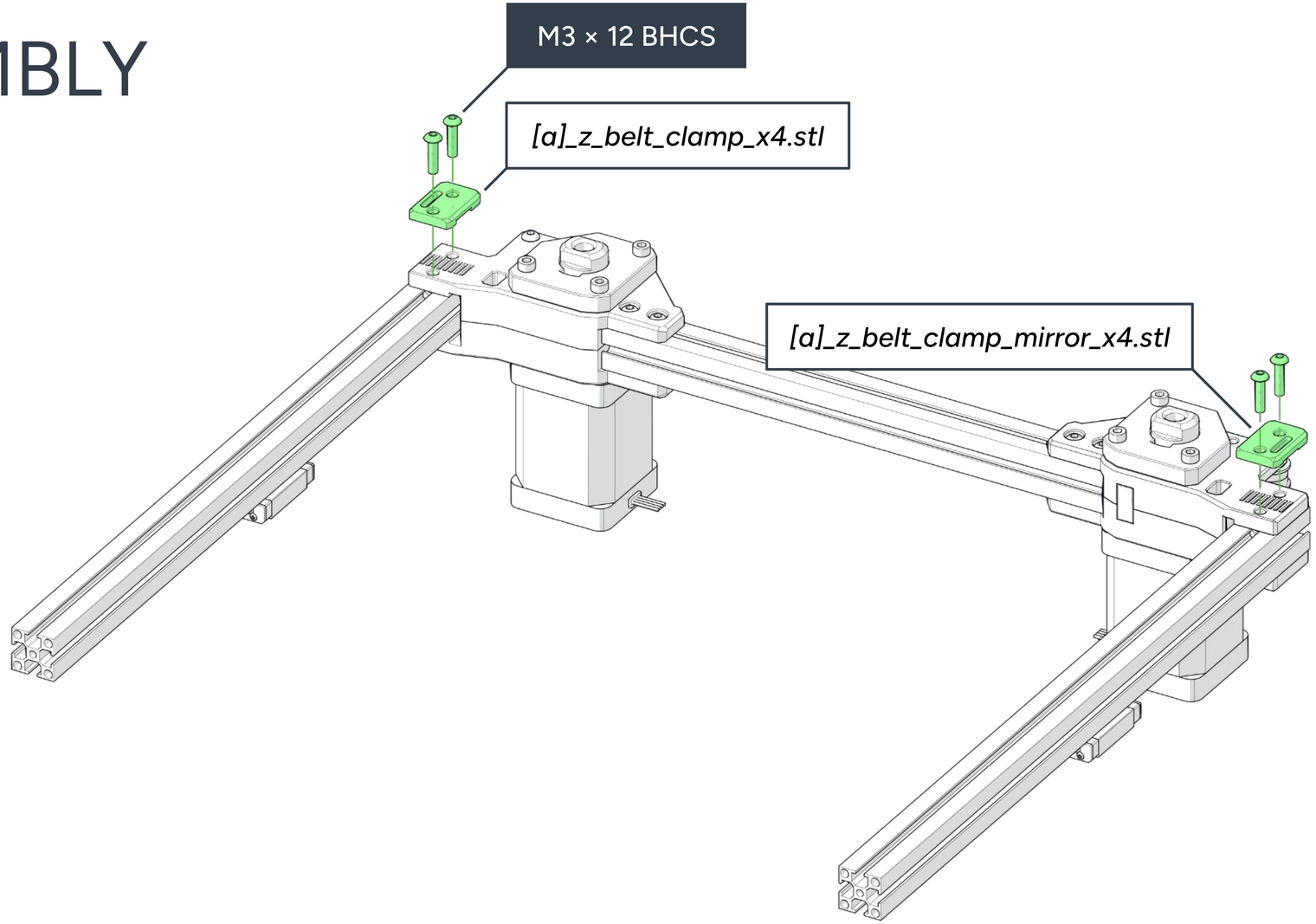
The ends of the C gantry Y extrusions should not protrude past the printed parts.



GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

Place the **top belt clamps** and secure with **four M3 × 12 BHCS** to the preloaded hex nuts.



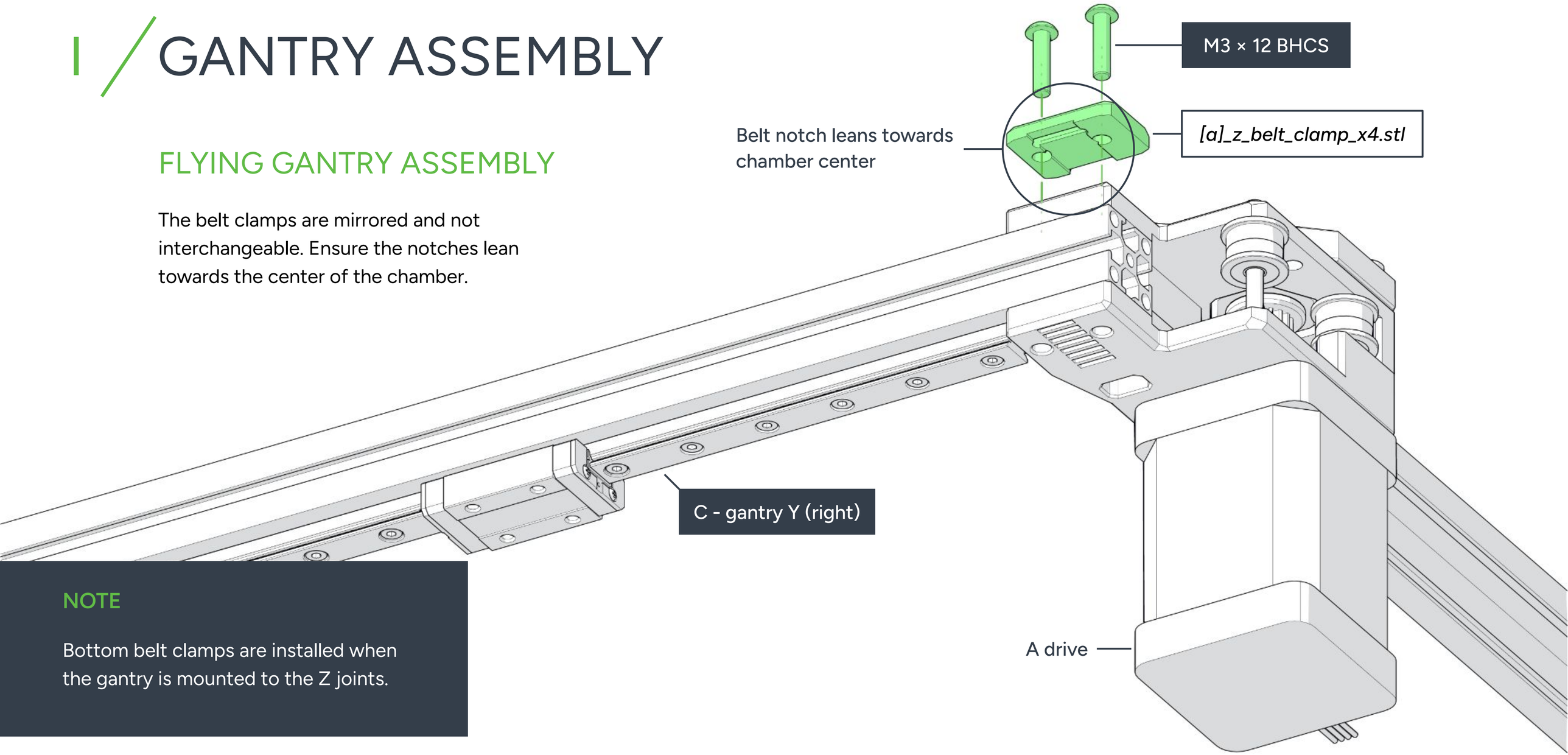
NOTE

Bottom belt clamps are installed when the gantry is mounted to the Z joints.

GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

The belt clamps are mirrored and not interchangeable. Ensure the notches lean towards the center of the chamber.



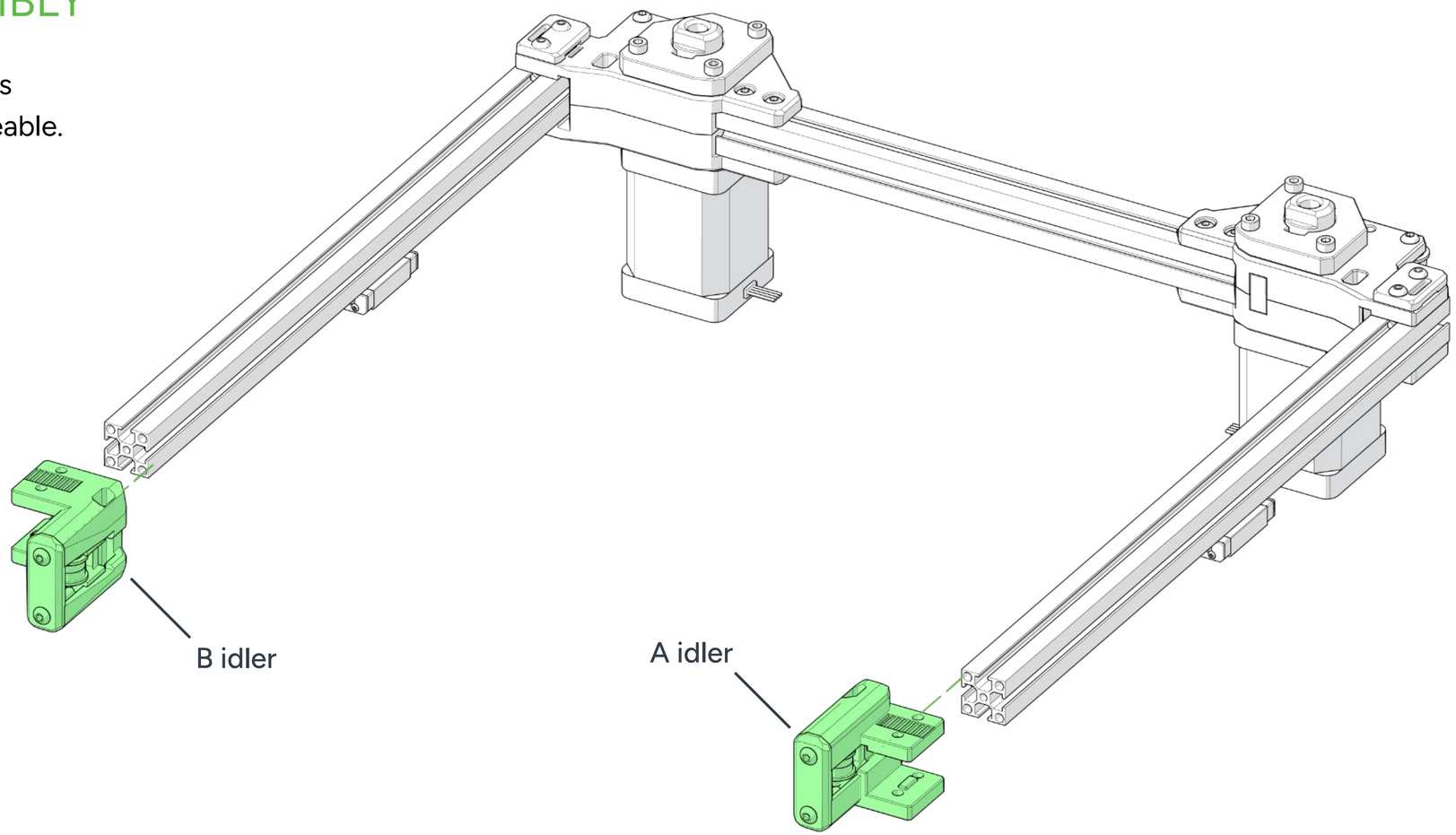
NOTE

Bottom belt clamps are installed when the gantry is mounted to the Z joints.

GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

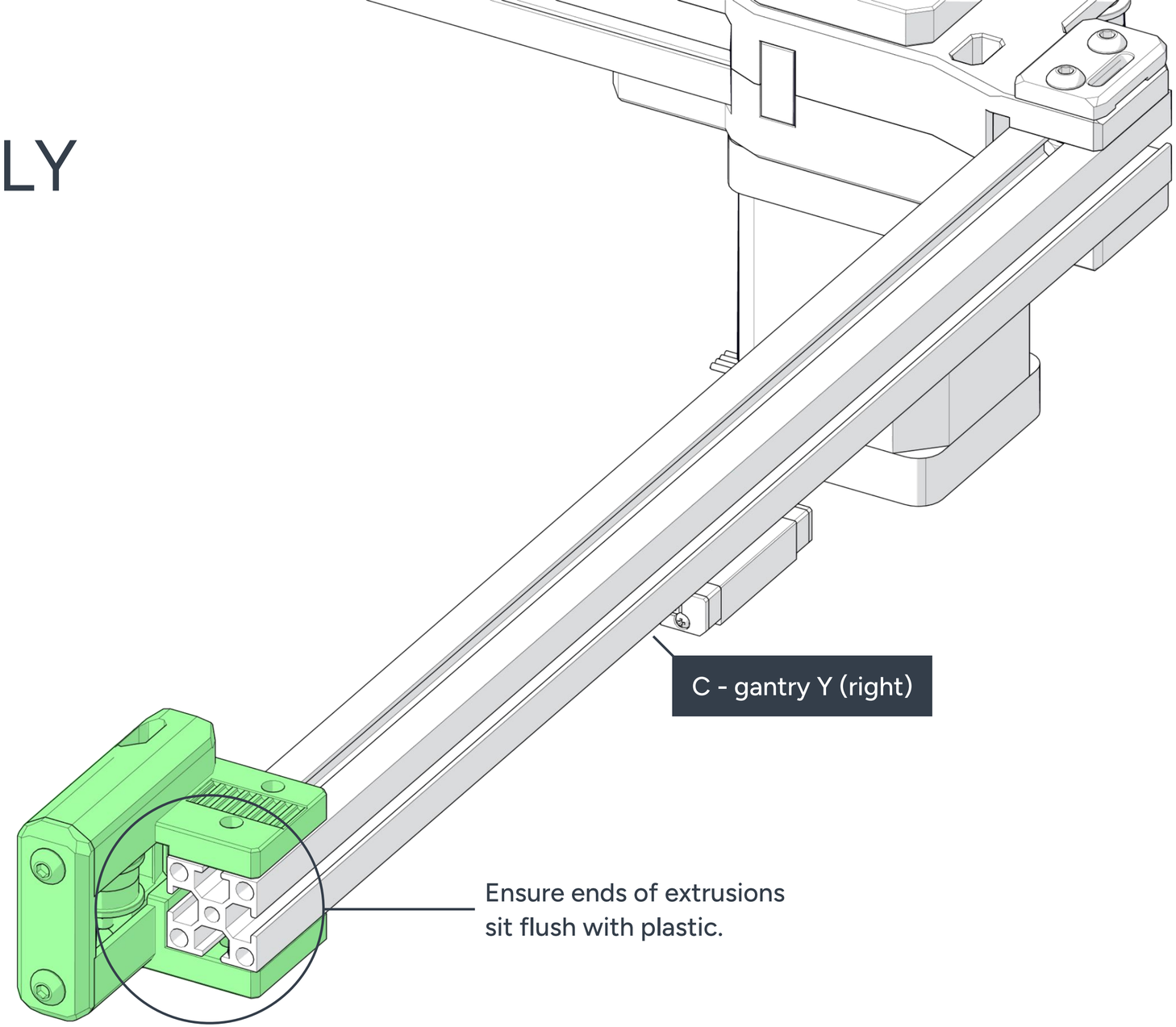
Place the front idlers on the fronts ends of the extrusions. They are interchangeable.



GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

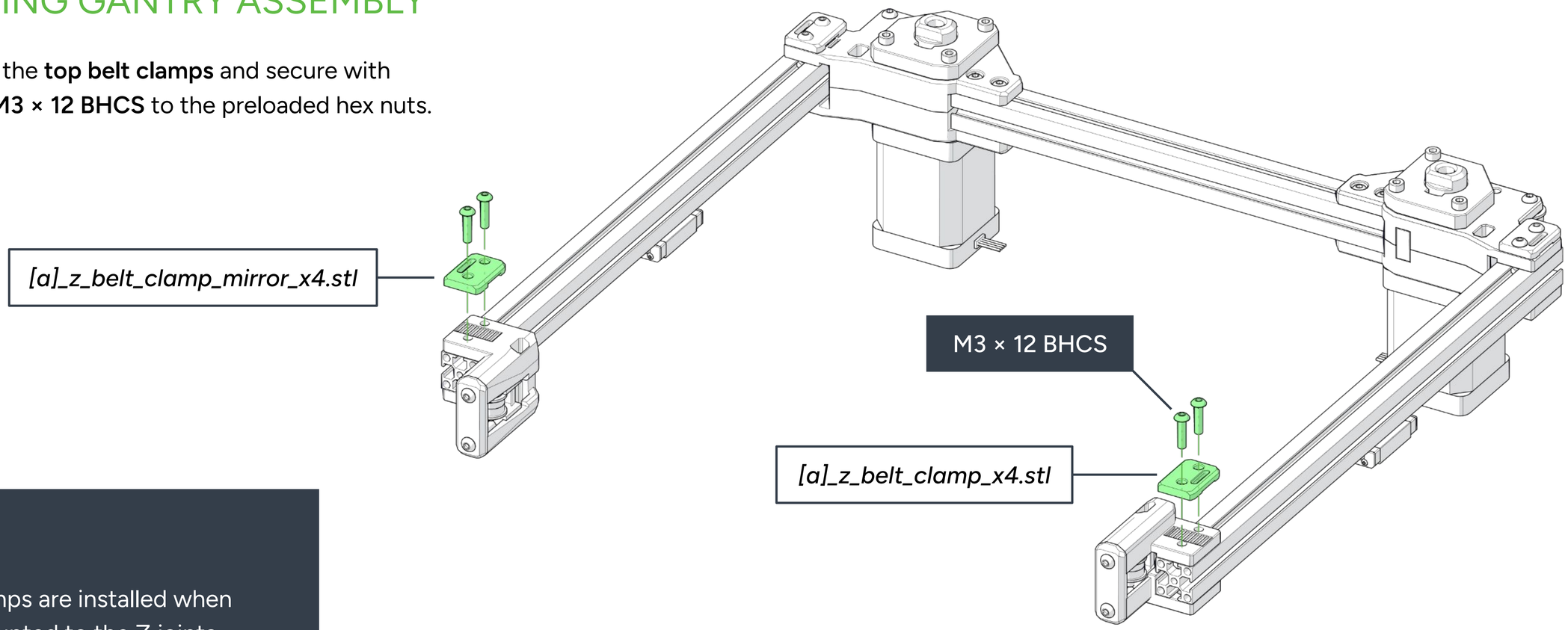
The front ends of the C gantry Y extrusions should not protrude past the printed parts.



GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

Place the **top belt clamps** and secure with **four M3 × 12 BHCS** to the preloaded hex nuts.



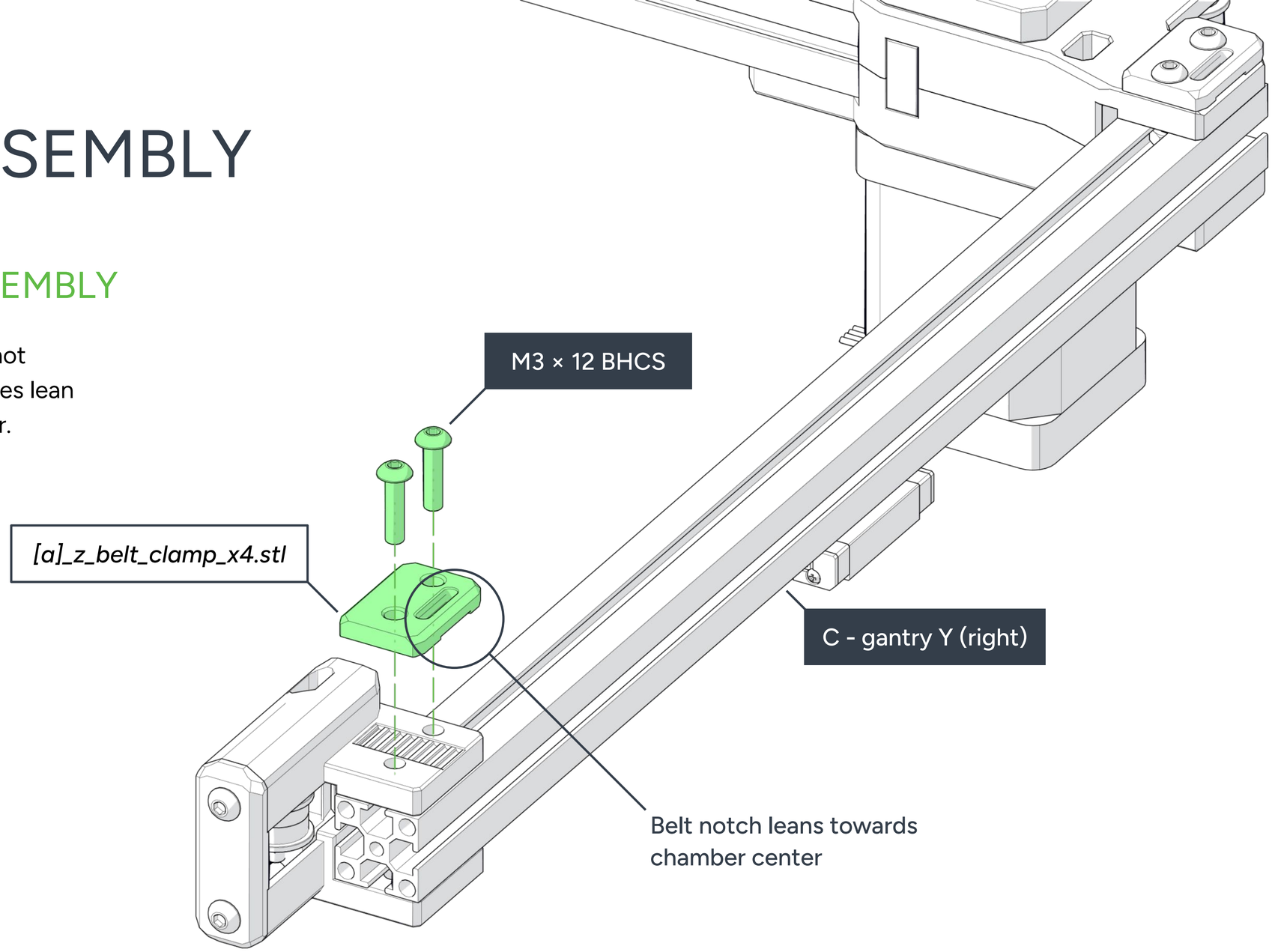
NOTE

Bottom belt clamps are installed when the gantry is mounted to the Z joints.

GANTRY ASSEMBLY

FLYING GANTRY ASSEMBLY

The belt clamps are mirrored and not interchangeable. Ensure the notches lean towards the center of the chamber.



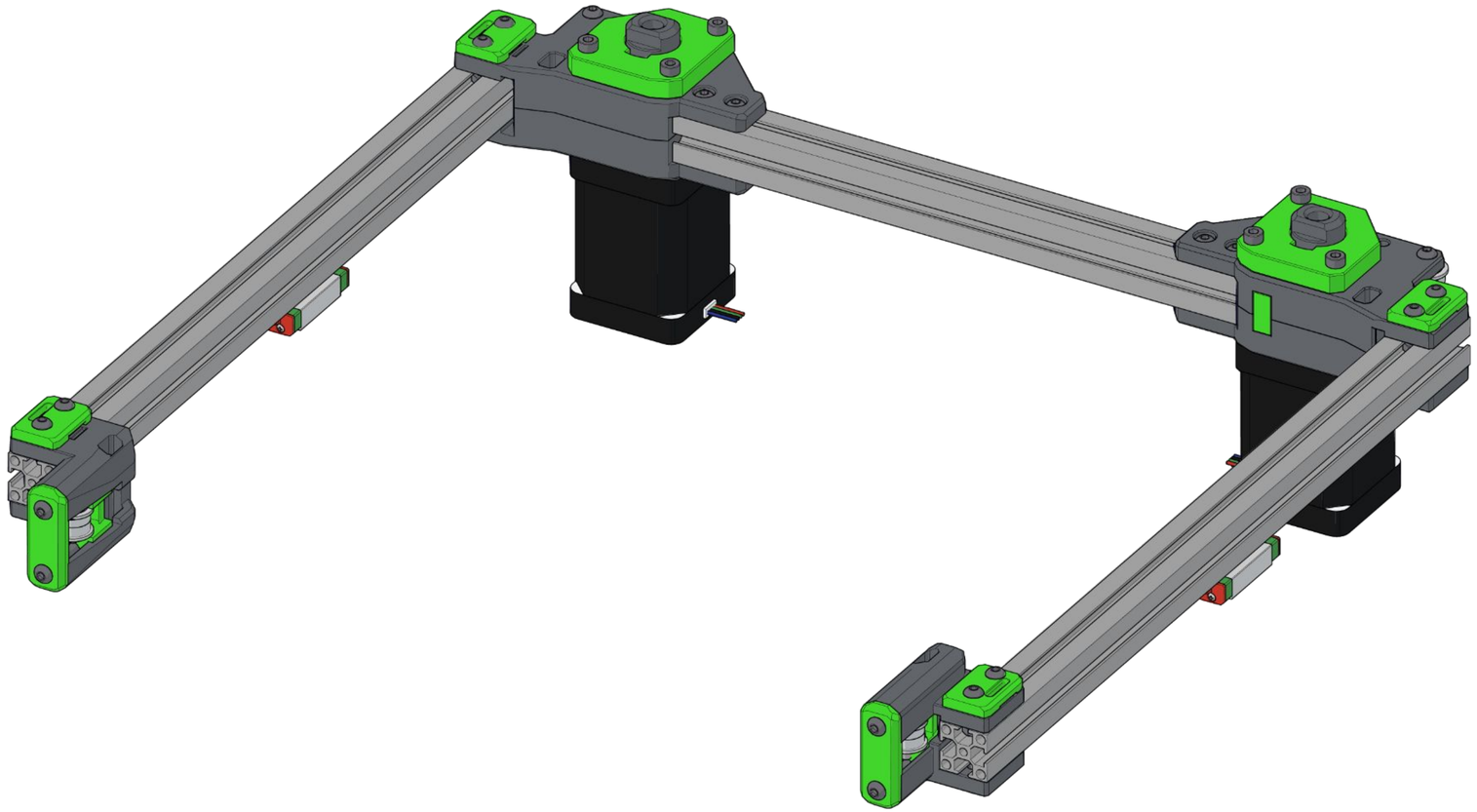
GANTRY ASSEMBLY

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

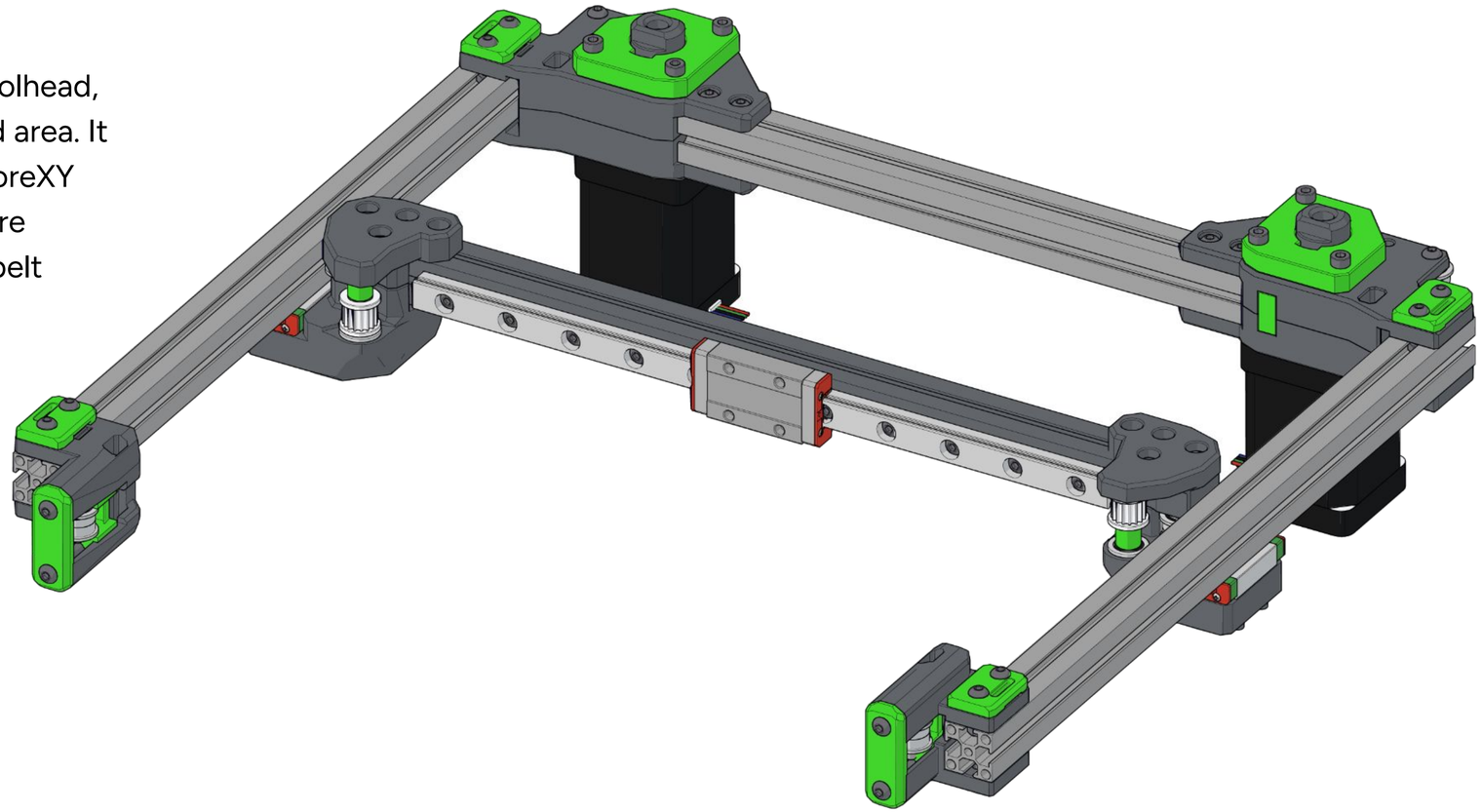
- ▢ Front idlers spin freely
- ▢ A/B drive bearings spin freely
- ▢ A drive pulley teeth lines up with the top bearings
- ▢ B drive pulley teeth lines up with the bottom bearings
- ▢ Both double shear twist locks are secured
- ▢ Ends of extrusions are flush with printed parts



GANTRY ASSEMBLY

X AXIS ASSEMBLY

The X-axis spans the gantry and carries the toolhead, providing motion across the width of the build area. It rides on the Y-axis rails and is driven by the CoreXY belt system. Proper alignment and assembly are critical to maintain smooth travel, consistent belt tracking, and accurate positioning.



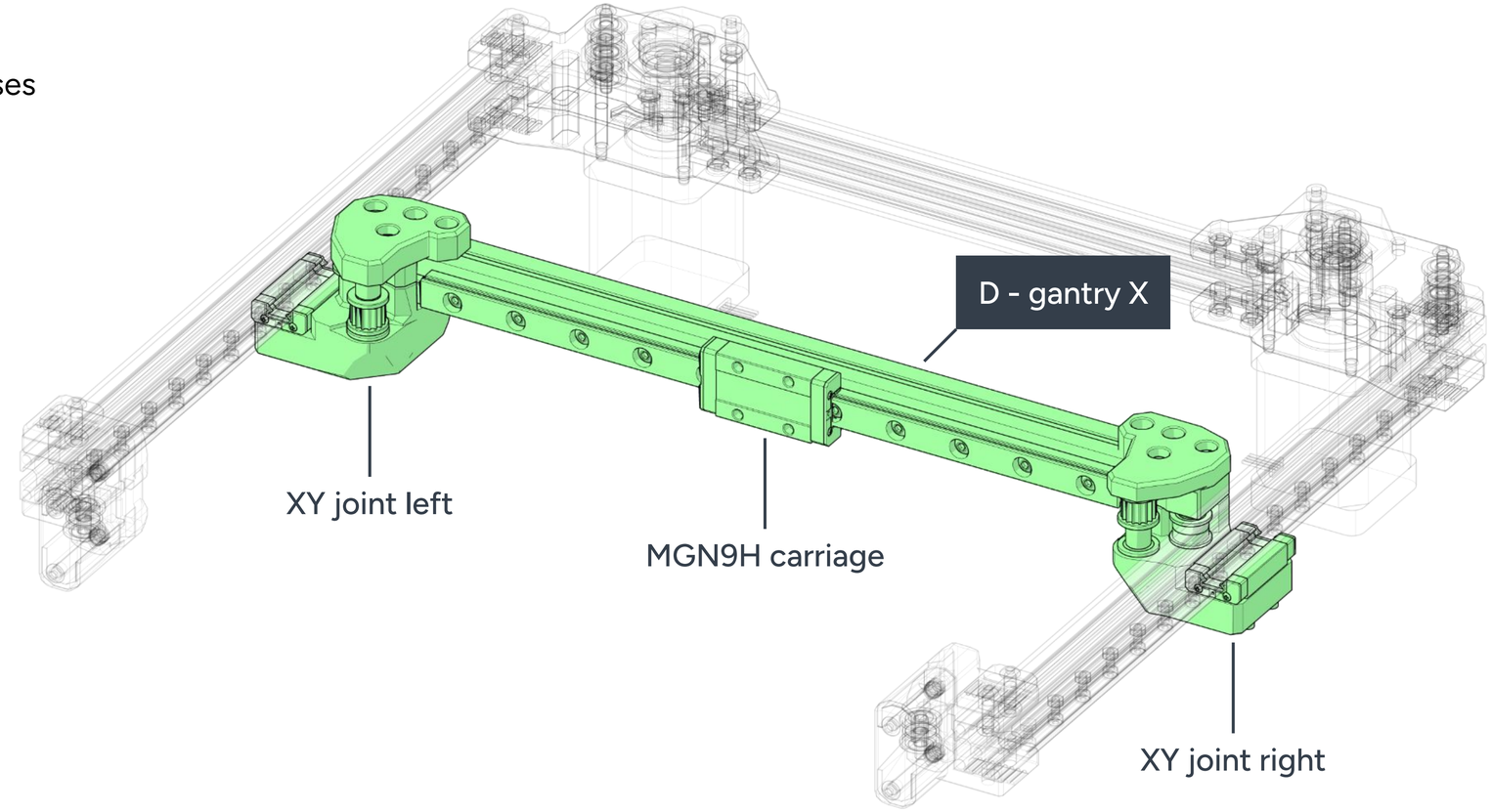
GANTRY ASSEMBLY

X AXIS ASSEMBLY

XY joint hardware varies by kit. The LDO kit uses live idler assemblies.

HARDWARE

1	MGN9H linear rail & carriage
4	3 mm × 25 mm pin
4	F623-2RS bearing
2	2GT 16T idler
4	M3 × 7 × 0.5 mm shim
15	M3 × 8 BHCS
2	M3 × 20 BHCS
4	M3 × 25 BHCS
4	M3 × 35 SHCS
8	M2 × 12 SHCS
17	M3 hex nuts



GANTRY ASSEMBLY

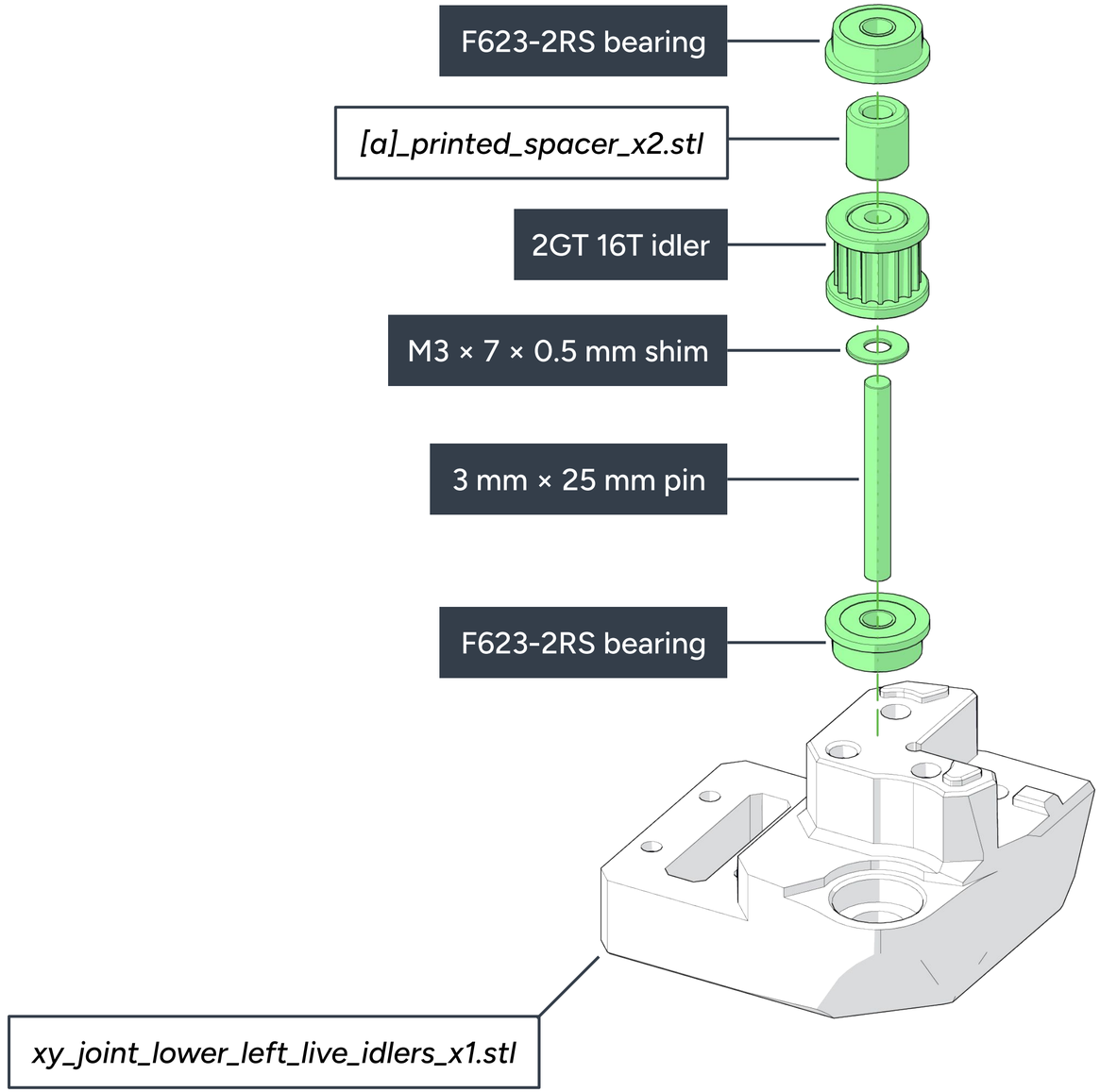
XY JOINT LEFT ASSEMBLY

The LDO kit uses live idlers, where the shaft and idler rotate together on support bearings, reducing friction compared to a fixed shaft design.

Press an **F623-2RS bearing (flange up)** into the recess in ***xy_joint_lower_left_live_idlers_x1.stl***. Then, assemble the following component stack on a **3 mm × 25 mm pin** and insert it into the bearing.

Live idler stack:

- **F623-2RS bearing (flange down)**
- ***[a]_printed_spacer_x2.stl***
- **2GT 16T idler**
- **M3 × 7 × 0.5 mm shim**
- F623-2RS bearing (flange up, already inserted)



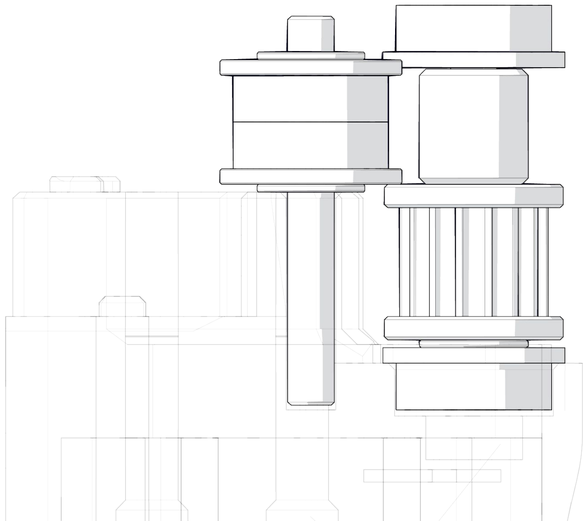
GANTRY ASSEMBLY

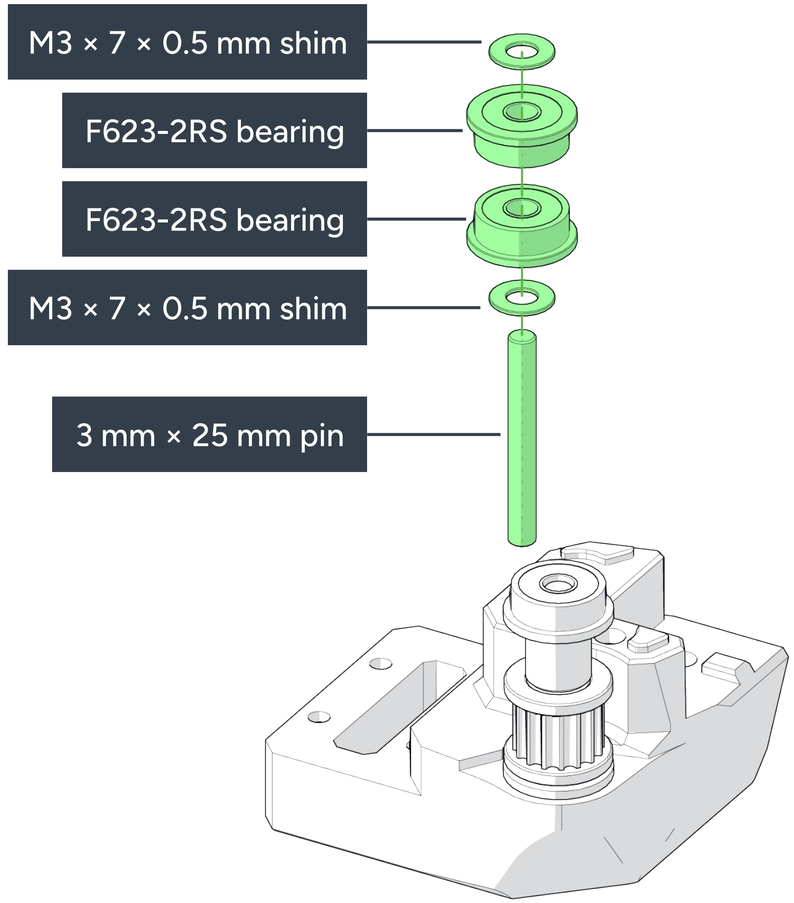
XY JOINT LEFT ASSEMBLY

The second stack for the other belt is a traditional bearing stack on a **3 mm × 25 mm pin**.

- **M3 × 7 × 0.5 mm shim**
- **F623-2RS bearing (flange up)**
- **F623-2RS bearing (flange down)**
- **M3 × 7 × 0.5 mm shim**

Ensure your stacks match the diagram.



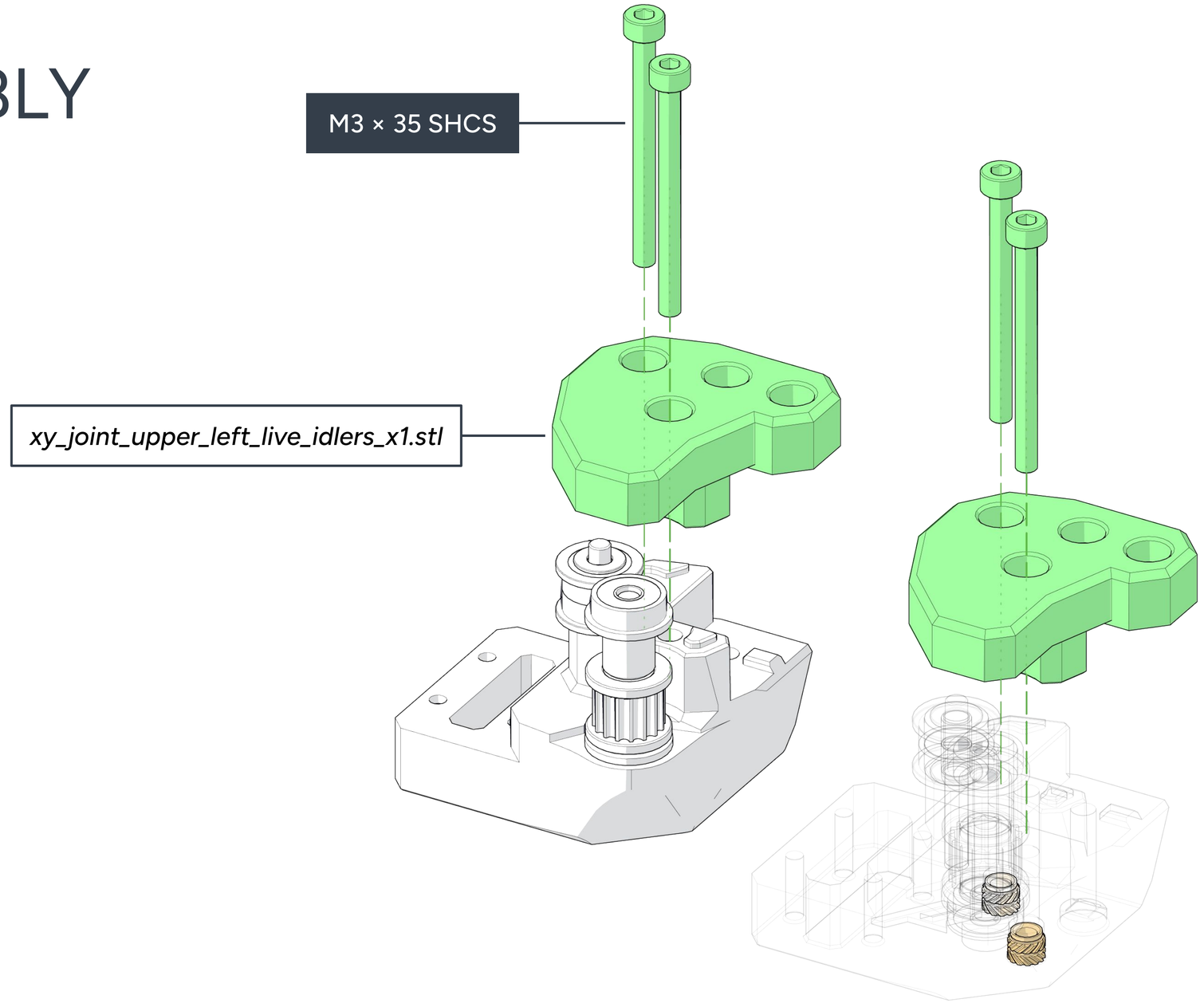


GANTRY ASSEMBLY

XY JOINT LEFT ASSEMBLY

Place ***xy_joint_upper_left_live_idlers_x1.stl*** and secure the two parts with **two M3 × 35 SHCS** into the M3 heatset inserts.

Ensure bearings and idlers spin freely.



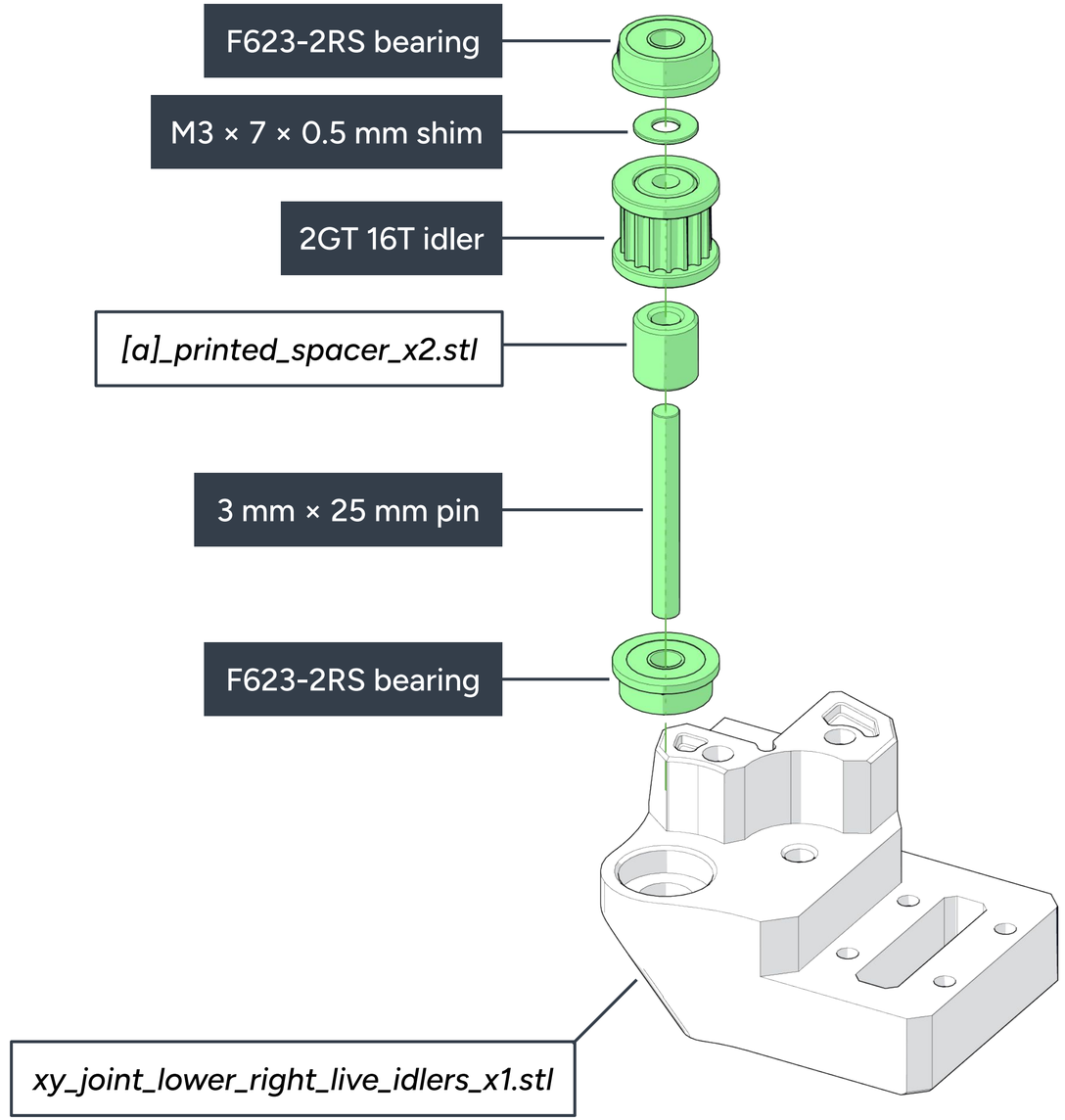
GANTRY ASSEMBLY

XY JOINT RIGHT ASSEMBLY

Press an **F623-2RS bearing (flange up)** into the recess in ***xy_joint_lower_right_live_idlers_x1.stl***. Then, assemble the following component stack on a **3 mm × 25 mm pin** and insert it into the bearing.

Live idler stack:

- **F623-2RS bearing (flange down)**
- ***M3 × 7 × 0.5 mm shim***
- ***2GT 16T idler***
- ***[a]_printed_spacer_x2.stl***
- F623-2RS bearing (flange up, already inserted)



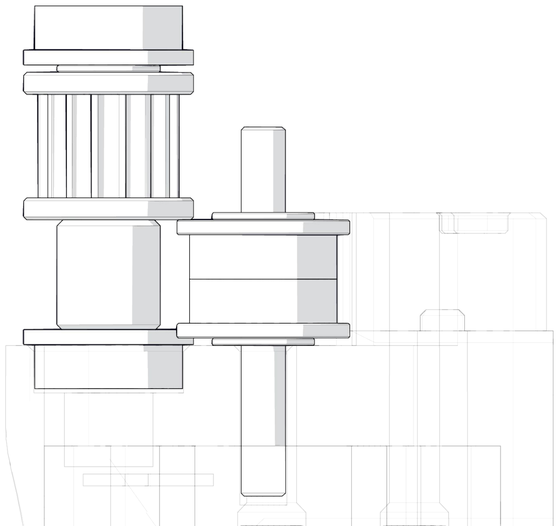
GANTRY ASSEMBLY

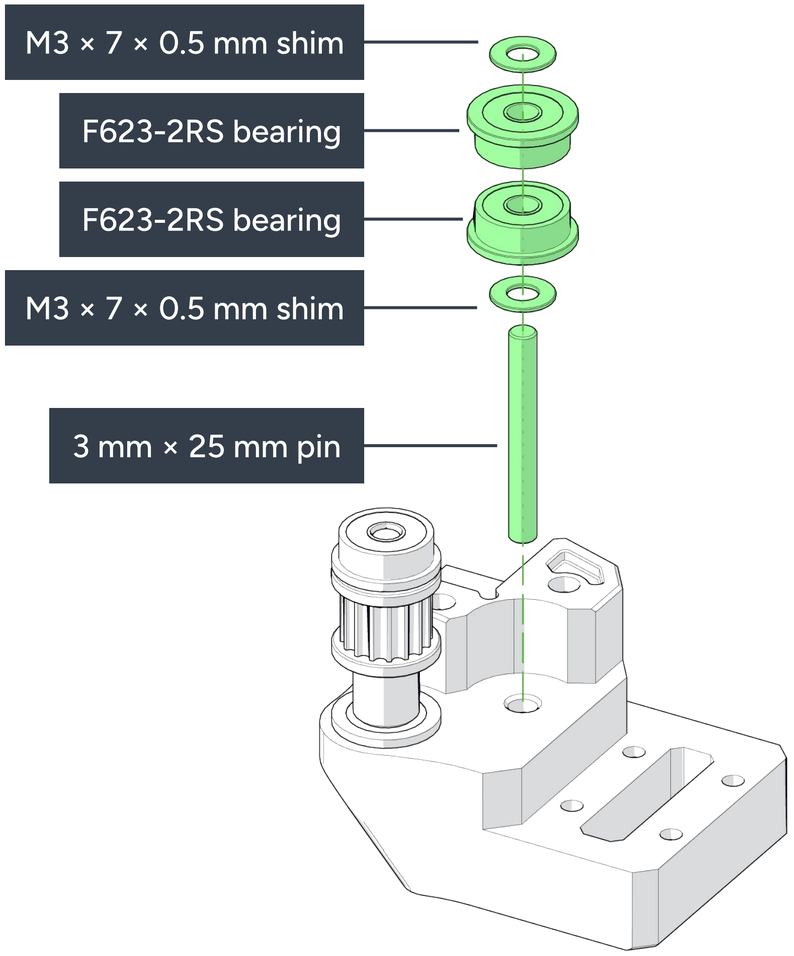
XY JOINT RIGHT ASSEMBLY

The second stack for the other belt is a traditional bearing stack on a **3 mm × 25 mm pin**.

- **M3 × 7 × 0.5 mm shim**
- **F623-2RS bearing (flange up)**
- **F623-2RS bearing (flange down)**
- **M3 × 7 × 0.5 mm shim**

Ensure your stacks match the diagram.



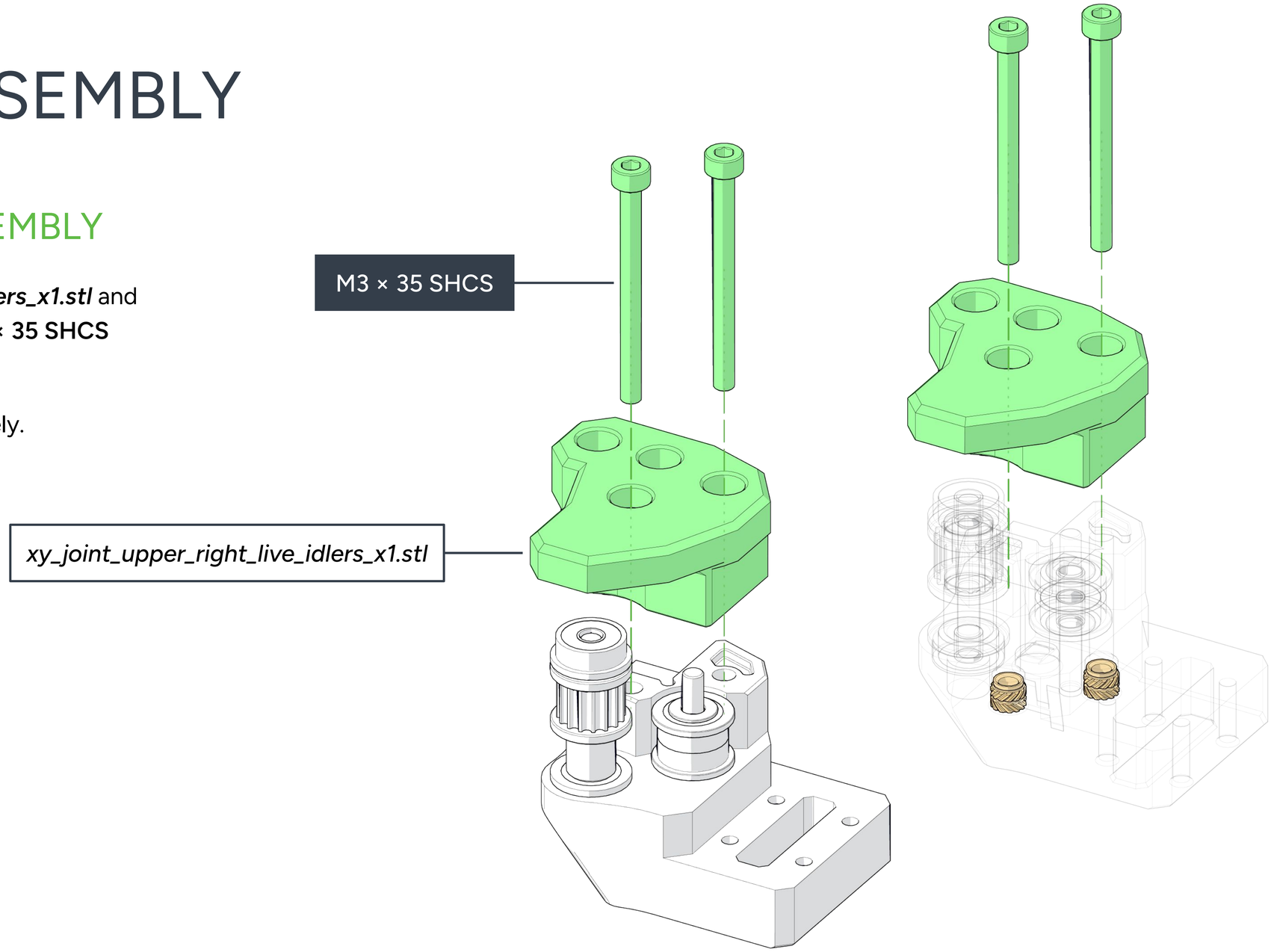


GANTRY ASSEMBLY

XY JOINT RIGHT ASSEMBLY

Place ***xy_joint_upper_right_live_idlers_x1.stl*** and secure the two parts with **two M3 × 35 SHCS** into the M3 heatset inserts.

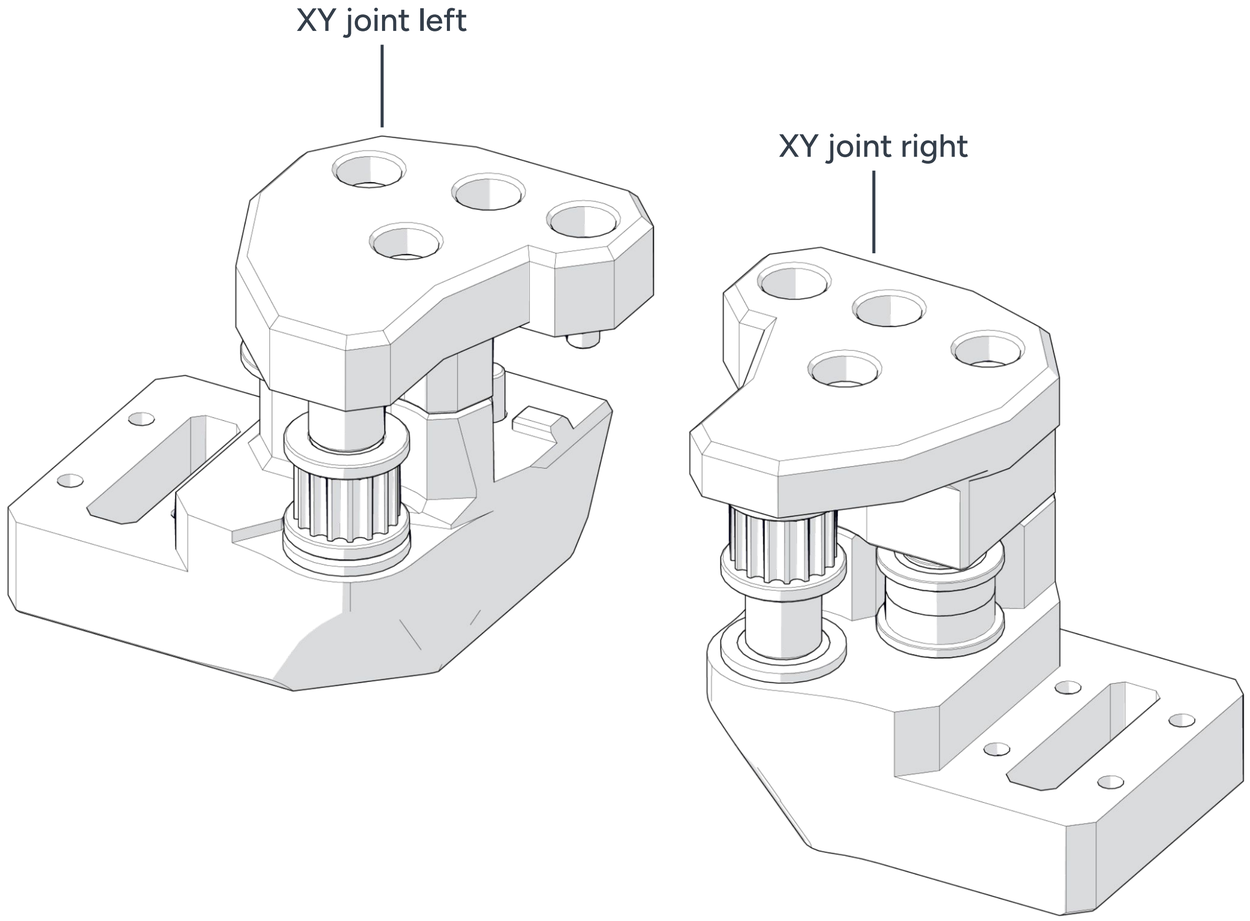
Ensure bearings and idlers spin freely.



GANTRY ASSEMBLY

XY JOINTS ASSEMBLIES

You should now have assembled XY left and XY right joints. Ensure all bearings and idlers spin freely before proceeding.



GANTRY ASSEMBLY

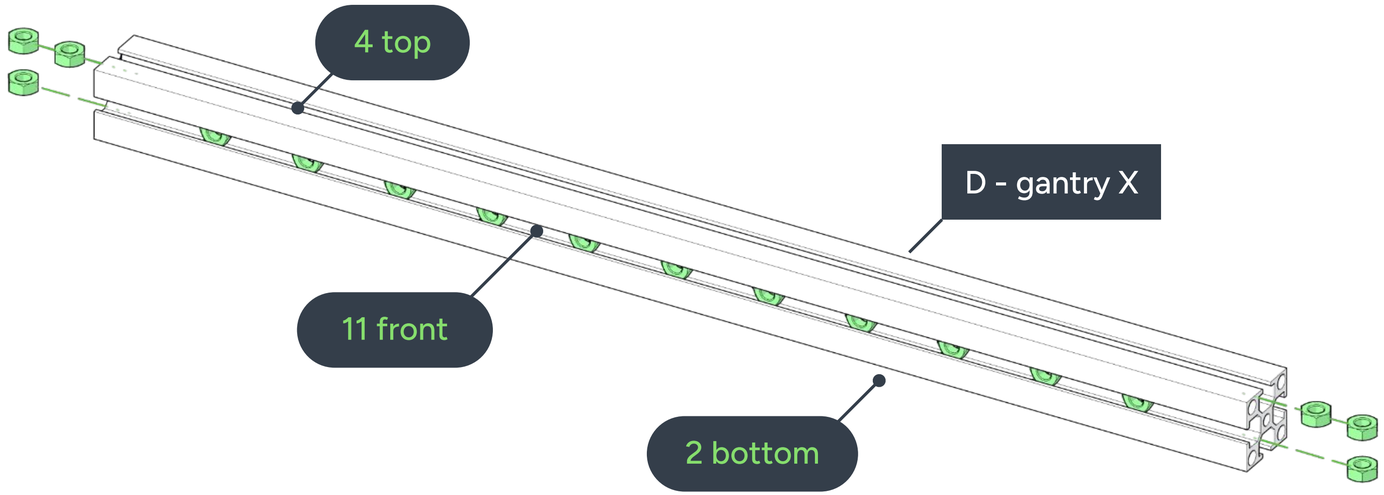
PREPPING GANTRY X EXTRUSIONS

Preloaded nuts for the **D gantry X extrusion**. Unlike the previous linear rails, the X rail does not use printed nut adapters, only preloaded M3 hex nuts. Printed “no-drop” nut holders can hold them in place with friction.

Front slot: **11 M3 hex nuts**

Top slot: **4 M3 hex nuts**

Bottom slot: **2 M3 hex nuts**



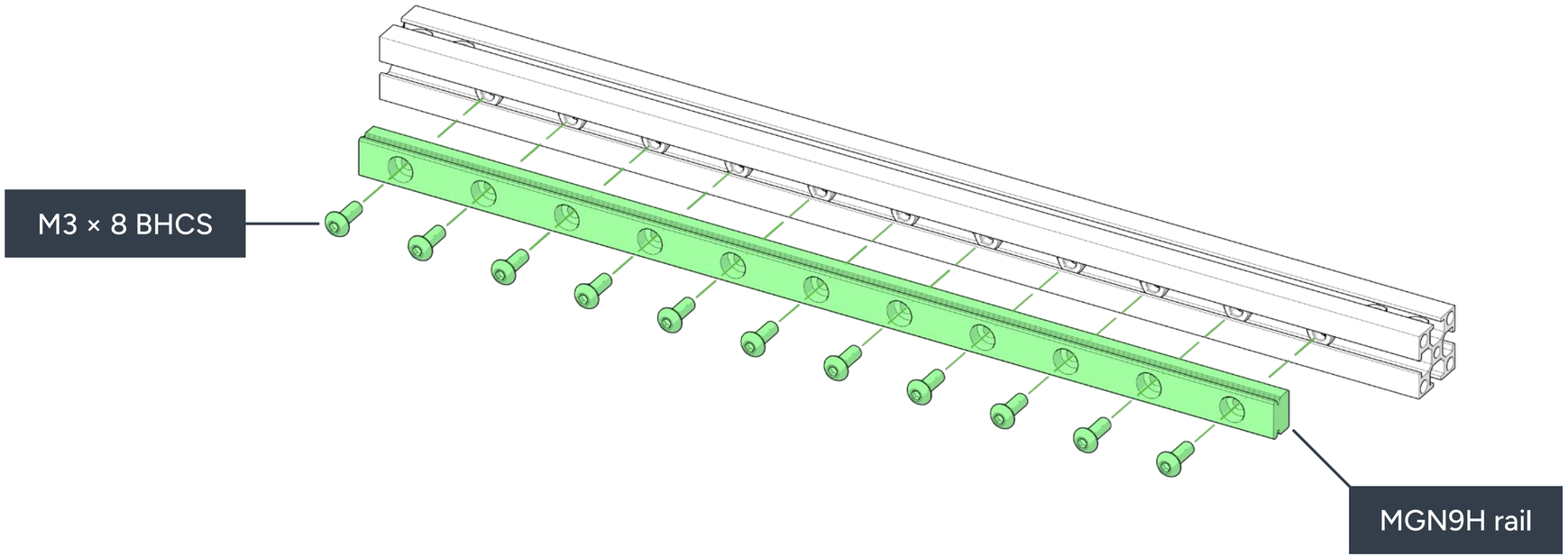
NOTE

Don’t forget the preloaded nuts.

GANTRY ASSEMBLY

PREPPING GANTRY X EXTRUSIONS

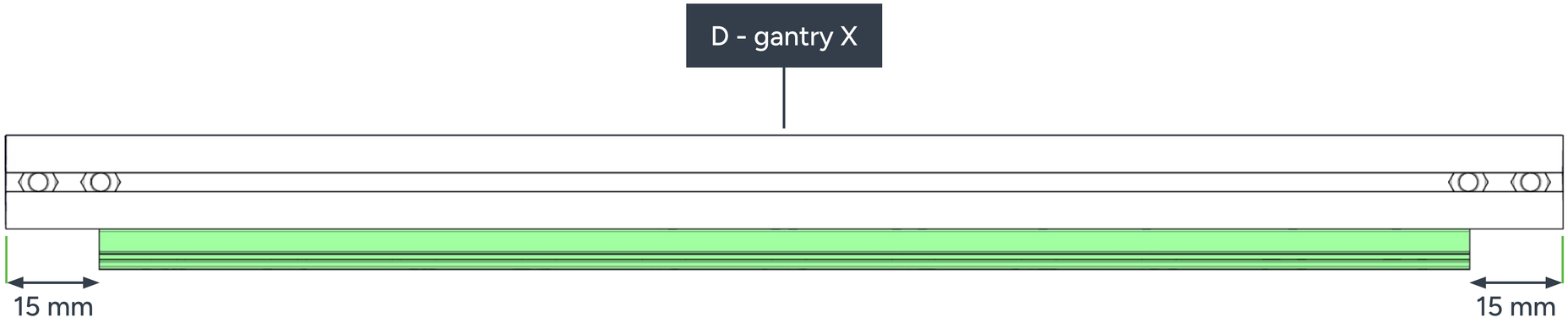
Line up the **MGN9H rail** over the nuts and lightly secure it with **11 M3 × 8 BHCS**. Do not tighten them down yet.



GANTRY ASSEMBLY

PREPPING GANTRY EXTRUSIONS

Position the rail 15 mm from the ends of the extrusion, then tighten down the 11 M3 × 8 BHCS.



GANTRY ASSEMBLY

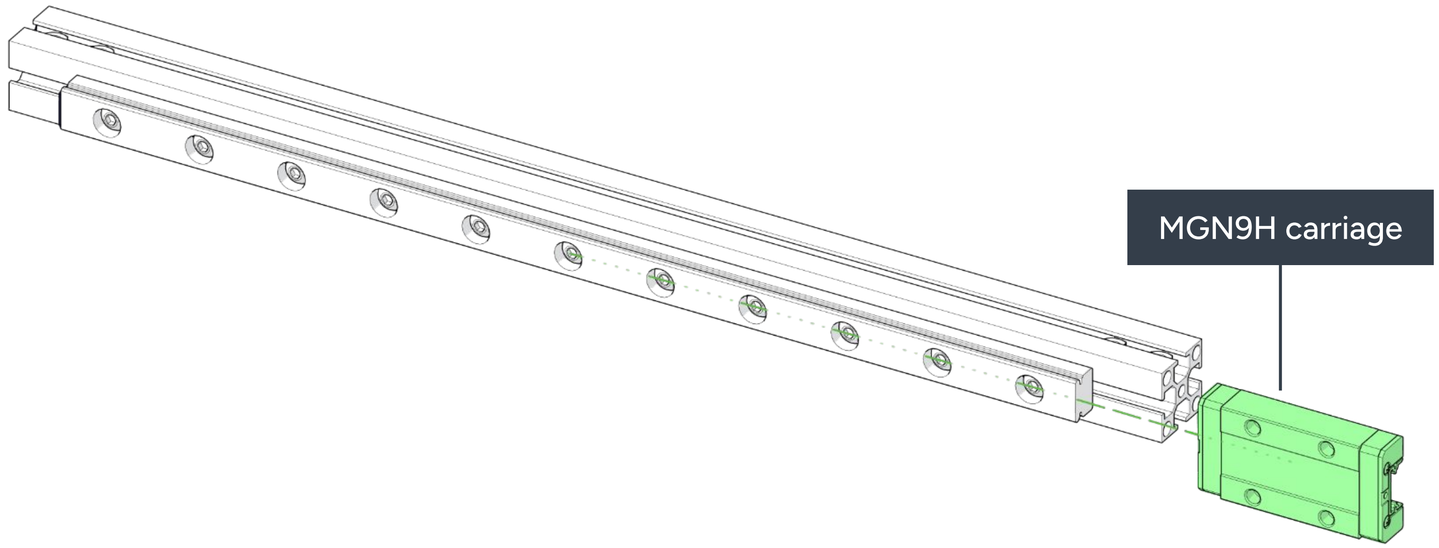
PREPPING GANTRY X EXTRUSIONS

Apply grease to the **MGN9H carriage** bearings.

Carefully reinstall the carriage by sliding it directly onto the rail. Keep the carriage aligned during installation to prevent ball loss.

Work the grease into the internal raceways by moving the carriage back and forth several times. Printable installation aids can assist with installation.

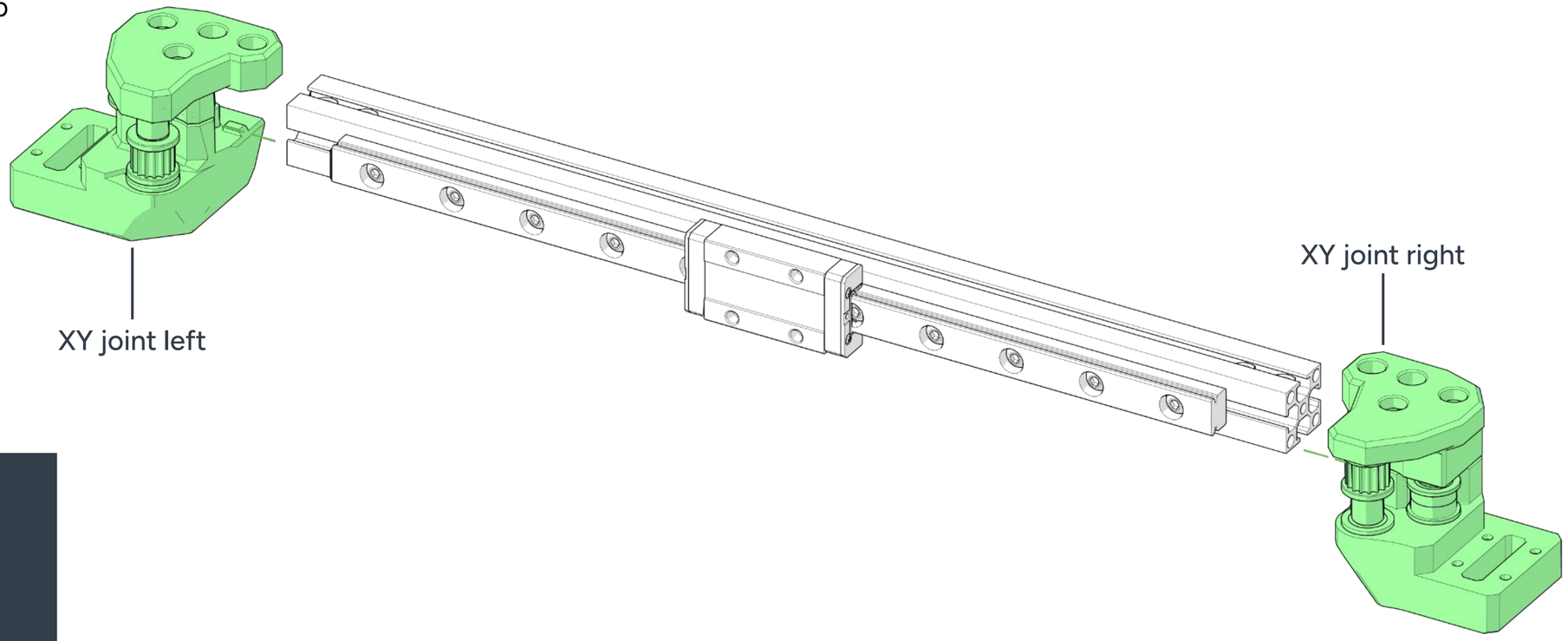
Once installed, do not allow the carriage to slide off the rail.



GANTRY ASSEMBLY

X-AXIS ASSEMBLY

Install the XY joint assemblies and line up the mounting holes with the preloaded nuts (four on top and two on bottom).



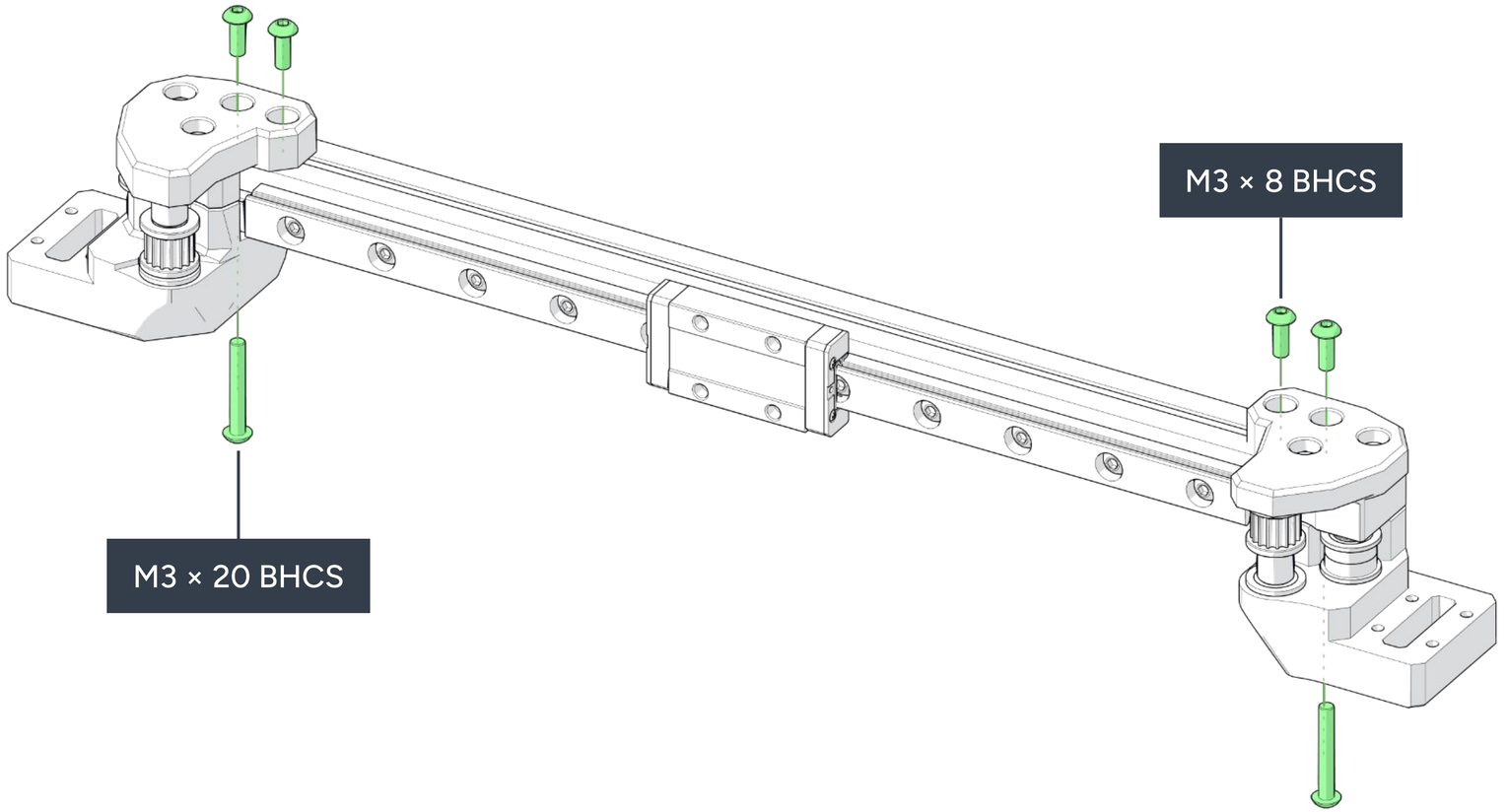
NOTE

Don’t forget the preloaded nuts.

GANTRY ASSEMBLY

X-AXIS ASSEMBLY

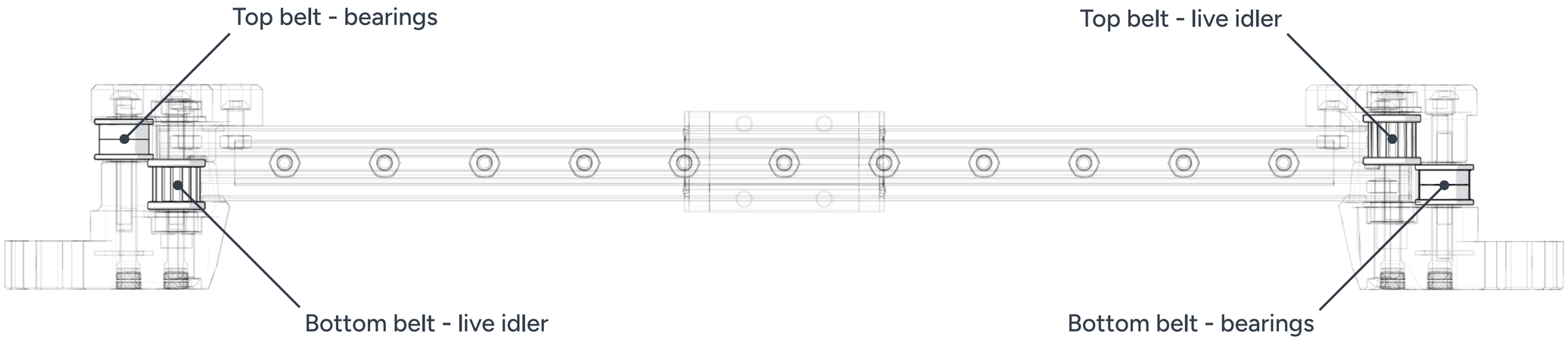
Secure the XY joint assemblies with **four M3 × 8 BHCS** on the top and **two M3 × 20 BHCS** on the bottom. Do not fully tighten just yet. Final tightening will be done when gantry is installed and squared.



GANTRY ASSEMBLY

X-AXIS ASSEMBLY

Ensure that the bearings and idlers match this diagram.



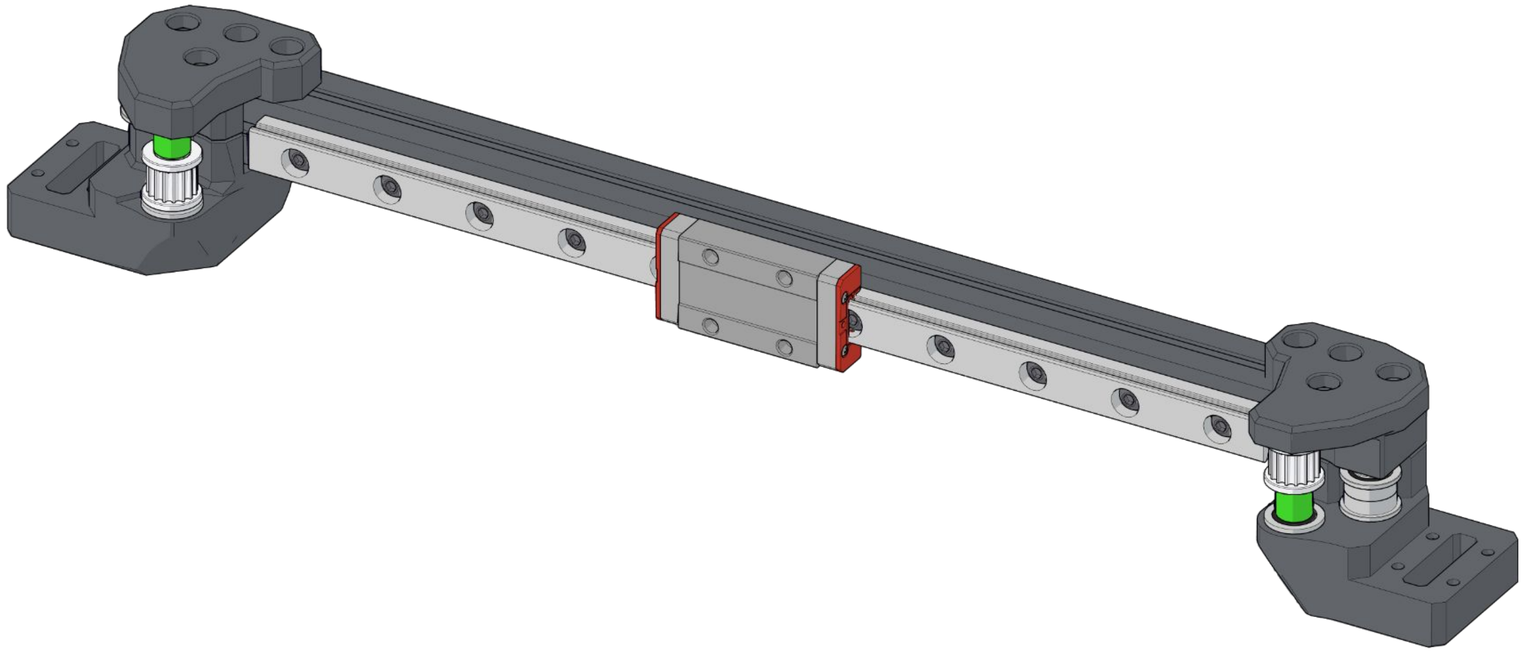
GANTRY ASSEMBLY

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

- ▢ Left live idler on bottom track
- ▢ Right live idler on top track
- ▢ Live idlers spin freely
- ▢ Bearings spin freely
- ▢ MGN9H carriage is greased

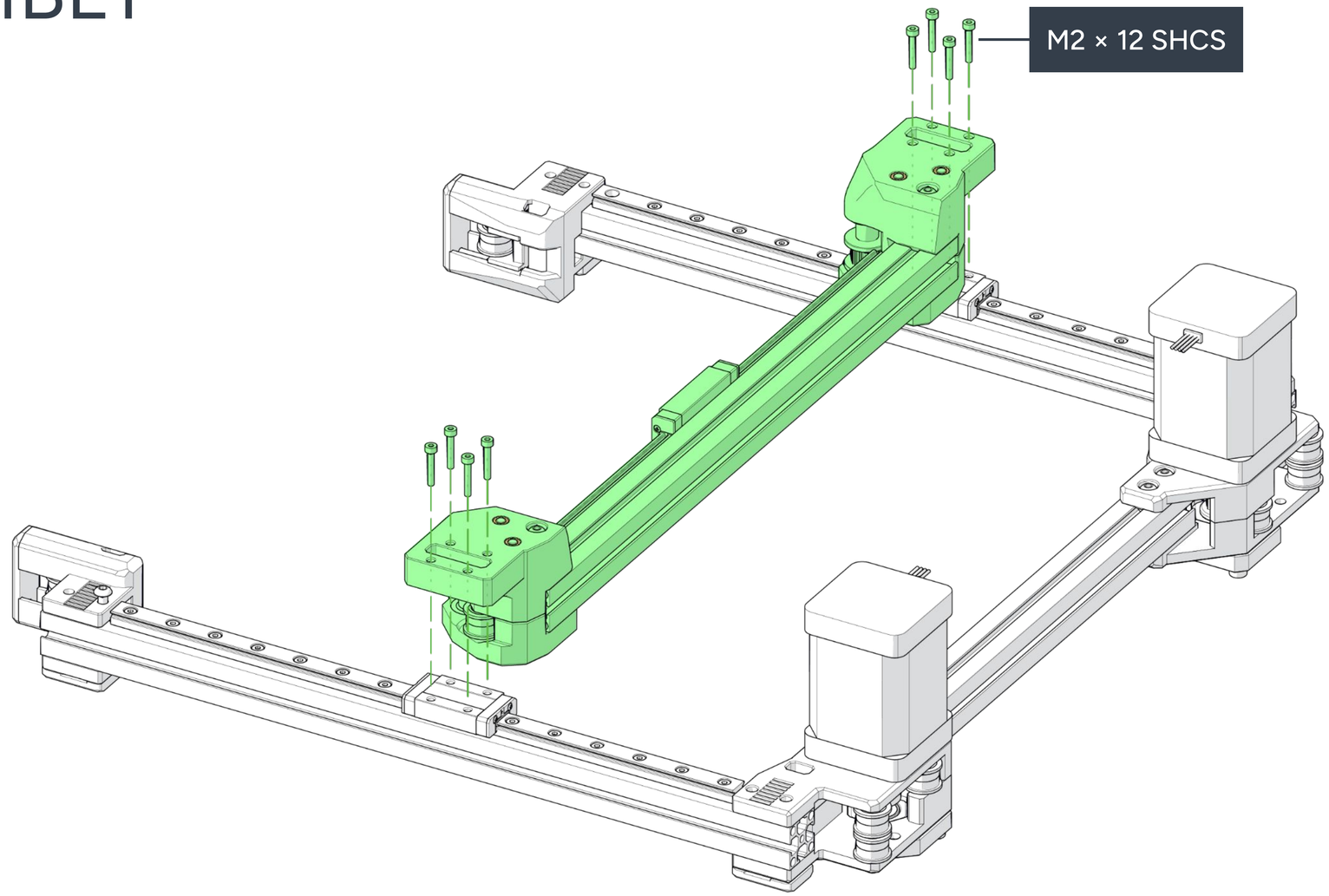


GANTRY ASSEMBLY

X-AXIS ASSEMBLY

Flip the entire gantry assembly upside down for easy access to the Y carriage screws.

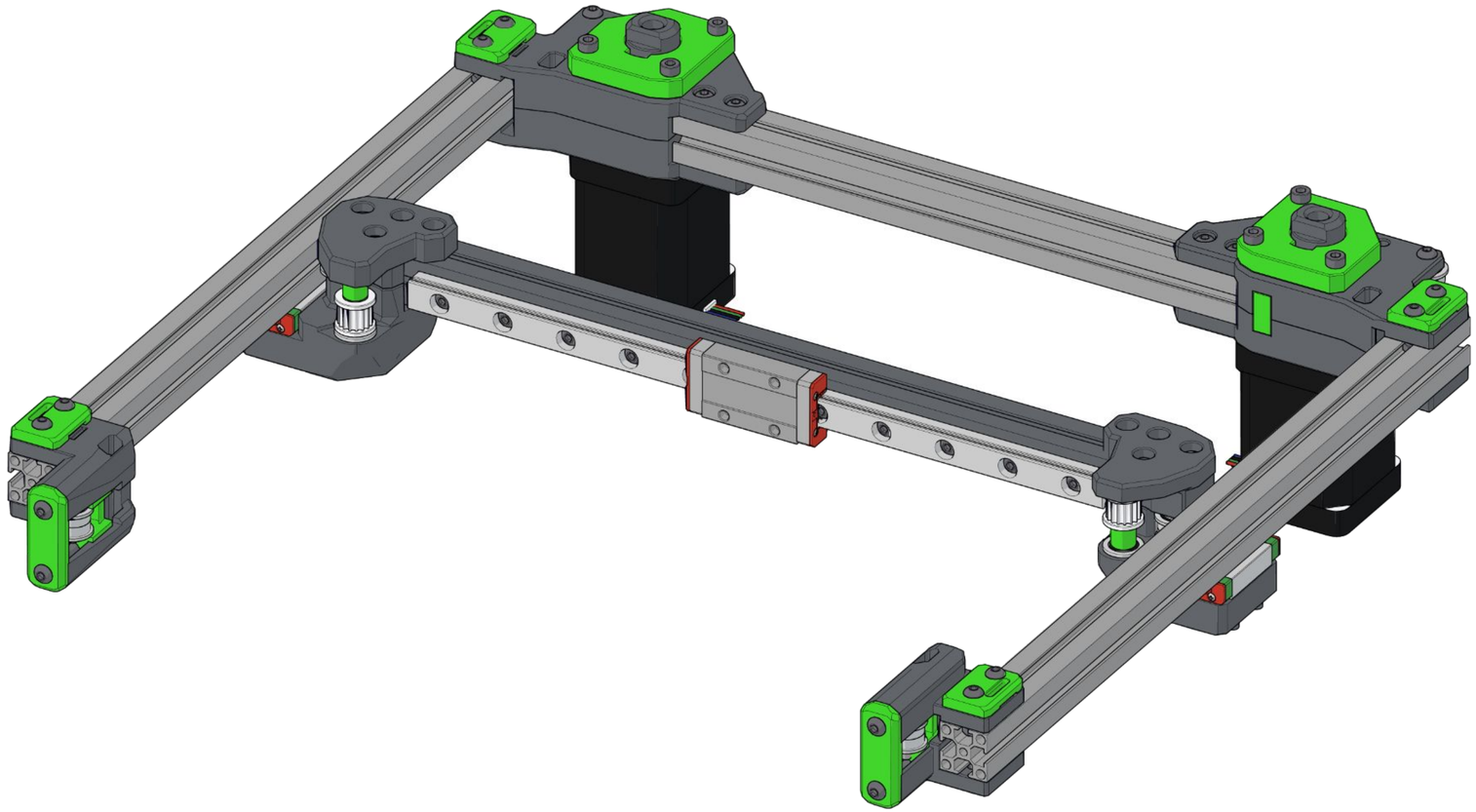
Place the X axis assembly on the Y carriages, lining up the XY joints' holes with the carriages' holes. Secure the X axis to the carriages with **8 M2 × 12 SHCS**.



GANTRY ASSEMBLY

BUILDER CHECK

Your assembly should match this view.



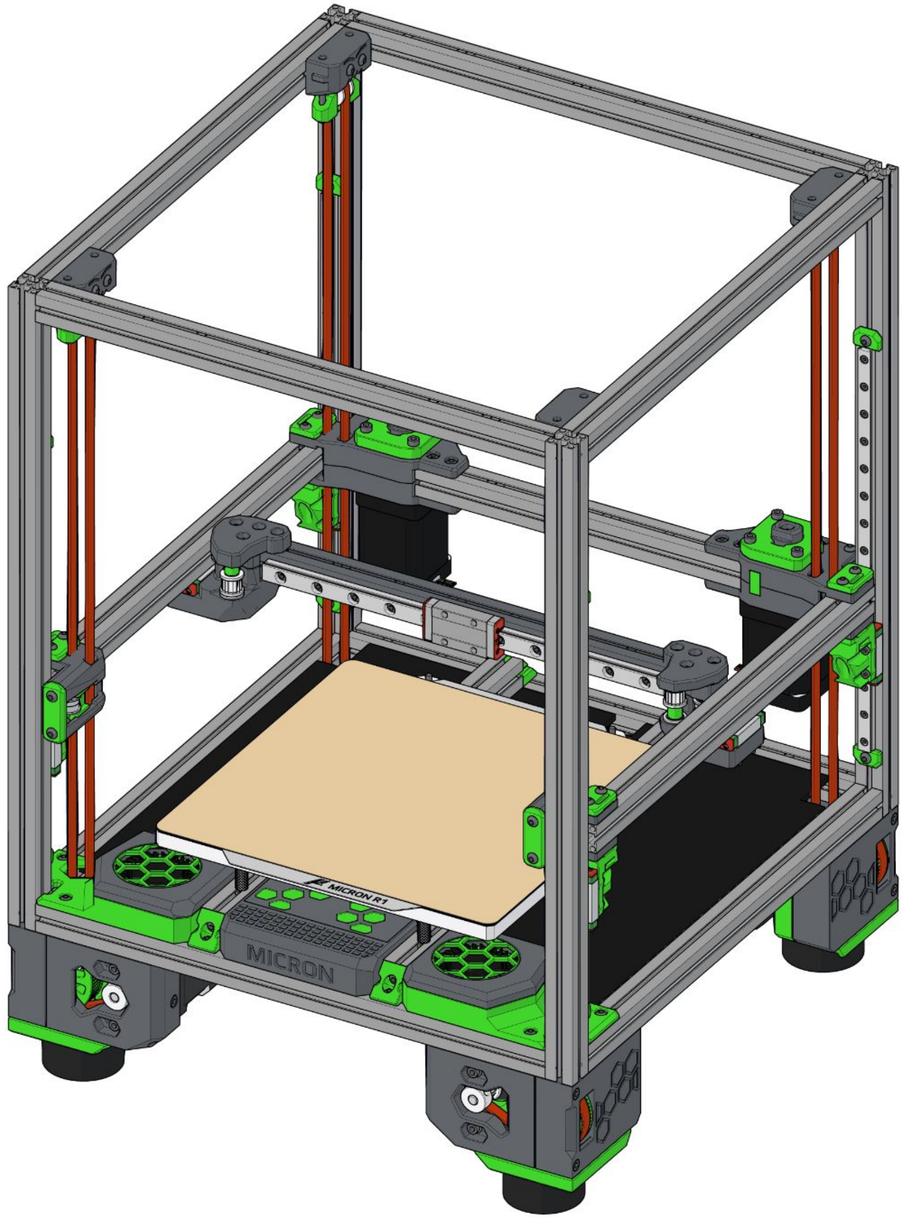
GANTRY ASSEMBLY

FLYING GANTRY INSTALL

The flying gantry is suspended from four independent Z drives, enabling vertical motion and automatic gantry tramming.

HARDWARE

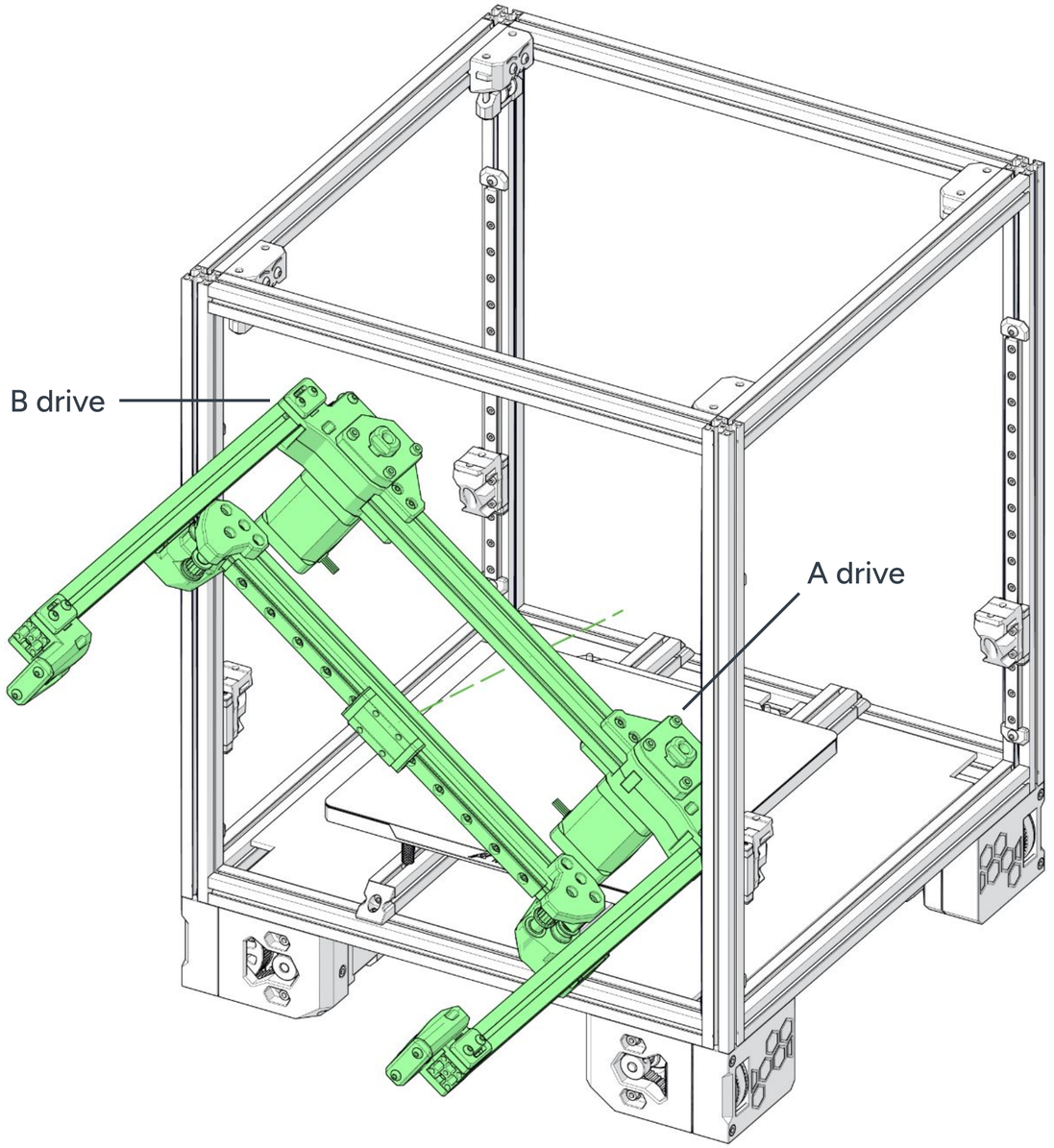
4	2GT belts, 6x152mm
4	Zip ties
8	M3 × 8 BHCS
8	M3 × 20 BHCS
4	M5 × 16 BHCS
4	M5 hex nut
4	Rubber feet



GANTRY ASSEMBLY

FLYING GANTRY INSTALL

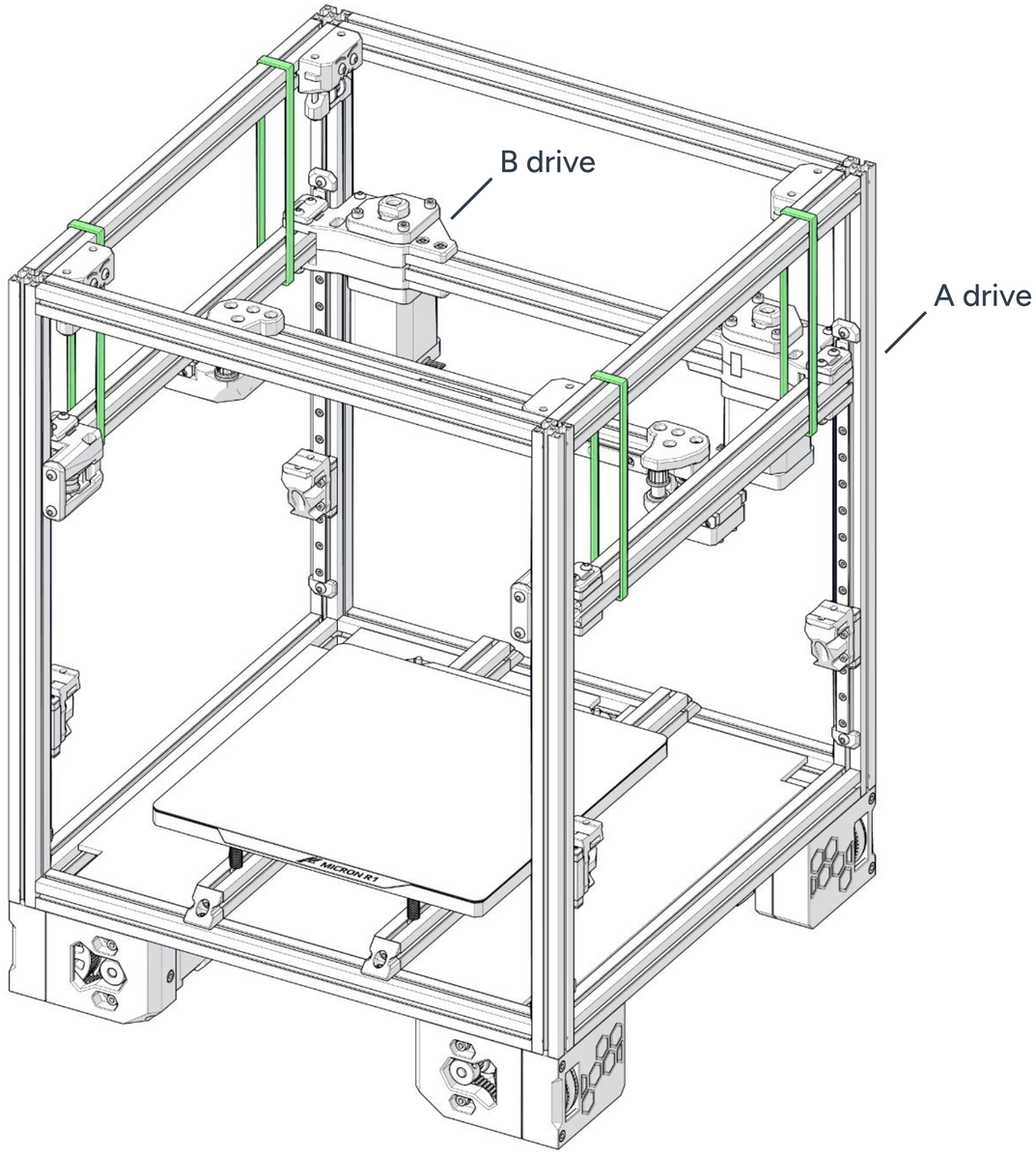
Tilt the gantry assembly and carefully slide it into the frame from the front. Verify the X gantry and A/B drives are properly oriented.



GANTRY ASSEMBLY

FLYING GANTRY INSTALL

Support all four gantry corners with **zip ties** or printed aids at roughly equal height. This stabilizes the assembly for Z joint installation.

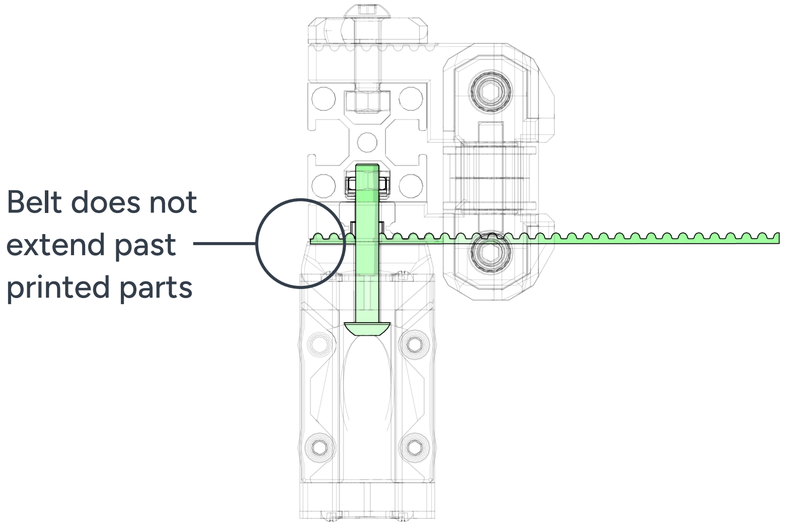


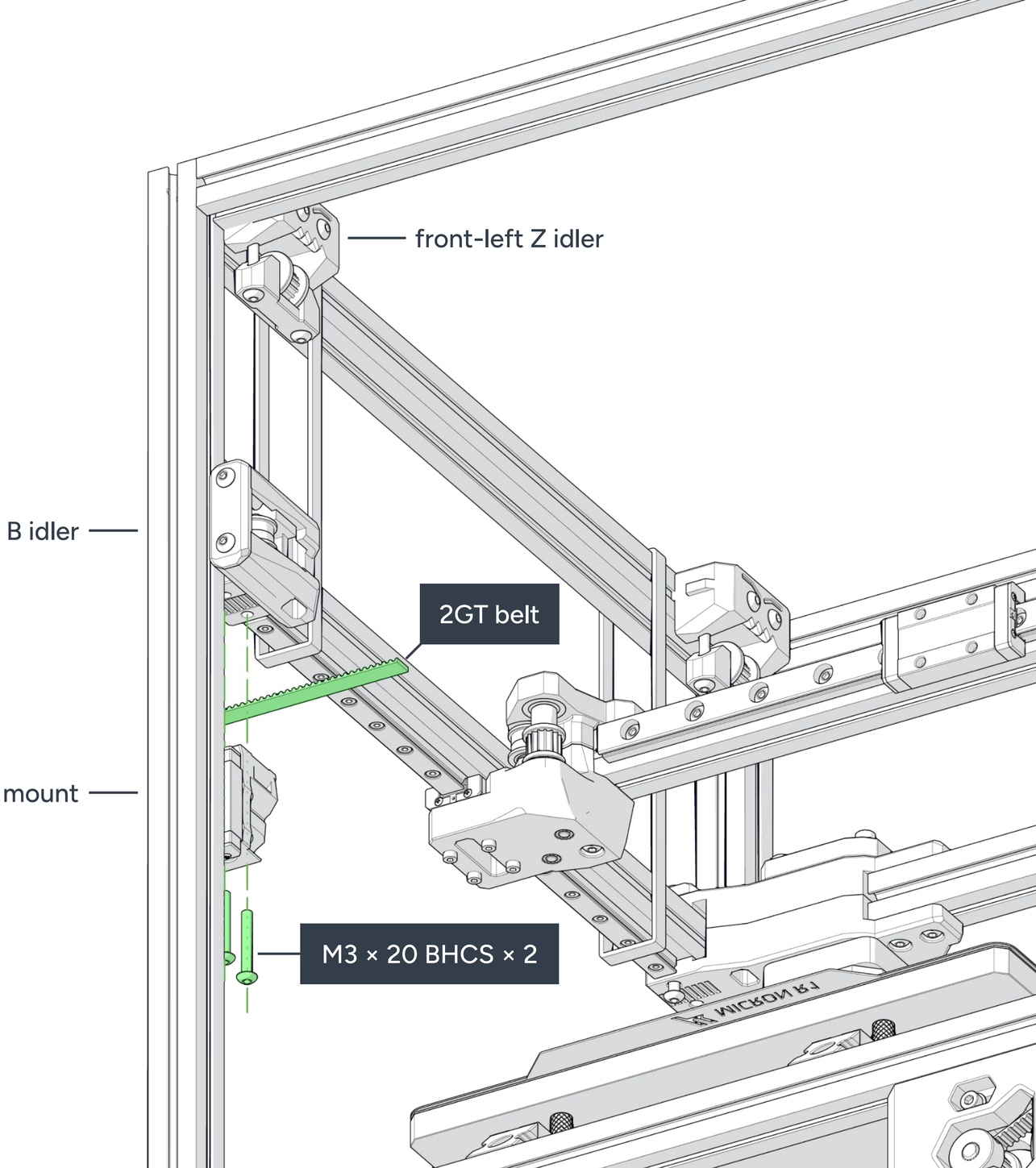
GANTRY ASSEMBLY

FLYING GANTRY INSTALL

Insert the end of a **2GT belt** into the B idler's lower clamp grooves with belt teeth facing upward. Slide the rigid mount up to clamp the belt and secure with **two M3 × 20 BHCS** to preloaded M3 nuts. Ensure the remaining belt length feeds inward toward the chamber for final Z routing.

Repeat for all four corners.





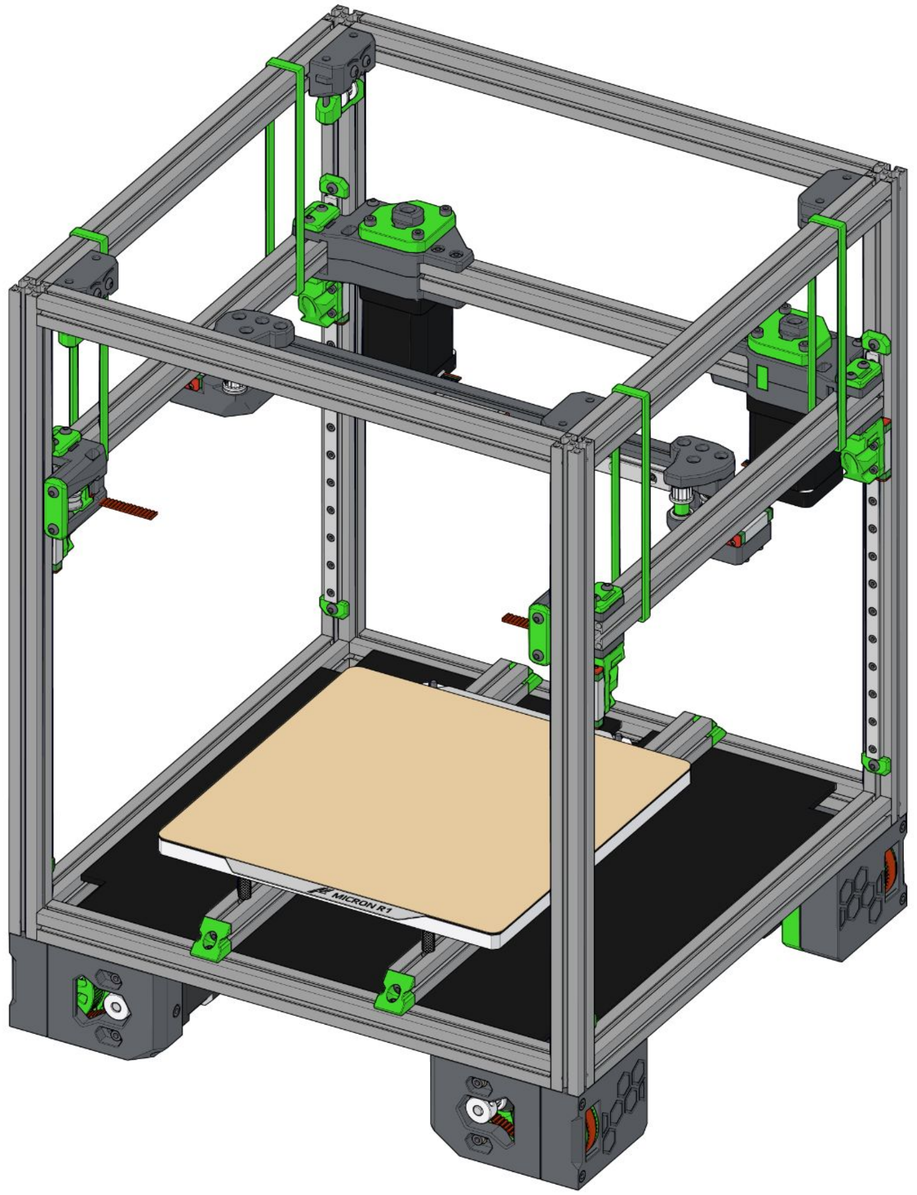
GANTRY ASSEMBLY

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

- ▢ Rigid mounts are securely clamping one end of the belts with teeth facing upward
- ▢ Ends of 2GT belts do not extend past printed parts
- ▢ All 2GT belt lengths feed inward toward the chamber

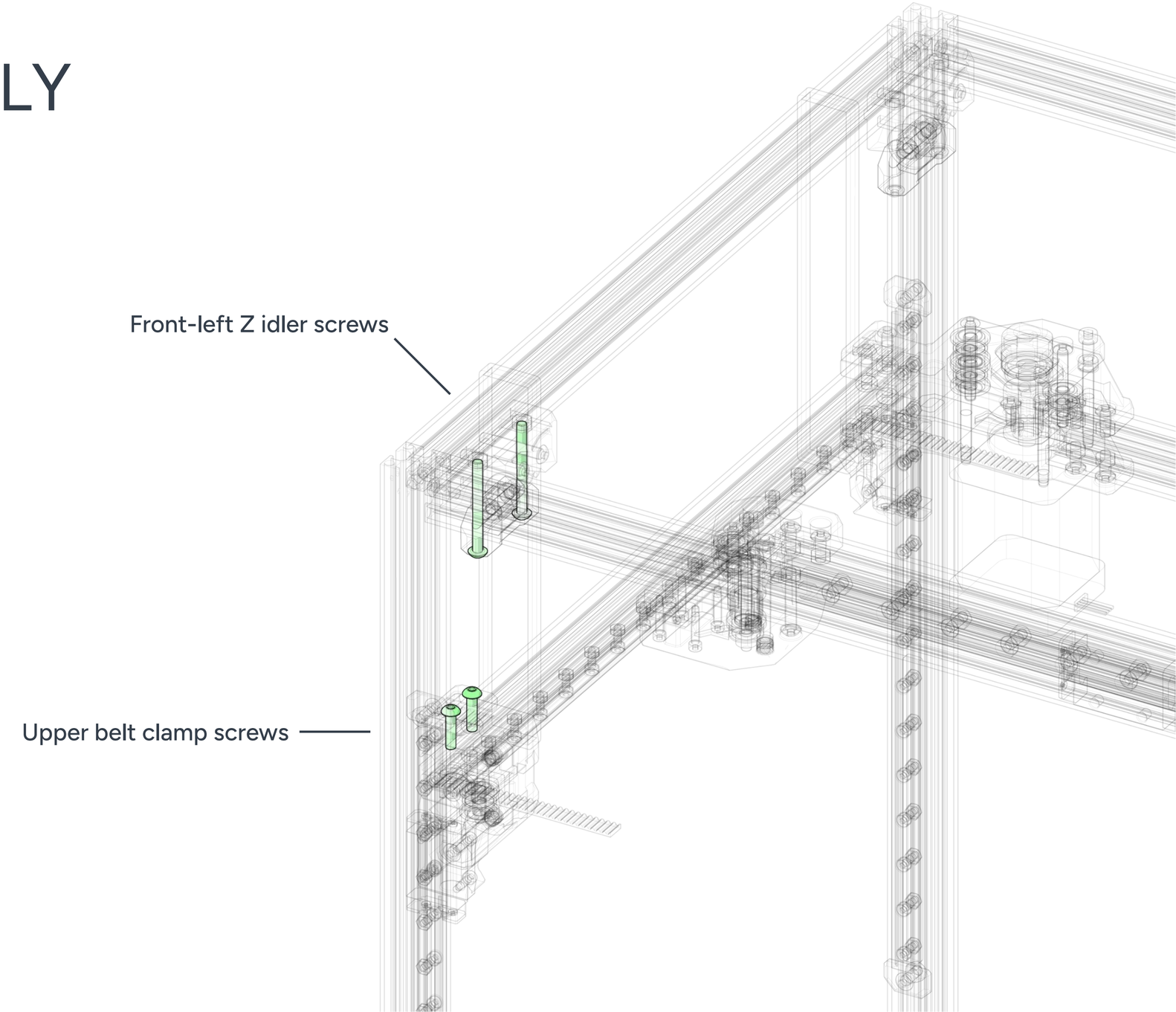


GANTRY ASSEMBLY

ROUTING THE Z BELTS

Loosen the Z idler screws to create clearance for belt routing and later tensioning. Do not fully remove the screws.

Loosen the upper belt clamp, which will secure the belt's final routed position.



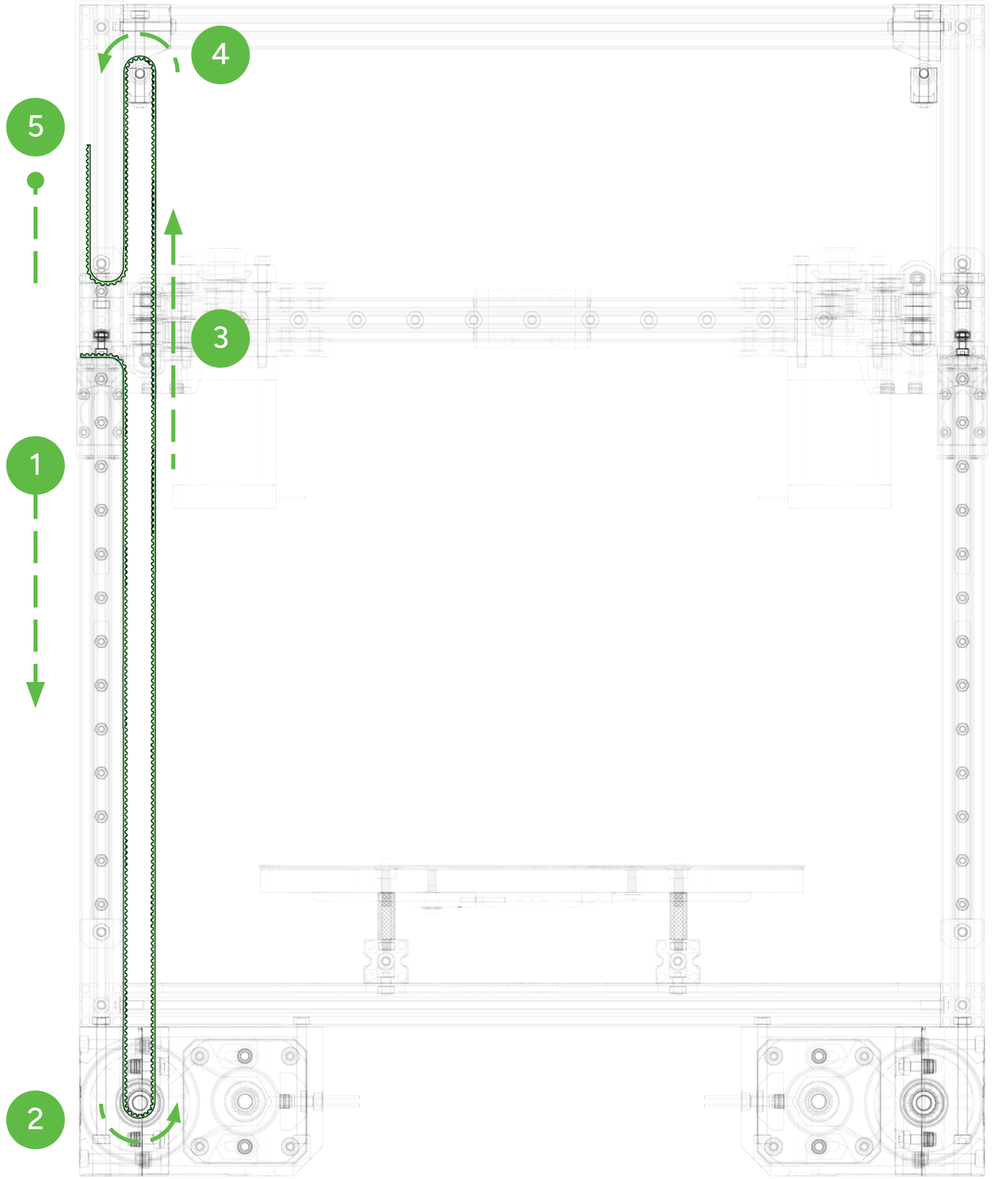
GANTRY ASSEMBLY

ROUTING THE Z BELTS

Follow the path in the diagram to route the belt.

1. Starting at the lower belt clamp, route the belt downward toward the Z drive.
2. Feed the belt around the Z drive's 2GT 16T pulley. The belt teeth should be engaging with the pulley teeth.
3. Feed the belt upward through the front idler housing slot and continue towards the upper Z idler.
4. Feed the belt around the upper Z idler. The belt teeth should be engaging with the idler teeth.
5. Insert the belt through the upper belt clamp, feed the excess through the top slot and pull to remove the slack. Verify tooth engagement, and secure the clamp.

Repeat for all four corners.

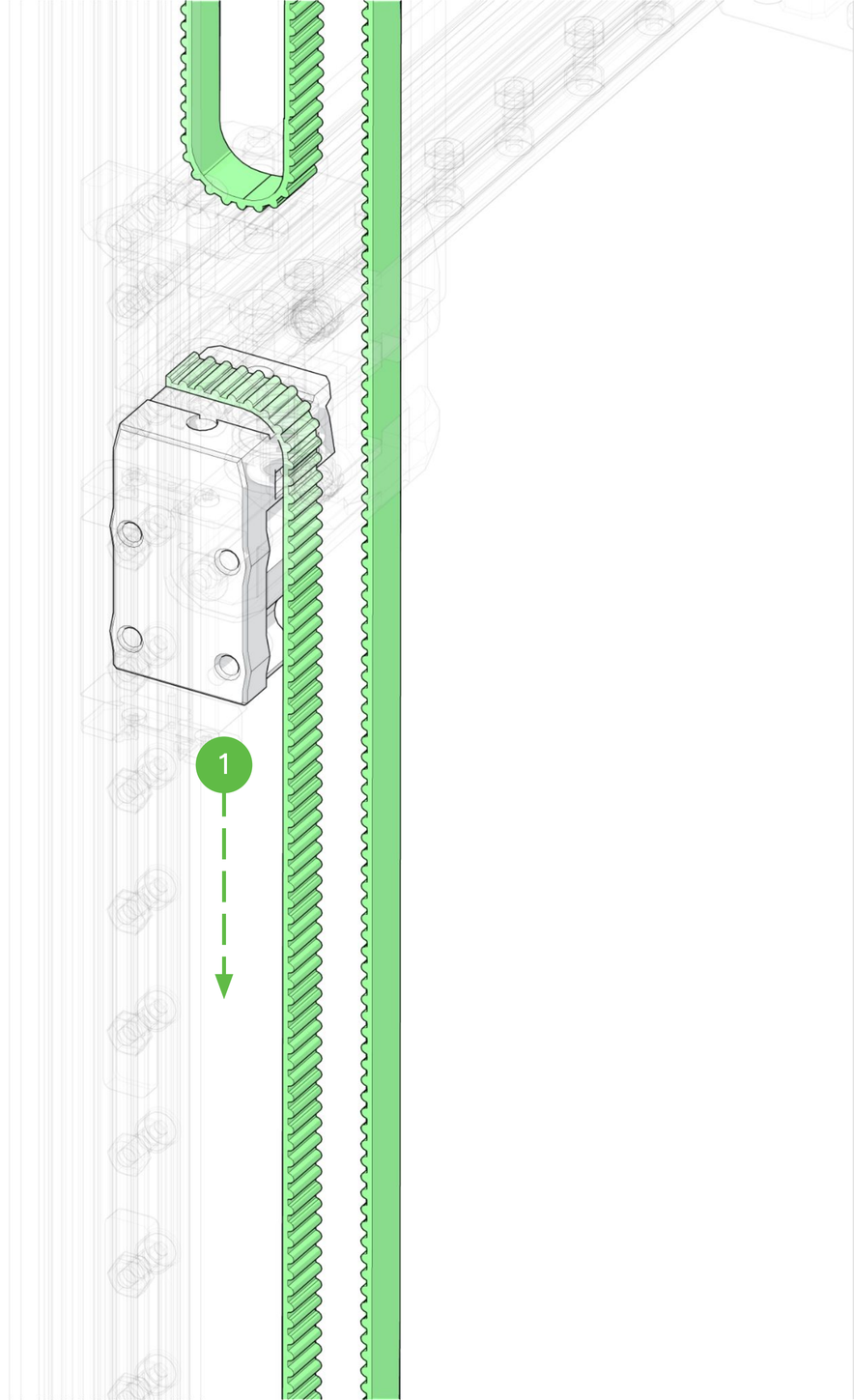


GANTRY ASSEMBLY

ROUTING THE Z BELTS (1 of 5)

Follow the path in the diagram to route the belt.

1. Starting at the lower belt clamp, route the belt downward toward the Z drive.

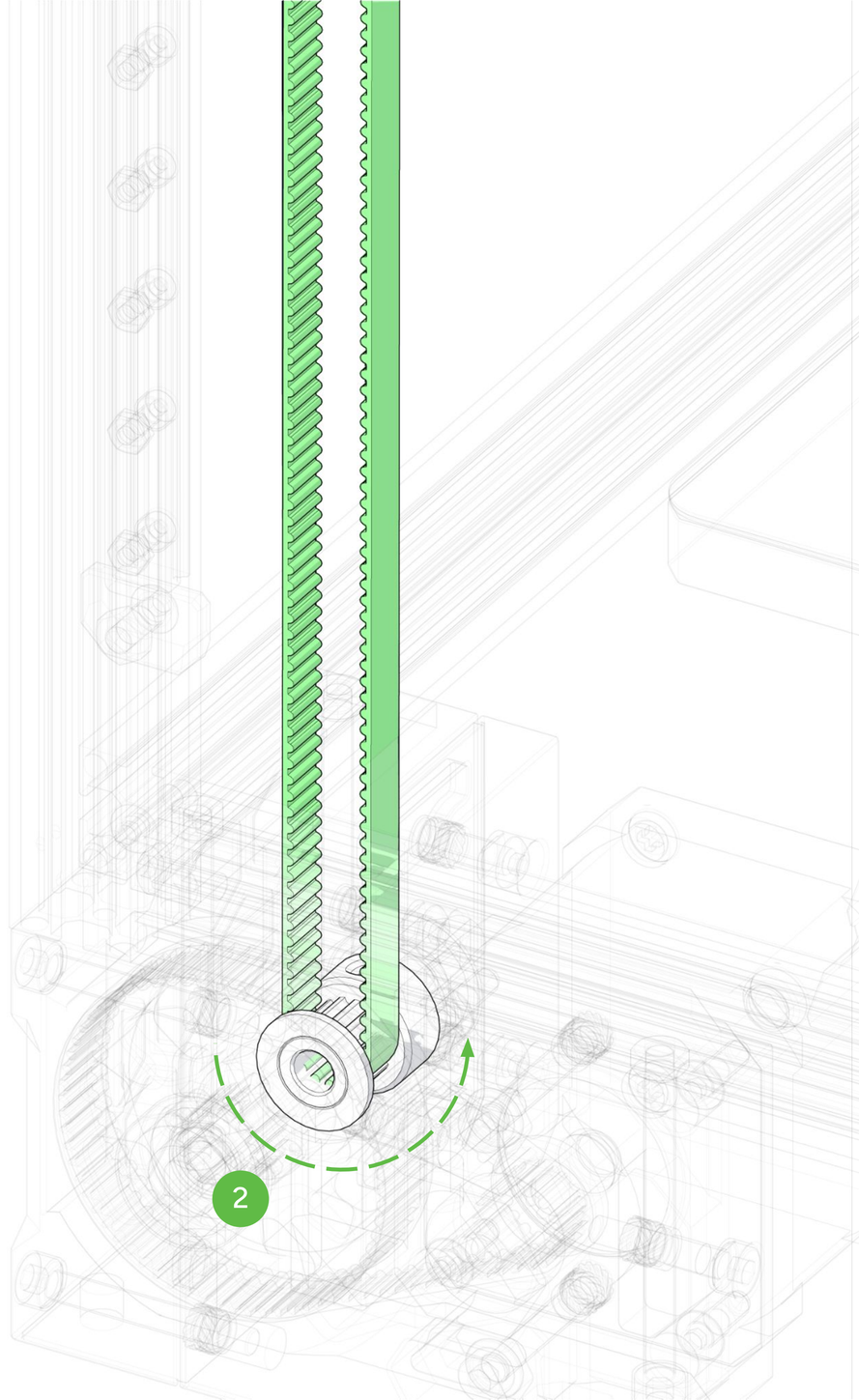


I / GANTRY ASSEMBLY

ROUTING THE Z BELTS (2 of 5)

Follow the path in the diagram to route the belt.

2. Feed the belt around the Z drive's 2GT 16T pulley.
   The belt teeth should be engaging with the pulley teeth.



NOTE

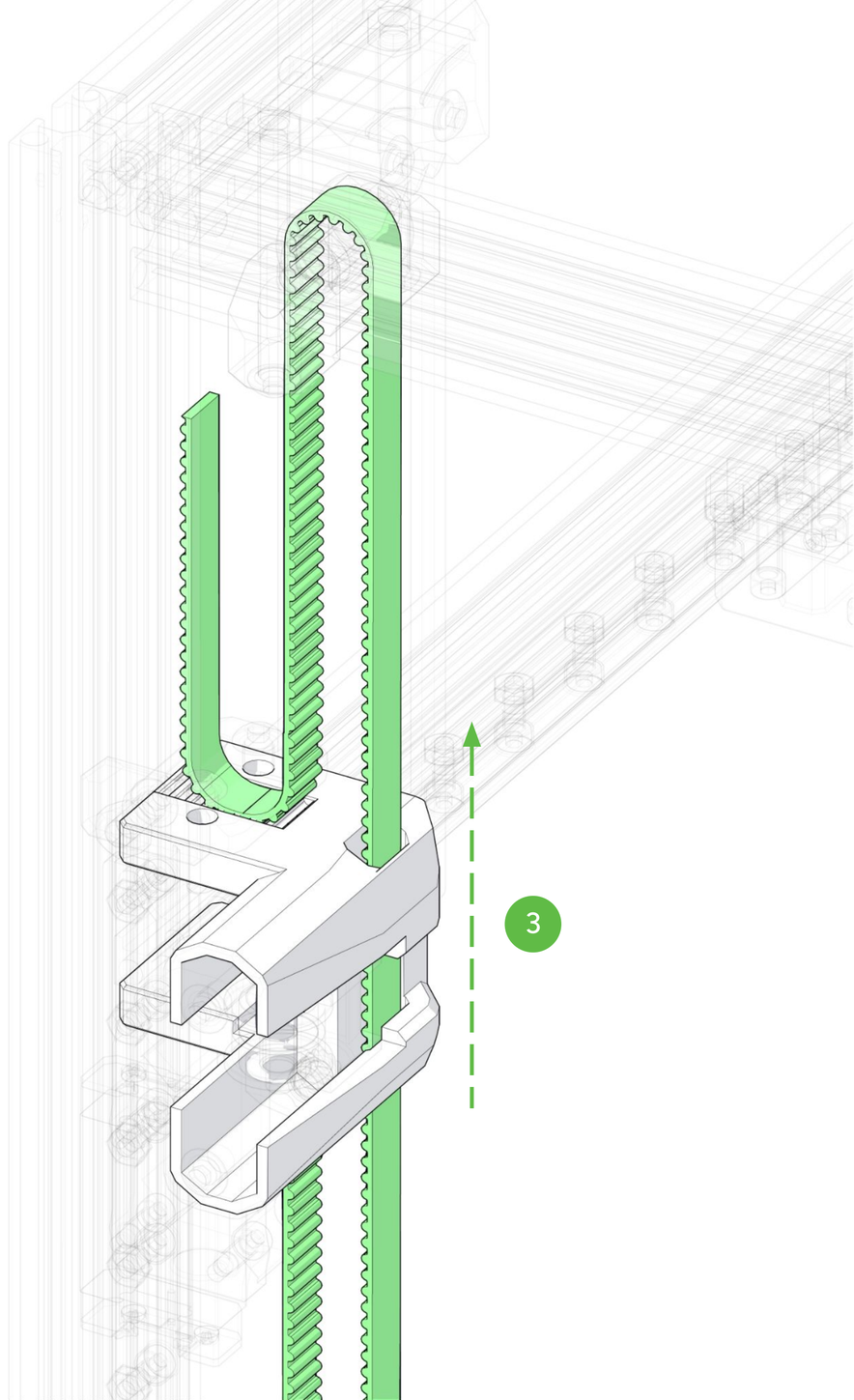
The open bottom frame provides access for tweezers, pliers, or an Allen wrench to help guide the belt around the pulley.

GANTRY ASSEMBLY

ROUTING THE Z BELTS (3 of 5)

Follow the path in the diagram to route the belt.

3. Feed the belt upward through the front idler housing slot and continue towards the upper Z idler.

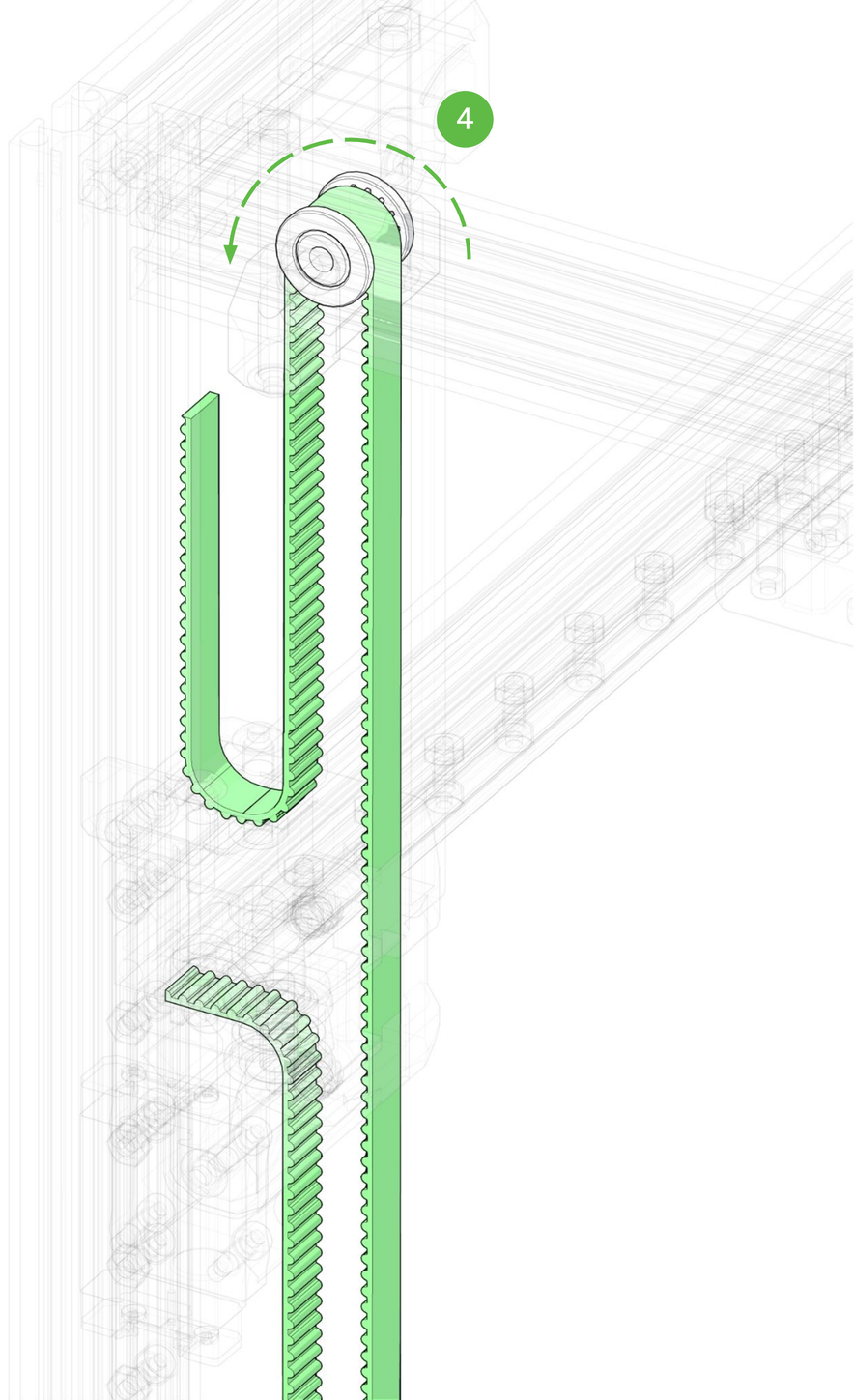


GANTRY ASSEMBLY

ROUTING THE Z BELTS (4 of 5)

Follow the path in the diagram to route the belt.

4. Feed the belt around the upper Z idler. The belt teeth should be engaging with the idler teeth.



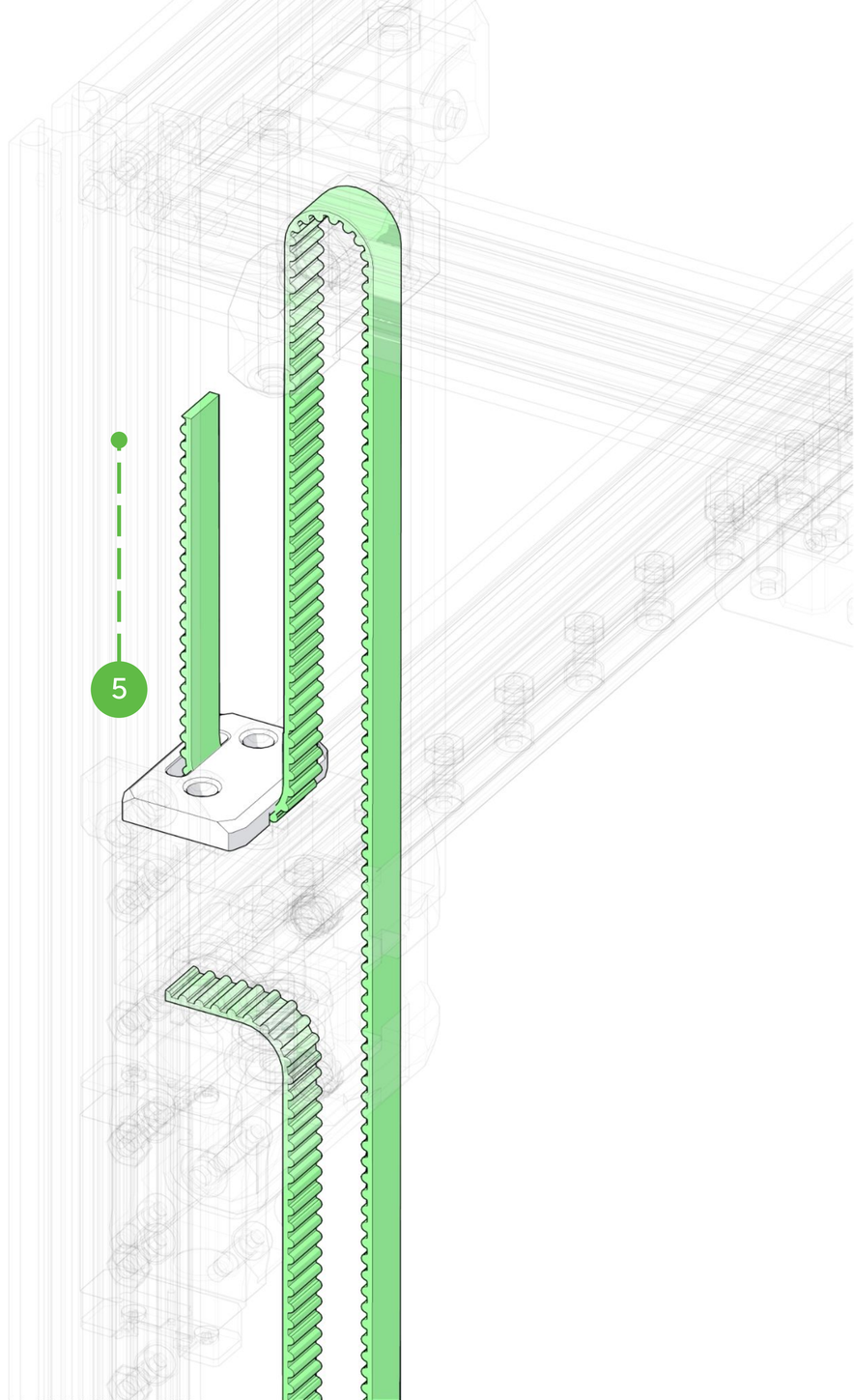
GANTRY ASSEMBLY

ROUTING THE Z BELTS (5 of 5)

Follow the path in the diagram to route the belt.

5. Insert the belt through the upper belt clamp, feed the excess through the top slot and pull to remove the slack. Verify tooth engagement, and secure the clamp.

Repeat for all four corners.



CONCEPT

Z BELT & TENSIONING

Each Z belt supports one corner of the gantry. Balanced tension across all four Z belts is required for level motion and consistent first layers.

Z belts do not need to be extremely tight, but they must be evenly tensioned.

Uneven tension can cause:

- Gantry tilt during movement
- Inconsistent Z height across the bed
- Increased wear on belts and bearings
- Binding or rough motion

Target state:

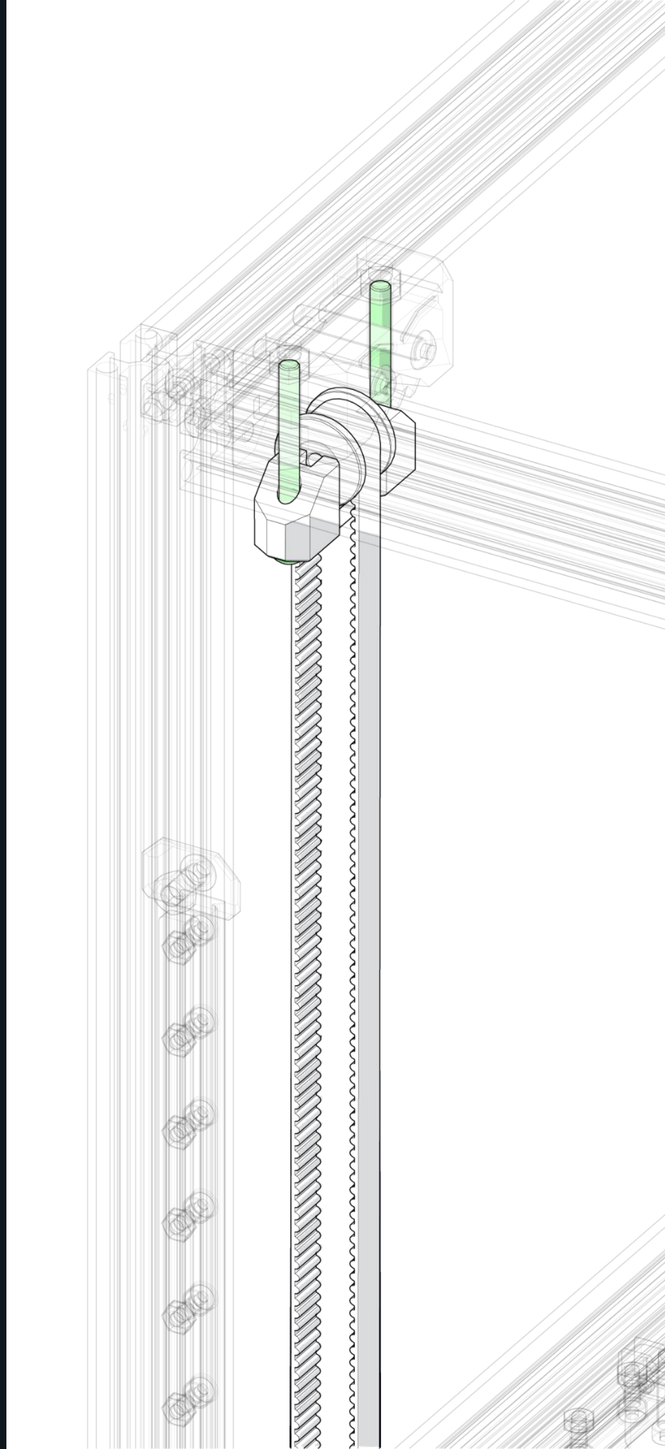
- Belts should feel even across all four corners
- When pressed lightly, each belt should deflect slightly with similar resistance

What to look for:

- Gantry remains level when moving up and down
- No corner leads or lags during Z movement
- One side of the first layer appears higher or lower

How to correct it:

1. Identify which corner feels looser or tighter
2. Adjust the corresponding Z belt tensioner and its two tensioning screws
3. Make adjustments in small ¼-turns and work both screws evenly to keep the belt centered on the idler
4. After each adjustment, move the gantry through full Z travel and observe if the gantry stays level
5. Repeat until all four belts feel even and motion is smooth



GANTRY ASSEMBLY

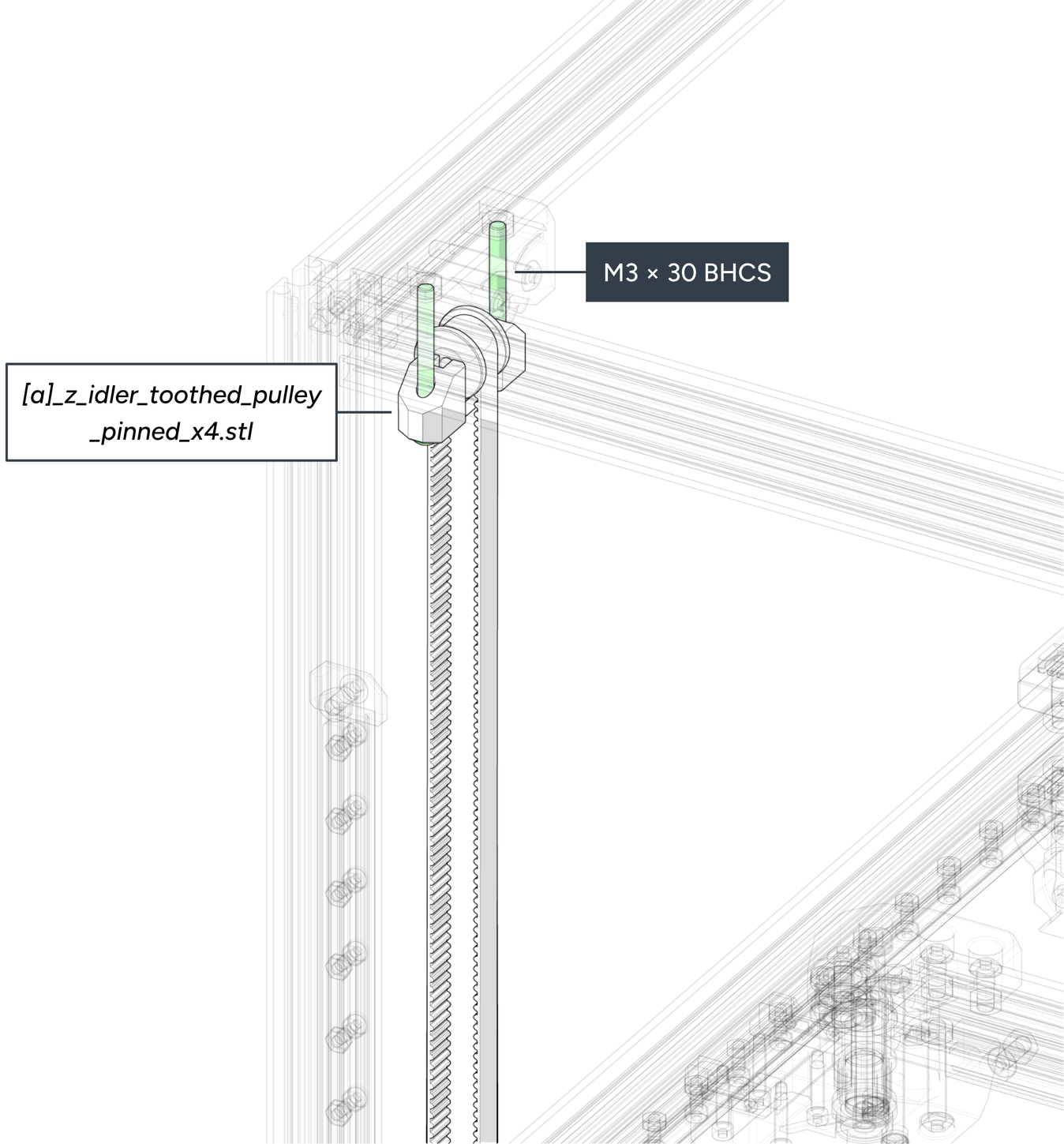
TENSIONING THE Z BELTS

Z belts do not need to be extremely tight, but they must be evenly tensioned.

1. Identify which corner feels looser or tighter
2. Adjust the corresponding Z belt tensioner and its two tensioning screws
3. Make adjustments in small ¼-turns and work both screws evenly to keep the belt centered on the idler
4. After each adjustment, move the gantry through full Z travel and observe if the gantry stays level
5. Repeat until all four belts feel even and motion is smooth

Target state:

- Belts should feel even across all four corners
- When pressed lightly, each belt should deflect slightly with similar resistance



GANTRY ASSEMBLY

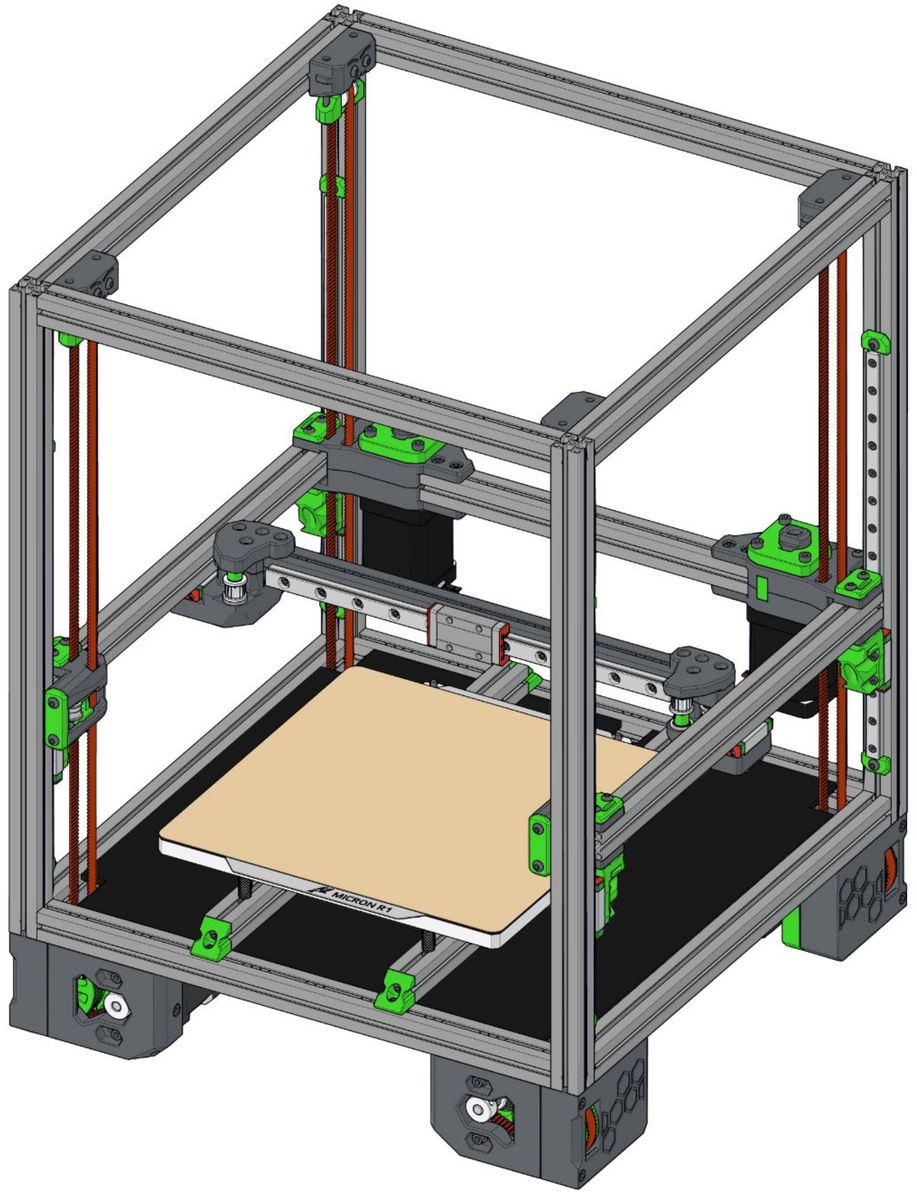
BUILDER CHECK

Once all four gantry corners have been belted, the zip tie supports can be removed.

Your assembly should match this view.

CHECKLIST

- ▢ All four gantry corners are belted
- ▢ Belts are secured at both ends via the rigid mount and the belt clamp
- ▢ Excess belt lengths are folded over and secured
- ▢ Ensure belts are centered on pulleys and idlers, and not rubbing on plastic
- ▢ All four belts are tensioned



CONCEPT

GANTRY SQUARENESS

The gantry must remain square to the frame for accurate XY motion. Misalignment affects belt paths and motion accuracy.

An out-of-square gantry can cause:

- Skewed or distorted prints
- Uneven belt tracking
- Binding or increased resistance
- Inconsistent layer alignment

Target state:

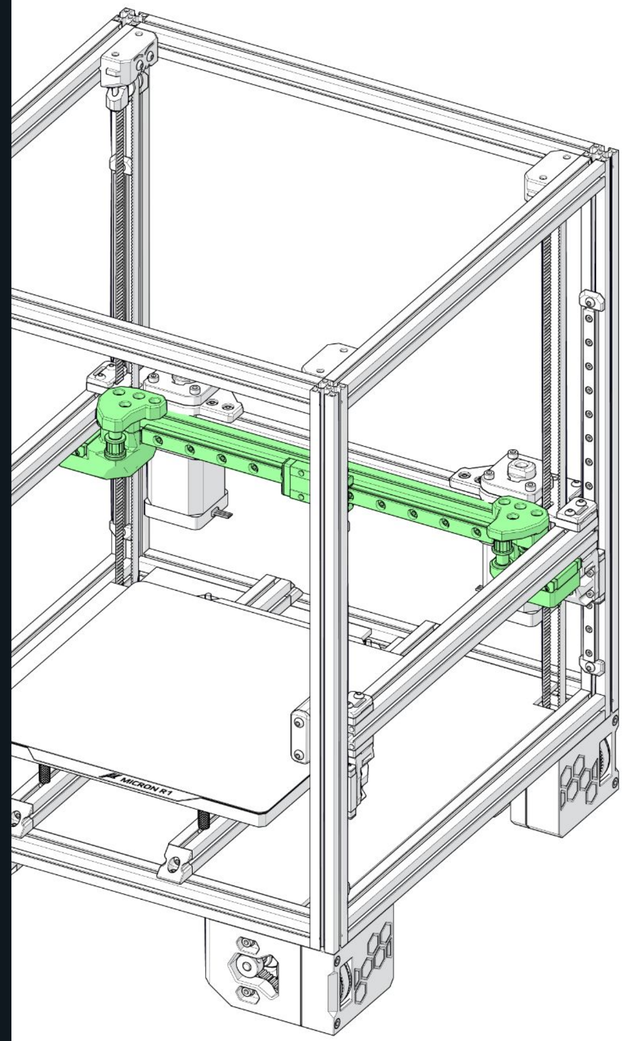
- X beam parallel to frame
- Smooth, even motion
- Symmetrical belt paths

What to look for:

- X beam not parallel front-to-back
- One side feels tighter
- Uneven belt path alignment
- Consistent XY skew in prints

How to correct it:

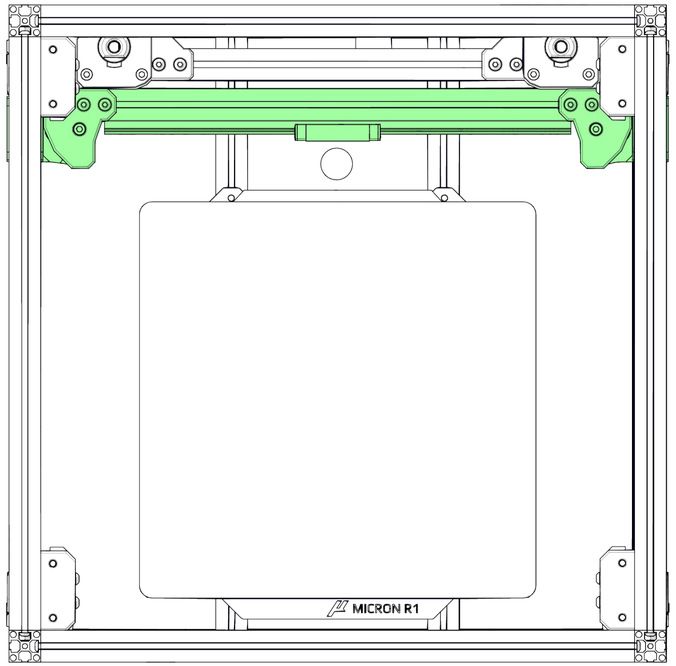
1. Loosen the gantry XY joints slightly
2. Set the gantry fully to the rear against the A/B drives
3. Apply light pressure to align the X beam parallel to the rear extrusion
4. Tighten fasteners gradually and evenly while maintaining alignment
5. Re-check through full travel

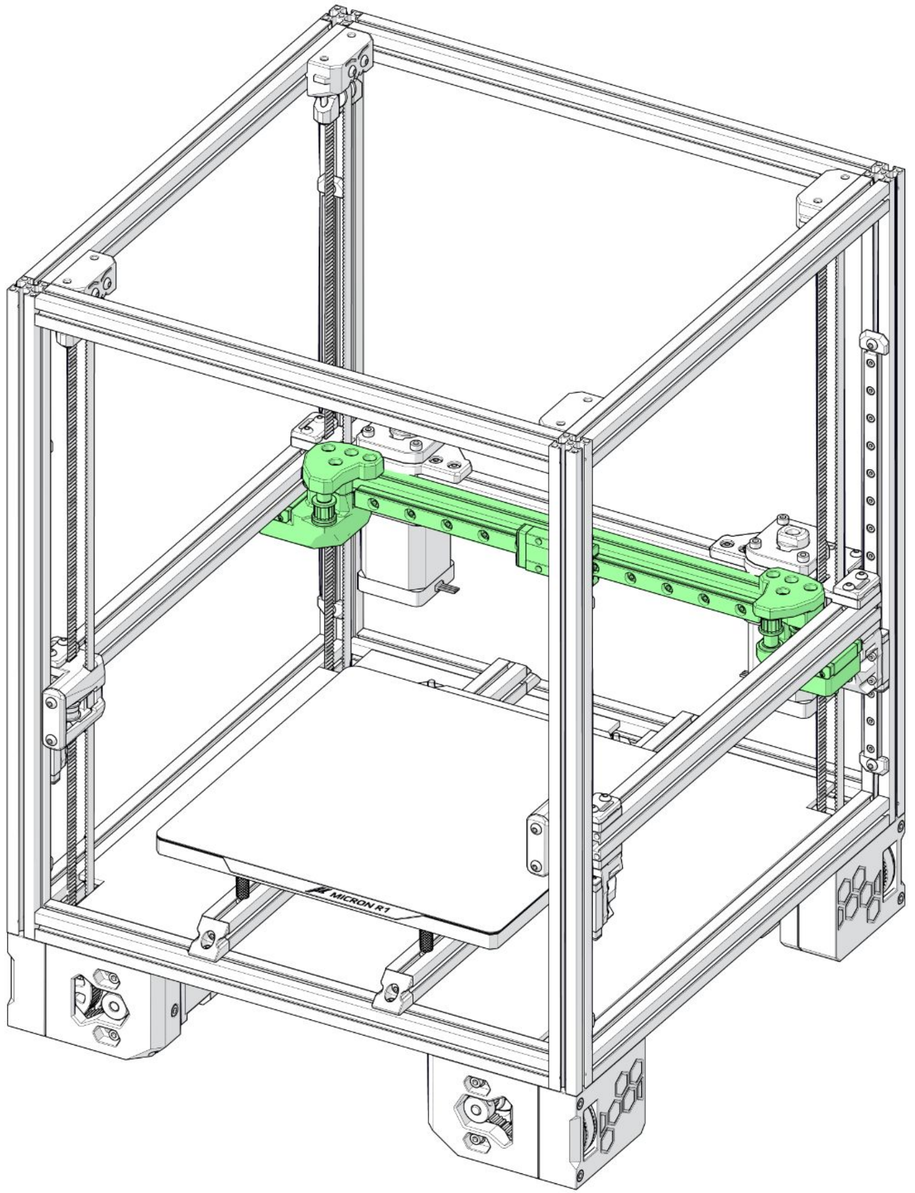


GANTRY ASSEMBLY

SQUARING THE GANTRY

Move the gantry fully to the rear until both gantry corners contact the A and B drives evenly.





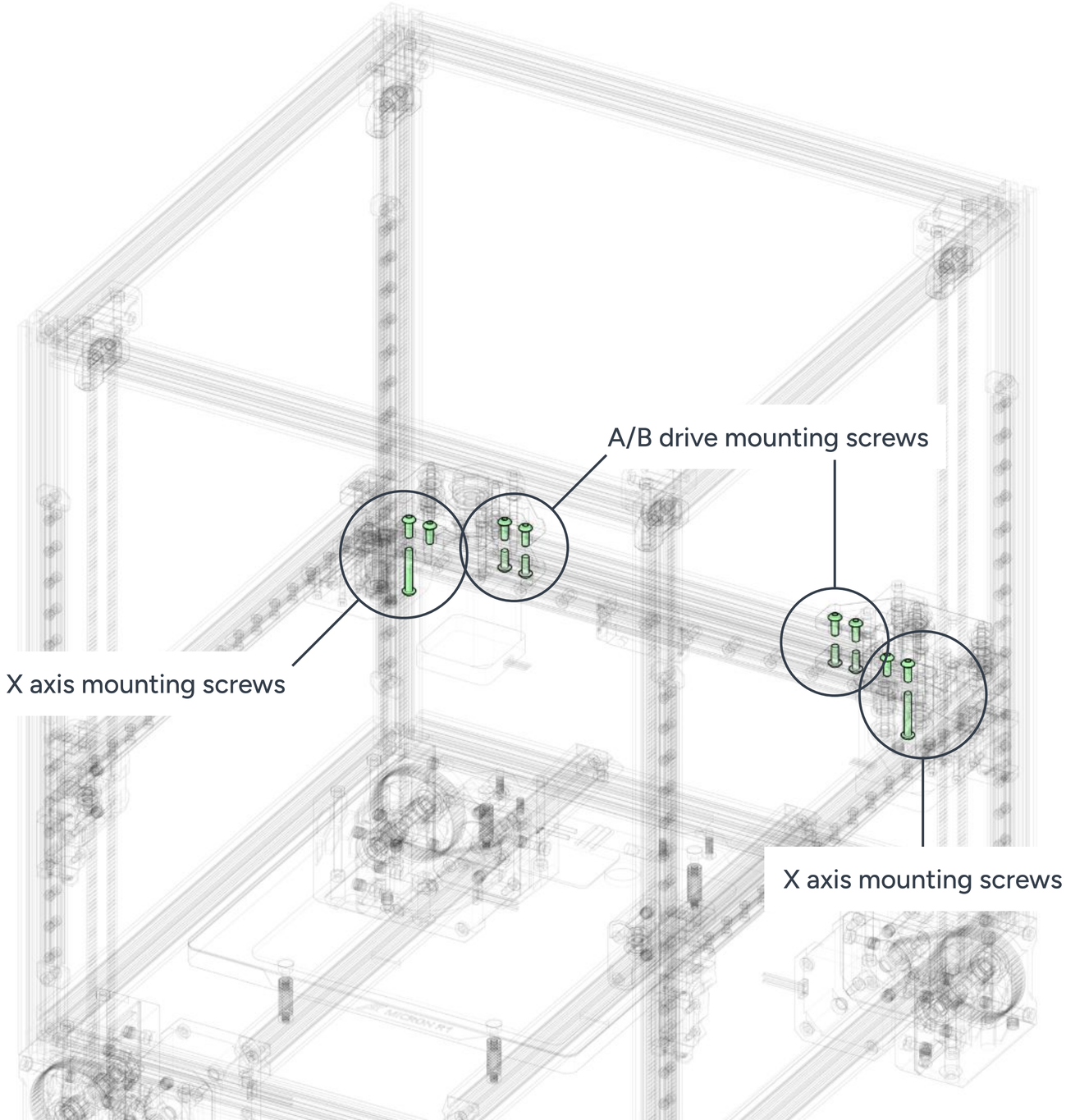
GANTRY ASSEMBLY

SQUARING THE GANTRY

With the gantry aligned against both rear drives, fully tighten the X axis mounting screws.

Move the gantry front-to-back by hand to verify smooth travel without binding.

If binding occurs or alignment is uneven, loosen the A/B drive mounting screws, adjust drive spacing until the gantry travels freely, and fully tighten the A/B drive mounting screws.



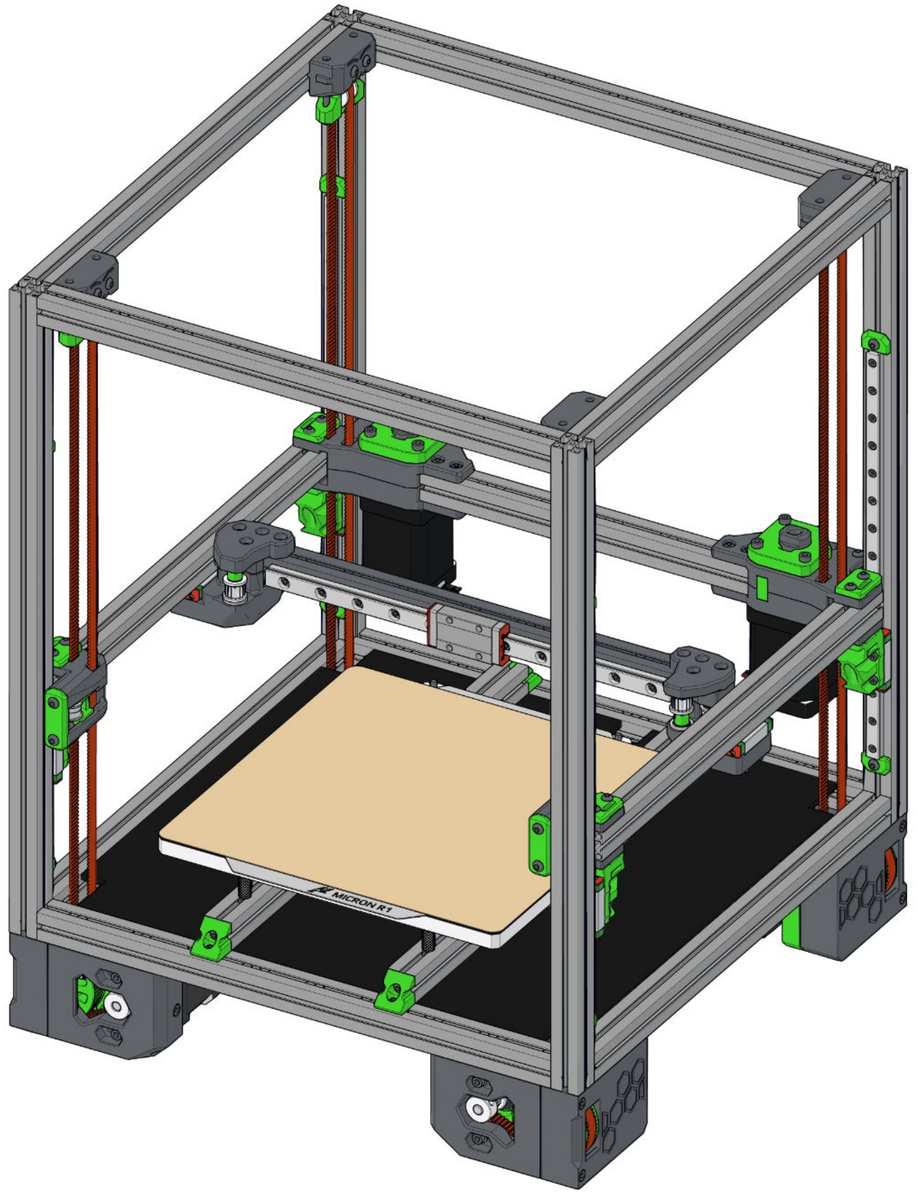
GANTRY ASSEMBLY

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

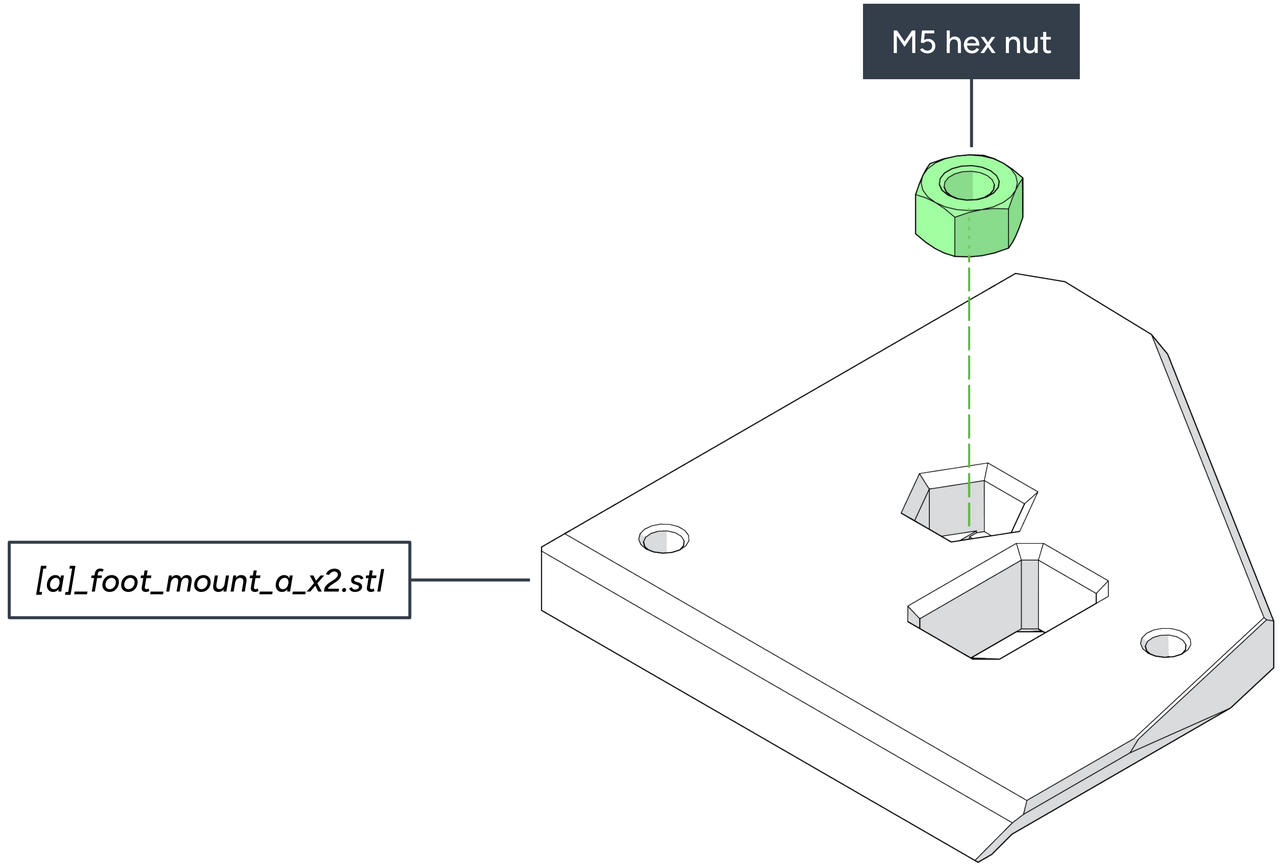
- ▢ Gantry contacts both rear drives evenly
- ▢ X axis screws fully tightened
- ▢ Gantry slides smoothly front-to-back



GANTRY ASSEMBLY

Z DRIVES - FOOT MOUNTS

Insert an **M5 hex nut** into the center recess of each ***[a]_foot_mount_a_x2.stl*** and ***[a]_foot_mount_b_x2.stl***.

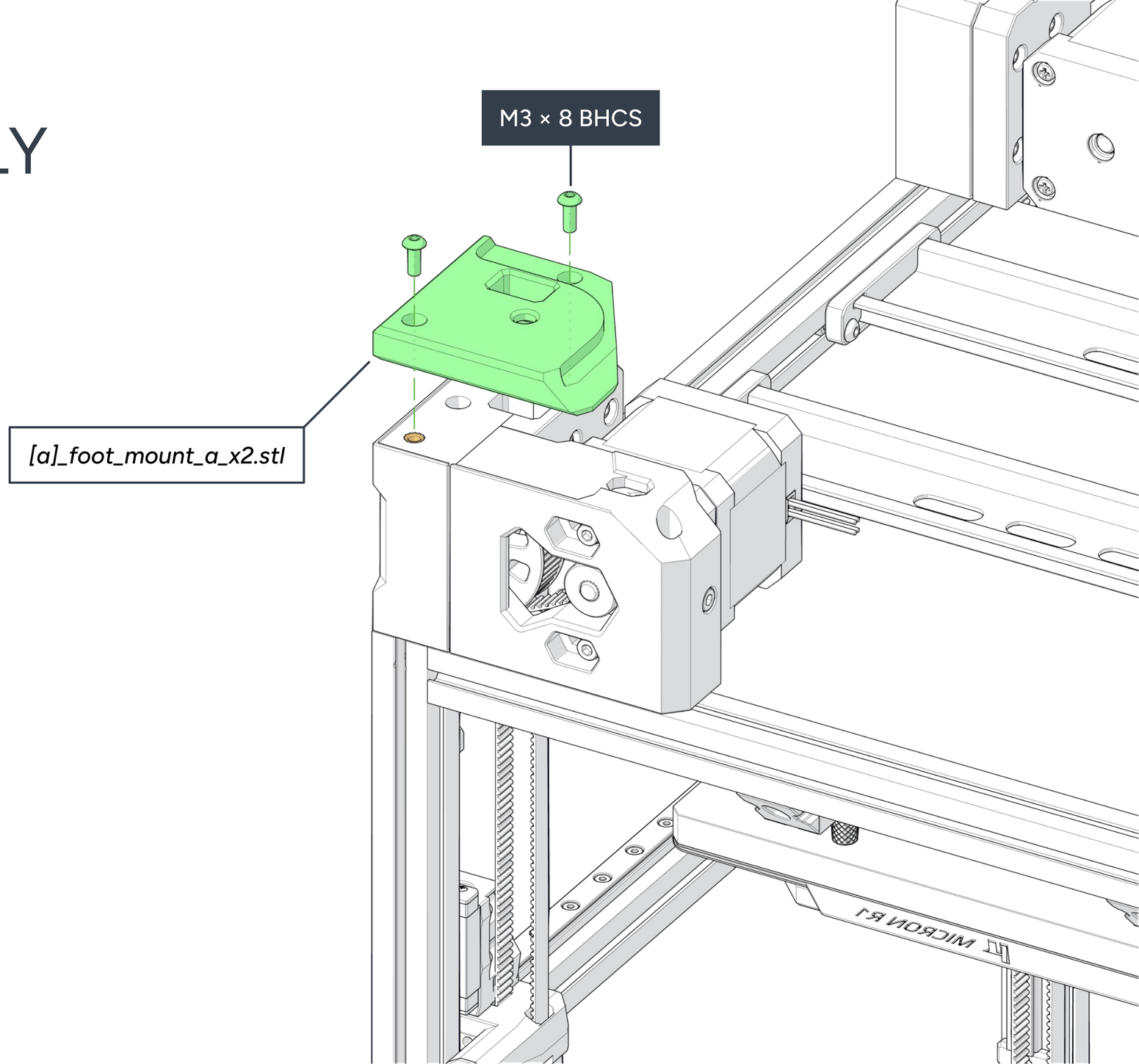


GANTRY ASSEMBLY

Z DRIVES - FOOT MOUNTS

Carefully flip the printer upside down.

Secure each foot mount to the Z drive assemblies with **two M3 × 8 BHCS**.

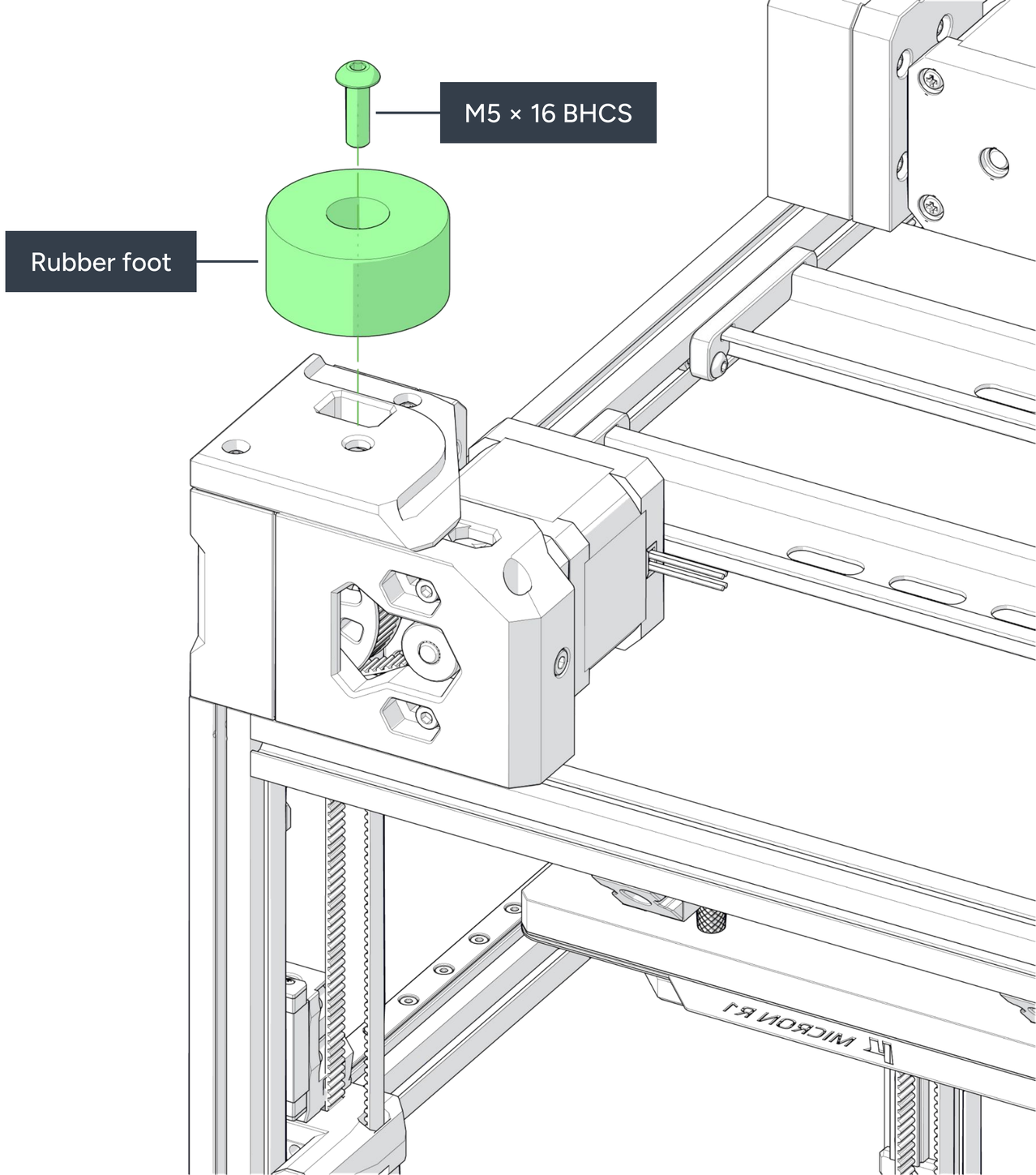


GANTRY ASSEMBLY

Z DRIVES - FOOT MOUNTS

Center and secure the **rubber foot** with an **M5 × 16 BHCS**.

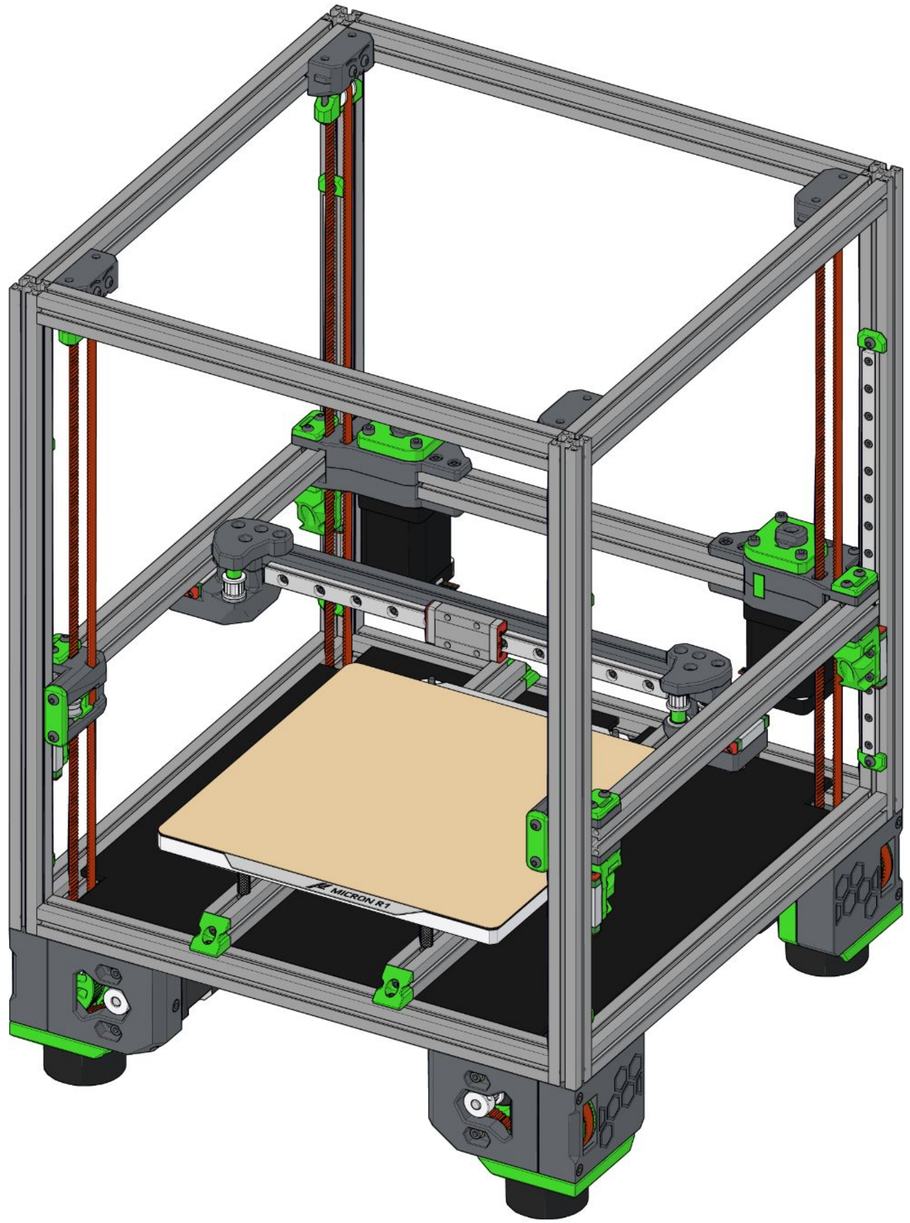
Repeat for all four Z drives.



GANTRY ASSEMBLY

BUILDER CHECK

Your assembly should match this view.



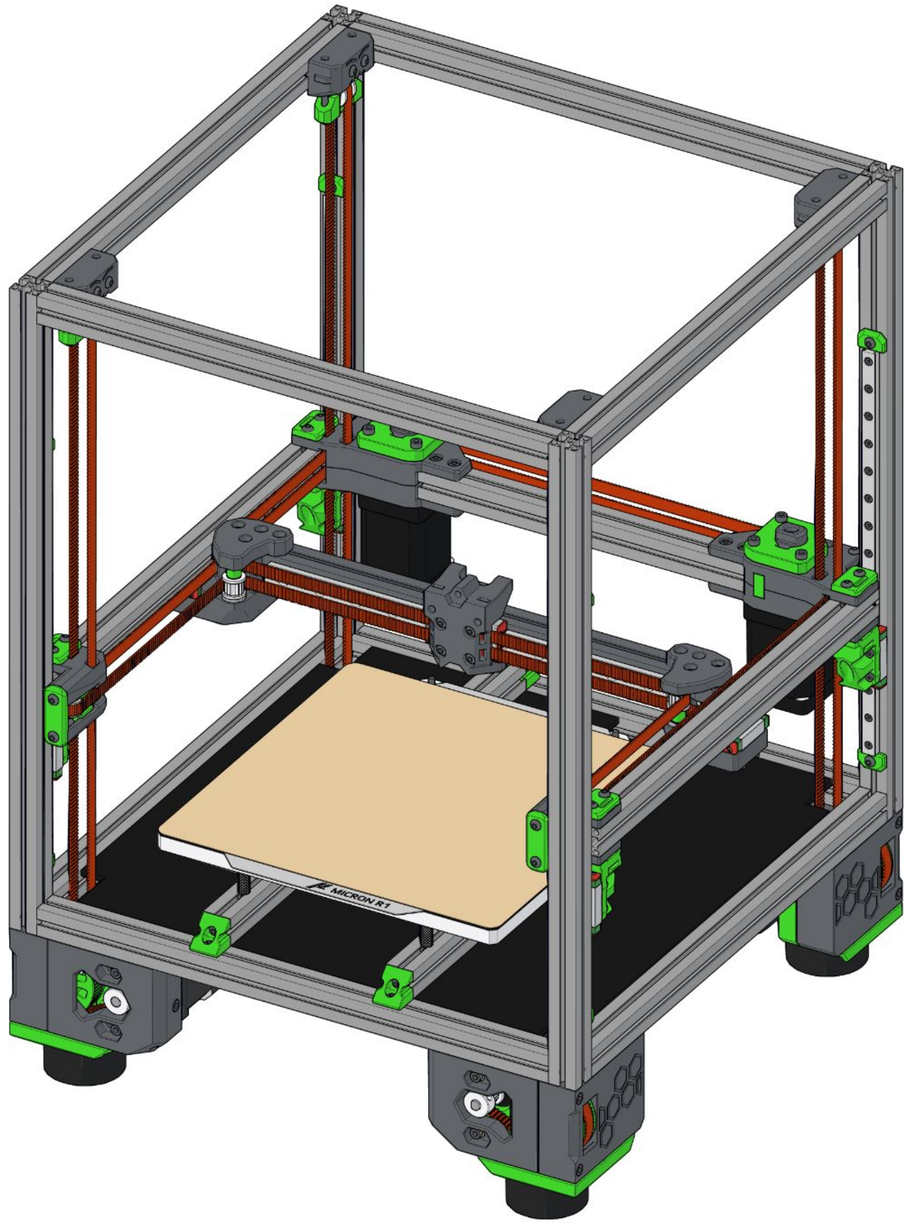
GANTRY ASSEMBLY

ROUTING THE A/B BELTS

Toolhead movement in the XY plane is controlled via two belts driven by the A/B drives. Proper belt routing and matched belt lengths are critical for smooth motion and balanced tension.

HARDWARE

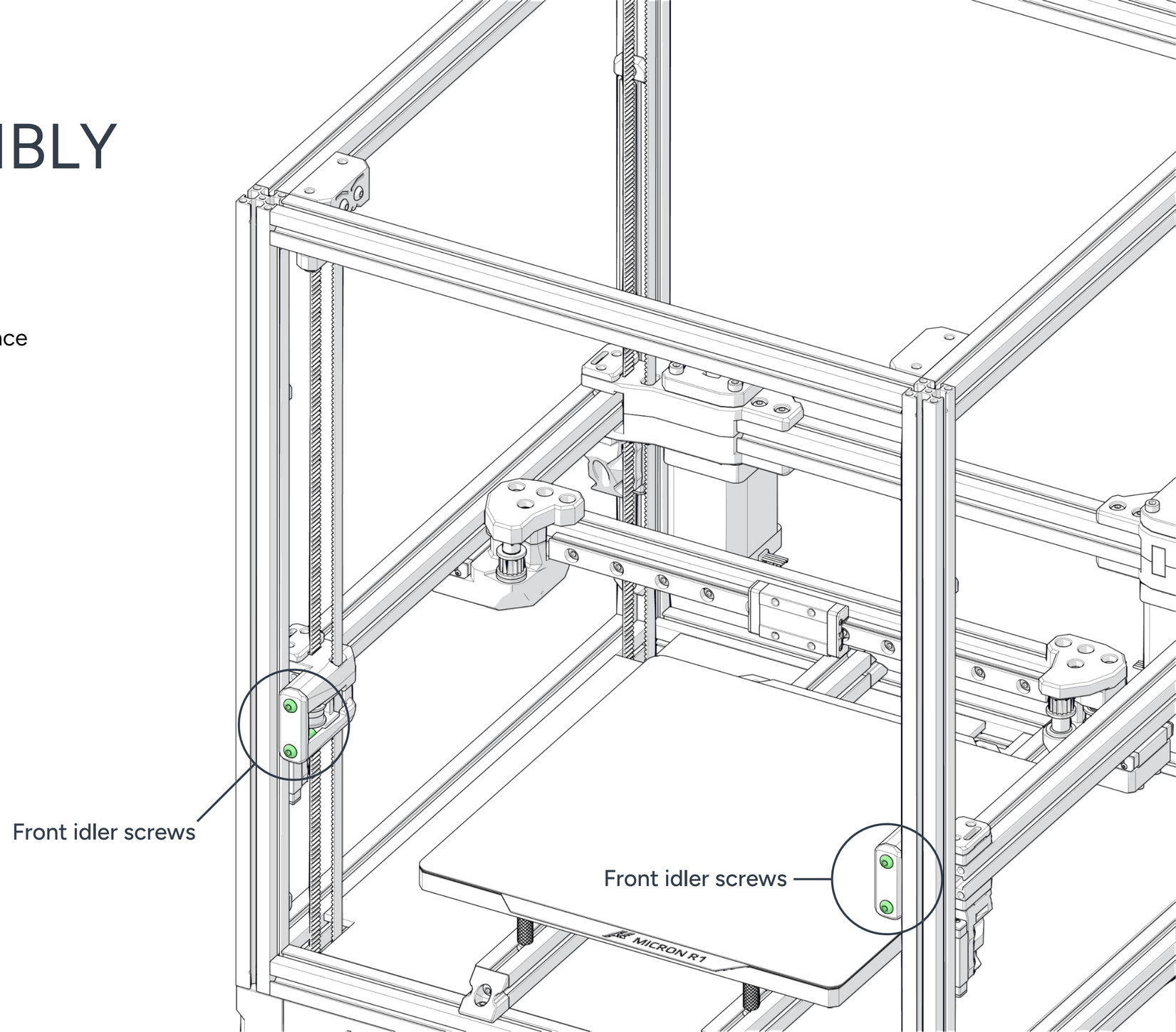
2	2GT belts, 6 mm
4	M3 × 8 BHCS



GANTRY ASSEMBLY

ROUTING THE A/B BELTS

Loosen the front idler screws to create clearance for belt routing and later tensioning.

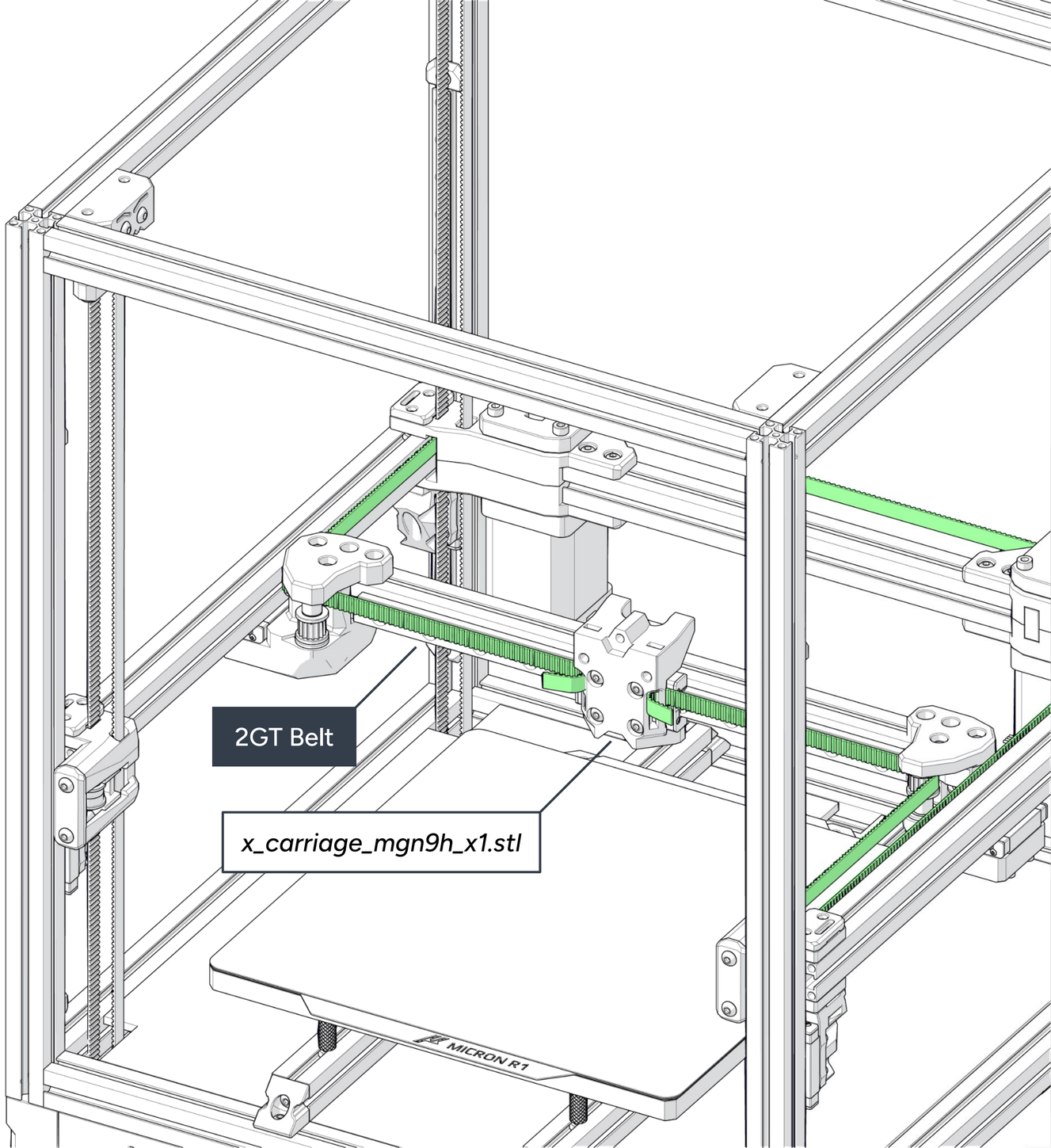


GANTRY ASSEMBLY

PREPPING THE UPPER A BELT

Route and trim the upper A belt first.

This belt will be used as the reference length for the lower B belt to ensure both belts are matched for equal tensioning.



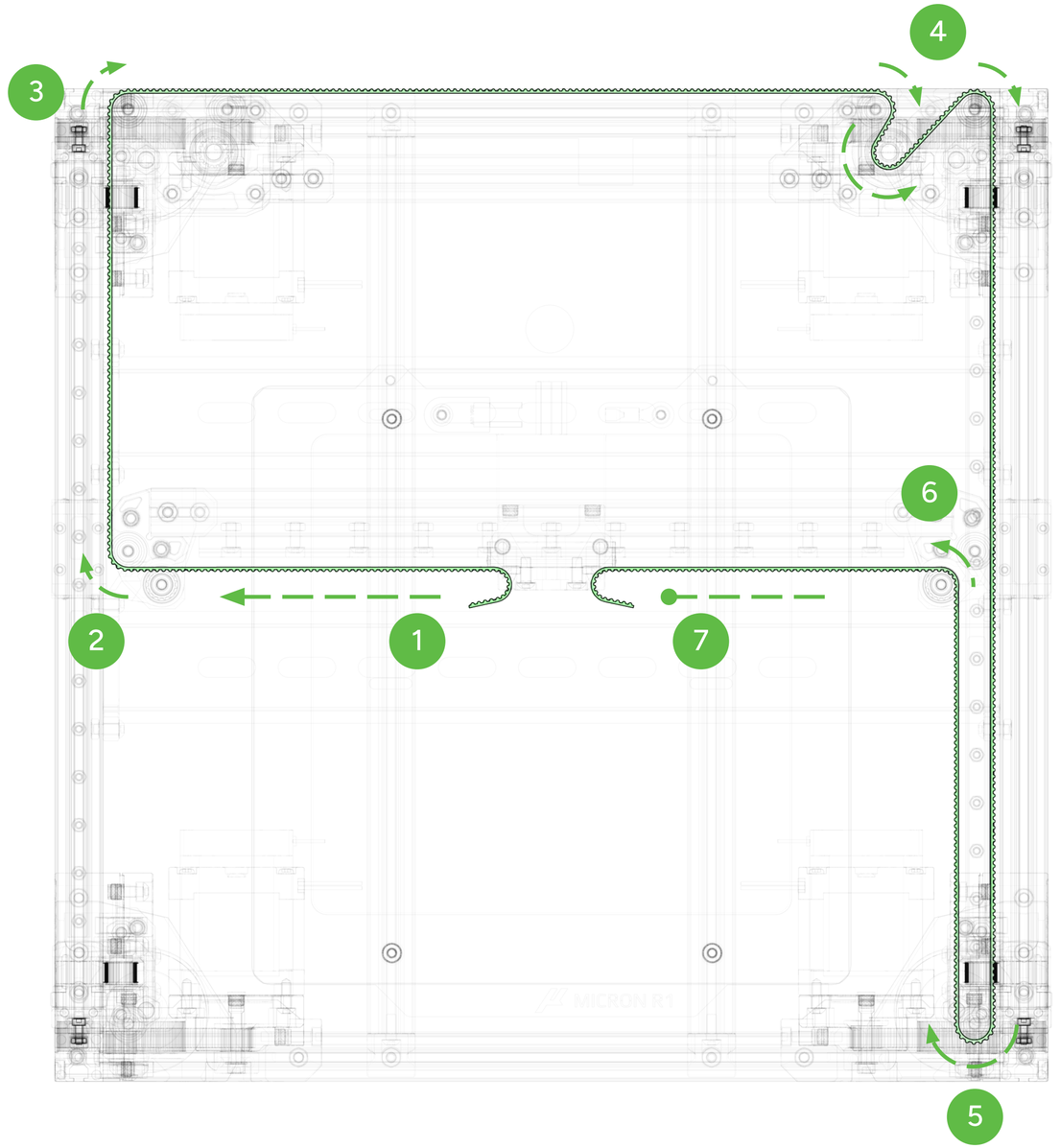
GANTRY ASSEMBLY

ROUTING THE UPPER A BELT

Follow the path in the diagram to route the belt. Ensure belt teeth engage all drive pulleys and run smooth-side over idlers.

1. Starting at the carriage, route the **2GT belt** to the left.
2. Feed the belt around the left XY joint's upper idler.
3. Continue to the B drive and route around the B drive's outer idler.
4. Route through the A drive assembly:
   a. Around the first idler.
   b. Around the A drive pulley ensuring tooth engagement.
   c. Around the second idler.
5. Route around the front-right idler.
6. Feed the belt around the right XY joint's upper idler and return to the carriage.
7. Insert the belt end into the carriage's upper toothed slot.

Once the end is secure (7), line up the starting point (1) against the upper toothed slot, leave a small amount of extra length for final tensioning, and trim the belt to final length.

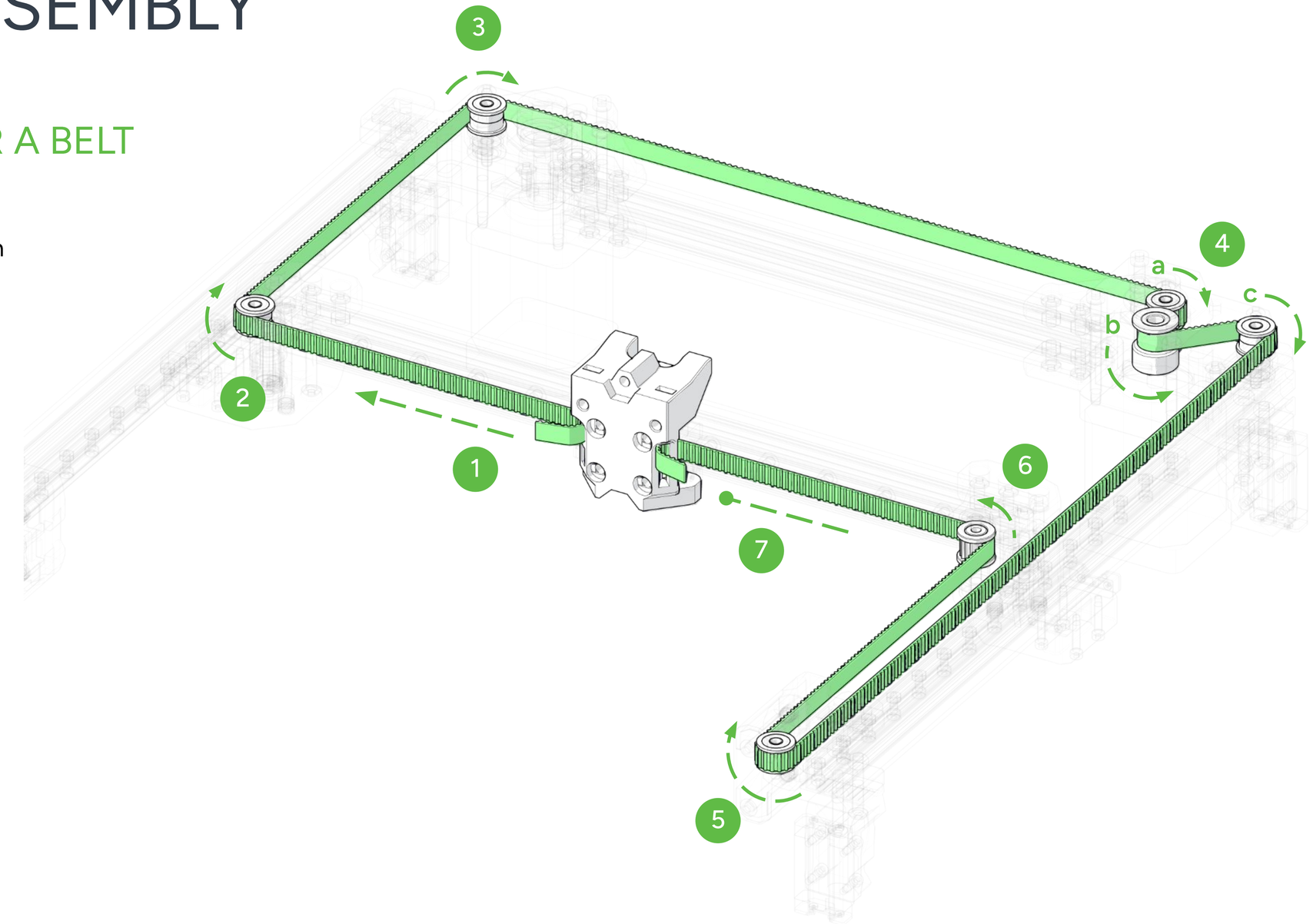


GANTRY ASSEMBLY

ROUTING THE UPPER A BELT

Follow the path in the diagram to route the belt. Ensure belt teeth engage all drive pulleys and run smooth-side over idlers.

Once the end is secure ⑦, line up the starting point ① against the upper toothed slot, leave a small amount of extra length for final tensioning, and trim the belt to final length.



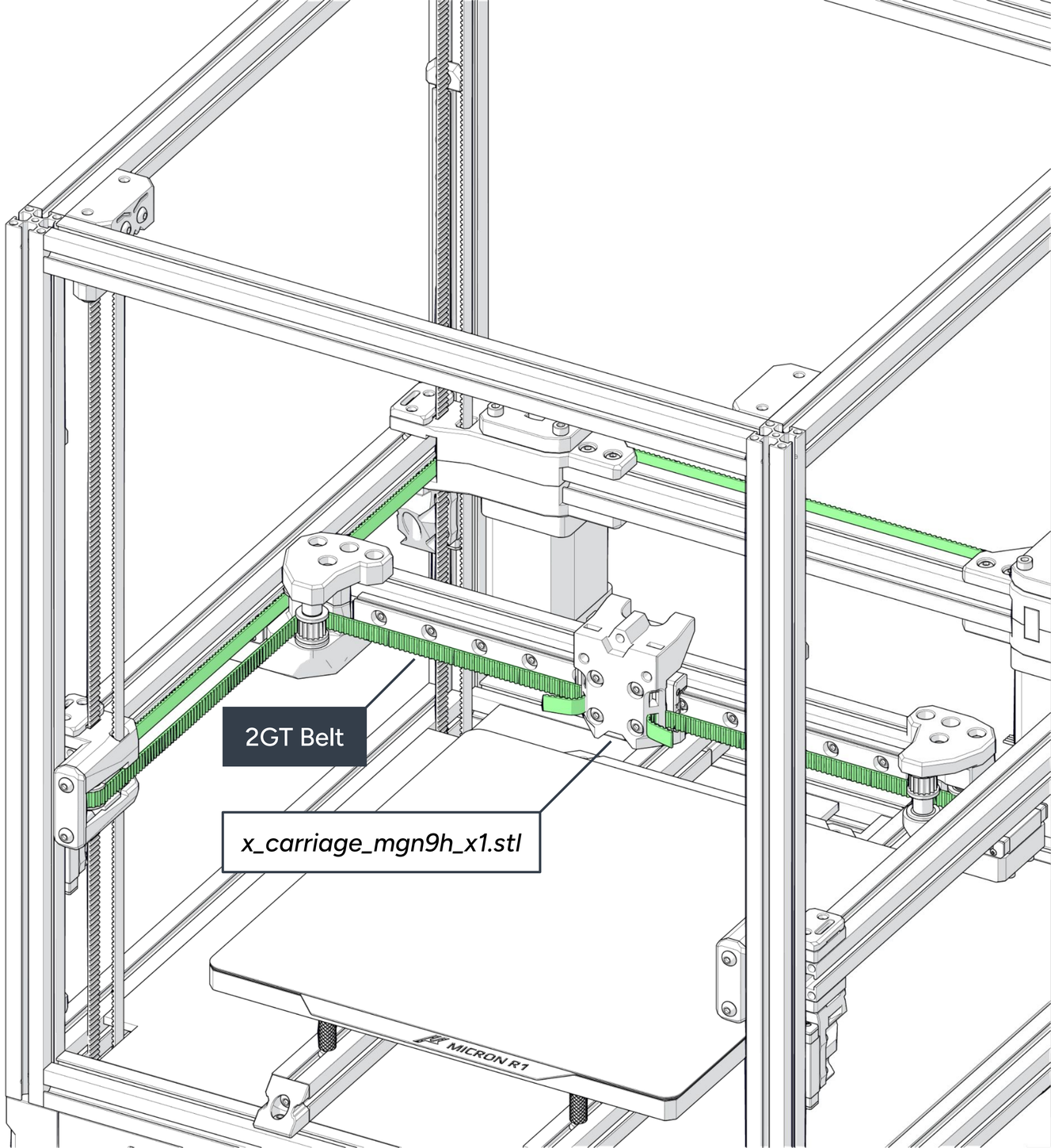
GANTRY ASSEMBLY

PREPPING THE LOWER B BELT

Remove the previously installed upper A belt from the printer.

Align one end of the trimmed belt with the **extra 2GT belt**, teeth engaged.

Using a “zipper” motion, run along the full length of the upper A belt to measure and cut a matching length for the lower B belt.



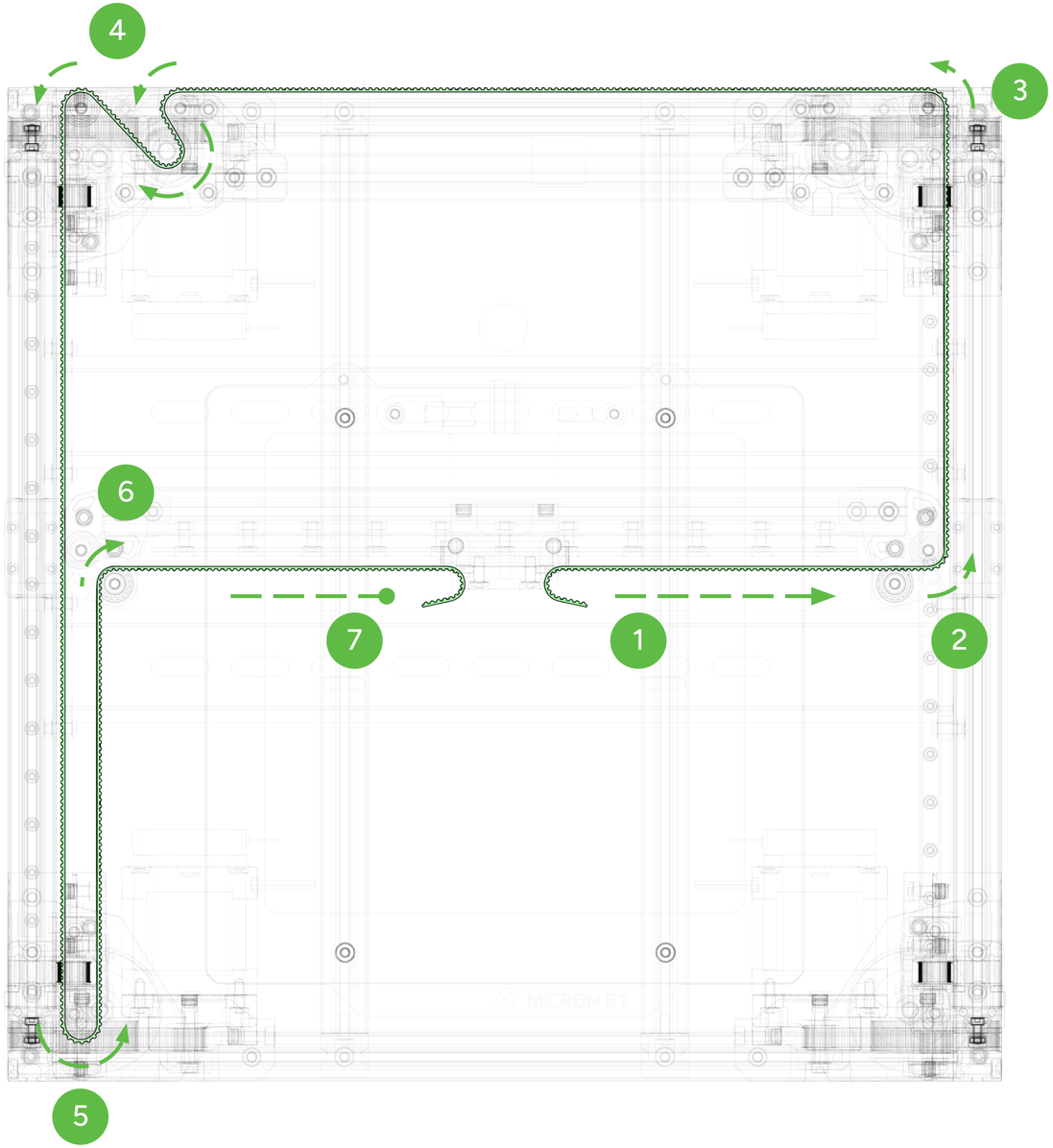
GANTRY ASSEMBLY

ROUTING THE LOWER B BELT

This path mirrors the upper A belt in the opposite direction. Follow the path in the diagram to route the belt. Ensure belt teeth engage all drive pulleys and run smooth-side over idlers.

1. Starting at the carriage, route the 2GT belt to the right.
2. Feed the belt around the right XY joint's lower idler.
3. Continue to the A drive and route around the A drive's outer idler.
4. Route through the B drive assembly.
   a. Around the first idler.
   b. Around the B drive pulley ensuring tooth engagement.
   c. Around the second idler.
5. Route around the front-left idler.
6. Feed the belt around the left XY joint's lower idler and return to the carriage.
7. Insert the belt end into the carriage's lower toothed slot.

The lower B belt should already be trimmed to final length.

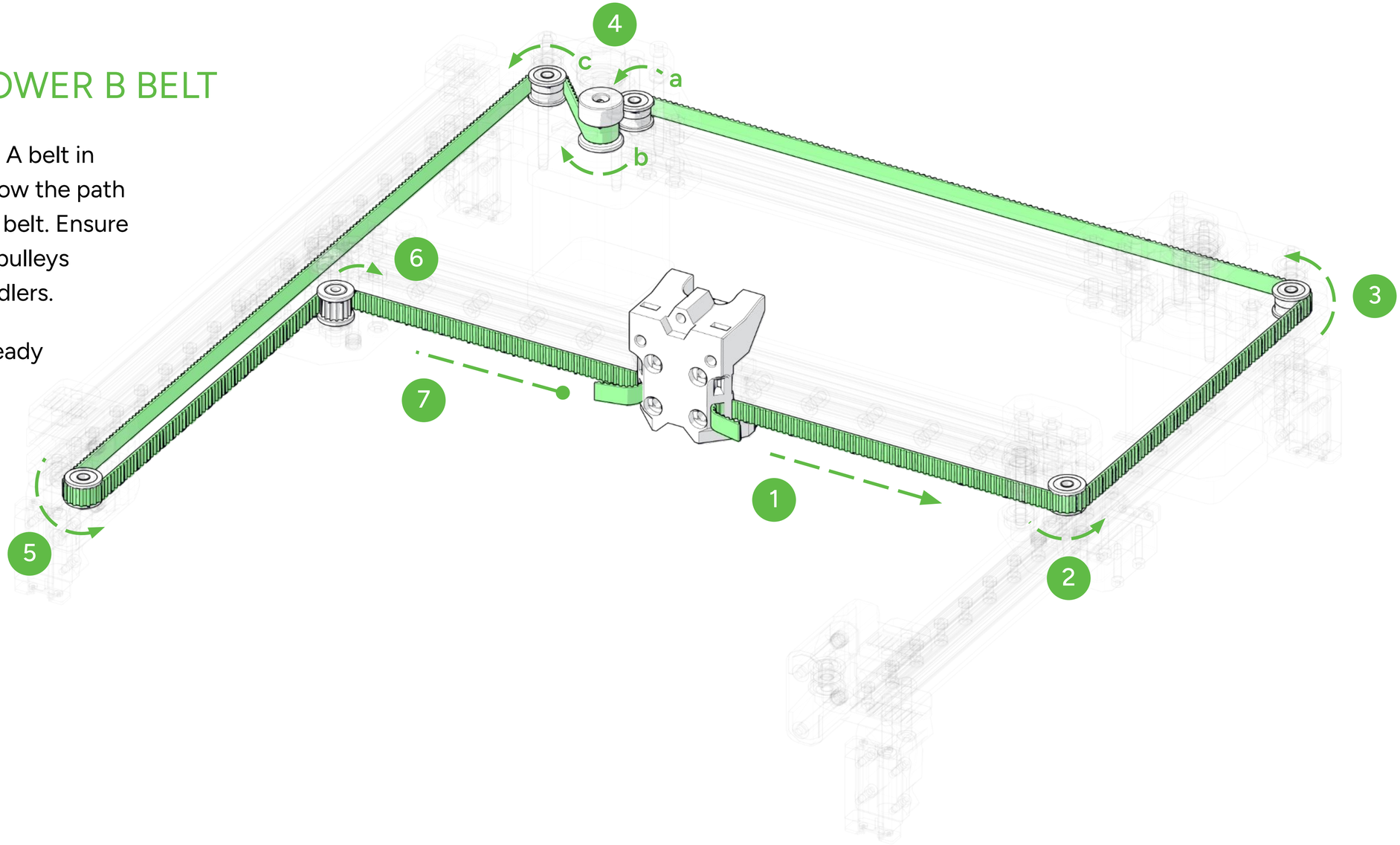


GANTRY ASSEMBLY

ROUTING THE LOWER B BELT

This path mirrors the upper A belt in the opposite direction. Follow the path in the diagram to route the belt. Ensure belt teeth engage all drive pulleys and run smooth-side over idlers.

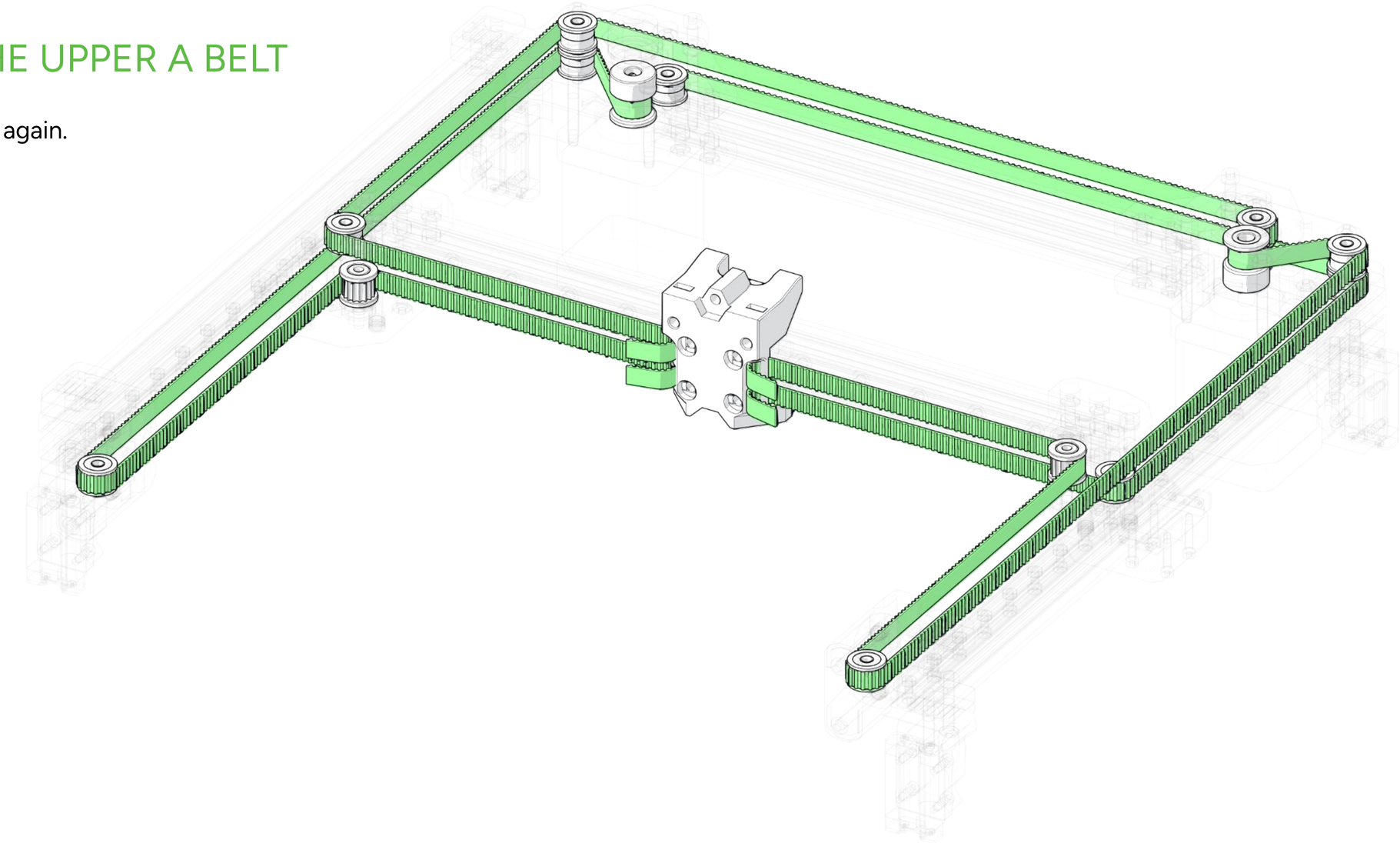
The lower B belt should already be trimmed to final length.



GANTRY ASSEMBLY

RE-ROUTING THE UPPER A BELT

Re-route the upper A belt again.



I / GANTRY ASSEMBLY

SECURING THE A/B BELTS

Once both belts are routed, feed the belt ends through their matching slots in ***x_carriage_mgn9h_x1.stl***. Ensure the belt teeth are engaged with the printed grooves.

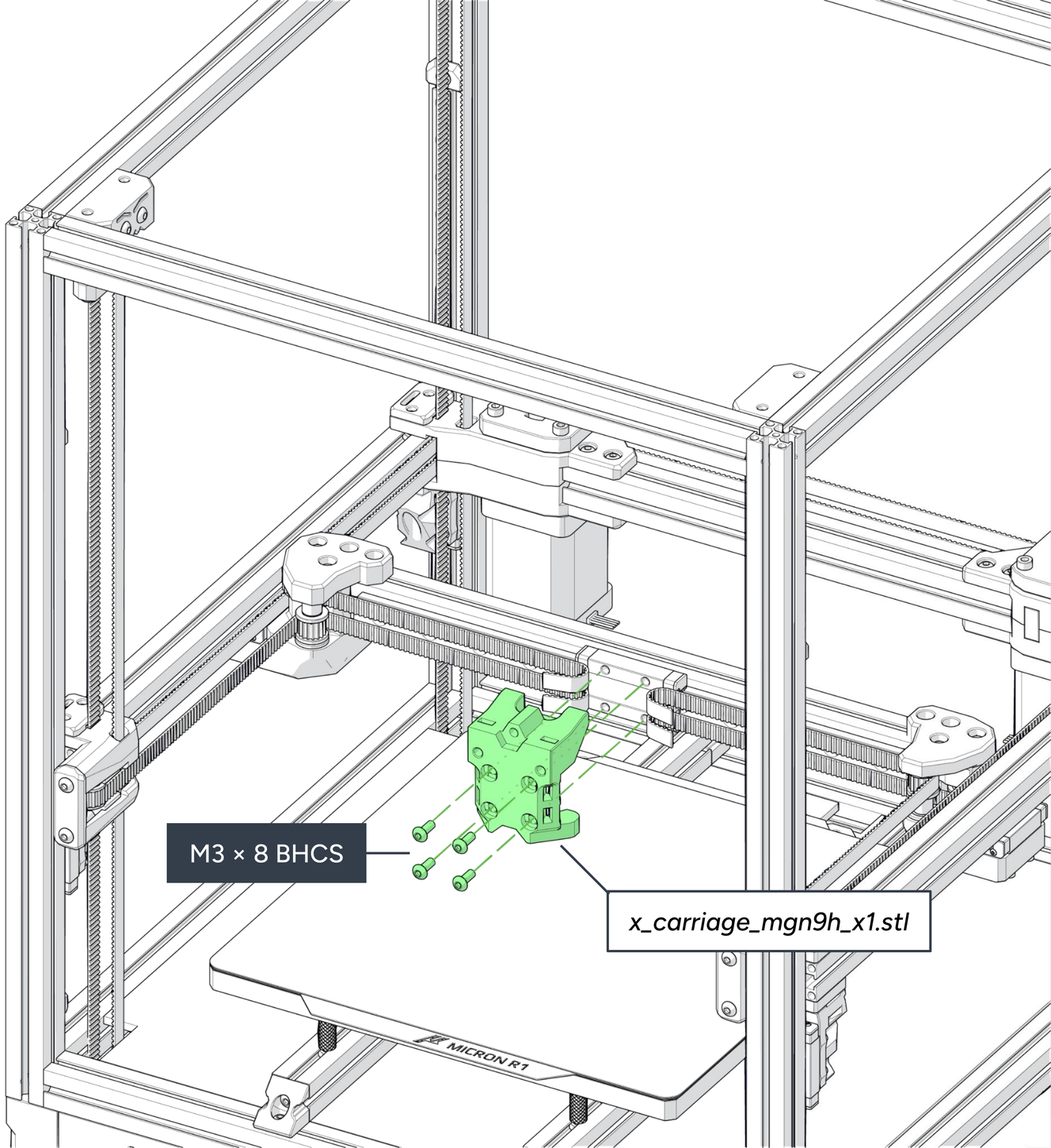
Pull belt ends enough to remove slack. Keep excess lengths even. Do not apply final tension.

Secure the carriage assembly to the MGN9H carriage using **four M3 × 8 BHCS**.

Final A/B belt tensioning is performed later.

NOTE

Remove belt slack only. Uneven pulling at this stage can introduce skew before tensioning. Count the exposed teeth.



CONCEPT

A/B BELT & TENSIONING

Movement in X and Y is achieved by both A/B belts moving in coordination, so their lengths and tension must match.

Uneven tension can cause:

- Skewed or distorted prints
- Belt tracking issues on pulleys/idlers
- Increased wear and noise
- Subtle motion inaccuracies

Target state:

- Belts should feel even and consistent
- When pressed lightly, each belt should deflect slightly with similar resistance
- When plucked, both belts should produce a similar pitch and response

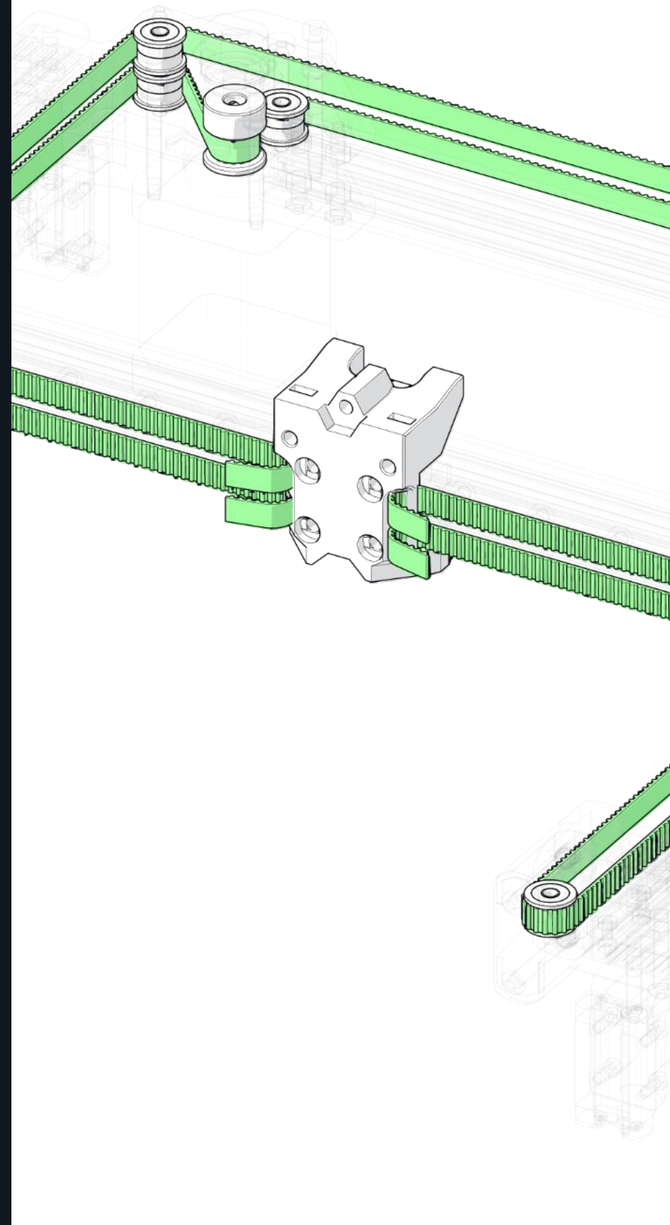
Avoid over-tensioning. Excess tension increases wear and does not improve accuracy.

What to look for:

- Uneven resistance in motion
- One side feels tighter
- Belt tracking differs side-to-side
- Consistent XY skew

How to adjust:

1. Compare belt feel at front idlers and identify which belt is tighter or looser
2. Adjust the corresponding front idler
3. Make adjustments in small ¼-turns and work both screws evenly to keep the belt centered on the idler
4. After each adjustment, move the gantry through full travel to re-check
5. Repeat until motion is smooth in all directions and both belts feel similar



GANTRY ASSEMBLY

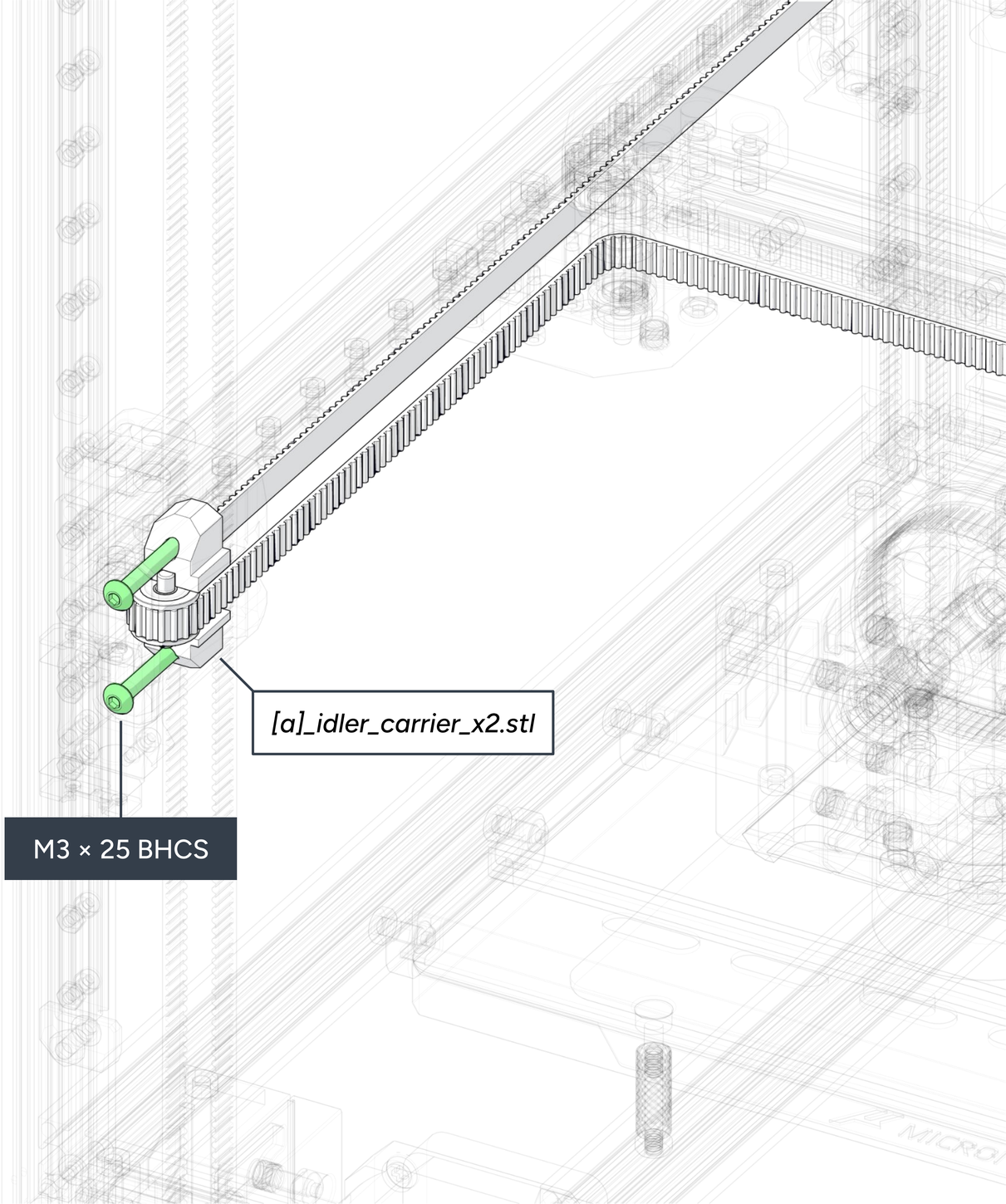
TENSIONING THE A/B BELTS

A/B belts do not need to be extremely tight, but they must be evenly tensioned.

1. Compare belt feel at front idlers and identify which belt is tighter or looser
2. Adjust the corresponding front idler
3. Make adjustments in small ¼-turns and work both screws evenly to keep the belt centered on the idler
4. After each adjustment, move the gantry through full travel to re-check
5. Repeat until motion is smooth in all directions and both belts feel similar

Target state:

- Belts should feel even and consistent
- When pressed lightly, each belt should deflect slightly with similar resistance
- When plucked, both belts should produce a similar pitch and response



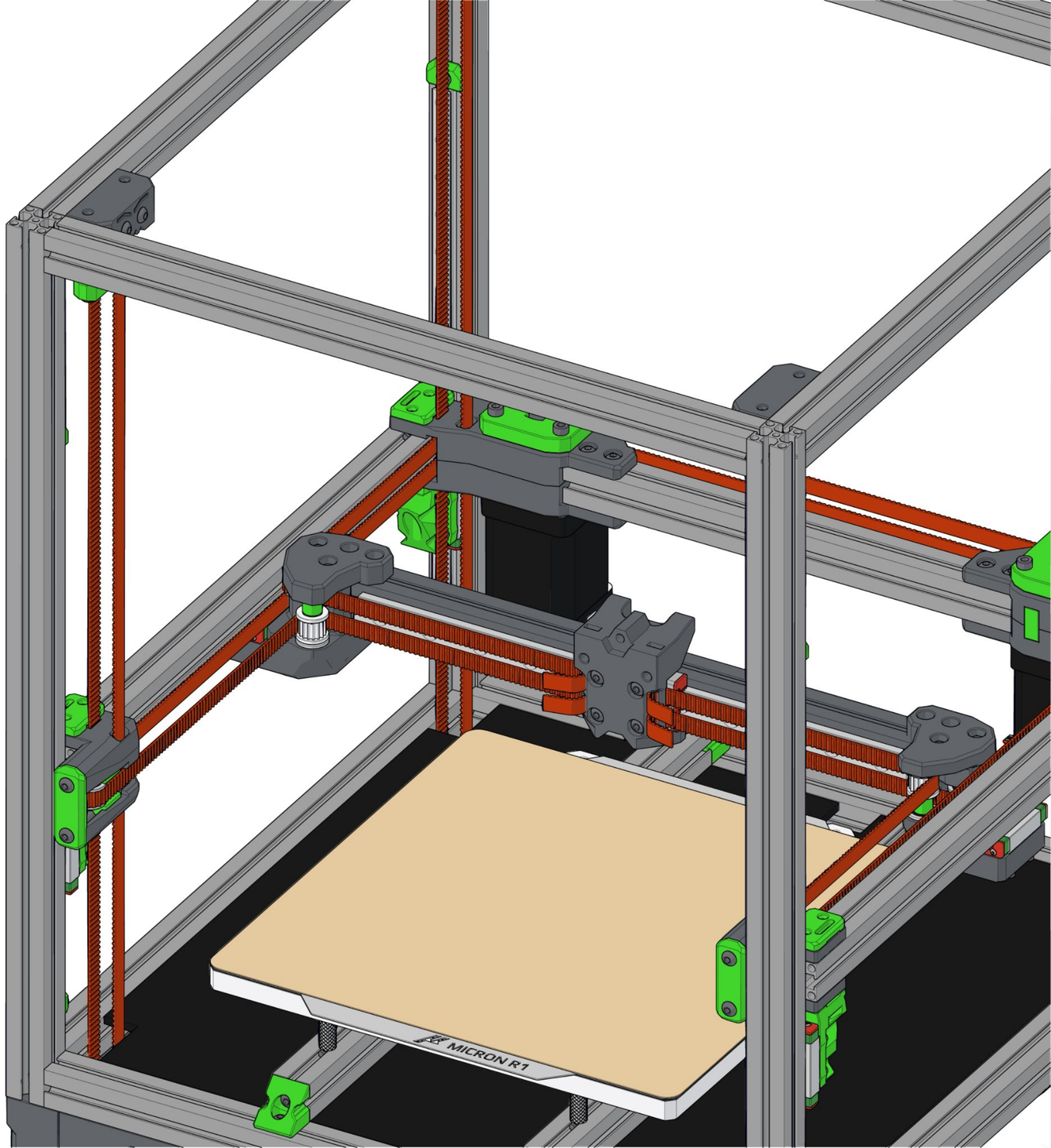
GANTRY ASSEMBLY

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

- ▢ Belts are the same length
- ▢ Belts have no slack
- ▢ Belts are centered on pulleys and idlers, and not rubbing on plastic
- ▢ Upper A belt is centered on A drive pulley and front-right idler
- ▢ Lower B belt is centered on B drive pulley and front-left idler
- ▢ Excess belt lengths have the same tooth count
- ▢ Excess belt lengths are folded over and secured



J / TOOLHEAD & EXTRUDER

Micron+ supports a wide range of toolheads and extruders. This guide is based on the LDO kit, which includes parts for **Anthead** and **WWG2**.

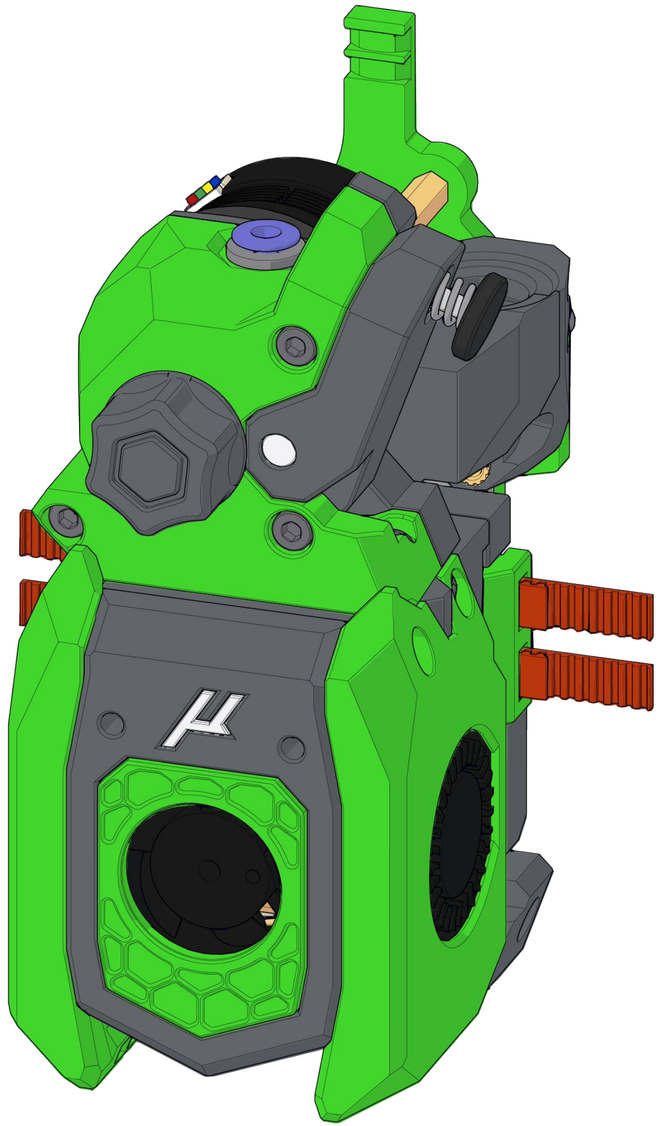
ANTHEAD

- Compact, low-mass toolhead optimized for CoreXY motion
- Dual 4010 part cooling fans for symmetrical airflow
- 2510 fan for hotend cooling
- Compatible with a wide range of hotends and extruders
- Integrated cable routing with CAN toolhead support

WWG2

- Planetary gear-driven extruder for consistent filament control
- High gear reduction for strong, precise extrusion
- Compact Orbiter 2 footprint designed for toolhead mounting
- Compatible with standard NEMA14 pancake steppers

Official documentation: https://github.com/PrintersForAnts/AntHead

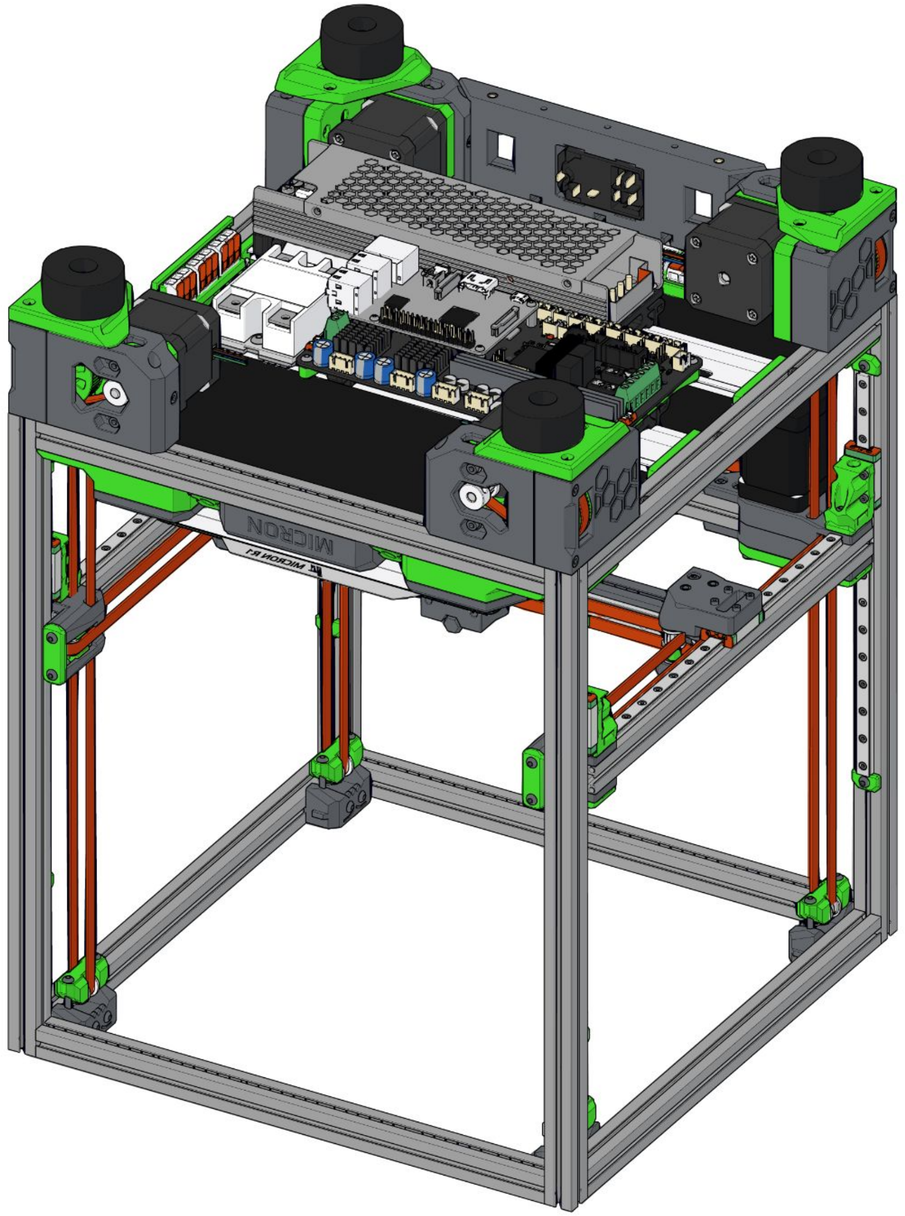


K / ELECTRONICS

When assembling the printer electronics, you will be working with both low-voltage DC and line-voltage AC (120V/220V). Proper wiring and secure connections are critical for safe and reliable operation.

Always ensure the printer is unplugged before working on any wiring. Verify that power supply capacitors have discharged before touching terminals. Take time to route wires cleanly and confirm all connections are secure before powering on.

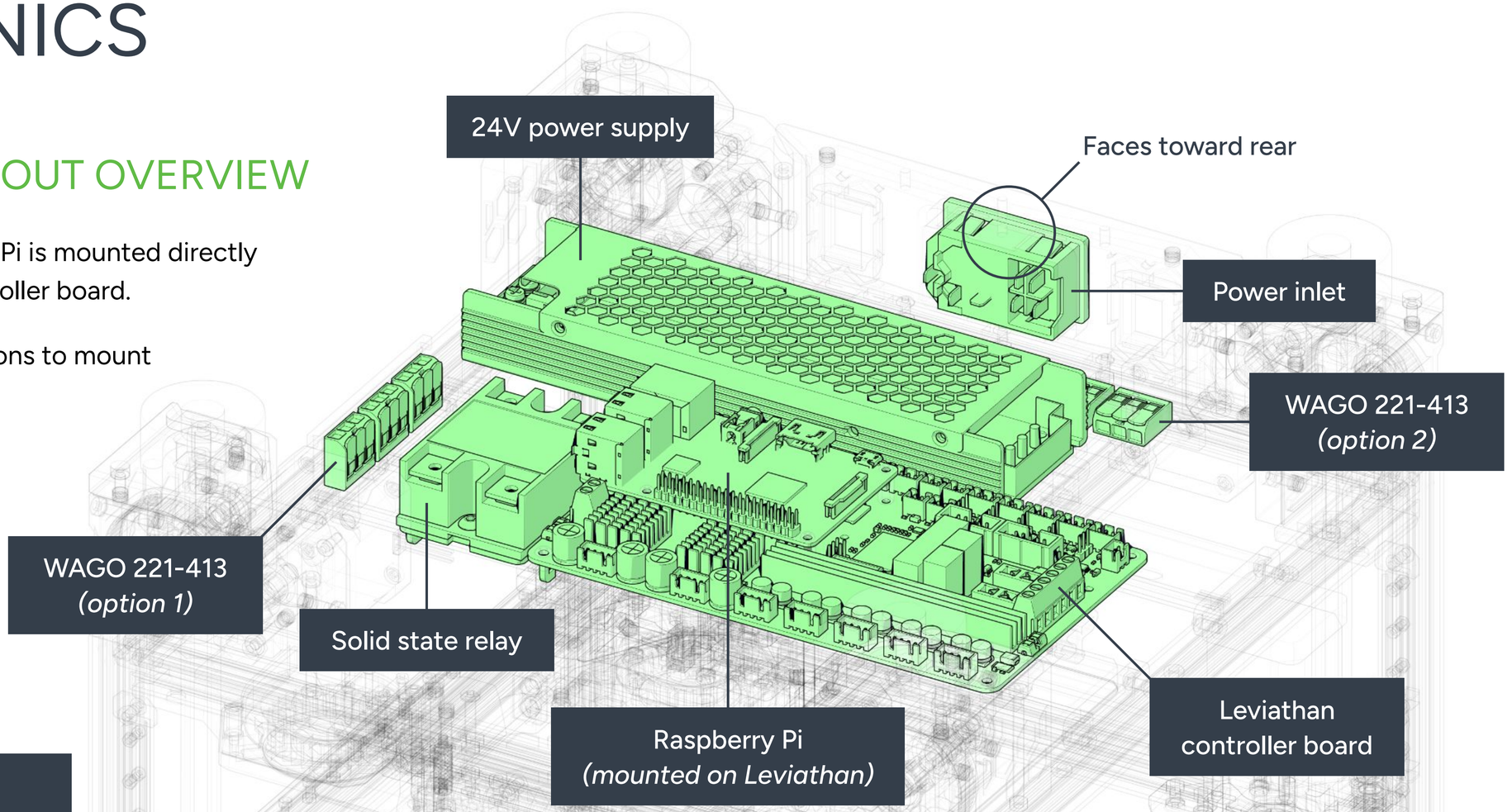
Poor wiring can lead to intermittent faults that are difficult to diagnose later.



COMPONENT LAYOUT OVERVIEW

In the LDO kit, the Raspberry Pi is mounted directly on top of the Leviathan controller board.

There are two possible positions to mount the WAGO 221-413 block.



NOTE

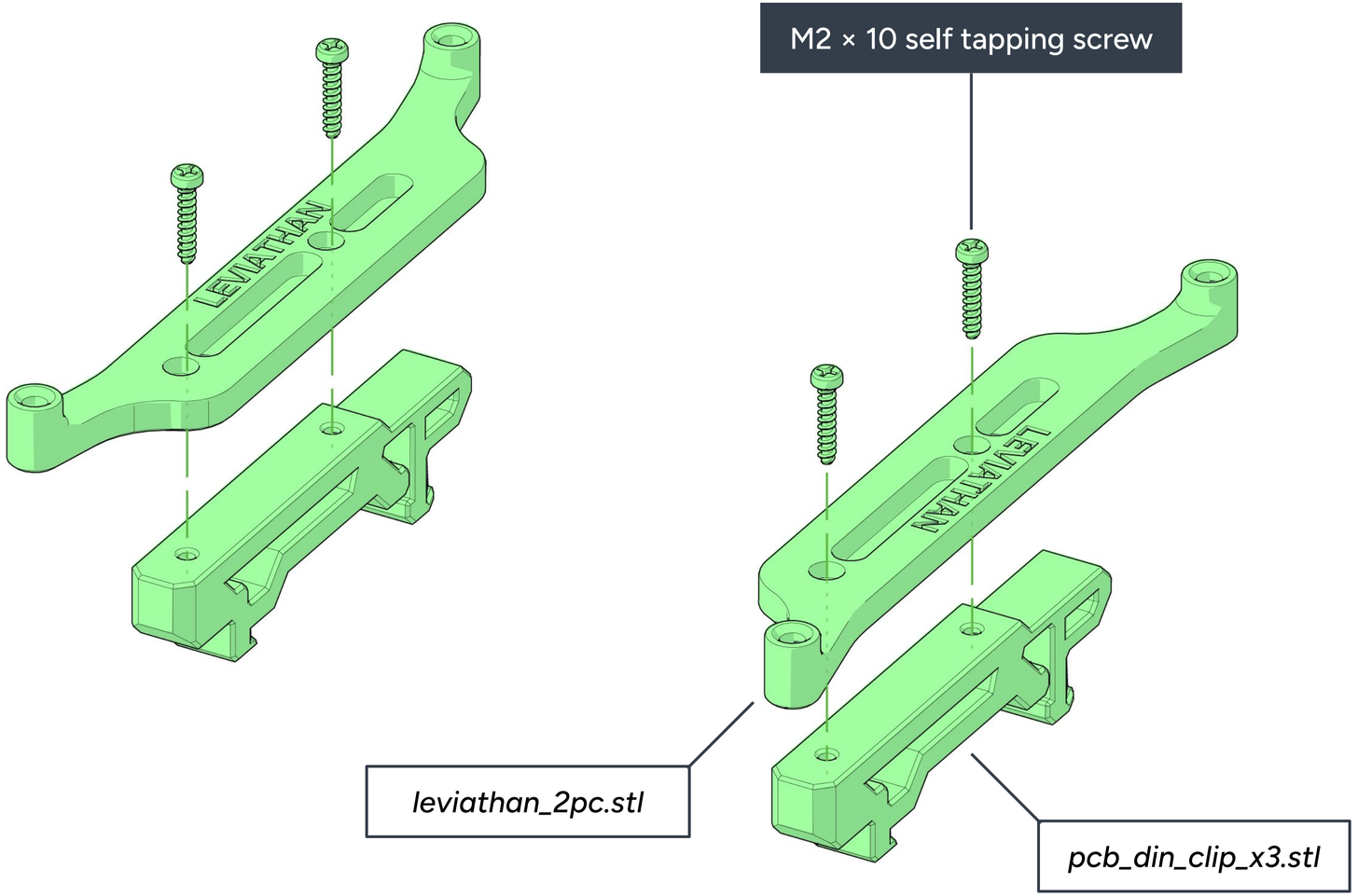
Included hardware will vary by kit.

K/ ELECTRONICS

CONTROLLER BOARD PREP

Secure ***leviathan_2pc.stl*** to **two *pcb_din_clip_x3.stl*** with **four M2 × 10 self tapping screws**.

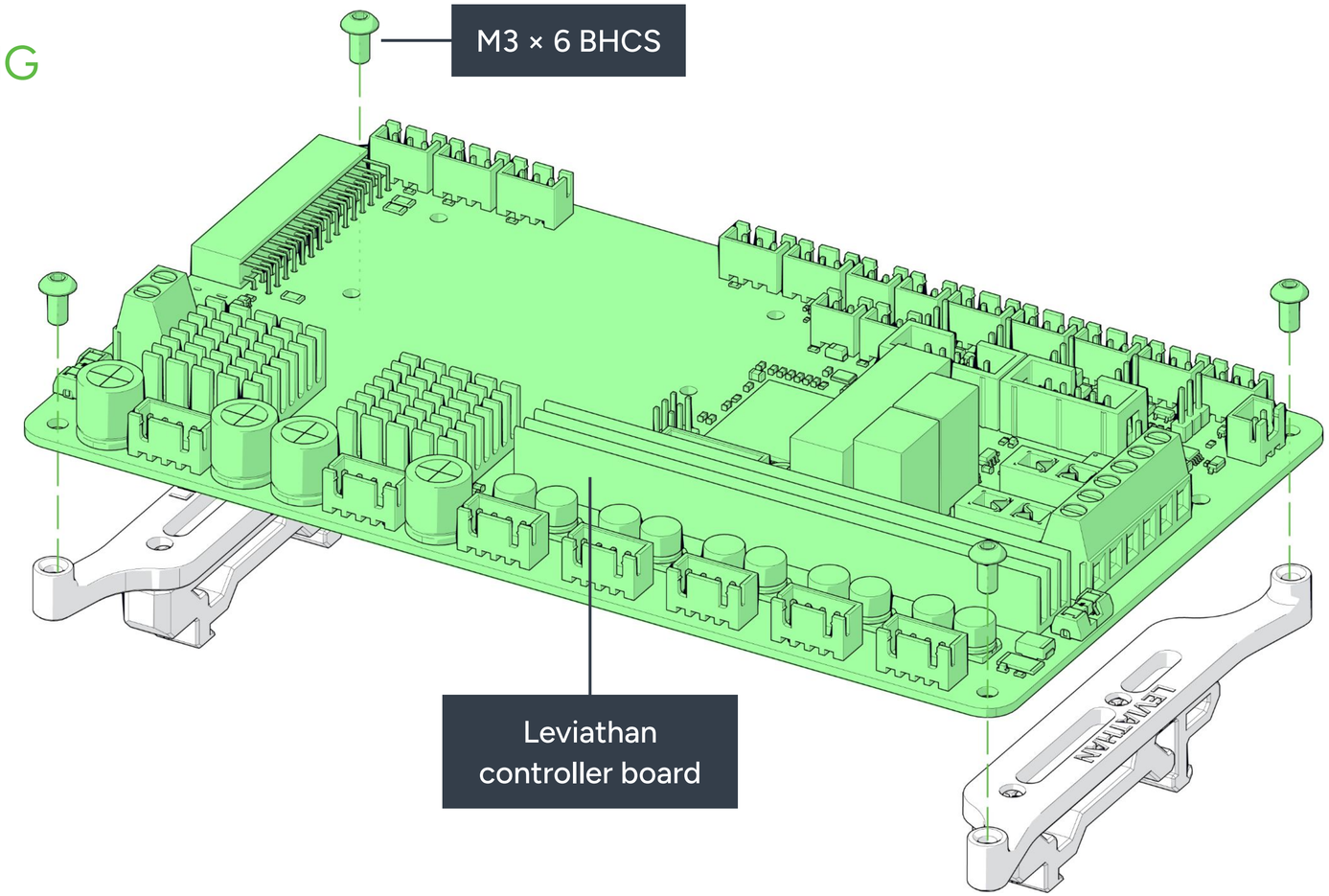
Assemble one pair, then mirror the second so the clips face opposite directions.



CONTROLLER BOARD MOUNTING

Attach the two DIN clip assemblies to the mounting holes of the **controller board** with **four M3 × 6 BHCS**.

The DIN clips are offset. Ensure both clips are oriented correctly so the board will align with the DIN rail during installation.

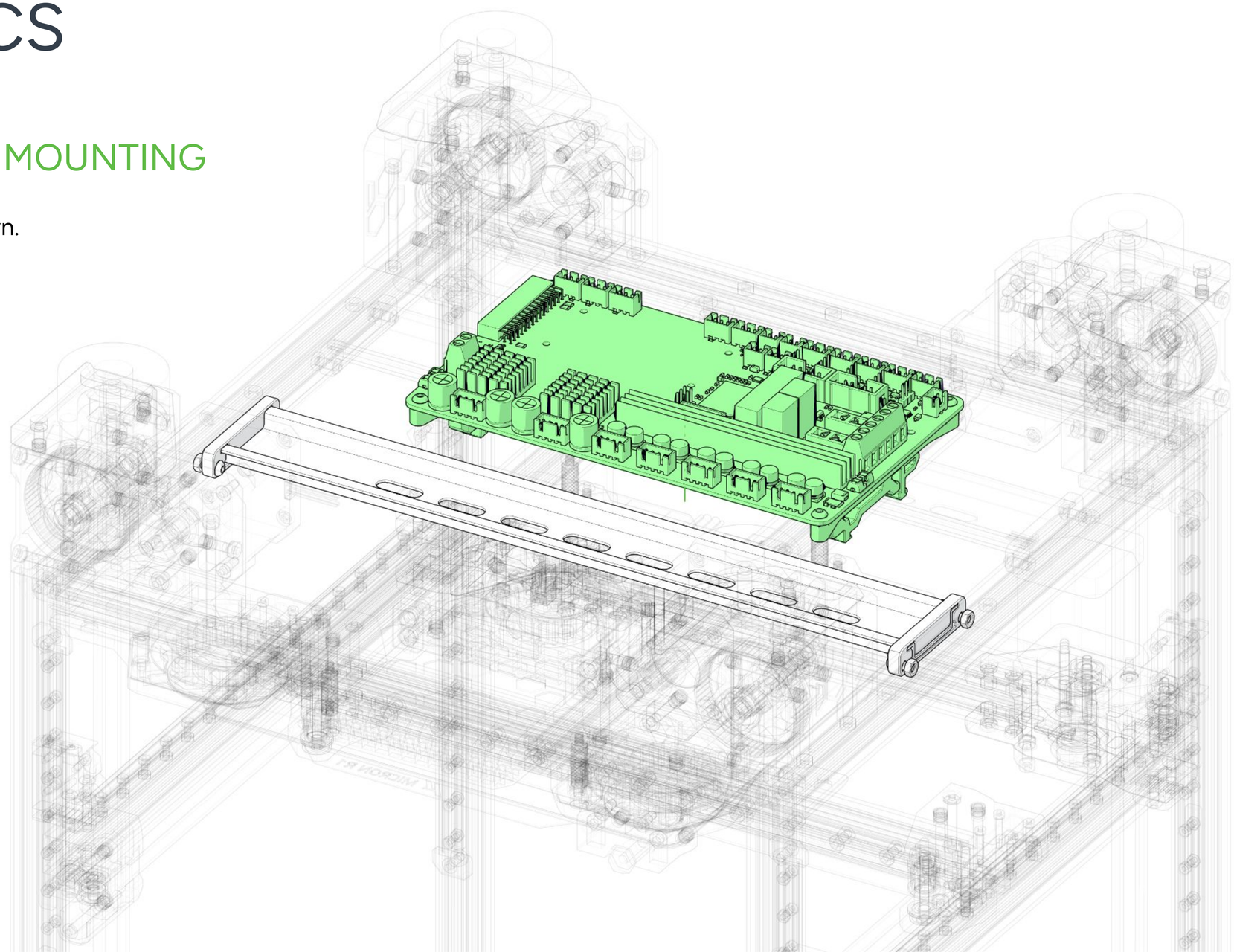


K/ ELECTRONICS

CONTROLLER BOARD MOUNTING

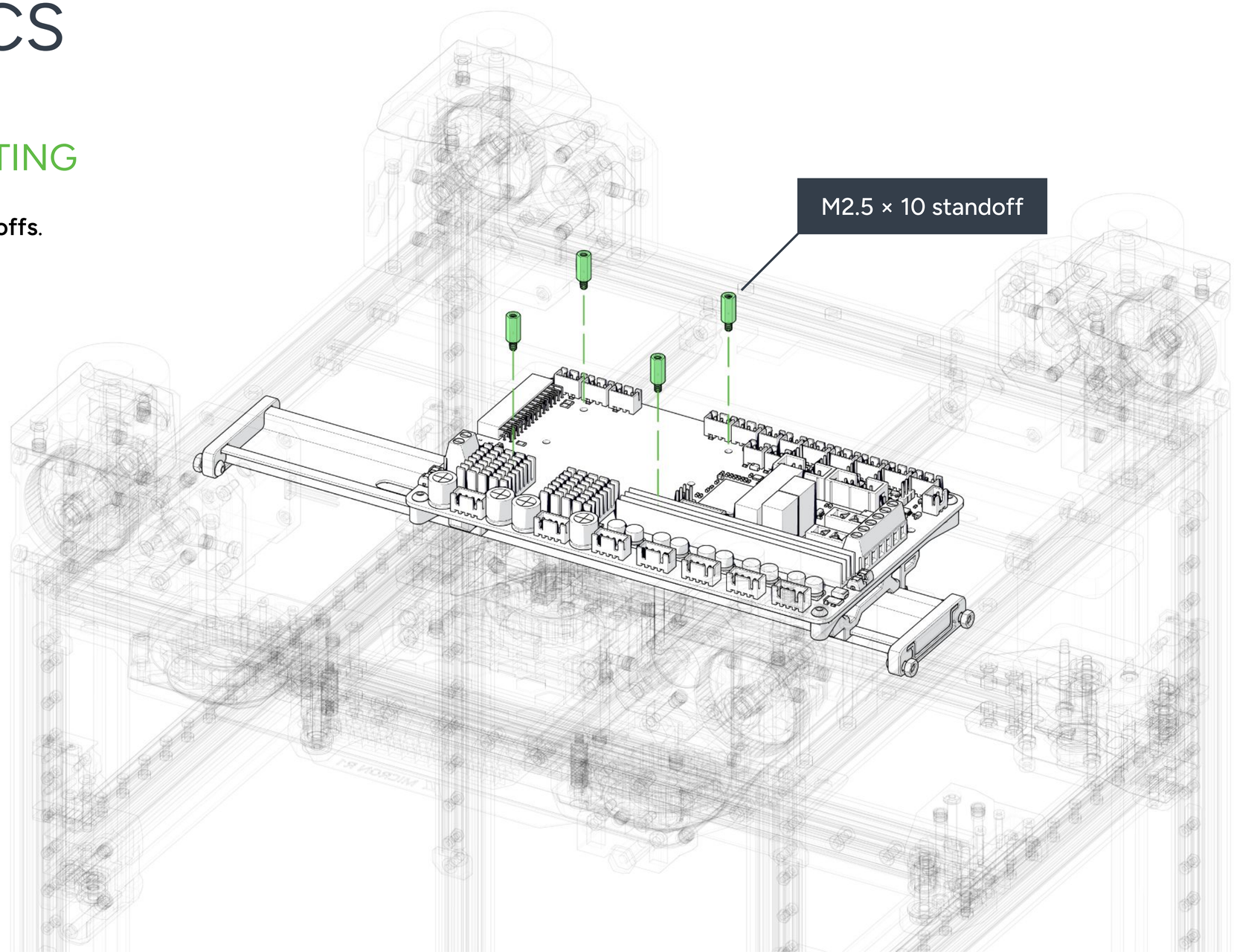
Carefully flip the printer upside down.

Clip the controller board onto the front DIN rail, ensuring both clips fully engage.



RASPBERRY PI MOUNTING

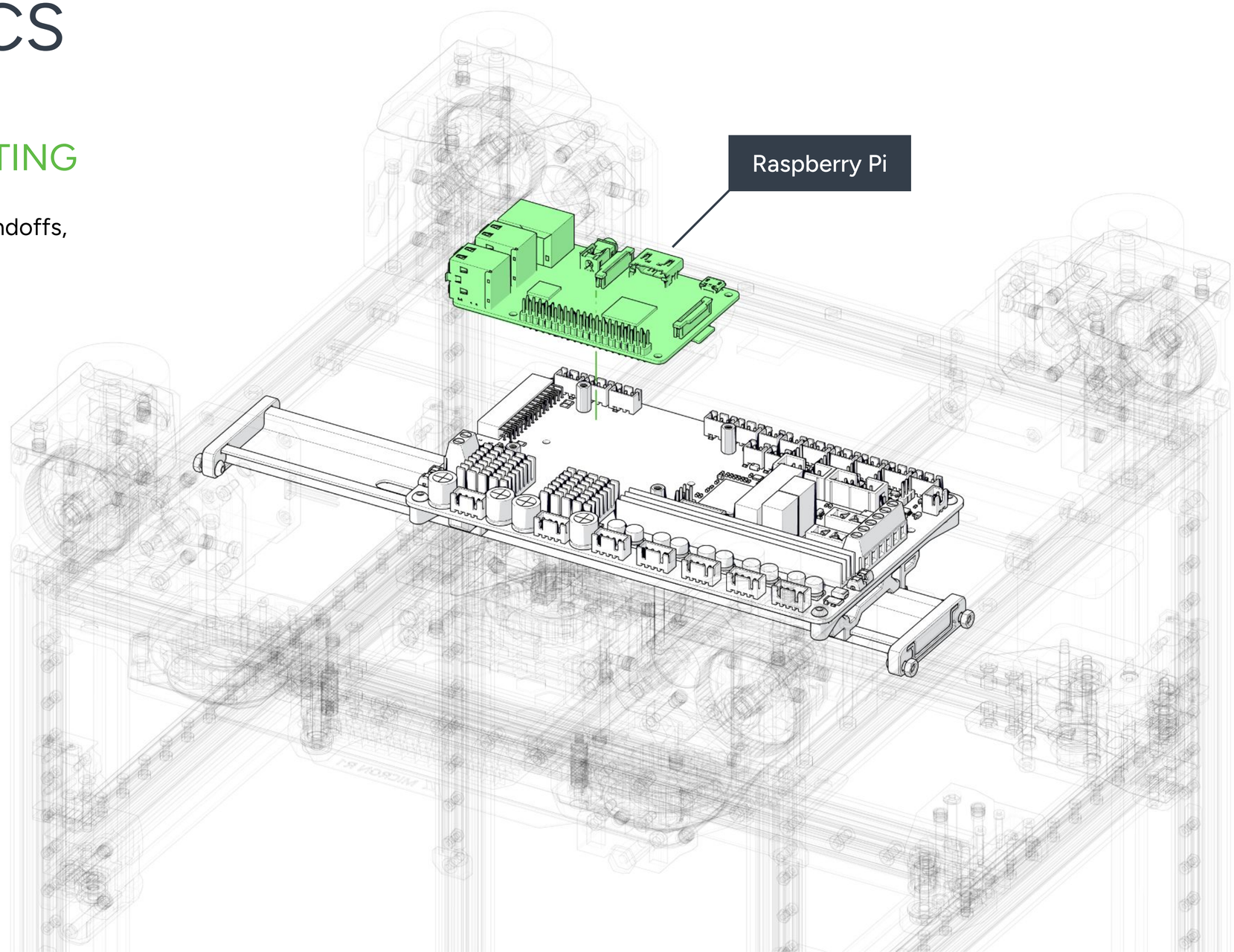
Install the **four M2.5 x 10 mm standoffs**.



K/ ELECTRONICS

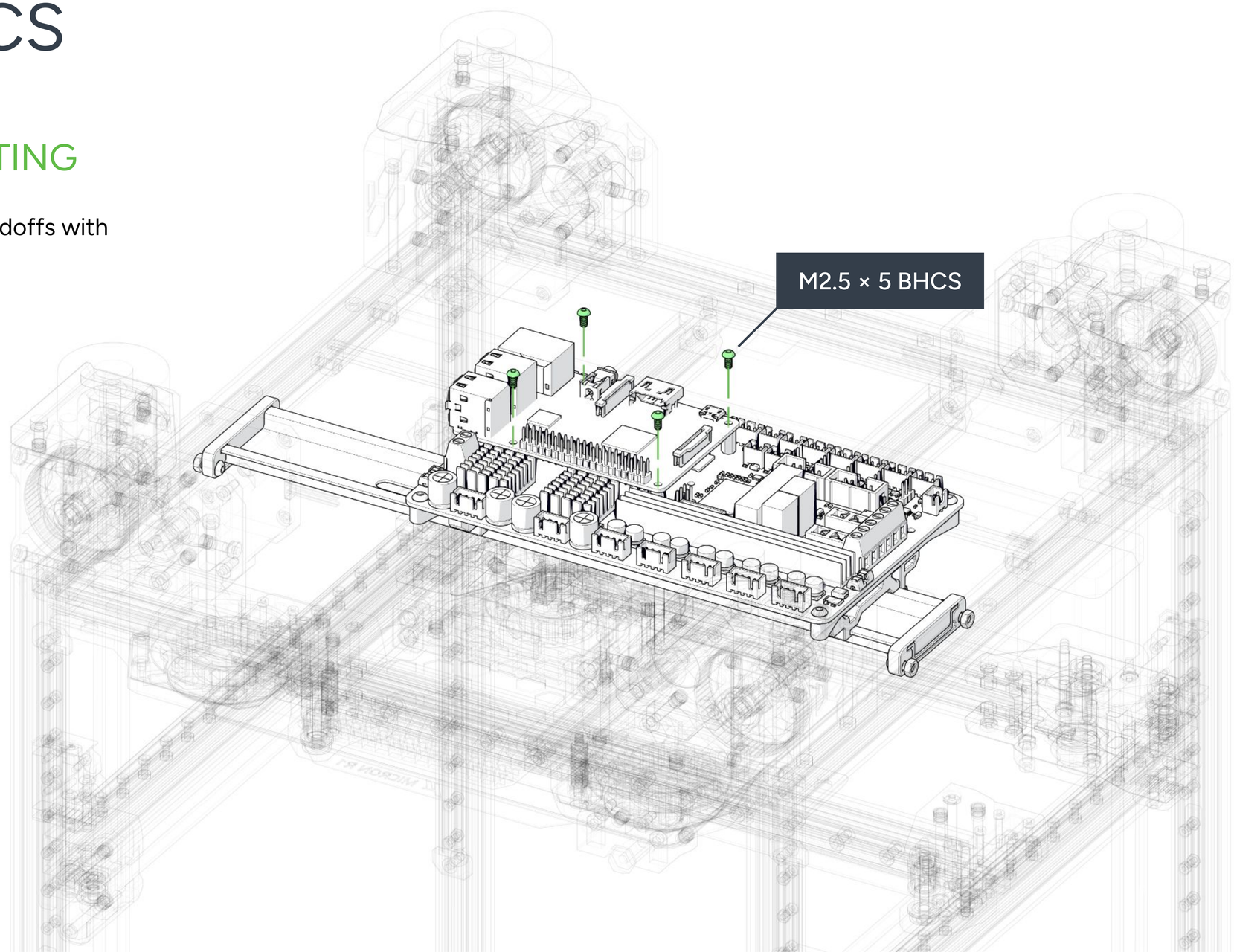
RASPBERRY PI MOUNTING

Place the **Raspberry Pi** over the standoffs, aligning the mounting holes.



RASPBERRY PI MOUNTING

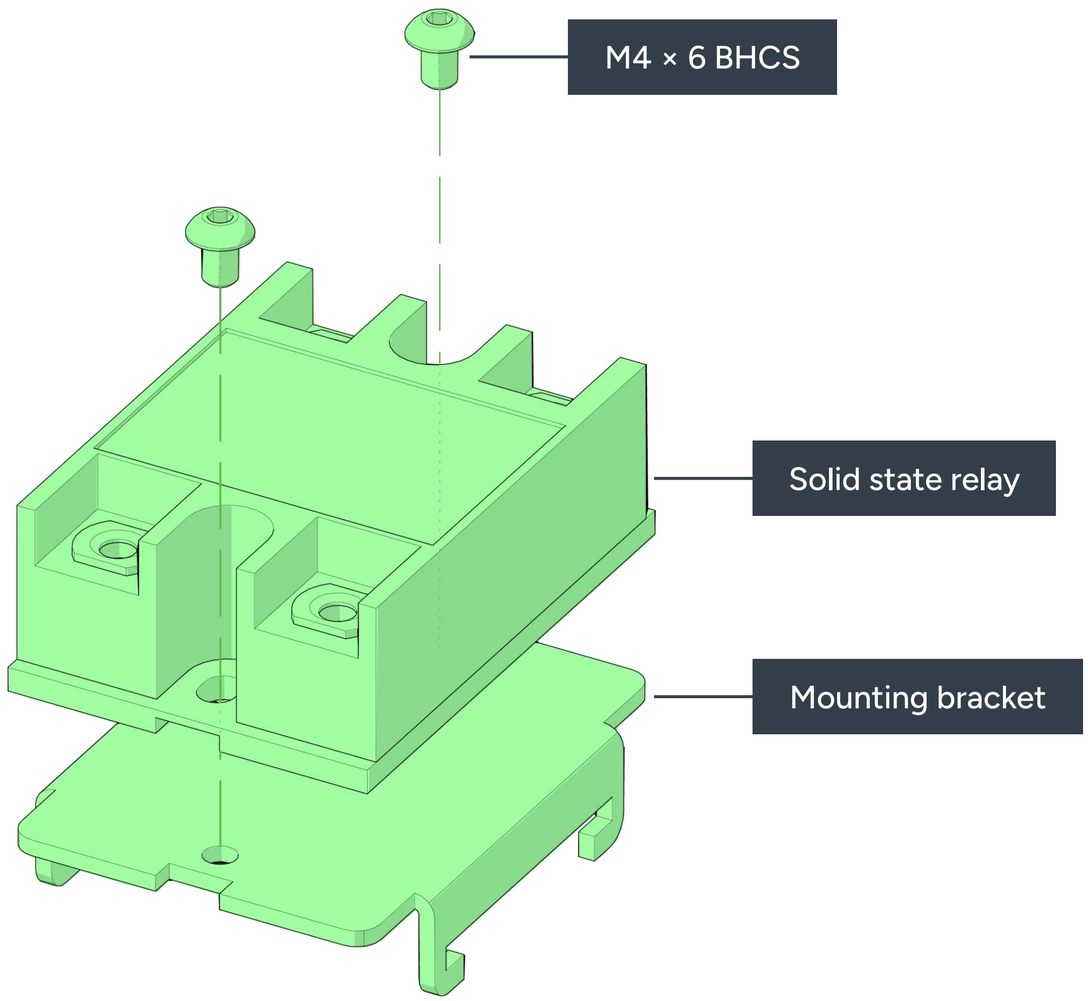
Secure the **Raspberry Pi** to the standoffs with **M2.5 × 5 BHCS**.



K/ ELECTRONICS

SSR PREP

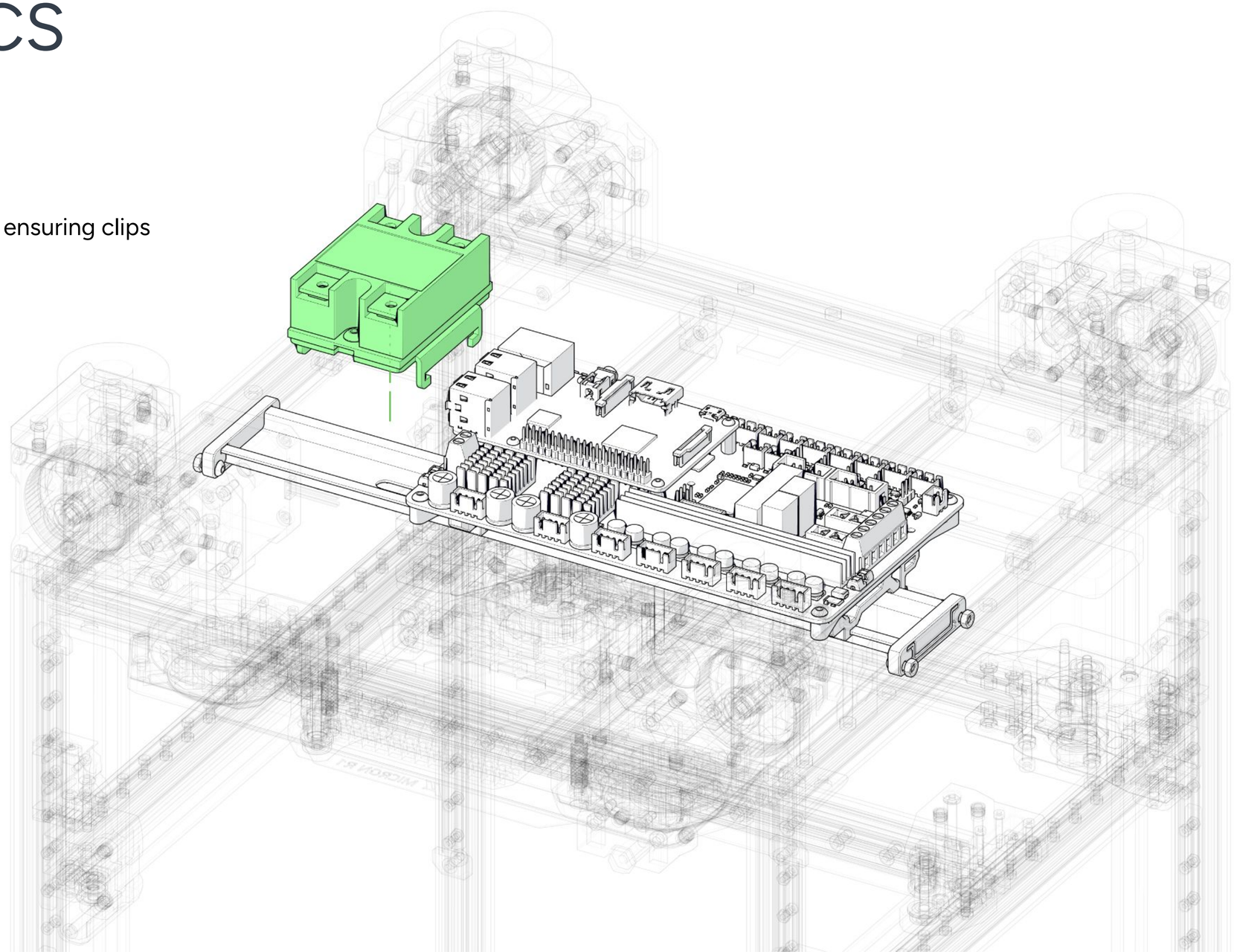
Line up the **solid state relay** with the **mounting bracket** and secure with **two M4 × 6 BHCS**.



K/ ELECTRONICS

SSR MOUNTING

Clip the SSR onto the front DIN rail, ensuring clips fully engage.

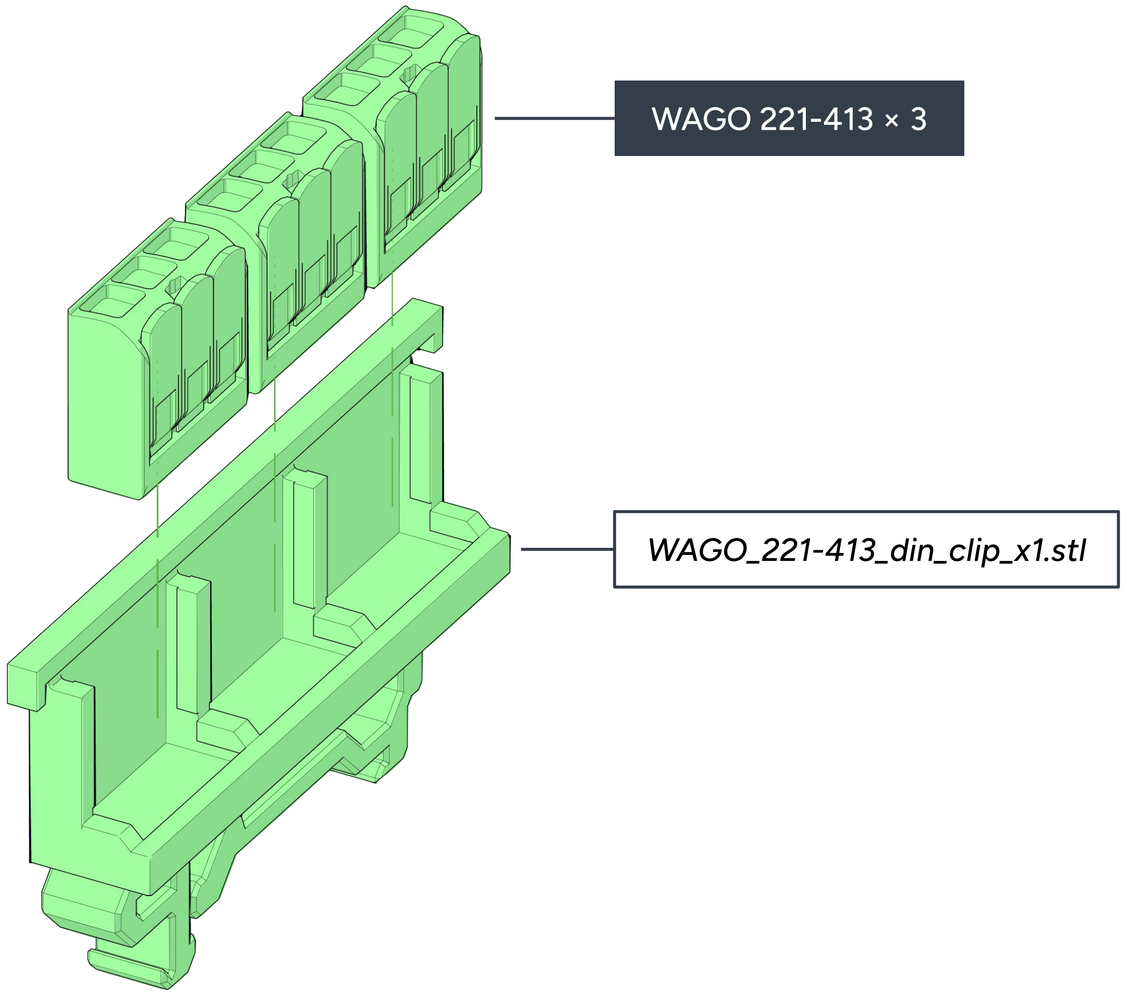


K/ ELECTRONICS

WAGO 221-413 (OPTION 1) PREP

The WAGO 221-413 block can be installed in one of two locations. This option mounts the block to a DIN rail.

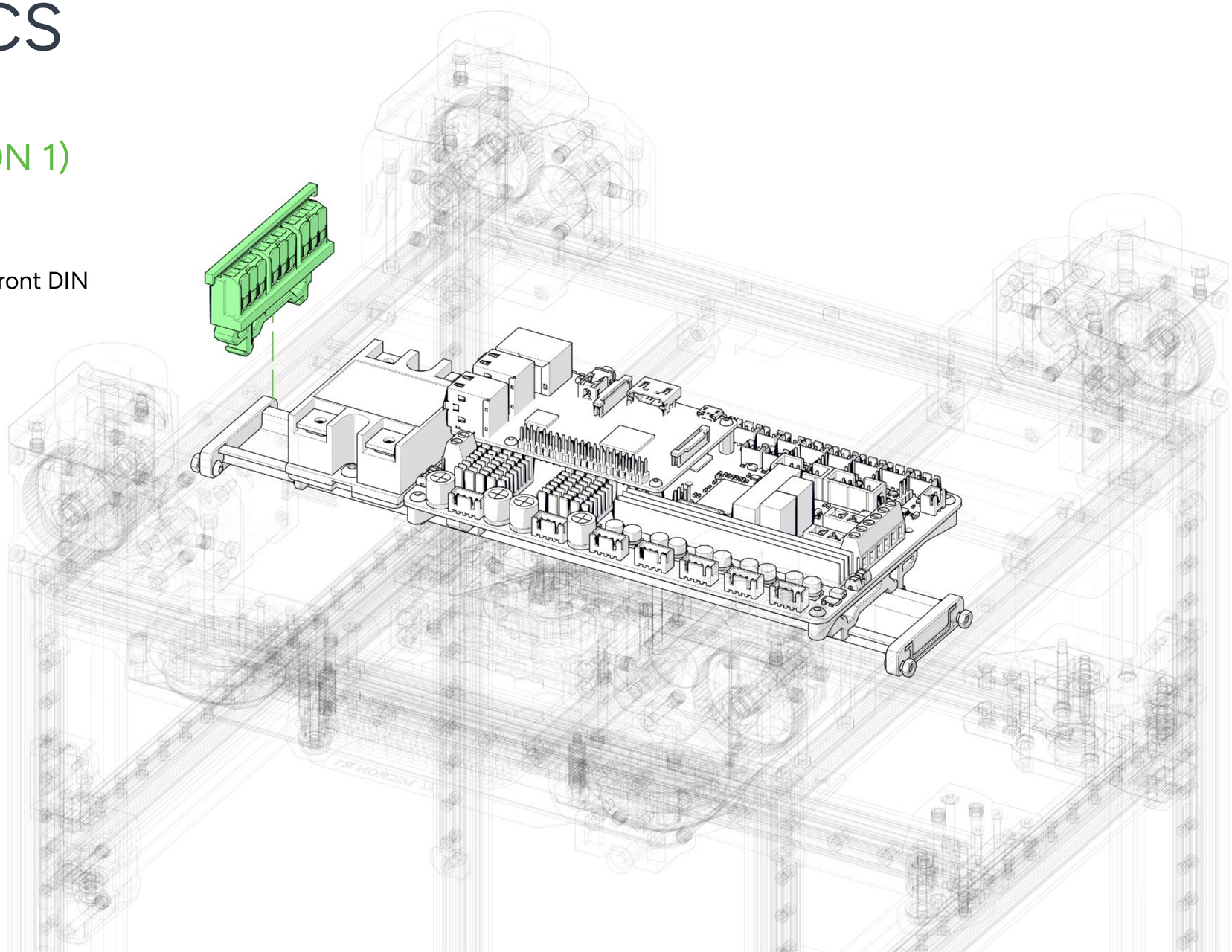
Insert **three WAGO 221-413 connectors** into ***WAGO_221-413_din_clip_x1.stl***. Ensure they click firmly into place with levers facing outwards and wire openings on top.



K/ ELECTRONICS

WAGO 221-413 (OPTION 1) MOUNTING

Clip the WAGO assembly onto the front DIN rail, ensuring the clip fully engages.

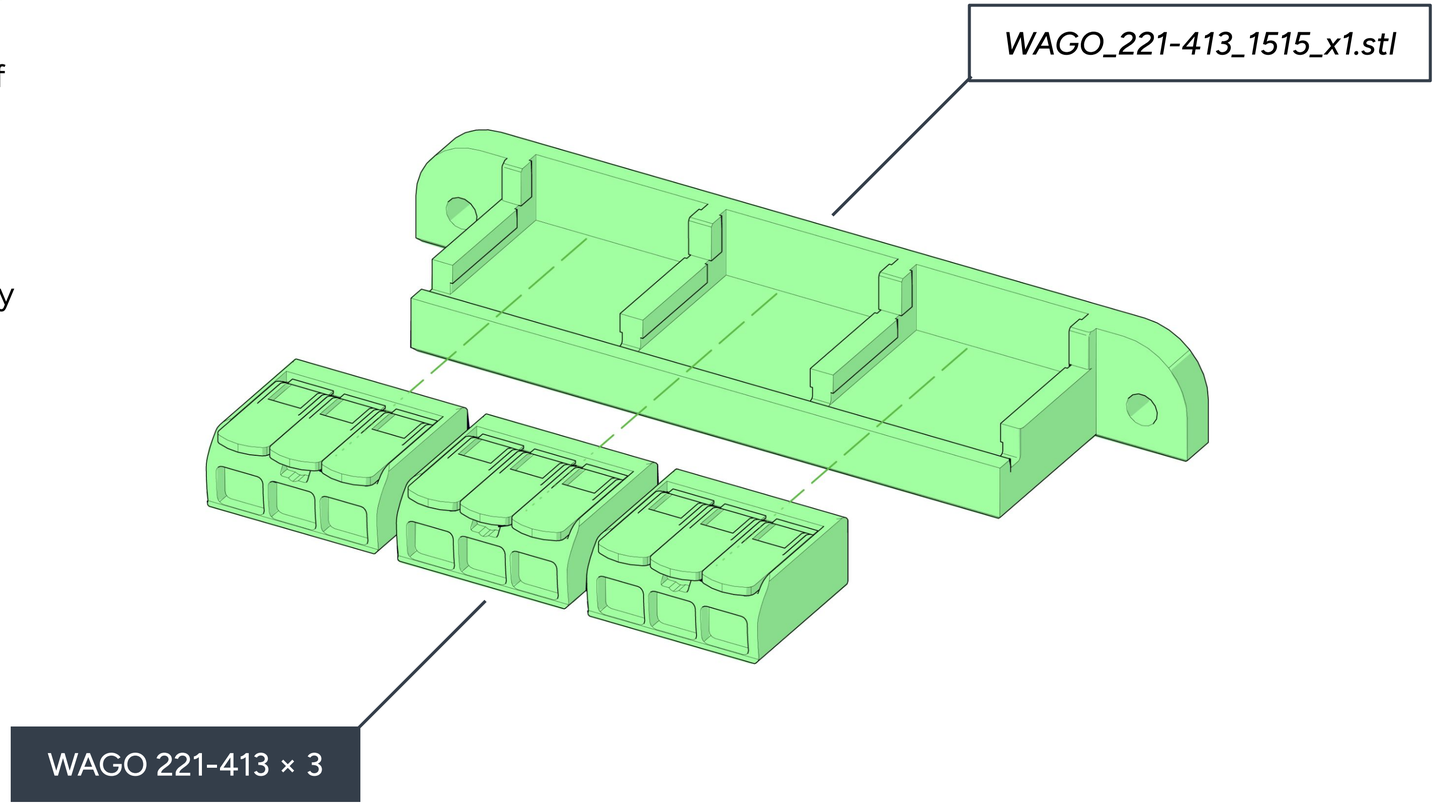


K/ ELECTRONICS

WAGO 221-413 (OPTION 2) PREP

The WAGO 221-413 block can be installed in one of two locations. This option mounts the block to the rear 1515 extrusion.

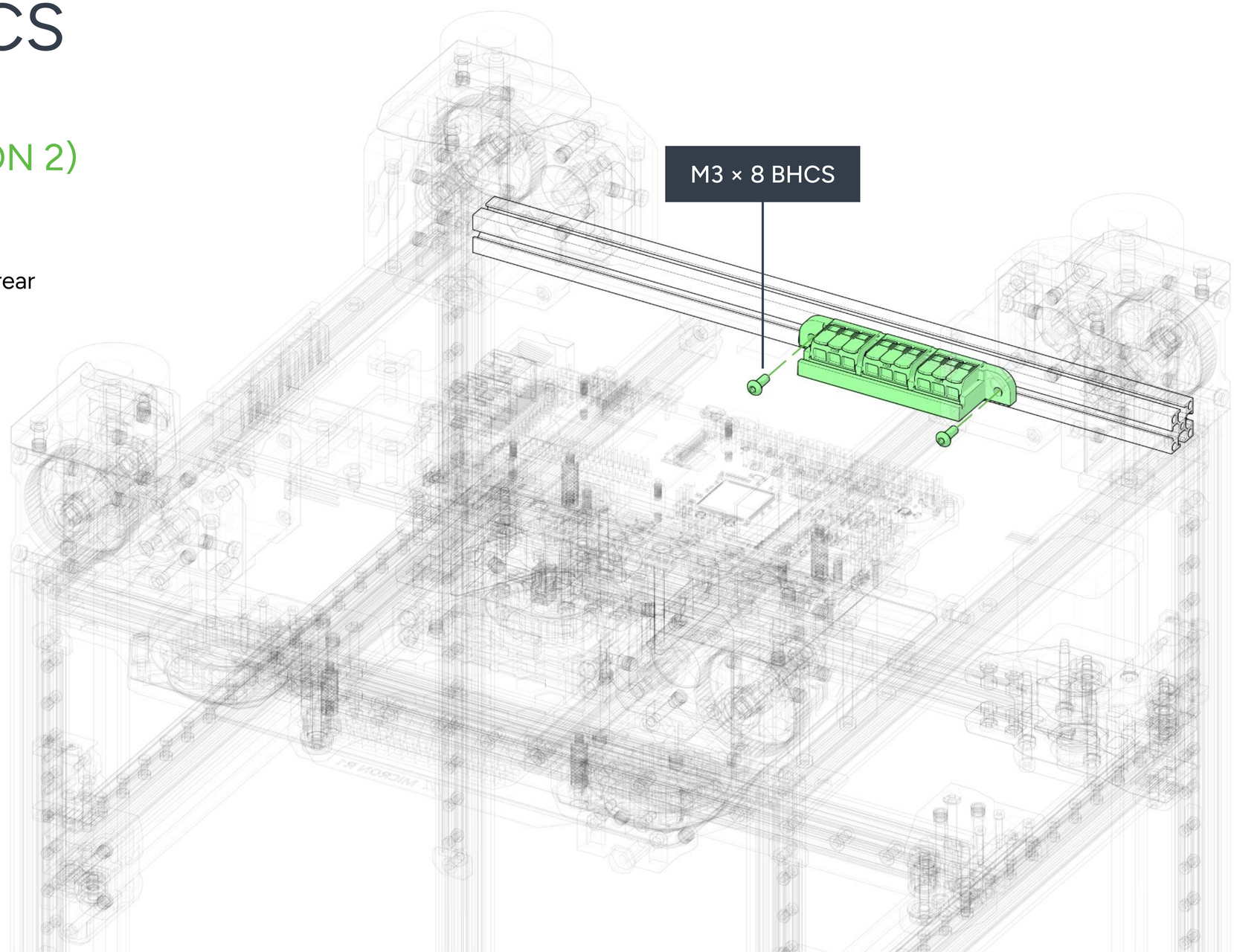
Insert **three WAGO 221-413 connectors** into ***WAGO_221-413_1515_x1.stl***. Ensure they click firmly into place with levers facing outwards and wire openings on top.



K/ ELECTRONICS

WAGO 221-413 (OPTION 2) MOUNTING

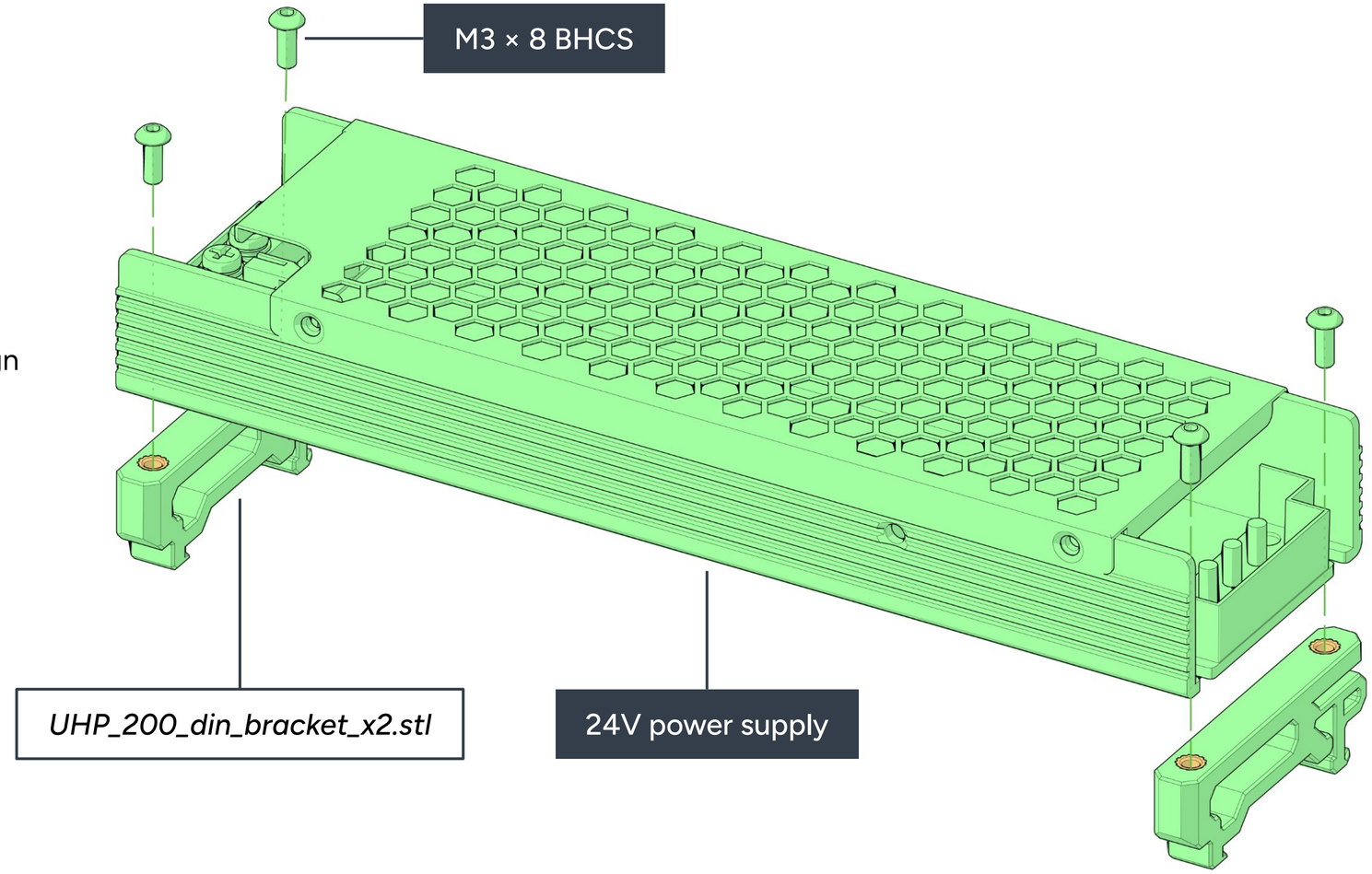
Attach the WAGO assembly to the rear extrusion with **two M3 × 8 BHCS**.



POWER SUPPLY PREP

Attach the two DIN clip assemblies to the mounting holes of the **24V power supply** with **four M3 × 8 BHCS**.

The DIN clips are offset. Ensure both clips are oriented correctly so the power supply will align with the DIN rail during installation.



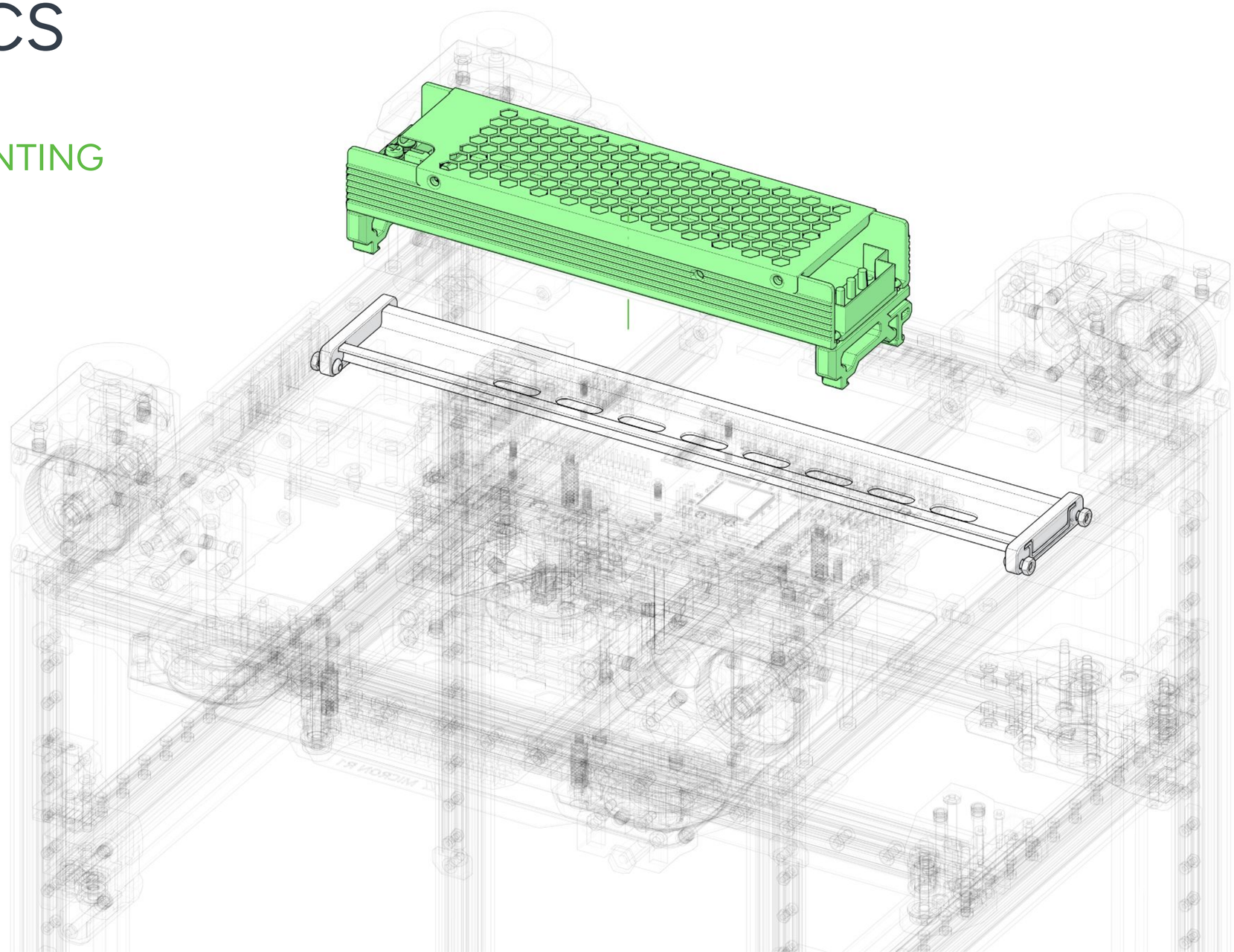
NOTE

Before mounting, verify the power supply voltage switch is set correctly for your mains voltage.

K/ ELECTRONICS

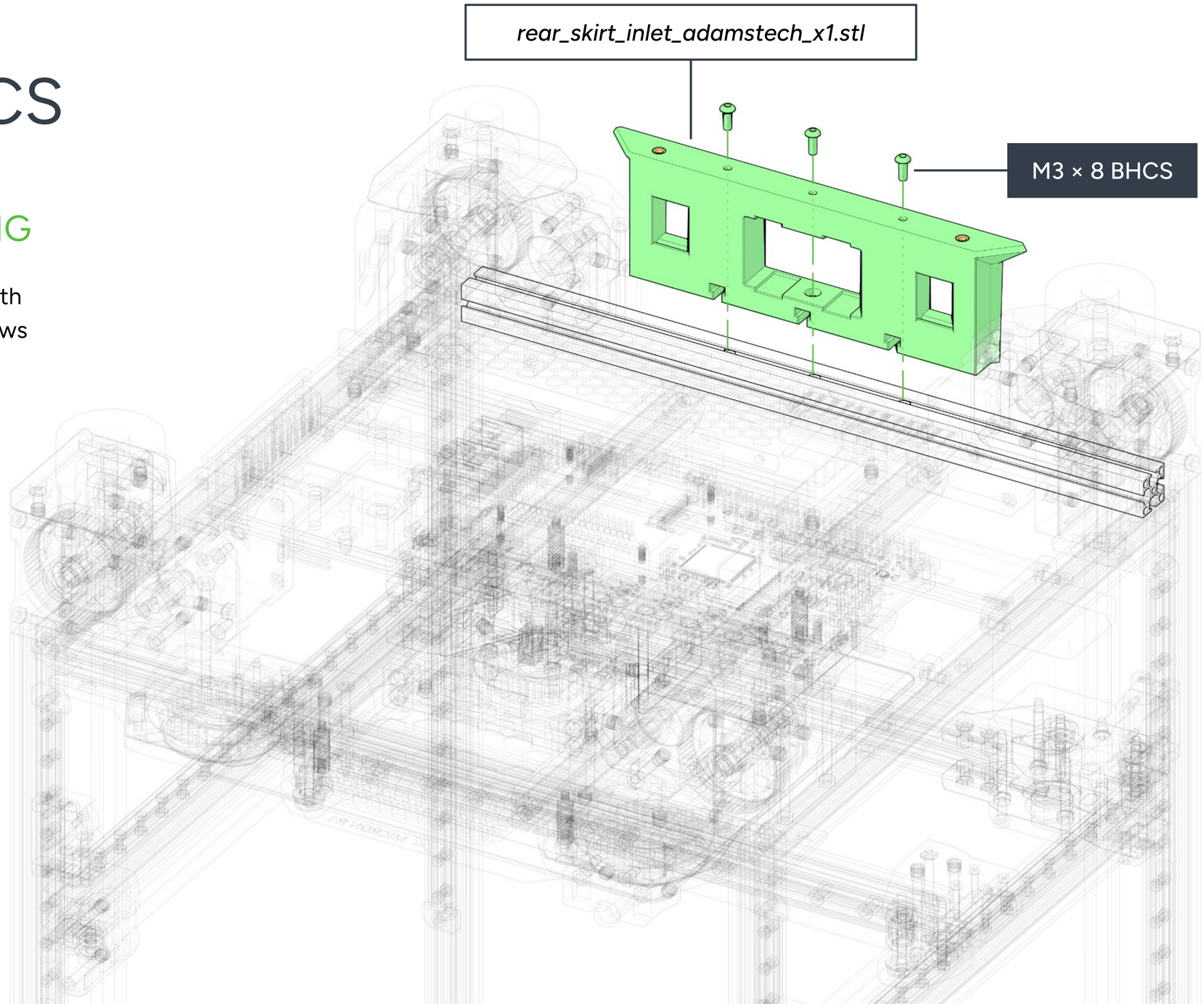
POWER SUPPLY MOUNTING

Clip the 24V power supply onto the rear DIN rail, ensuring both clips fully engage.



REAR SKIRT MOUNTING

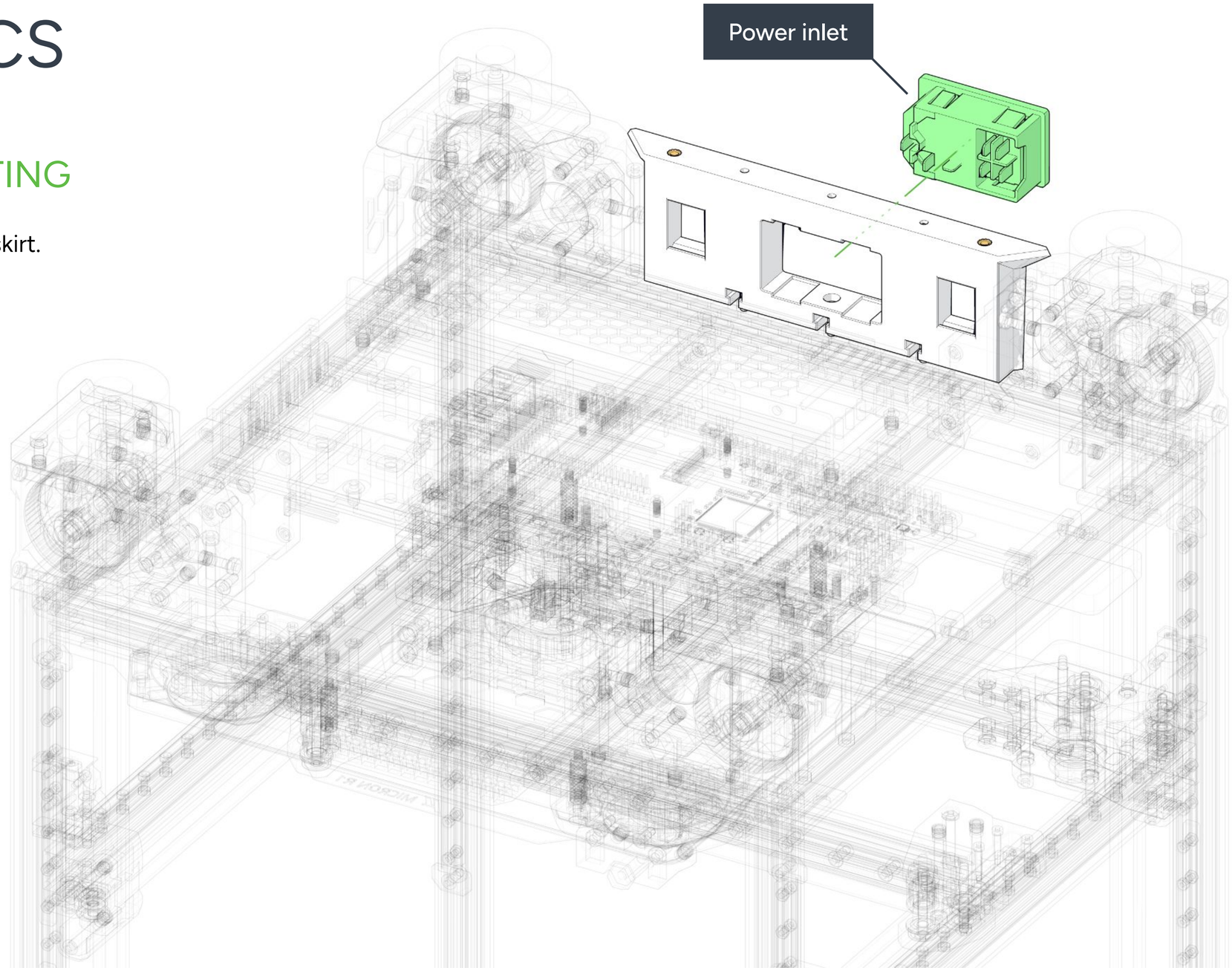
Mount the **rear skirt** to the frame with **three M3 × 8 BHCS**, sliding the screws into the rear skirt from the side and threading them into the three preloaded M3 hex nuts.



K/ ELECTRONICS

POWER INLET MOUNTING

Insert the **power inlet** into the rear skirt.
Ensure it fully clicks into place.



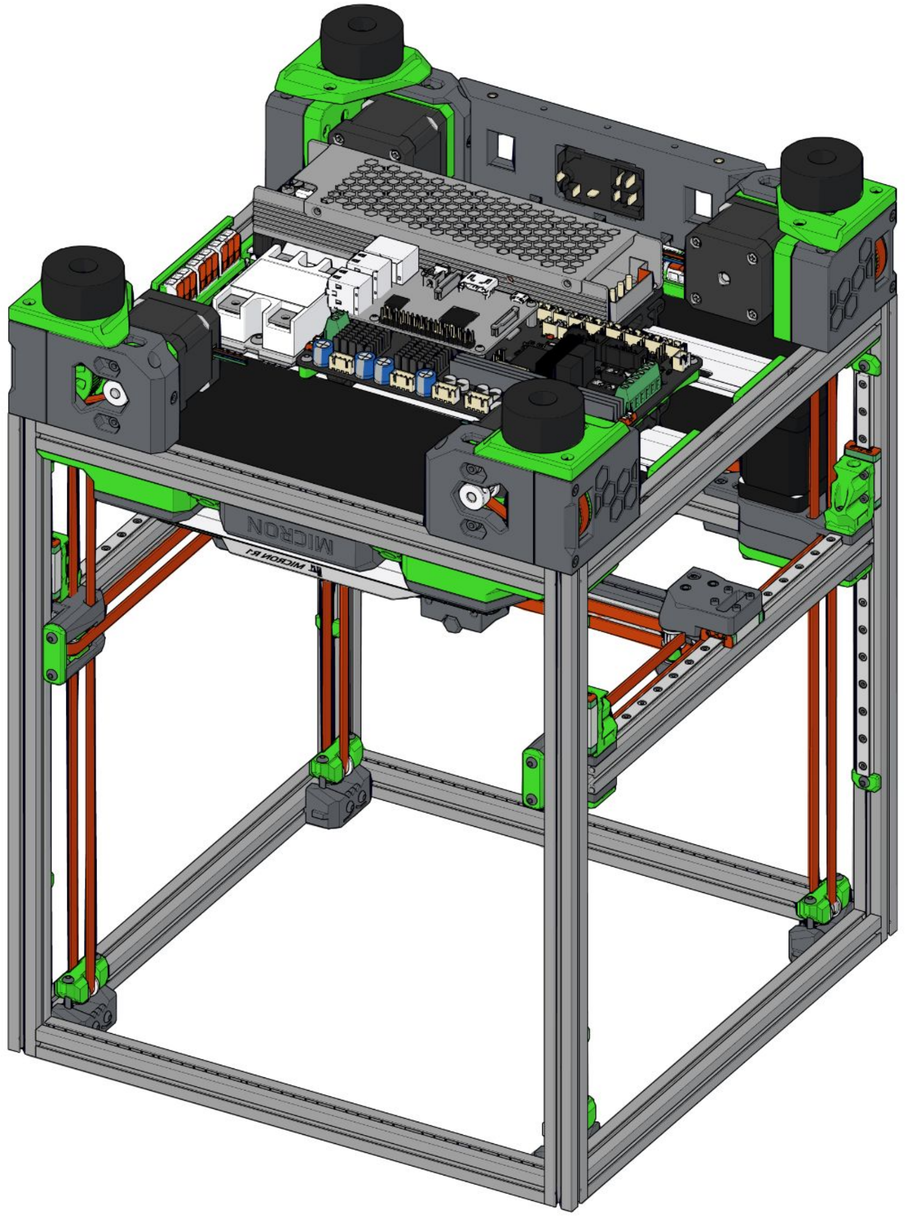
K/ ELECTRONICS

BUILDER CHECK

Your assembly should match this view.

CHECKLIST

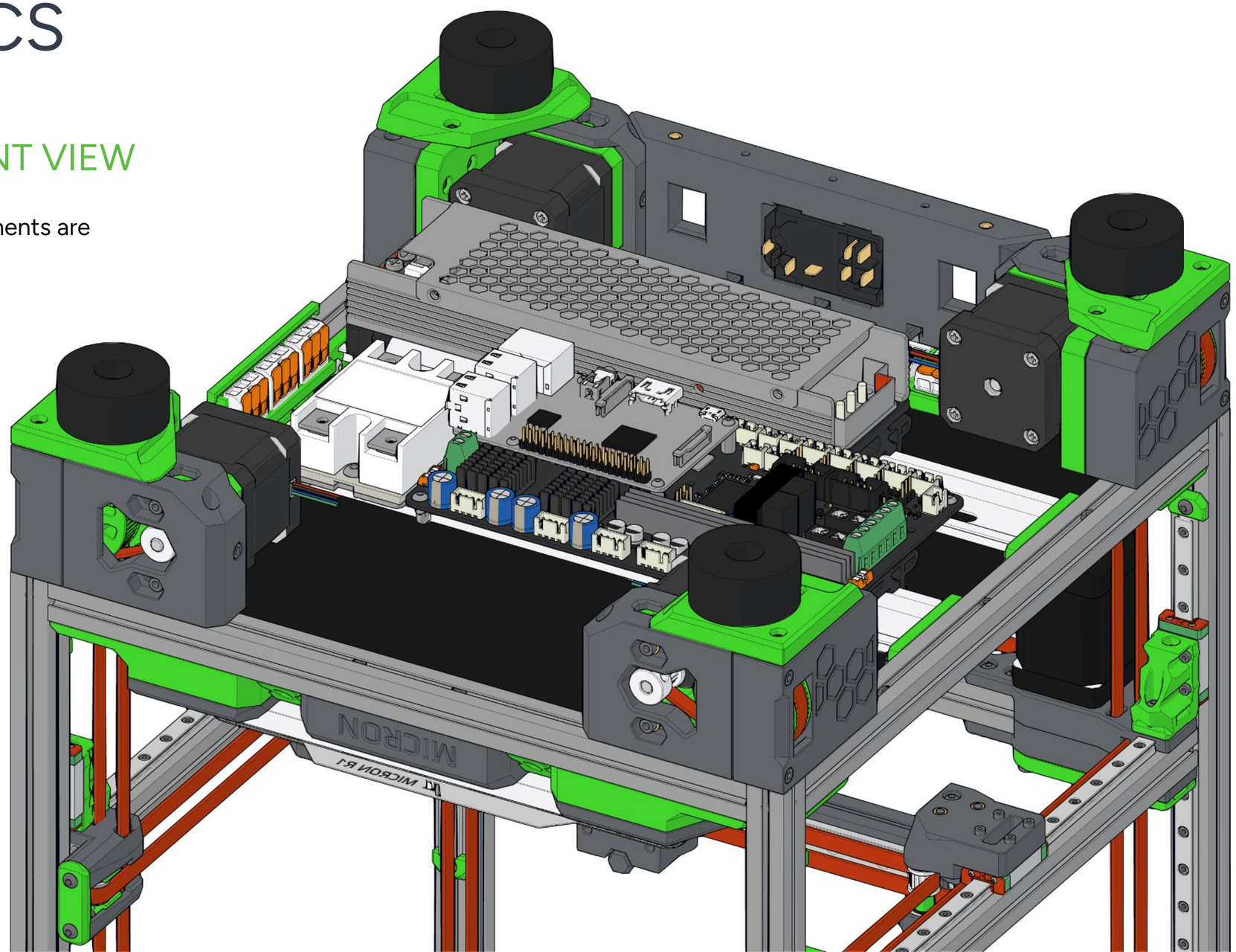
- ▢ Controller board, power supply, SSR, and WAGO assembly are fully clipped onto the DIN rails
- ▢ WAGO connectors are properly inserted and oriented
- ▢ Rear skirt is secured to the frame
- ▢ Power inlet is fully clicked into the rear skirt
- ▢ **Power supply voltage switch is set correctly for your mains voltage**



K / ELECTRONICS

DETAILED COMPONENT VIEW

Before wiring, verify that all components are mounted as shown.



L / WIRING

Micron+ wiring is based on the LDO electronics kit and follows a compact Voron-style CoreXY architecture. This section covers the stock wiring path, controller connections, power wiring, and pre-power checks before firmware setup.

SAFETY

This section includes both low-voltage DC wiring and line-voltage AC wiring. Always work with the printer unplugged, verify the power supply is discharged, and do not power on the printer until all wiring checks are complete.

Route wires cleanly before final termination. Keep wire paths organized, avoid sharp bends, and leave enough service loop for moving assemblies. Confirm connector orientation and polarity before making connections.

Wiring is divided into smaller sections with checks along the way. Do not continue if a connector location, polarity, or wire path is unclear.

POWER PATH

AC power enters through the rear inlet, passes through the power supply, and is converted to 24V DC for the controller, motors, heaters, fans, and accessories.

CONTROLLER

The controller board manages motion, heaters, fans, sensors, and communication with the Raspberry Pi. Connections should match the wiring diagrams before power is applied.

TOOLHEAD

The toolhead wiring carries heater, thermistor, fan, probe, and extruder motor signals. Route these wires so they do not bind through the full XY travel range.

CONCEPT

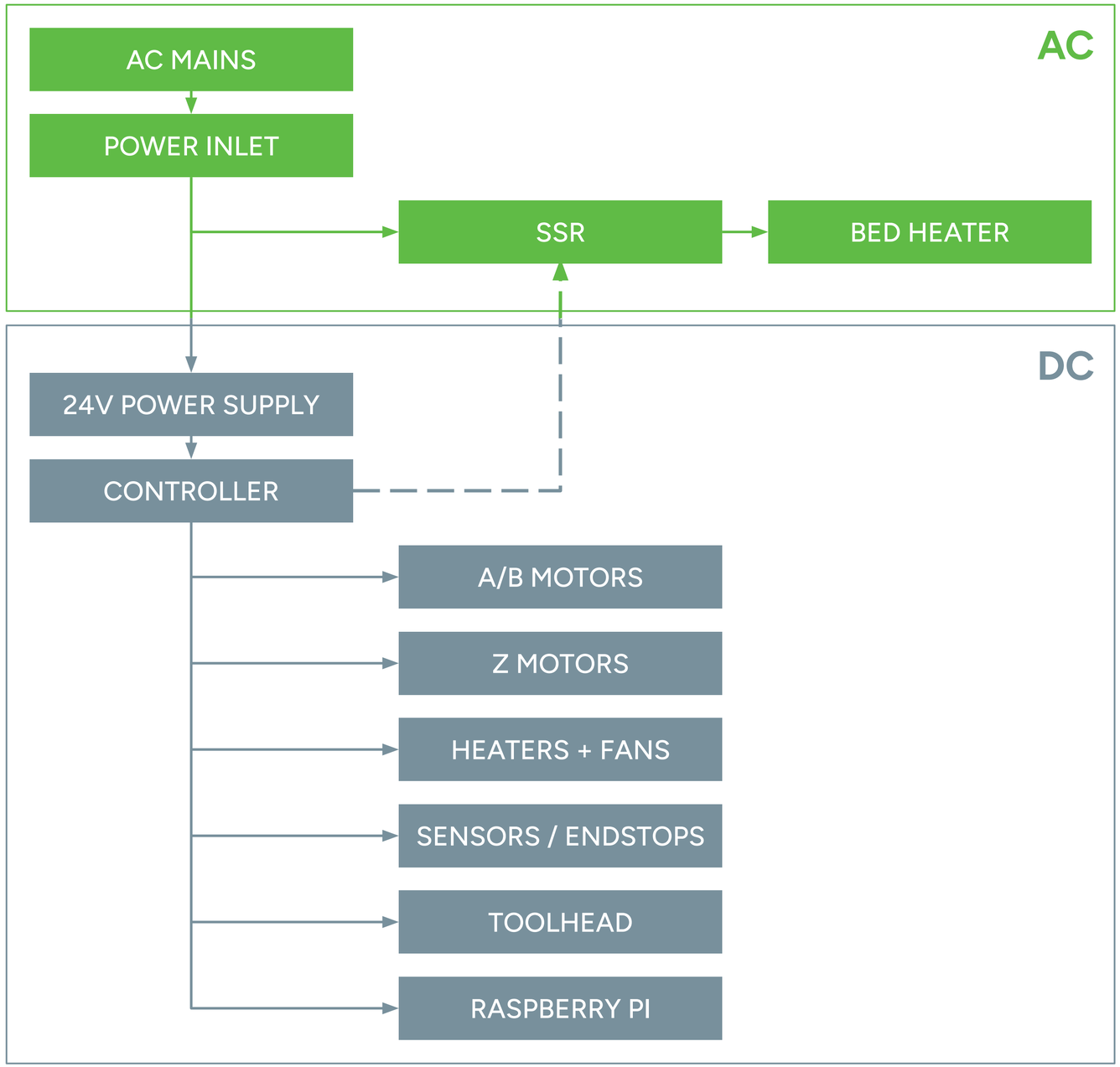
WIRING ARCHITECTURE

Before making connections, understand how power and control are distributed through the printer.

AC mains enters through the rear inlet and is distributed to the power supply and line-voltage devices. The power supply provides 24V DC for the controller and low-voltage systems. The controller distributes power and control signals to the printer's motors, heaters, fans, sensors, and toolhead.

NOTE

Follow the wiring diagrams. Do not use this overview as a substitute for terminal-level wiring instructions.



L / WIRING

MAINS INLET PREP

Text